Recapture Analsyis: Final

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Recapture Analysis: Combined

In this markdown, we'll combine the results from COLONY and PLINK2 to identify recaptures across the samples processed by RAPiD Genomics / LGC and GTSeek. We'll also estimate naive genotyping error rates, assess kinship coefficients between technical replicates, evaluate concordance in sex inferences from genotypic data, assign clone IDs for all samples, and assign final sexes for all unique genotypes. The outputs from this step will be used subsequently to create a capture history object for SCR modeling and to subsample to one sample per unique genotype for population genetic analyses.

The strategy used to combine inferences from COLONY and PLINK2 is as follows:

1. Start with COLONY clones as recaptures - most (if not all) of these are also recovered by PLINK
2. For each unique COLONY clone, ask if PLINK provides compelling evidence to move the clone to a larger group
    - Mean kinship across pairwise comparisons of clone A and clone B needs to be > 0.45 (based on the range of kinship coefficients seen in technical replicates)
    - At least one pairwise comparison between the clones needs to be based on a substantial number of shared markers (> 50)

Load COLONY and PLINK2 Results

First, load the files with results from COLONY and PLINK2.

```
#Read the metadata
smd <- read.csv("../sampleinfo/sampleMetadata_sexGT.csv",
                           header = TRUE)

#Read the COLONY output
col.out <- read.table("CGA.BestClone", header = TRUE)

#Read the PLINK output
kingrel <- read.table("recapPLINK.kin0", header = FALSE)
colnames(kingrel) <- c("id1","id2","nsnp","hehet","ibs0","kinship")

nrow(kingrel) #Number of rows (pairwise comparisons) in PLINK output...
```

```
## [1] 1485226
```

```
choose(sum(!smd$FILT),2) # should match choose(# retained samples, 2)
```

```
## [1] 1485226
```

Evaluate Technical Replicates - Error Rate, Kinship

First pull the technical replicate samples out of the larger data file. Start by extracting genotypes (in 012 format from the combined file that we used for the recapture analysis.

```
ulimit -n 3000
vcftools --vcf ../snpfiltering/finalSCR.recode.vcf --012 --out finalSCR
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf ../snpfiltering/finalSCR.recode.vcf
    --012
    --out finalSCR

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
```

```
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
After filtering, kept 1724 out of 1724 Individuals
Writing 012 matrix files ... Done.
After filtering, kept 1325 out of a possible 1325 Sites
Run Time = 1.00 seconds
```

Now read this into R and subset to the technical replicate samples

```r
reps <- smd[which(smd$is.rep == TRUE),]
length(unique(sub("R$","",reps$ugaID)))
```

```
## [1] 60
```

```r
nrow(reps) #60 replicated samples = 120 rows
```

```
## [1] 120
```

```r
genos <- read.table("finalSCR.012", header = FALSE)
genos.ind <- read.table("finalSCR.012.indv", header = FALSE)
genos[,1] <- genos.ind[,1]

repl.out <- data.frame(ugaID = unique(sub("R$","",reps$ugaID)),
                       vcfID1=NA, vcfID2=NA,
                       nSNPgt1=NA, nSNPgt2=NA,
                       filt1=NA, filt2=NA,
                       colclone1=NA, colclone2=NA,
                       is.clonePLINK=NA, is.cloneCOLONY=NA,
                       nSNPshared=NA, nSNPmatch=NA,
                       kinship=NA, GTservice=NA)
for(i in 1:nrow(repl.out)) {
  repl.out$GTservice[i] <- reps$GTservice[which(reps$ugaID == repl.out$ugaID[i])][1]

  #Replicates use the same UGA ID in LGC data...
  if(repl.out$GTservice[i] == "LGC") {
    #Record vcfIDs
    repl.out$vcfID1[i] <- reps$vcfID[which(reps$ugaID==repl.out$ugaID[i])][1]
    repl.out$vcfID2[i] <- reps$vcfID[which(reps$ugaID==repl.out$ugaID[i])][2]
  #But append an "R" in GTSeek data
  } else if(repl.out$GTservice[i] == "GTSeek") {
    repl.out$vcfID1[i] <- reps$vcfID[which(reps$ugaID==repl.out$ugaID[i])]
    repl.out$vcfID2[i] <- reps$vcfID[which(reps$ugaID==paste0(repl.out$ugaID[i],"R"))]
  }


  #Record number of SNPs genotyped
  repl.out$nSNPgt1[i] <- smd$nSNPgt[which(smd$vcfID == repl.out$vcfID1[i])]
  repl.out$nSNPgt2[i] <- smd$nSNPgt[which(smd$vcfID == repl.out$vcfID2[i])]
```

```r
#Record filtered status - if TRUE, no need to look for matches
repl.out$filt1[i] <- smd$FILT[which(smd$vcfID == repl.out$vcfID1[i])]
repl.out$filt2[i] <- smd$FILT[which(smd$vcfID == repl.out$vcfID2[i])]

if(repl.out$GTservice[i] == "LGC") {
  #Record clone IDs
  if(repl.out$filt1[i] == FALSE) {
    repl.out$colclone1[i] <- col.out$CloneIndex[grep(repl.out$vcfID1[i],
                                                     col.out[,3], fixed=TRUE)]
  }
  if(repl.out$filt2[i] == FALSE) {
    repl.out$colclone2[i] <- col.out$CloneIndex[grep(repl.out$vcfID2[i],
                                                     col.out[,3], fixed = TRUE)]
  }
} else if(repl.out$GTservice[i] == "GTSeek") { #Need to account for the overlap in vcfIDs for reps
  if(repl.out$filt2[i] == FALSE) {
    repl.out$colclone2[i] <- col.out$CloneIndex[grep(repl.out$vcfID2[i],
                                                     col.out[,3], fixed = TRUE)]
  }
  if(repl.out$filt1[i] == FALSE) {
    tmp <- col.out$CloneIndex[grep(repl.out$vcfID1[i],
                                   col.out[,3], fixed = TRUE)]
    if(length(tmp) > 1) {
      repl.out$colclone1[i] <- tmp[which(!tmp %in% repl.out$colclone2[i])]
      rm(tmp)
    } else {
      repl.out$colclone1[i] <- tmp
      rm(tmp)
    }
  }
}

if(repl.out$filt1[i] == repl.out$filt2[i] & repl.out$filt1[i] == FALSE) {
  #Record kinship estimate
  repl.out$kinship[i] <- kingrel$kinship[kingrel$id1 %in% c(repl.out$vcfID1[i],
                                                            repl.out$vcfID2[i]) &
                                         kingrel$id2 %in% c(repl.out$vcfID1[i],
                                                            repl.out$vcfID2[i])]

  repl.out$nSNPshared[i] <- kingrel$nsnp[kingrel$id1 %in% c(repl.out$vcfID1[i],
                                                            repl.out$vcfID2[i]) &
                                         kingrel$id2 %in% c(repl.out$vcfID1[i],
                                                            repl.out$vcfID2[i])]

  #Check if identified as recapture via PLINK
  if(!is.na(repl.out$kinship[i])) {
    if(repl.out$kinship[i] > 0.354) {
      repl.out$is.clonePLINK[i] <- TRUE
    } else {
      repl.out$is.clonePLINK[i] <- FALSE
    }
  } else {
    repl.out$is.clonePLINK[i] <- NA
```

```r
    }

    #Check if identified as recapture via COLONY
    if(repl.out$colclone1[i] == repl.out$colclone2[i]) {
      repl.out$is.cloneCOLONY[i] <- TRUE
    } else {
      repl.out$is.cloneCOLONY[i] <- FALSE
    }

    tmp.geno <- genos[genos$V1 %in% c(repl.out$vcfID1[i], repl.out$vcfID2[i]),-1]
    ovrlp <- which(colSums(tmp.geno == -1) == 0)
    if(length(ovrlp) > 0) {
      #If as.matrix(), even single SNP overlaps are OK!
      tmp.geno <- as.matrix(tmp.geno[,ovrlp])
      repl.out$nSNPmatch[i] <- sum(tmp.geno[1,] == tmp.geno[2,])
    } else {
      repl.out$nSNPmatch[i] <- NA
    }

    rm(tmp.geno, ovrlp)
  }

}

#Some high-level summaries for the pairs of replicated samples
sum(is.na(repl.out$is.clonePLINK))
```

```
## [1] 17
```

```r
sum(is.na(repl.out$is.cloneCOLONY))
```

```
## [1] 17
```

```r
sum(is.na(repl.out$is.clonePLINK) & is.na(repl.out$is.cloneCOLONY))
```

```
## [1] 17
```

```r
repl.out[is.na(repl.out$is.clonePLINK),]
```

```
##      ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1
## 2   121-23 UGA_173101_P015_WB11 UGA_173101_P016_WA12       1      83  TRUE
## 7   306-23 UGA_173101_P015_WH09 UGA_173101_P015_WH12       0       0  TRUE
## 14    14-C UGA_173101_P018_WA02 UGA_173101_P018_WG06    1199       1 FALSE
## 17  280-23 UGA_173101_P018_WD05 UGA_173101_P019_WC10      22      20  TRUE
## 21  134-24 UGA_173101_P019_WE07 UGA_173101_P019_WE09       0    1059  TRUE
## 26  434-24 UGA_173101_P019_WG10 UGA_173101_P019_WH01       1      53  TRUE
## 37  517-24           GTS_517-24          GTS_517-24R      NA      NA  TRUE
## 38  716-23           GTS_716-23          GTS_716-23R      NA      NA  TRUE
## 39  726-24           GTS_726-24          GTS_726-24R      NA      NA  TRUE
## 46 1384-24          GTS_1384-24         GTS_1384-24R      NA      NA  TRUE
## 52 2926-24          GTS_2926-24         GTS_2926-24R      NA      NA  TRUE
```

```
## 54  396-24          GTS_396-24          GTS_396-24R     113      NA FALSE
## 55  424-23          GTS_424-23          GTS_424-23R      NA      NA  TRUE
## 56  191-24          GTS_191-24          GTS_191-24R      NA      NA  TRUE
## 57  466-23          GTS_466-23          GTS_466-23R      NA      NA  TRUE
## 58  286-23          GTS_286-23          GTS_286-23R       3      NA  TRUE
## 60  311-23          GTS_311-23          GTS_311-23R      69      NA FALSE
##    filt2 colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 2   TRUE        NA        NA            NA             NA         NA        NA
## 7   TRUE        NA        NA            NA             NA         NA        NA
## 14  TRUE        16        NA            NA             NA         NA        NA
## 17  TRUE        NA        NA            NA             NA         NA        NA
## 21 FALSE        NA        33            NA             NA         NA        NA
## 26  TRUE        NA        NA            NA             NA         NA        NA
## 37  TRUE        NA        NA            NA             NA         NA        NA
## 38  TRUE        NA        NA            NA             NA         NA        NA
## 39  TRUE        NA        NA            NA             NA         NA        NA
## 46  TRUE        NA        NA            NA             NA         NA        NA
## 52  TRUE        NA        NA            NA             NA         NA        NA
## 54  TRUE       265        NA            NA             NA         NA        NA
## 55  TRUE        NA        NA            NA             NA         NA        NA
## 56  TRUE        NA        NA            NA             NA         NA        NA
## 57  TRUE        NA        NA            NA             NA         NA        NA
## 58  TRUE        NA        NA            NA             NA         NA        NA
## 60  TRUE       321        NA            NA             NA         NA        NA
##    kinship GTservice
## 2       NA       LGC
## 7       NA       LGC
## 14      NA       LGC
## 17      NA       LGC
## 21      NA       LGC
## 26      NA       LGC
## 37      NA    GTSeek
## 38      NA    GTSeek
## 39      NA    GTSeek
## 46      NA    GTSeek
## 52      NA    GTSeek
## 54      NA    GTSeek
## 55      NA    GTSeek
## 56      NA    GTSeek
## 57      NA    GTSeek
## 58      NA    GTSeek
## 60      NA    GTSeek
```

```r
#NAs come from cases where one or both members of a pair is filtered from the dataset

#PLINK recognizes samples as clones?
table(repl.out$is.clonePLINK)
```

```
##
## FALSE  TRUE
##     1    42
```

```r
repl.out[which(repl.out$is.clonePLINK == FALSE),]
```

```
##    ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1 35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     612     875 FALSE FALSE
##   colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 1        77       126         FALSE          FALSE        516       389
##    kinship GTservice
## 1 0.300898       LGC
```

```r
#All but 35-SGA

#COLONY recognizes samples as clones?
table(repl.out$is.cloneCOLONY)
```

```
##
## FALSE  TRUE
##     4    39
```

```r
repl.out[which(repl.out$is.cloneCOLONY == FALSE),]
```

```
##     ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1  35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     612     875 FALSE FALSE
## 29 227-23 UGA_173101_P021_WH06 UGA_173101_P023_WC12    1194    1164 FALSE FALSE
## 41 935-23           GTS_935-23          GTS_935-23R     123     123 FALSE FALSE
## 53 151-24           GTS_151-24          GTS_151-24R     122     118 FALSE FALSE
##    colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 1         77       126         FALSE          FALSE        516       389
## 29        14       205          TRUE          FALSE       1110       959
## 41       307       339          TRUE          FALSE        122       122
## 53       307        34          TRUE          FALSE        117       117
##     kinship GTservice
## 1  0.300898       LGC
## 29 0.425689       LGC
## 41 0.500000    GTSeek
## 53 0.500000    GTSeek
```

```r
#35-SGA, 227-23, 935-23, and 151-24

#How often do the programs agree??
table(repl.out$is.clonePLINK[which(repl.out$is.cloneCOLONY == TRUE)])
```

```
##
## TRUE
##   39
```

```r
table(repl.out$is.cloneCOLONY[which(repl.out$is.clonePLINK == TRUE)])
```

```
##
## FALSE  TRUE
##     3    39
```

```r
repl.out[which(repl.out$is.cloneCOLONY == FALSE &
                 repl.out$is.clonePLINK == TRUE),]
```

```
##     ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 29 227-23 UGA_173101_P021_WH06 UGA_173101_P023_WC12    1194    1164 FALSE FALSE
## 41 935-23           GTS_935-23          GTS_935-23R     123     123 FALSE FALSE
## 53 151-24           GTS_151-24          GTS_151-24R     122     118 FALSE FALSE
##    colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 29        14       205          TRUE          FALSE       1110       959
## 41       307       339          TRUE          FALSE        122       122
## 53       307        34          TRUE          FALSE        117       117
##     kinship GTservice
## 29 0.425689       LGC
## 41 0.500000    GTSeek
## 53 0.500000    GTSeek
```

So all `COLONY` matches in this set are confirmed by `PLINK`, but `PLINK` finds a handful of matches that COLONY does not. Obviously, all (or almost all) of the matches in this set are real. These come from duplicated samples. It's somewhat worrisome that `COLONY` does not match pairs of samples with matching genotypes at 117 of 117 and 122 of 122 mutually genotyped loci (see the comparisons above). It's less surprising that COLONY doesn't match the `227-23` sample, given the number of genotype mismatches there. Note that `PLINK` *does* recover the former two sample pairs (`151-24` and `935-23`) as matches, so the combined approach we're using would classify them as members of the same clone!

With the mismatches we see for `227-23` and `35-SGA`, it seems best to exclude those samples from the dataset.

Mismatching Genotypes?

Now let's look at the number of mismatches between replicated genotype calls. This will give us a crude estimate of the genotyping error rate.

```r
repl.out$mismatch <- repl.out$nSNPshared-repl.out$nSNPmatch
repl.out$errorRate <- repl.out$mismatch/repl.out$nSNPshared

range(repl.out$mismatch, na.rm = TRUE)
```

```
## [1]   0 151
```

```r
range(repl.out$errorRate, na.rm = TRUE)
```

```
## [1] 0.000000 0.246124
```

```r
#Naive estimate of genotyping error rate is the average proportion of mismatches?
mean(repl.out$errorRate, na.rm = TRUE)
```

```
## [1] 0.01975158
```

```r
# 1.98% including all samples
mean(repl.out$errorRate[which(repl.out$GTservice == "LGC")], na.rm=TRUE)
```

```
## [1] 0.03364751
```

```r
# 3.36% for LGC samples
mean(repl.out$errorRate[which(repl.out$GTservice == "GTSeek")], na.rm = TRUE)
```

```
## [1] 0.0004516712
```

```r
# < 0.1% for GTSeek

#Excluding samples that are suspicious (where COLONY doesn't find a match)
mean(repl.out$errorRate[which(repl.out$is.cloneCOLONY==TRUE)])
```

```
## [1] 0.01197841
```

```r
# 1.2% with only "good" pairs
mean(repl.out$errorRate[which(repl.out$is.cloneCOLONY==TRUE &
                              repl.out$GTservice == "LGC")])
```

```
## [1] 0.01995773
```

```r
# 2.0% for LGC matching samples
mean(repl.out$errorRate[which(repl.out$is.cloneCOLONY==TRUE &
                              repl.out$GTservice == "GTSeek")])
```

```
## [1] 0.0005081301
```

```r
# again <0.1% for GTSeek
```

Error rates are ~2% for the capture-seq (LGC) dataset, but substantially lower for GT-seq (GTSeek) data.

Kinship?

The range of kinship coefficients for (good) technical replicates will be used below as a threshold for combining `COLONY` and `PLINK` inferences. Let's calculate that now.

```r
#Range of kinship values between replicates
range(repl.out$kinship, na.rm = TRUE)
```

```
## [1] 0.300898 0.500000
```

```r
#Average kinship of "good" pairs
mean(repl.out$kinship[which(repl.out$is.cloneCOLONY==TRUE)])
```

```
## [1] 0.492365
```

```r
range(repl.out$kinship[which(repl.out$is.cloneCOLONY==TRUE)])
```

```
## [1] 0.452489 0.500000
```

```r
#So we can set the filter pretty high... >0.45 for recaps
```

Inferred as Recaptures?

Ideally, all technical replicates would be identified as such and called recaptures in our analysis. This isn't the case, a couple of the samples don't seem all that similar. Perhaps some contamination?

Suggest removing these samples from the dataset:

i. `35-SGA` - high error rate in replicated sample (24.6%), not identified as a match by `COLONY` *or* `PLINK2`. Unsure which genotype is accurate so best to drop them both.

ii. `227-23` - error rate his high again (13.6%), sample not identified as a match by `COLONY`. Kinship estimates are high, but the lowest of all pairs of replicated samples identified by `PLINK` as recaptures. Note that this pair of samples showed up in our `sexSamples` markdown too. One replicate is inferred to be male, the other inferred to be female. Best to exclude both samples.

Meanwhile, we should keep the match from `PLINK` for `935-23` and `151-24` and their replicates. They match at all 122 and 117 shared SNP loci, surprising that `COLONY` would not identify this pair as a match!

```r
table(repl.out$is.cloneCOLONY)
```

```
##
## FALSE  TRUE
##     4    39
```

```r
table(repl.out$is.clonePLINK)
```

```
##
## FALSE  TRUE
##     1    42
```

```r
repl.out[which(repl.out$is.cloneCOLONY == FALSE),]
```

```
##     ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1  35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     612     875 FALSE FALSE
## 29 227-23 UGA_173101_P021_WH06 UGA_173101_P023_WC12    1194    1164 FALSE FALSE
## 41 935-23           GTS_935-23          GTS_935-23R     123     123 FALSE FALSE
## 53 151-24           GTS_151-24          GTS_151-24R     122     118 FALSE FALSE
##    colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 1         77       126         FALSE          FALSE        516       389
## 29        14       205          TRUE          FALSE       1110       959
## 41       307       339          TRUE          FALSE        122       122
## 53       307        34          TRUE          FALSE        117       117
##     kinship GTservice mismatch errorRate
## 1  0.300898       LGC      127  0.246124
## 29 0.425689       LGC      151  0.136036
## 41 0.500000    GTSeek        0  0.000000
## 53 0.500000    GTSeek        0  0.000000
```

```r
repl.out[which(repl.out$is.cloneCOLONY == FALSE & repl.out$is.clonePLINK == FALSE),]
```

```
##    ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1 35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     612     875 FALSE FALSE
##   colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 1        77       126         FALSE          FALSE        516       389
##    kinship GTservice mismatch errorRate
## 1 0.300898       LGC      127  0.246124
```

```r
#Drop samples by setting FILT to TRUE
which(smd$ugaID == "35-SGA")
```

```
## [1] 13 15
```

```r
smd$FILT[which(smd$ugaID == "35-SGA")] <- TRUE
which(smd$ugaID == "227-23")
```

```
## [1] 648 786
```

```r
smd$FILT[which(smd$ugaID == "227-23")] <- TRUE
```

Check for Odd Patterns

Beyond the technical replicates, we have some other information that will be useful for evaluating the quality of the inferred recaptures. For instance, some bears were directly captured and have known sex information. We wouldn't want to mistake males for females in these samples. Space is also useful, a CGA bear is unlikely to be sampled in NGA and vice versa.

```r
nSex <- nSexGT <- nRegion <- c()
for(i in 1:nrow(col.out)) {
  tmpclone <- strsplit(col.out[i,3], split = ",")[[1]]
  tmp <- smd[which(smd$vcfID %in% tmpclone),]
  nSex[i] <- length(unique(tmp$knownSEX[tmp$knownSEX != ""]))
  nSexGT[i] <- length(unique(tmp$sexGT[tmp$sexGT != "undet"]))
  nRegion[i] <- length(unique(tmp$region))
  rm(tmp, tmpclone)
}
```

Known Sex

We should not have multiple known sexes per clone

```r
nSex
```

```
##   [1] 0 0 1 2 2 0 0 0 0 1 1 1 1 0 2 1 1 1 1 2 1 0 1 0 1 1 0 1 0 0 1 0 1 1 0 1 0
##  [38] 1 0 1 1 1 0 0 0 2 2 1 1 1 0 1 1 1 1 1 0 1 0 1 1 1 0 1 0 0 0 1 0 1 0 2 2 1
##  [75] 2 0 1 1 1 1 0 1 1 1 1 2 1 1 0 1 0 1 1 1 1 1 1 1 1 1 2 1 0 1 1 1 0 1 1 1 0
## [112] 1 1 1 0 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 2 0 1 0 1
## [149] 1 1 1 1 1 1 1 1 0 1 1 1 1 0 1 2 1 1 1 1 0 0 0 1 1 0 1 1 0 1 1 1 0 1 1 0 1
```

```
## [186] 1 0 0 1 0 1 2 0 0 1 0 1 0 1 1 2 1 1 0 0 1 0 0 0 1 0 0 0 0 0 1 0 0 0 1 0 0
## [223] 1 0 0 0 0 0 0 0 0 1 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 0 0 1 1 0 1 0 0 0 0 0 1
## [260] 0 0 0 0 0 1 0 0 0 0 0 2 1 0 1 1 1 0 1 0 0 1 1 0 0 0 0 0 0 0 0 0 1 1 1 1 1
## [297] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [334] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [371] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```r
max(nSex)
```

```
## [1] 2
```

```r
#Does genotype-inferred sex match known sex of samples?
table(smd$sexGT[smd$knownSEX == "F"])
```

```
##
##     F undet
##    64     3
```

```r
table(smd$sexGT[smd$knownSEX == "M"])
```

```
##
##     F     M undet
##     1    82     4
```

```r
#The single case of a mismatch (within a sample) between a known sex
# and genotype-inferred sex
smd[which(smd$knownSEX=="M" & smd$sexGT == "F"),]
```

```
##                   extID ugaID is.SCR is.rep week year snare utmSnare
## 71 UGA_173101_P015_WF11 63-DC  FALSE  FALSE <NA> 2024  <NA>     <NA>
##                                                                     fqfile
## 71 RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
##    knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice sexGT
## 71        M  Den Check    CGA   1240 UGA_173101_P015_WF11 FALSE       LGC     F
```

```r
bigMD <- read.csv("../input/Finalized Hair Samples w Metadata_Carr.csv", header = TRUE)
bigMD[bigMD$Hair.ID == smd$ugaID[which(smd$knownSEX=="M" & smd$sexGT == "F")],]
```

```
##      Sample.. RandomID Hair.ID    Rapid.Genomics.ID Site.ID Week Check.Date
## 1112     1063     1063   63-DC UGA_173101_P012_WC09
##      Strand Sample.Quality DNR.ID          Database.ID     PIT.Tag     Bear Sex
## 1112                              CGBB_M_2024_C_87815B 606-306-842 87815 C2   M
##      Year   Date  County Region      What Plate Follicles Abnormalitiy.
## 1112 2024 3/2/24 Houston    CGA Den Check    12        14
##                     Comments        Address Latitude Longitude X X.1
## 1112 87815 Biological Cub 2 87815 Den 2024 32.52144 -83.44958
```

```r
#This was a captured cub, identified as male in the field
#  we saw this individual before - see the `sexSamples` markdown for more detail
```

Genotype-inferred Sex

Similarly, our genotype-inferred sexes should not vary within clones.

```
nSexGT
```

```
##   [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [112] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1
## [149] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 1 1 1
## [186] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [223] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [260] 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 0 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 2 1 1 1 1
## [297] 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [334] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [371] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1
```

```
max(nSexGT)
```

```
## [1] 2
```

```
which(nSexGT > 1)
```

```
## [1]  14  31  47 292 307
```

```
#They do in a few cases...
# 14 31 47 292 307

#Clone 14 - 33 samples, 29 female, 2 male, 2 undet - no direct capture to confirm majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 14,3], ",")[[1]]])
```

```
##
##     F     M undet
##    29     2     2
```

```
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 14,3], ",")[[1]],]
```

```
##                    extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 83  UGA_173101_P015_WG11  11-23   TRUE  FALSE    6 2023 HS 029 259736x3594236
## 91  UGA_173101_P015_WH07 432-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 125 UGA_173101_P016_WC05  73-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 138 UGA_173101_P016_WD06 179-23   TRUE  FALSE    7 2023 HS 027 264464x3592495
## 186 UGA_173101_P016_WH06 312-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 188 UGA_173101_P016_WH08 330-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 194 UGA_173101_P017_WA02 393-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 233 UGA_173101_P017_WD05 417-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 256 UGA_173101_P017_WF04 468-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 260 UGA_173101_P017_WF08 409-23   TRUE   TRUE    5 2023 HS 027 264464x3592495
## 285 UGA_173101_P017_WH09 409-23   TRUE   TRUE    5 2023 HS 027 264464x3592495
## 338 UGA_173101_P018_WE02  52-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
```

```
## 345 UGA_173101_P018_WE09 206-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 351 UGA_173101_P018_WF03  35-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 392 UGA_173101_P019_WC02 188-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 396 UGA_173101_P019_WC06 200-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 406 UGA_173101_P019_WD04 362-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 467 UGA_173101_P020_WA05 209-23   TRUE  FALSE    7 2023 HS 025 265276x3591787
## 469 UGA_173101_P020_WA07  38-23   TRUE  FALSE    3 2023 HS 027 264464x3592495
## 471 UGA_173101_P020_WA09 319-23   TRUE  FALSE    6 2023 HS 032 263672x3593292
## 540 UGA_173101_P020_WG06 277-23   TRUE  FALSE    5 2023 HS 025 265276x3591787
## 541 UGA_173101_P020_WG07 443-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 549 UGA_173101_P020_WH03 212-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 550 UGA_173101_P020_WH04 271-23   TRUE  FALSE    6 2023 HS 025 265276x3591787
## 557 UGA_173101_P020_WH11 447-23   TRUE  FALSE    7 2023 HS 027 264464x3592495
## 583 UGA_173101_P021_WC01  91-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 648 UGA_173101_P021_WH06 227-23   TRUE   TRUE    5 2023 HS 027 264464x3592495
## 666 UGA_173101_P022_WA12 160-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 735 UGA_173101_P022_WG09 471-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 766 UGA_173101_P023_WB04  37-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 768 UGA_173101_P023_WB06 190-23   TRUE  FALSE    5 2023 HS 025 265276x3591787
## 773 UGA_173101_P023_WB11   7-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 826 UGA_173101_P023_WG04 136-23   TRUE  FALSE    6 2023 HS 025 265276x3591787
##                                                                          fqfile
## 83   RAPiD-Genomics_F534_UGA_173101_P015_WG11_i5-502_i7-96_S14821_L003_R1_001.fastq
## 91    RAPiD-Genomics_F534_UGA_173101_P015_WH07_i5-502_i7-1_S14829_L003_R1_001.fastq
## 125 RAPiD-Genomics_F534_UGA_173101_P016_WC05_i5-504_i7-125_S14863_L003_R1_001.fastq
## 138 RAPiD-Genomics_F534_UGA_173101_P016_WD06_i5-504_i7-138_S14876_L003_R1_001.fastq
## 186 RAPiD-Genomics_F534_UGA_173101_P016_WH06_i5-504_i7-186_S14923_L003_R1_001.fastq
## 188 RAPiD-Genomics_F534_UGA_173101_P016_WH08_i5-504_i7-188_S14925_L003_R1_001.fastq
## 194  RAPiD-Genomics_F534_UGA_173101_P017_WA02_i5-512_i7-98_S14931_L003_R1_001.fastq
## 233 RAPiD-Genomics_F534_UGA_173101_P017_WD05_i5-512_i7-137_S14970_L003_R1_001.fastq
## 256 RAPiD-Genomics_F534_UGA_173101_P017_WF04_i5-512_i7-160_S14993_L003_R1_001.fastq
## 260 RAPiD-Genomics_F534_UGA_173101_P017_WF08_i5-512_i7-164_S14997_L003_R1_001.fastq
## 285 RAPiD-Genomics_F534_UGA_173101_P017_WH09_i5-512_i7-189_S15022_L003_R1_001.fastq
## 338 RAPiD-Genomics_F534_UGA_173101_P018_WE02_i5-537_i7-146_S15075_L003_R1_001.fastq
## 345 RAPiD-Genomics_F534_UGA_173101_P018_WE09_i5-537_i7-153_S15084_L003_R1_001.fastq
## 351 RAPiD-Genomics_F534_UGA_173101_P018_WF03_i5-537_i7-159_S15090_L003_R1_001.fastq
## 392 RAPiD-Genomics_F534_UGA_173101_P019_WC02_i5-1102_i7-67_S15131_L003_R1_001.fastq
## 396  RAPiD-Genomics_F534_UGA_173101_P019_WC06_i5-1102_i7-3_S15135_L003_R1_001.fastq
## 406 RAPiD-Genomics_F534_UGA_173101_P019_WD04_i5-1102_i7-91_S15145_L003_R1_001.fastq
## 467   RAPiD-Genomics_F566_UGA_173101_P020_WA05_i5-505_i7-38_S3895_L001_R1_001.fastq
## 469   RAPiD-Genomics_F566_UGA_173101_P020_WA07_i5-505_i7-77_S3897_L001_R1_001.fastq
## 471   RAPiD-Genomics_F566_UGA_173101_P020_WA09_i5-505_i7-36_S3899_L001_R1_001.fastq
## 540   RAPiD-Genomics_F566_UGA_173101_P020_WG06_i5-505_i7-14_S3968_L001_R1_001.fastq
## 541   RAPiD-Genomics_F566_UGA_173101_P020_WG07_i5-505_i7-18_S3969_L001_R1_001.fastq
## 549   RAPiD-Genomics_F566_UGA_173101_P020_WH03_i5-505_i7-88_S3977_L001_R1_001.fastq
## 550   RAPiD-Genomics_F566_UGA_173101_P020_WH04_i5-505_i7-73_S3978_L001_R1_001.fastq
## 557   RAPiD-Genomics_F566_UGA_173101_P020_WH11_i5-505_i7-95_S3985_L001_R1_001.fastq
## 583     RAPiD-Genomics_F566_UGA_173101_P021_WC01_i5-501_i7-65_S25_L003_R1_001.fastq
## 648     RAPiD-Genomics_F566_UGA_173101_P021_WH06_i5-501_i7-49_S90_L003_R1_001.fastq
## 666  RAPiD-Genomics_F566_UGA_173101_P022_WA12_i5-1098_i7-23_S7594_L003_R1_001.fastq
## 735  RAPiD-Genomics_F566_UGA_173101_P022_WG09_i5-1098_i7-41_S7663_L003_R1_001.fastq
## 766  RAPiD-Genomics_F566_UGA_173101_P023_WB04_i5-1099_i7-20_S7694_L003_R1_001.fastq
## 768   RAPiD-Genomics_F566_UGA_173101_P023_WB06_i5-1099_i7-6_S7696_L003_R1_001.fastq
## 773   RAPiD-Genomics_F566_UGA_173101_P023_WB11_i5-1099_i7-5_S7701_L003_R1_001.fastq
```

```
## 826  RAPiD-Genomics_F566_UGA_173101_P023_WG04_i5-1099_i7-10_S7754_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 83          Hair Snare    CGA    333 UGA_173101_P015_WG11 FALSE       LGC
## 91          Hair Snare    CGA   1231 UGA_173101_P015_WH07 FALSE       LGC
## 125         Hair Snare    CGA    460 UGA_173101_P016_WC05 FALSE       LGC
## 138         Hair Snare    CGA   1018 UGA_173101_P016_WD06 FALSE       LGC
## 186         Hair Snare    CGA    940 UGA_173101_P016_WH06 FALSE       LGC
## 188         Hair Snare    CGA   1103 UGA_173101_P016_WH08 FALSE       LGC
## 194         Hair Snare    CGA   1274 UGA_173101_P017_WA02 FALSE       LGC
## 233         Hair Snare    CGA    486 UGA_173101_P017_WD05 FALSE       LGC
## 256         Hair Snare    CGA   1256 UGA_173101_P017_WF04 FALSE       LGC
## 260         Hair Snare    CGA    863 UGA_173101_P017_WF08 FALSE       LGC
## 285         Hair Snare    CGA    948 UGA_173101_P017_WH09 FALSE       LGC
## 338         Hair Snare    CGA   1272 UGA_173101_P018_WE02 FALSE       LGC
## 345         Hair Snare    CGA   1110 UGA_173101_P018_WE09 FALSE       LGC
## 351         Hair Snare    CGA   1174 UGA_173101_P018_WF03 FALSE       LGC
## 392         Hair Snare    CGA   1280 UGA_173101_P019_WC02 FALSE       LGC
## 396         Hair Snare    CGA   1282 UGA_173101_P019_WC06 FALSE       LGC
## 406         Hair Snare    CGA   1288 UGA_173101_P019_WD04 FALSE       LGC
## 467         Hair Snare    CGA   1252 UGA_173101_P020_WA05 FALSE       LGC
## 469         Hair Snare    CGA   1146 UGA_173101_P020_WA07 FALSE       LGC
## 471         Hair Snare    CGA   1059 UGA_173101_P020_WA09 FALSE       LGC
## 540         Hair Snare    CGA   1270 UGA_173101_P020_WG06 FALSE       LGC
## 541         Hair Snare    CGA    676 UGA_173101_P020_WG07 FALSE       LGC
## 549         Hair Snare    CGA   1301 UGA_173101_P020_WH03 FALSE       LGC
## 550         Hair Snare    CGA   1299 UGA_173101_P020_WH04 FALSE       LGC
## 557         Hair Snare    CGA   1303 UGA_173101_P020_WH11 FALSE       LGC
## 583         Hair Snare    CGA   1203 UGA_173101_P021_WC01 FALSE       LGC
## 648         Hair Snare    CGA   1194 UGA_173101_P021_WH06  TRUE       LGC
## 666         Hair Snare    CGA   1103 UGA_173101_P022_WA12 FALSE       LGC
## 735         Hair Snare    CGA   1293 UGA_173101_P022_WG09 FALSE       LGC
## 766         Hair Snare    CGA   1262 UGA_173101_P023_WB04 FALSE       LGC
## 768         Hair Snare    CGA   1200 UGA_173101_P023_WB06 FALSE       LGC
## 773         Hair Snare    CGA   1230 UGA_173101_P023_WB11 FALSE       LGC
## 826         Hair Snare    CGA   1280 UGA_173101_P023_WG04 FALSE       LGC
##     sexGT
## 83      F
## 91      F
## 125     F
## 138     F
## 186     F
## 188     F
## 194     F
## 233     F
## 256 undet
## 260     F
## 285     F
## 338     F
## 345     F
## 351     F
## 392     F
## 396     F
## 406     F
## 467     F
```

```
## 469     F
## 471     F
## 540     F
## 541     F
## 549     M
## 550     M
## 557     F
## 583     F
## 648     F
## 666 undet
## 735     F
## 766     F
## 768     F
## 773     F
## 826     F
```

```
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 14,3], ",")[[1]]]
```

```
##  [1] "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "
## [26] "" "" "" "" "" "" "" ""
```

```
#Clone 31 - 6 samples, 5 female, 1 male - direct capture confirms majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 31,3], ",")[[1]]])
```

```
##
## F M
## 5 1
```

```
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 31,3], ",")[[1]],]
```

```
##                    extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 86  UGA_173101_P015_WH02   45-C  FALSE  FALSE <NA> 2024   <NA>           <NA>
## 169 UGA_173101_P016_WG01 448-24   TRUE  FALSE    5 2024 HS 051 260705x3599596
## 209 UGA_173101_P017_WB05 497-24   TRUE  FALSE    2 2024 HS 048 261267x3598930
## 426 UGA_173101_P019_WE12 154-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 698 UGA_173101_P022_WD08 341-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 784 UGA_173101_P023_WC10 205-23   TRUE  FALSE    7 2023 HS 051 260705x3599596
##                                                                          fqfile
## 86   RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169 RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209 RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426 RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
## 698  RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
## 784  RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 86         F    Capture    CGA   1153 UGA_173101_P015_WH02 FALSE       LGC
## 169          Hair Snare    CGA    709 UGA_173101_P016_WG01 FALSE       LGC
## 209          Hair Snare    CGA   1299 UGA_173101_P017_WB05 FALSE       LGC
## 426          Hair Snare    CGA   1294 UGA_173101_P019_WE12 FALSE       LGC
## 698          Hair Snare    CGA   1109 UGA_173101_P022_WD08 FALSE       LGC
## 784          Hair Snare    CGA   1250 UGA_173101_P023_WC10 FALSE       LGC
##     sexGT
```

```
## 86      F
## 169     F
## 209     M
## 426     F
## 698     F
## 784     F
```

```
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 31,3], ",")[[1]]]
```

```
## [1] "F" ""  ""  ""  ""  ""
```

```
#Clone 47 - 6 samples, 5 female, 1 male - direct capture confirms majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 47,3], ",")[[1]]])
```

```
##
## F M
## 5 1
```

```
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 47,3], ",")[[1]],]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 49                     UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                    UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                    UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>           <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                        fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                          <NA>
## 1598                                                                          <NA>
## 1819                                                                          <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F     Capture    CGA   1126 UGA_173101_P015_WE01 FALSE       LGC
## 492            Hair Snare    CGA   1259 UGA_173101_P020_WC06 FALSE       LGC
## 629            Hair Snare    CGA   1257 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120           GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123          GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124          GTS_2920-24 FALSE    GTSeek
##      sexGT
## 49       F
## 492      F
## 629      M
```

```
## 1048     F
## 1598     F
## 1819     F
```

```
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 47,3], ",")[[1]]]
```

```
## [1] "F" ""  ""  NA  NA  NA
```

```
#Clone 292 - 9 samples, 8 male, 1 female - no direct capture to confirm majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 292,3], ",")[[1]]])
```

```
##
## F M
## 1 8
```

```
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 292,3], ",")[[1]],]
```

```
##                                     extID   ugaID is.SCR is.rep week year
## 828     GTseek_024_384i5_001_P098.001_5-23    5-23   TRUE  FALSE    6 2023
## 1213  GTseek_025_384i5_003_P098.005_738-24  738-24   TRUE  FALSE    7 2024
## 1451 GTseek_025_384i5_241_P098.006_2962-24 2962-24   TRUE  FALSE    4 2024
## 1615 GTseek_026_384i5_025_P098.010_2748-24 2748-24   TRUE  FALSE    7 2024
## 1715 GTseek_026_384i5_163_P098.009_2320-24 2320-24   TRUE  FALSE    7 2024
## 1760 GTseek_026_384i5_253_P098.010_2864-24 2864-24   TRUE  FALSE    7 2024
## 1792 GTseek_026_384i5_317_P098.010_2894-24 2894-24   TRUE  FALSE    4 2024
## 1816 GTseek_026_384i5_365_P098.009_2729-24 2729-24   TRUE  FALSE    7 2024
## 2095 GTseek_028_384i5_270_P098.UGA3_260-24  260-24   TRUE  FALSE    4 2024
##       snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt
## 828  HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    104
## 1213 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    119
## 1451 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    119
## 1615 HS 008 256165x3589588   <NA>     <NA> Hair Sample    CGA    122
## 1715 HS 008 256165x3589588   <NA>     <NA> Hair Sample    CGA    124
## 1760 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    124
## 1792 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    124
## 1816 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    121
## 2095 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    112
##            vcfID  FILT GTservice sexGT
## 828     GTS_5-23 FALSE    GTSeek     F
## 1213  GTS_738-24 FALSE    GTSeek     M
## 1451 GTS_2962-24 FALSE    GTSeek     M
## 1615 GTS_2748-24 FALSE    GTSeek     M
## 1715 GTS_2320-24 FALSE    GTSeek     M
## 1760 GTS_2864-24 FALSE    GTSeek     M
## 1792 GTS_2894-24 FALSE    GTSeek     M
## 1816 GTS_2729-24 FALSE    GTSeek     M
## 2095  GTS_260-24 FALSE    GTSeek     M
```

```
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 292,3], ",")[[1]]]
```

```
## [1] NA NA NA NA NA NA NA NA NA
```

```r
#Clone 307 - 28 samples, 25 male, 1 female, 2 undet - no direct capture to confirm majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 307,3], ",")[[1]]])
```

```
##
##     F     M undet
##     1    25     2
```

```r
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 307,3], ",")[[1]],]
```

```
##                                      extID   ugaID is.SCR is.rep week year
## 873   GTseek_024_384i5_046_P098.003_949-23  949-23   TRUE  FALSE    6 2023
## 936   GTseek_024_384i5_109_P098.001_646-23  646-23   TRUE  FALSE    7 2023
## 962   GTseek_024_384i5_135_P098.001_655-23  655-23   TRUE  FALSE    6 2023
## 974   GTseek_024_384i5_148_P098.004_630-24  630-24   TRUE  FALSE    3 2024
## 993   GTseek_024_384i5_167_P098.001_665-23  665-23   TRUE  FALSE    7 2023
## 998   GTseek_024_384i5_172_P098.003_997-23  997-23   TRUE  FALSE    7 2023
## 1013  GTseek_024_384i5_187_P098.002_818-23  818-23   TRUE  FALSE    6 2023
## 1029  GTseek_024_384i5_203_P098.001_677-23  677-23   TRUE  FALSE    3 2023
## 1030 GTseek_024_384i5_204_P098.003_1012-23 1012-23   TRUE  FALSE    7 2023
## 1374  GTseek_025_384i5_164_P098.007_935-23  935-23   TRUE   TRUE    7 2023
## 1411  GTseek_025_384i5_201_P098.005_855-24  855-24   TRUE  FALSE    3 2024
## 1426 GTseek_025_384i5_216_P098.008_1344-24 1344-24   TRUE  FALSE    3 2024
## 1473  GTseek_025_384i5_263_P098.005_886-24  886-24   TRUE  FALSE    3 2024
## 1490 GTseek_025_384i5_280_P098.008_1403-24 1403-24   TRUE  FALSE    3 2024
## 1513  GTseek_025_384i5_303_P098.005_931-24  931-24   TRUE  FALSE    3 2024
## 1518 GTseek_025_384i5_308_P098.008_1428-24 1428-24   TRUE  FALSE    2 2024
## 1527 GTseek_025_384i5_317_P098.006_3012-24 3012-24   TRUE  FALSE    6 2024
## 1548 GTseek_025_384i5_338_P098.008_1451-24 1451-24   TRUE  FALSE    3 2024
## 1563  GTseek_025_384i5_353_P098.005_955-24  955-24   TRUE  FALSE    2 2024
## 1641 GTseek_026_384i5_061_P098.010_2777-24 2777-24   TRUE  FALSE    3 2024
## 1691 GTseek_026_384i5_129_P098.009_1672-24 1672-24   TRUE  FALSE    3 2024
## 1717 GTseek_026_384i5_167_P098.009_2376-24 2376-24   TRUE  FALSE    6 2024
## 1747 GTseek_026_384i5_227_P098.009_2669-24 2669-24   TRUE  FALSE    6 2024
## 1756 GTseek_026_384i5_245_P098.010_2857-24 2857-24   TRUE  FALSE    3 2024
## 1782 GTseek_026_384i5_297_P098.009_2697-24 2697-24   TRUE  FALSE    2 2024
## 1805 GTseek_026_384i5_343_P098.010_2906-24 2906-24   TRUE  FALSE    6 2024
## 1903 GTseek_028_384i5_078_P098.UGA3_151-24  151-24   TRUE   TRUE    3 2024
## 1920 GTseek_028_384i5_095_P098.UGA2_426-23  426-23   TRUE  FALSE    6 2023
##       snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt
## 873  HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    124
## 936  HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    120
## 962  HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    116
## 974  HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    119
## 993  HS 002 255814x3587061   <NA>     <NA> Hair Sample    CGA    122
## 998  HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    123
## 1013 HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    123
## 1029 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    119
## 1030 HS 002 255814x3587061   <NA>     <NA> Hair Sample    CGA    121
## 1374 HS 002 255814x3587061   <NA>     <NA> Hair Sample    CGA    123
## 1411 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    123
## 1426 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    124
## 1473 HS 017 260417x3588089   <NA>     <NA> Hair Sample    CGA    124
## 1490 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    122
```

```
## 1513 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    119
## 1518 HS 014 261409x3586557   <NA>     <NA> Hair Sample    CGA    123
## 1527 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    122
## 1548 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    123
## 1563 HS 014 261409x3586557   <NA>     <NA> Hair Sample    CGA    122
## 1641 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    121
## 1691 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    121
## 1717 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    123
## 1747 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    123
## 1756 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    124
## 1782 HS 014 261409x3586557   <NA>     <NA> Hair Sample    CGA    122
## 1805 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    124
## 1903 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    122
## 1920 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    119
##            vcfID  FILT GTservice sexGT
## 873   GTS_949-23 FALSE    GTSeek     M
## 936   GTS_646-23 FALSE    GTSeek undet
## 962   GTS_655-23 FALSE    GTSeek     F
## 974   GTS_630-24 FALSE    GTSeek     M
## 993   GTS_665-23 FALSE    GTSeek undet
## 998   GTS_997-23 FALSE    GTSeek     M
## 1013  GTS_818-23 FALSE    GTSeek     M
## 1029  GTS_677-23 FALSE    GTSeek     M
## 1030 GTS_1012-23 FALSE    GTSeek     M
## 1374  GTS_935-23 FALSE    GTSeek     M
## 1411  GTS_855-24 FALSE    GTSeek     M
## 1426 GTS_1344-24 FALSE    GTSeek     M
## 1473  GTS_886-24 FALSE    GTSeek     M
## 1490 GTS_1403-24 FALSE    GTSeek     M
## 1513  GTS_931-24 FALSE    GTSeek     M
## 1518 GTS_1428-24 FALSE    GTSeek     M
## 1527 GTS_3012-24 FALSE    GTSeek     M
## 1548 GTS_1451-24 FALSE    GTSeek     M
## 1563  GTS_955-24 FALSE    GTSeek     M
## 1641 GTS_2777-24 FALSE    GTSeek     M
## 1691 GTS_1672-24 FALSE    GTSeek     M
## 1717 GTS_2376-24 FALSE    GTSeek     M
## 1747 GTS_2669-24 FALSE    GTSeek     M
## 1756 GTS_2857-24 FALSE    GTSeek     M
## 1782 GTS_2697-24 FALSE    GTSeek     M
## 1805 GTS_2906-24 FALSE    GTSeek     M
## 1903  GTS_151-24 FALSE    GTSeek     M
## 1920  GTS_426-23 FALSE    GTSeek     M
```

```
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 307,3], ",")[[1]]]
```

```
##  [1] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## [26] NA NA NA
```

So sex is not inferred perfectly. That's not surprising, given the pretty lenient threshold we're using to identify males (just 5 reads required at 2 SNPs within 3 Y-specific scaffolds). Contamination could easily lead to us calling a sample a male erroneously. Nonetheless, instances with multiple sexes are relatively rare, and in all cases there is either a clear majority from the set of samples or a direct capture that confirms the consensus from the set.

Region

We shouldn't have multiple regions for any clones...

```
nRegion
```

```
##   [1] 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [112] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [149] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [186] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [223] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [260] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [297] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1
## [334] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [371] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```
which(nRegion > 1)
```

```
## [1]   3  63 332
```

```
smd$region[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex==3,3],",")[[1]]]
```

```
##  [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA-H" "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
```

```
#Not really a separate region, just a historic hair sample (from 2012)

smd$region[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex==63,3],",")[[1]]]
```

```
## [1] "CGA-H" "CGA"
```

```
#Not really a separate region, just a historic hair sample (from 2012)

smd$region[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex==332,3],",")[[1]]]
```

```
## [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA-H"
```

```
#Not really a separate region, just a historic hair sample (from 2012)
```

So no instances of recaptures identified across regions. That's good!

Known vs. Genotype-inferred Sex

Finally, comparing the known sex of unique genotypes (from direct captures) with that inferred for all samples within a clone...

```r
#check for discrepancies between known and genotyped sex for all clones
discrepancy <- multknown <- c()
#for(i in 1:max(sampleMetadata$cloneID, na.rm = TRUE)) {
for(i in 1:nrow(col.out)) {
  tmpclone <- strsplit(col.out[i,3], split = ",")[[1]]
  tmp <- smd[which(smd$vcfID %in% tmpclone),]
  tmp.knownSEX <- unique(tmp$knownSEX[tmp$knownSEX != "" & !is.na(tmp$knownSEX)])
  if(length(tmp.knownSEX) > 0) {
    if(length(tmp.knownSEX) > 1) {
      multknown[i] <- TRUE
      discrepancy[i] <- NA
    } else {
      multknown[i] <- FALSE
      if(sum(tmp$sexGT[tmp$sexGT != "undet"] != tmp.knownSEX) == 0) {
        discrepancy[i] <- FALSE
      } else {
        discrepancy[i] <- TRUE
      }
    }
  }
  rm(tmpclone, tmp, tmp.knownSEX)
}

sum(discrepancy, na.rm = TRUE) #3 cases with discrepancies...
```

```
## [1] 3
```

```r
sum(multknown, na.rm = TRUE) #0 cases with multiple "known" sexes
```

```
## [1] 0
```

```r
#Check the discrepancies...
which(discrepancy == TRUE)
```

```
## [1]  31  47 154
```

```r
#Clone 31 - seen above, capture agrees with majority of samples
smd[smd$vcfID %in% strsplit(col.out[which(col.out$CloneIndex == 31),3],",")[[1]],]
```

```
##                    extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 86  UGA_173101_P015_WH02   45-C  FALSE  FALSE <NA> 2024   <NA>           <NA>
## 169 UGA_173101_P016_WG01 448-24   TRUE  FALSE    5 2024 HS 051 260705x3599596
## 209 UGA_173101_P017_WB05 497-24   TRUE  FALSE    2 2024 HS 048 261267x3598930
## 426 UGA_173101_P019_WE12 154-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 698 UGA_173101_P022_WD08 341-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 784 UGA_173101_P023_WC10 205-23   TRUE  FALSE    7 2023 HS 051 260705x3599596
##                                                                          fqfile
## 86   RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169 RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209 RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426 RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
```

```
## 698  RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
## 784  RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 86         F    Capture    CGA   1153 UGA_173101_P015_WH02 FALSE       LGC
## 169          Hair Snare    CGA    709 UGA_173101_P016_WG01 FALSE       LGC
## 209          Hair Snare    CGA   1299 UGA_173101_P017_WB05 FALSE       LGC
## 426          Hair Snare    CGA   1294 UGA_173101_P019_WE12 FALSE       LGC
## 698          Hair Snare    CGA   1109 UGA_173101_P022_WD08 FALSE       LGC
## 784          Hair Snare    CGA   1250 UGA_173101_P023_WC10 FALSE       LGC
##     sexGT
## 86      F
## 169     F
## 209     M
## 426     F
## 698     F
## 784     F
```

```r
#Clone 47 - seen above, capture agrees with majority of samples
smd[smd$vcfID %in% strsplit(col.out[which(col.out$CloneIndex == 47),3],",")[[1]],]
```

```
##                                     extID   ugaID is.SCR is.rep week year
## 49                    UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                   UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                   UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>           <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                        fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                          <NA>
## 1598                                                                          <NA>
## 1819                                                                          <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F     Capture    CGA   1126 UGA_173101_P015_WE01 FALSE       LGC
## 492            Hair Snare    CGA   1259 UGA_173101_P020_WC06 FALSE       LGC
## 629            Hair Snare    CGA   1257 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120           GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123          GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124          GTS_2920-24 FALSE    GTSeek
##      sexGT
## 49       F
## 492      F
## 629      M
## 1048     F
## 1598     F
```

```
## 1819     F
```

```
#Clone 154 - noted previously, cub recorded as M in the field, genotype looks F
smd[smd$vcfID %in% strsplit(col.out[which(col.out$CloneIndex == 154),3],",")[[1]],]
```

```
##                    extID ugaID is.SCR is.rep week year snare utmSnare
## 71 UGA_173101_P015_WF11 63-DC  FALSE  FALSE <NA> 2024  <NA>     <NA>
##                                                                     fqfile
## 71 RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
##    knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice sexGT
## 71        M  Den Check    CGA   1240 UGA_173101_P015_WF11 FALSE       LGC     F
```

Compare Inferred Recaptures

Now we create a new data frame to compare inferences (same individual / not) for each pair of samples. Note that, for PLINK, we're using 0.354 as a threshold for KING relatedness coefficients of duplicated samples. This is based off of information provided in the PLINK2 manual. Quoting directly (emphasis added)...

Note that KING kinship coefficients are scaled such that duplicate samples have kinship 0.5, not 1. First-degree relations (parent-child, full siblings) correspond to ~0.25, second-degree relations correspond to ~0.125, etc. ***It is conventional to use a cutoff of ~0.354 (the geometric mean of 0.5 and 0.25) to screen for monozygotic twins and duplicate samples****, ~0.177 to add first-degree relations, etc.*

However, based on our replicated genotypes, we will increase this threshold substantially when combining inferences from PLINK and COLONY (to a mean kinship - across samples within clones - of at least 0.45).

Overall summaries of the matches

First let's see how many pairs of samples are identified as the same individual in our anlayses...

```
#Create a new data frame with info for each pair...
compare.out <- data.frame(id1 = kingrel$id1,
                          id2 = kingrel$id2,
                          kinship = kingrel$kinship,
                          nsnp = kingrel$nsnp,
                          colclone1 = NA,
                          colclone2 = NA,
                          isclonePLINK = NA,
                          iscloneCOL = NA)

#This loop takes a little bit of time (~50 minutes)
T1 <- Sys.time()
for(r in 1:nrow(kingrel)) {
  tmp.grep1 <- grep(compare.out$id1[r], col.out[,3])
  if(length(tmp.grep1) > 1) {
    for(i in 1:length(tmp.grep1)) {
      tmp.split <- strsplit(col.out[tmp.grep1[i],3], ",")[[1]]
      if(compare.out$id1[r] %in% tmp.split) {
        compare.out$colclone1[r] <- col.out$CloneIndex[tmp.grep1[i]]
      }
      rm(tmp.split)
    }
  } else {
```

```r
    compare.out$colclone1[r] <- col.out$CloneIndex[tmp.grep1]
  }

  tmp.grep2 <- grep(compare.out$id2[r], col.out[,3])
  if(length(tmp.grep2) > 1) {
    for(i in 1:length(tmp.grep2)) {
      tmp.split <- strsplit(col.out[tmp.grep2[i],3], ",")[[1]]
      if(compare.out$id2[r] %in% tmp.split) {
        compare.out$colclone2[r] <- col.out$CloneIndex[tmp.grep2[i]]
      }
      rm(tmp.split)
    }
  } else {
    compare.out$colclone2[r] <- col.out$CloneIndex[tmp.grep2]
  }

  rm(tmp.grep1, tmp.grep2)

  if(compare.out$colclone1[r] == compare.out$colclone2[r]) {
    compare.out$iscloneCOL[r] <- TRUE
  } else {
    compare.out$iscloneCOL[r] <- FALSE
  }

  if(!is.na(compare.out$kinship[r])) {
    if(compare.out$kinship[r] >= 0.354) {
      compare.out$isclonePLINK[r] <- TRUE
    } else {
      compare.out$isclonePLINK[r] <- FALSE
    }
  } else {
    compare.out$isclonePLINK[r] <- FALSE
  }

}
Sys.time()-T1
```

```
## Time difference of 3.907655 hours
```

```r
#How many pairs identified by COLONY are also identified by PLINK?
table(compare.out$isclonePLINK[compare.out$iscloneCOL == TRUE])
```

```
##
##  TRUE
## 16176
```

```r
#How many pairs identified by PLINK are also identified by COLONY?
table(compare.out$iscloneCOL[compare.out$isclonePLINK == TRUE])
```

```
##
## FALSE  TRUE
## 20688 16176
```

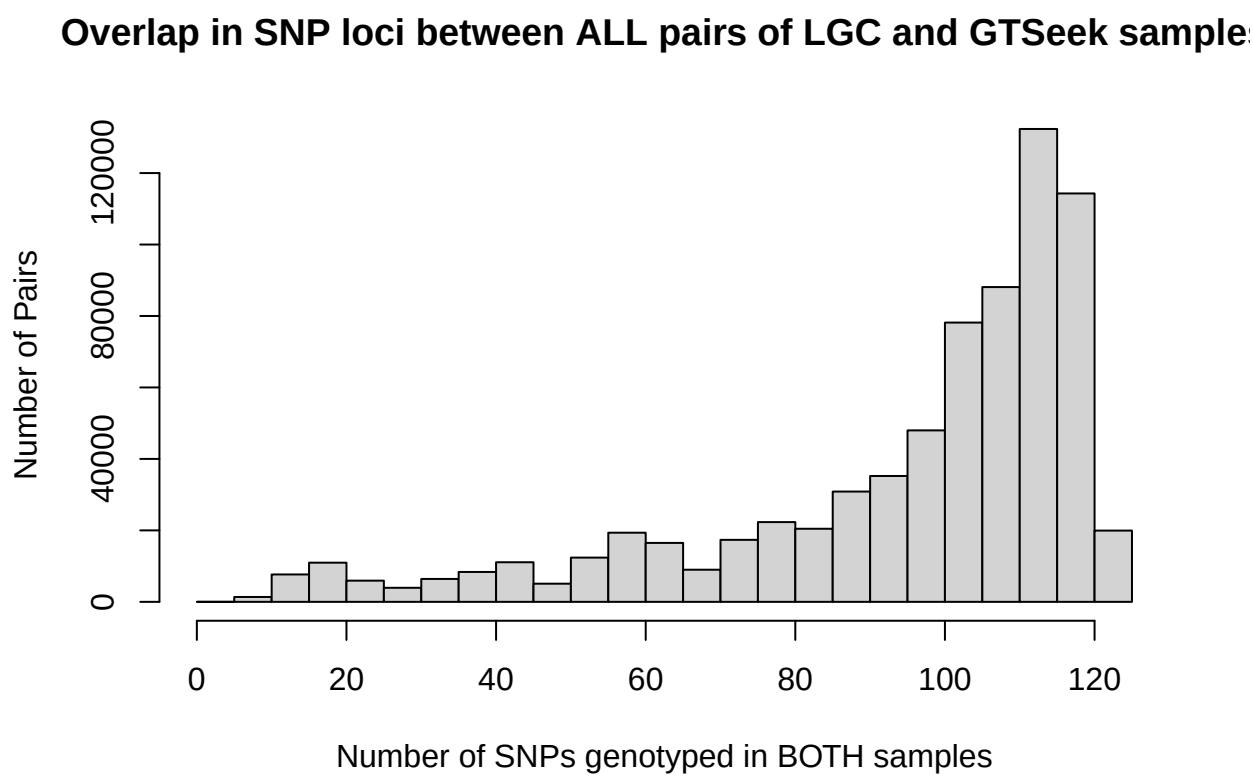
From a total of 1485226 pairwise comparisons, COLONY identifies 16176 matches, while PLINK identifies 36864 matches. There are 20688 comparisons where PLINK finds a match but COLONY does not and 0 comparisons where COLONY finds a match but PLINK does not. In other words 100% of COLONY matches are confirmed by PLINK, while 43.9% of PLINK matches are confirmed by COLONY.

Let's look more closely at some of the discrepancies between the two analyses. First those from the COLONY analysis *only*..

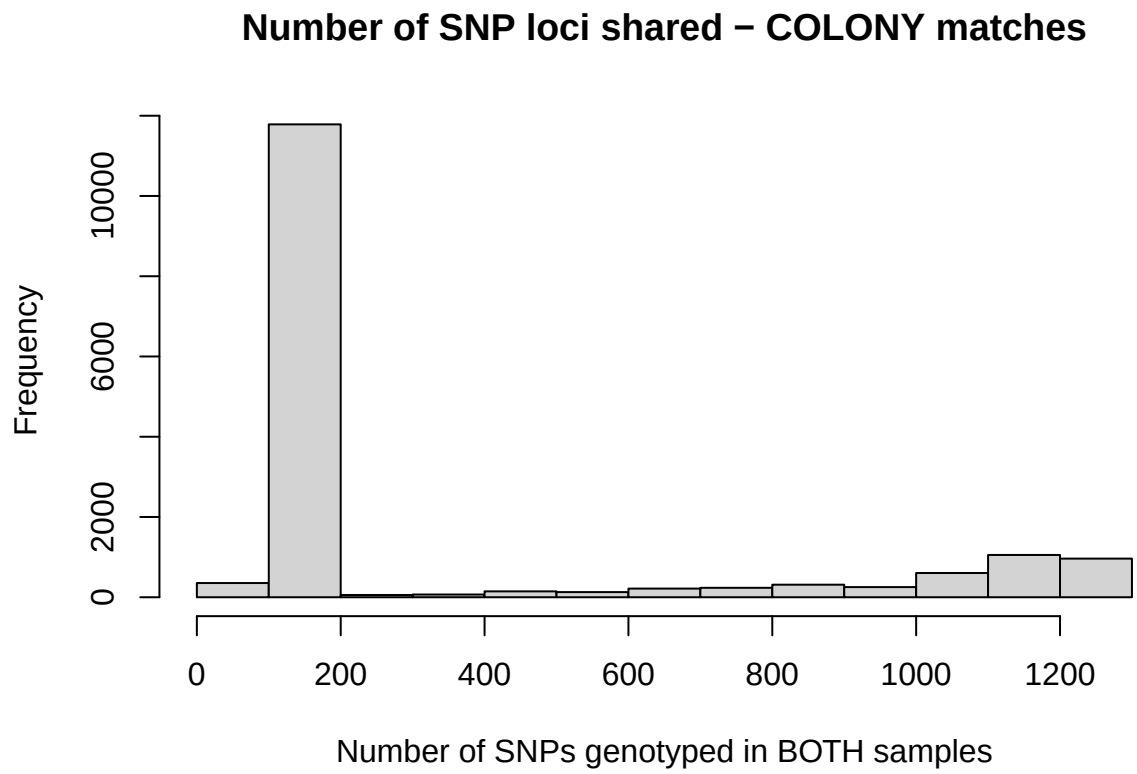
```
compare.out[which(compare.out$iscloneCOL == TRUE & compare.out$isclonePLINK == FALSE),]
```

```
## [1] id1          id2          kinship      nsnp         colclone1
## [6] colclone2    isclonePLINK iscloneCOL
## <0 rows> (or 0-length row.names)
```

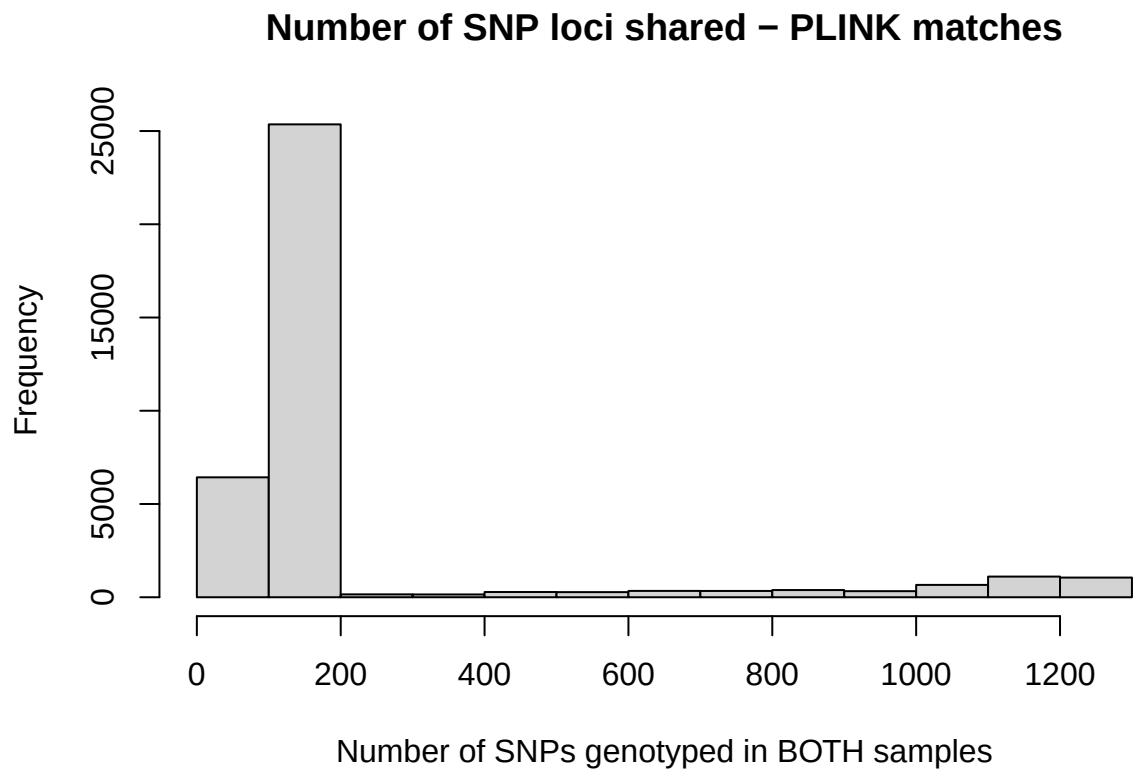
```
hist(compare.out$nsnp[which(compare.out$id1 %in% smd$vcfID[which(smd$GTservice == "GTSeek")] &
                        compare.out$id2 %in% smd$vcfID[which(smd$GTservice == "LGC")] |
                        compare.out$id1 %in% smd$vcfID[which(smd$GTservice == "LGC")] &
                        compare.out$id2 %in% smd$vcfID[which(smd$GTservice == "GTSeek")])],
     main = "Overlap in SNP loci between ALL pairs of LGC and GTSeek samples",
     xlab = "Number of SNPs genotyped in BOTH samples", ylab = "Number of Pairs")
```




```
hist(compare.out$nsnp[which(compare.out$iscloneCOL == TRUE)],
     main = "Number of SNP loci shared - COLONY matches",
     xlab = "Number of SNPs genotyped in BOTH samples")
```




```
hist(compare.out$nsnp[which(compare.out$isclonePLINK == TRUE)],
     main = "Number of SNP loci shared - PLINK matches",
     xlab = "Number of SNPs genotyped in BOTH samples")
```




```
#Range of shared SNPs for COLONY matches b/w GTSeek and LGC
colMatches <- compare.out[which(compare.out$iscloneCOL==TRUE),]
range(colMatches$nsnp[which(colMatches$id1 %in% smd$vcfID[which(smd$GTservice == "GTSeek")] &
                          colMatches$id2 %in% smd$vcfID[which(smd$GTservice == "LGC")] |
                          colMatches$id1 %in% smd$vcfID[which(smd$GTservice == "LGC")] &
                          colMatches$id2 %in% smd$vcfID[which(smd$GTservice == "GTSeek")])])
```

```
## [1]  16 122
```

```
#Same for PLINK
plinkMatches <- compare.out[which(compare.out$isclonePLINK == TRUE),]
range(plinkMatches$nsnp[which(plinkMatches$id1 %in% smd$vcfID[which(smd$GTservice == "GTSeek")] &
                          plinkMatches$id2 %in% smd$vcfID[which(smd$GTservice == "LGC")] |
                          plinkMatches$id1 %in% smd$vcfID[which(smd$GTservice == "LGC")] &
                          plinkMatches$id2 %in% smd$vcfID[which(smd$GTservice == "GTSeek")])])
```

```
## [1]   8 125
```

The filters for LGC samples (at least 100 SNPs genotyped) keep us from matching samples based on *very* few loci.

Final Recapture Calls

Now to combine the output from the two programs using the scheme described above. `COLONY` must be missing a few genotype matches. For each `COLONY` clone, are there samples that would have been combined

by PLINK? If so, are *all* members of those clones also identified by PLINK as recaptures of the focal group? What is the mean kinship coefficient estimated between the samples associated with the two COLONY clones?

Below, we set two thresholds for re-assigning a sample to a different COLONY clone. These include:

i. Mean kinship to samples in the recipient clone ≥ 0.45 (based on range of kinship estimates for technical replicates)
ii. At least one of the comparisons to samples in the other clone needs to be based on a minimum of 50 SNP loci (this number will show up again below)

Now to reassign samples to clones...

```
## Set up the object
totClones <- max(c(compare.out$colclone1, compare.out$colclone2))
finalCloneDF <- data.frame(initial = c(1:totClones),
                           init.size = NA,
                           prob = NA,
                           final = c(1:totClones),
                           possibles = NA,
                           is.moved = FALSE,
                           is.recip = FALSE)

## Fill in init.size & prob
for(i in 1:totClones) {
  tmpclone <- strsplit(col.out[which(col.out$CloneIndex == i),3], ",")[[1]]
  finalCloneDF$init.size[i] <- length(tmpclone)
  finalCloneDF$prob[i] <- col.out$Prob[which(col.out$CloneIndex == i)]
  rm(tmpclone)
}

## Loop over the clones and identify all possible matches
for(i in 1:totClones) {
  tmp <- compare.out[which(compare.out$colclone1 == i & compare.out$colclone2 != i |
                             compare.out$colclone2 == i & compare.out$colclone1 != i),]
  ## If no other clones are a possible match for the focal clone
  ##  set the final clone ID, it doesn't move, and it's not a recipient
  if(sum(tmp$isclonePLINK) == 0) {
    finalCloneDF$final[i] <- i
  } else {
    ## If some clones do match...
    ##  identify the set of clones that match the focal clone
    tmp.match <- unique(c(tmp$colclone1[tmp$isclonePLINK==TRUE],
                          tmp$colclone2[tmp$isclonePLINK==TRUE]))
    tmp.match <- tmp.match[-which(tmp.match == i)]
    ##  then continue adding possible matches to all of these other clones
    ##  do this until the list of possibilities stops growing!!
    growing <- TRUE
    while(growing) {
      tmp.tmp.match <- unique(
        c(compare.out$colclone1[which(compare.out$colclone1 %in% tmp.match &
                                      compare.out$colclone2 %in% tmp.match &
                                      compare.out$iscloneCOL == FALSE &
                                      compare.out$isclonePLINK == TRUE)],
          compare.out$colclone2[which(compare.out$colclone1 %in% tmp.match &
```

```
                                    compare.out$colclone2 %in% tmp.match &
                                    compare.out$iscloneCOL == FALSE &
                                    compare.out$isclonePLINK == TRUE)]))
  tmp.tmp.match <- tmp.tmp.match[-which(tmp.tmp.match == i)]
  if(length(tmp.tmp.match) > length(tmp.match)) {
    tmp.match <- tmp.tmp.match
  } else {
    growing <- FALSE
  }
  rm(tmp.tmp.match)
}

#Record the possible matches...
finalCloneDF$possibles[i] <- paste0(tmp.match, collapse="_")

## If any of these clones have been moved to a new clone...
##  replace them in the tmp.match list with their new clone
for(j in tmp.match) {
  if(finalCloneDF$final[which(finalCloneDF$initial == j)] != j) {
    tmp.match[which(tmp.match == j)] <- finalCloneDF$final[which(finalCloneDF$initial == j)]
  }
}
tmp.match <- unique(tmp.match[which(tmp.match != i)])
## This means we'll need to refer to the final column when calculating
##  mean kinship values for the pairs below...

## Now check each of these to ask if it should be moved...
real.matches <- i #The focal clone, which can not have moved yet...

for(j in tmp.match) {
  i.set <- unique(c(
    real.matches, #The focal clone plus any earlier in the list that matched
    #Other clones previously moved into the focal clone
    finalCloneDF$initial[which(finalCloneDF$final == i)]
  ))
  j.set <- unique(c(
    j, #The possible match
    #If j has moved, it's final clone
    finalCloneDF$final[which(finalCloneDF$initial == j)],
    #Other clones that previously moved into j's final clone
    finalCloneDF$initial[which(finalCloneDF$final ==
                                 finalCloneDF$final[which(finalCloneDF$initial ==
                                                          j)])]
  ))

  subtmp <- compare.out[which(compare.out$colclone1 %in% i.set &
                              compare.out$colclone2 %in% j.set |
                            compare.out$colclone1 %in% j.set &
                              compare.out$colclone2 %in% i.set),]

  ## Focus on comparisons involving the possible mover - j
  #subtmp <- subtmp[which(subtmp$colclone1 == j | subtmp$colclone2 == j),]
```

```r
      ## Check mean kinship
      if(mean(subtmp$kinship) >= 0.45 & max(subtmp$nsnp) >= 50) {
        real.matches <- c(real.matches, j)
      }
      rm(subtmp, i.set, j.set)
    }

    ## Now if there are real matches, figure out where to put the set of samples
    if(length(real.matches) > 1) {
      tmp.sizes <- data.frame(finalClone=unique(finalCloneDF$final[real.matches]),
                              finalSize = NA)
      for(r in 1:nrow(tmp.sizes)) {
        tmp.sizes$finalSize[r] <- sum(
          finalCloneDF$init.size[which(finalCloneDF$final ==
                                       tmp.sizes$finalClone[r])]
          )
      }
      bigclone <- tmp.sizes$finalClone[which(tmp.sizes$finalSize ==
                                             max(tmp.sizes$finalSize))]
      if(length(bigclone) > 1) {
        bigclone <- min(bigclone)
      }

      finalCloneDF$final[real.matches] <- bigclone

      rm(tmp.sizes, bigclone)
    }
    rm(tmp.match)
  }
}

#Did clone move? Was it a recipient of movers?
for(i in 1:totClones) {
  if(finalCloneDF$initial[i] != finalCloneDF$final[i]) {
    finalCloneDF$is.moved[i] <- TRUE
  }
  if(length(which(finalCloneDF$final==i)) > 1) {
    finalCloneDF$is.recip[i] <- TRUE
  }
}

#Structure of the object
head(finalCloneDF)
```

```
##   initial init.size  prob final                          possibles is.moved
## 1       1         6 1.000     1                                297    FALSE
## 2       2         1 0.221   324  33_107_171_234_260_277_296_300_324     TRUE
## 3       3        15 0.591     3                        223_288_363    FALSE
## 4       4        42 0.924     4 6_67_71_225_242_263_273_287_294_313    FALSE
## 5       5         5 0.955     5                               <NA>    FALSE
## 6       6         1 1.000     6         4_67_71_174_226_252_294_313    FALSE
##   is.recip
## 1     TRUE
```

```
## 2    FALSE
## 3     TRUE
## 4     TRUE
## 5    FALSE
## 6    FALSE
```

```r
#How many clones total from COLONY
nrow(finalCloneDF)
```

```
## [1] 407
```

```r
#How many survive?
length(unique(finalCloneDF$final))
```

```
## [1] 345
```

```r
#Difference
length(unique(finalCloneDF$final)) - nrow(finalCloneDF)
```

```
## [1] -62
```

```r
#How many moved?
sum(finalCloneDF$initial != finalCloneDF$final)
```

```
## [1] 62
```

```r
sum(finalCloneDF$is.moved)
```

```
## [1] 62
```

```r
#Into how many recipients?
sum(finalCloneDF$is.recip)
```

```
## [1] 37
```

```r
#Shouldn't be any recipients that were moved
table(finalCloneDF$is.moved[which(finalCloneDF$is.recip == TRUE)])
```

```
##
## FALSE
##    37
```

```r
#Did clones reassigned to other clones tend to be sparse?
boxplot(finalCloneDF$init.size,
        finalCloneDF$init.size[finalCloneDF$is.moved],
        finalCloneDF$init.size[!finalCloneDF$is.moved],
        names = c("ALL","YES","NO"),
        xlab = "Clone Moved?",
        ylab = "Number of samples in clone")
```

```
#It doesn't appear that they were

#Did clones that were moved tend to have lower probabilities from COLONY?
boxplot(finalCloneDF$prob,
        finalCloneDF$prob[finalCloneDF$is.moved],
        finalCloneDF$prob[!finalCloneDF$is.moved],
        names = c("ALL","YES","NO"),
        xlab = "Clone Moved?",
        ylab = 'Clone "Probability" (from COLONY)')
```

```
#Yes, COLONY tended to be less confident about clones that were reassigned...
```

Interested to look more closely at clones that were moved by this combination. In those cases, there should be more than one row of the dataset (initial `COLONY` clone) that gets assigned to a particular final clone.

```
#37 clones were the recipients of other clones
table(finalCloneDF$final)
```

```
##
##   1   3   4   5   6   7   8  11  14  15  17  18  19  20  21  22  24  25  30  35
##   2   2   6   1   1   3   2   1   3   1   1   1   1   1   1   1   1   1   1   1
##  36  38  39  40  42  43  44  45  46  47  49  50  51  52  53  54  55  56  58  59
##   1   1   1   1   1   1   1   2   1   1   1   1   1   1   1   1   1   1   1   1
##  60  61  62  64  66  68  69  70  71  72  73  74  75  76  77  78  79  80  82  83
##   1   1   1   1   1   1   1   1   1   2   1   1   1   1   1   1   1   1   1   1
##  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98 100 101 102 104 105
##   1   1   1   1   2   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 108 109 110 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 169 170 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 189 190
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
```

```
## 192 194 195 197 198 199 200 201 202 204 205 206 207 208 209 210 211 212 213 214
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 215 216 217 218 220 221 224 226 227 228 229 230 231 232 233 234 235 236 237 238
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 259 260 261 262 264 265 266 267 268 269 270 271 272 273 274 275 276 278 279 280
##   1   1   1   1   1   5   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 281 283 284 285 286 287 288 289 290 291 293 295 298 299 302 303 306 307 308 310
##   1   1   1   1   1   1   1   1   1   1   2   2   1   2   2   1   2   6   2   2
## 312 315 316 317 318 319 321 322 323 324 327 328 329 330 333 334 335 336 337 338
##   2   2   2   2   2   2   2   1   4   8   3   3   1   1   2   3   2   1   1   1
## 340 342 343 344 345 346 347 348 349 350 353 354 355 356 357 358 359 360 361 362
##   3   1   2   1   1   2   1   1   1   1   2   1   1   1   1   2   1   1   1   1
## 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 403 404 405 406 407
##   1   1   1   1   1
```

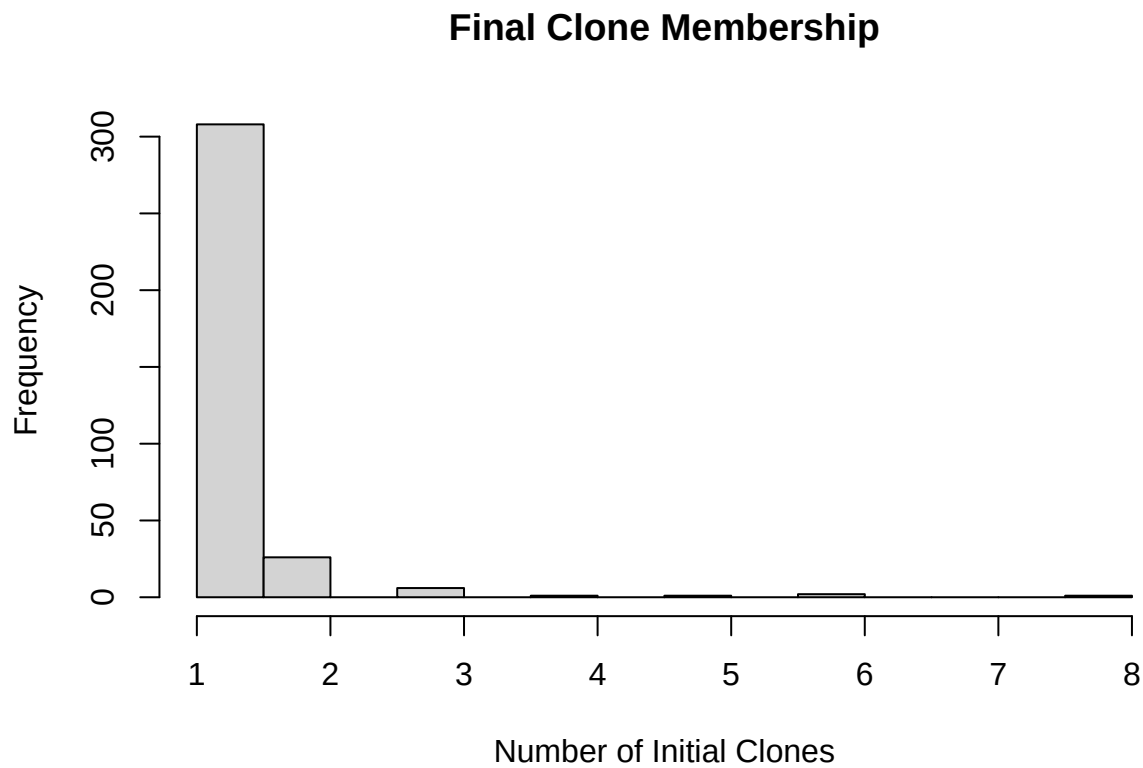
```r
sum(table(finalCloneDF$final) > 1)
```

```
## [1] 37
```

```r
names(which(table(finalCloneDF$final) > 1))
```

```
##  [1] "1"   "3"   "4"   "7"   "8"   "14"  "45"  "72"  "88"  "265" "293" "295"
## [13] "299" "302" "306" "307" "308" "310" "312" "315" "316" "317" "318" "319"
## [25] "321" "323" "324" "327" "328" "333" "334" "335" "340" "343" "346" "353"
## [37] "358"
```

```r
hist(table(finalCloneDF$final), main = "Final Clone Membership",
     xlab = "Number of Initial Clones")
```




```
#Spot checking moved clones...
#Final Clone 4 - 5 other clones moved into clone 4
cloneInQ <- 4
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

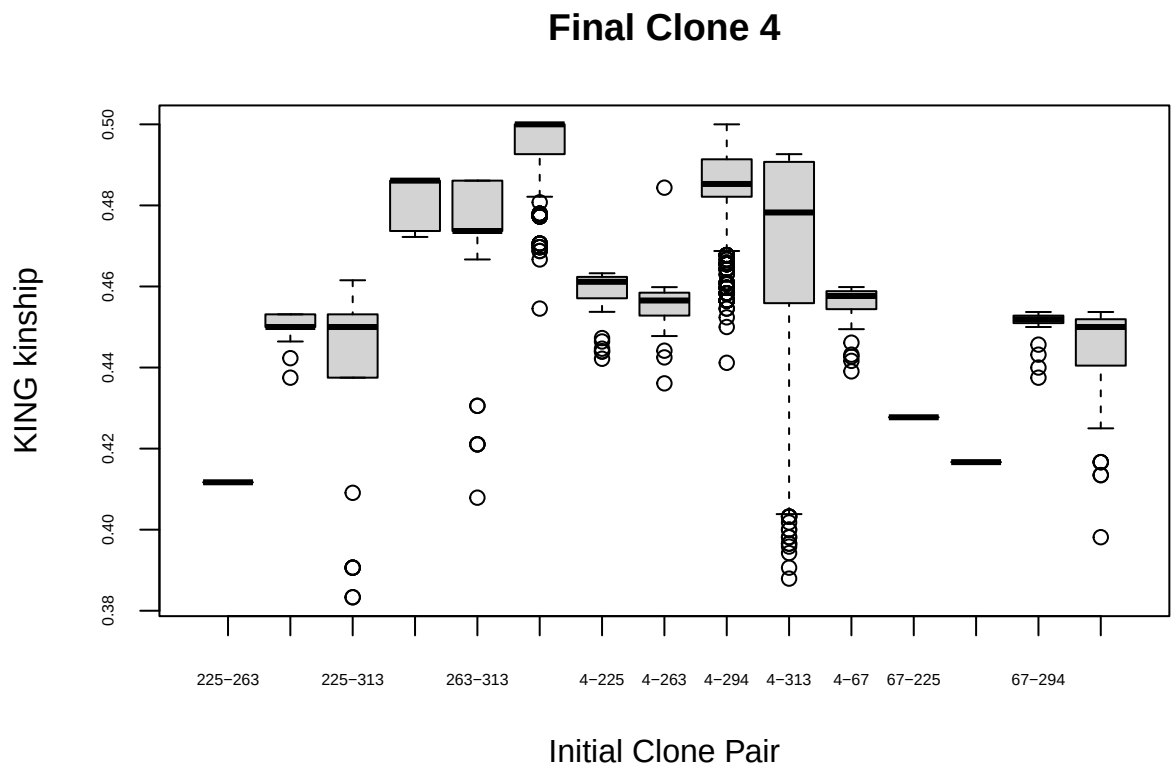
```
##     initial init.size  prob final                          possibles is.moved
## 4         4        42 0.924     4 6_67_71_225_242_263_273_287_294_313    FALSE
## 67       67         1 0.571     4  4_71_225_242_263_273_287_294_313_6     TRUE
## 225     225         1 0.469     4            4_242_263_273_294_313_67     TRUE
## 263     263         1 0.260     4        273_4_287_294_313_67_225_242     TRUE
## 294     294        35 0.493     4       313_4_6_67_71_225_242_263_273     TRUE
## 313     313        29 0.493     4   4_294_67_71_225_242_263_273_6_287     TRUE
##     is.recip
## 4       TRUE
## 67     FALSE
## 225    FALSE
## 263    FALSE
## 294    FALSE
## 313    FALSE
```

```
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final                          possibles is.moved
```

```
## 4          4        42 0.924     4 6_67_71_225_242_263_273_287_294_313    FALSE
## 6          6         1 1.000     6         4_67_71_174_226_252_294_313    FALSE
## 67        67         1 0.571     4  4_71_225_242_263_273_287_294_313_6     TRUE
## 71        71         1 1.000    71              4_174_252_294_313_6_67    FALSE
## 225      225         1 0.469     4            4_242_263_273_294_313_67     TRUE
## 242      242         1 0.611   242        4_263_273_287_294_313_67_225    FALSE
## 263      263         1 0.260     4        273_4_287_294_313_67_225_242     TRUE
## 273      273         1 0.451   273            4_294_313_67_225_242_263    FALSE
## 287      287         1 0.812   287                    313_4_67_242_263    FALSE
## 294      294        35 0.493     4       313_4_6_67_71_225_242_263_273     TRUE
## 313      313        29 0.493     4   4_294_67_71_225_242_263_273_6_287     TRUE
##     is.recip
## 4       TRUE
## 6      FALSE
## 67     FALSE
## 71     FALSE
## 225    FALSE
## 242    FALSE
## 263    FALSE
## 273    FALSE
## 287    FALSE
## 294    FALSE
## 313    FALSE
```

```r
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```




```
#Final Clone 7 - 2 other clones moved into clone 7
cloneInQ <- 7
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

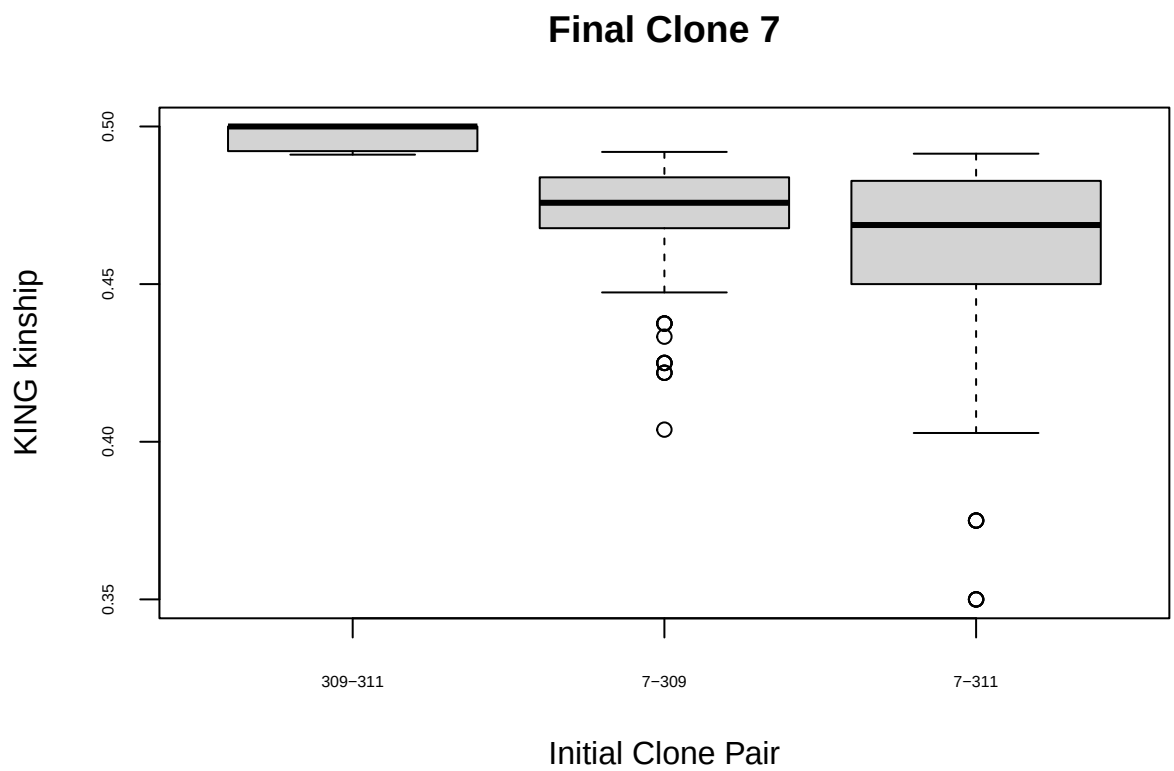
```
##     initial init.size  prob final possibles is.moved is.recip
## 7         7        31 0.992     7   309_311    FALSE     TRUE
## 309     309         9 0.992     7     311_7     TRUE    FALSE
## 311     311        15 0.992     7     309_7     TRUE    FALSE
```

```
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final possibles is.moved is.recip
## 7         7        31 0.992     7   309_311    FALSE     TRUE
## 309     309         9 0.992     7     311_7     TRUE    FALSE
## 311     311        15 0.992     7     309_7     TRUE    FALSE
```

```
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
```

```r
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```



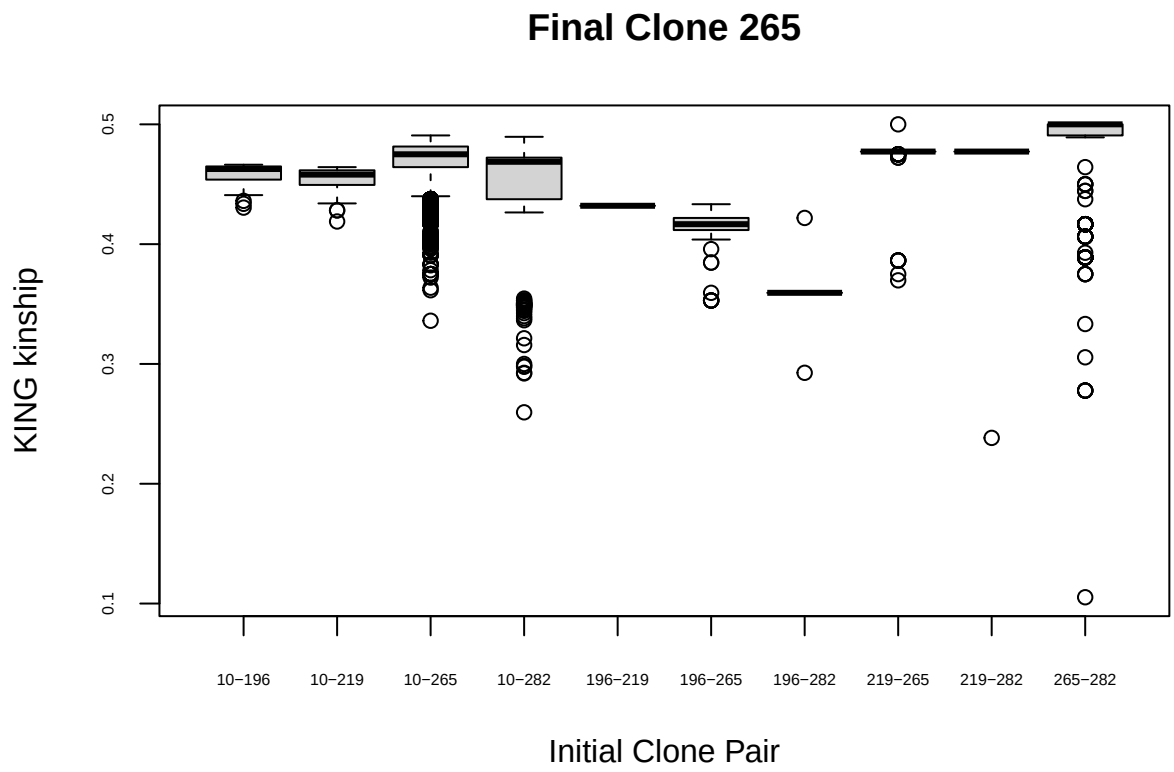

```r
#Final Clone 265 - 4 other clones moved into clone 265
cloneInQ <- 265
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

```
##     initial init.size  prob final                     possibles is.moved
## 10       10        32 0.757   265    196_219_233_257_265_282_290     TRUE
## 196     196         1 0.653   265     10_219_233_257_265_290_282     TRUE
## 219     219         1 0.045   265         10_257_265_290_282_196     TRUE
## 265     265       108 0.500   265 10_282_196_219_229_233_257_290    FALSE
## 282     282         5 0.500   265     265_10_196_219_233_257_290     TRUE
##     is.recip
## 10     FALSE
## 196    FALSE
## 219    FALSE
## 265     TRUE
## 282    FALSE
```

```r
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final                      possibles is.moved
## 10       10        32 0.757   265    196_219_233_257_265_282_290     TRUE
## 196     196         1 0.653   265     10_219_233_257_265_290_282     TRUE
## 219     219         1 0.045   265         10_257_265_290_282_196     TRUE
## 229     229         1 0.135   229                       257_265    FALSE
## 233     233         1 0.558   233         10_257_290_265_282_196    FALSE
## 257     257         1 0.439   257 10_290_265_282_196_219_229_233    FALSE
## 265     265       108 0.500   265 10_282_196_219_229_233_257_290    FALSE
## 282     282         5 0.500   265     265_10_196_219_233_257_290     TRUE
## 290     290         1 0.529   290     265_282_10_196_219_233_257    FALSE
##     is.recip
## 10     FALSE
## 196    FALSE
## 219    FALSE
## 229    FALSE
## 233    FALSE
## 257    FALSE
## 265     TRUE
## 282    FALSE
## 290    FALSE
```

```r
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```



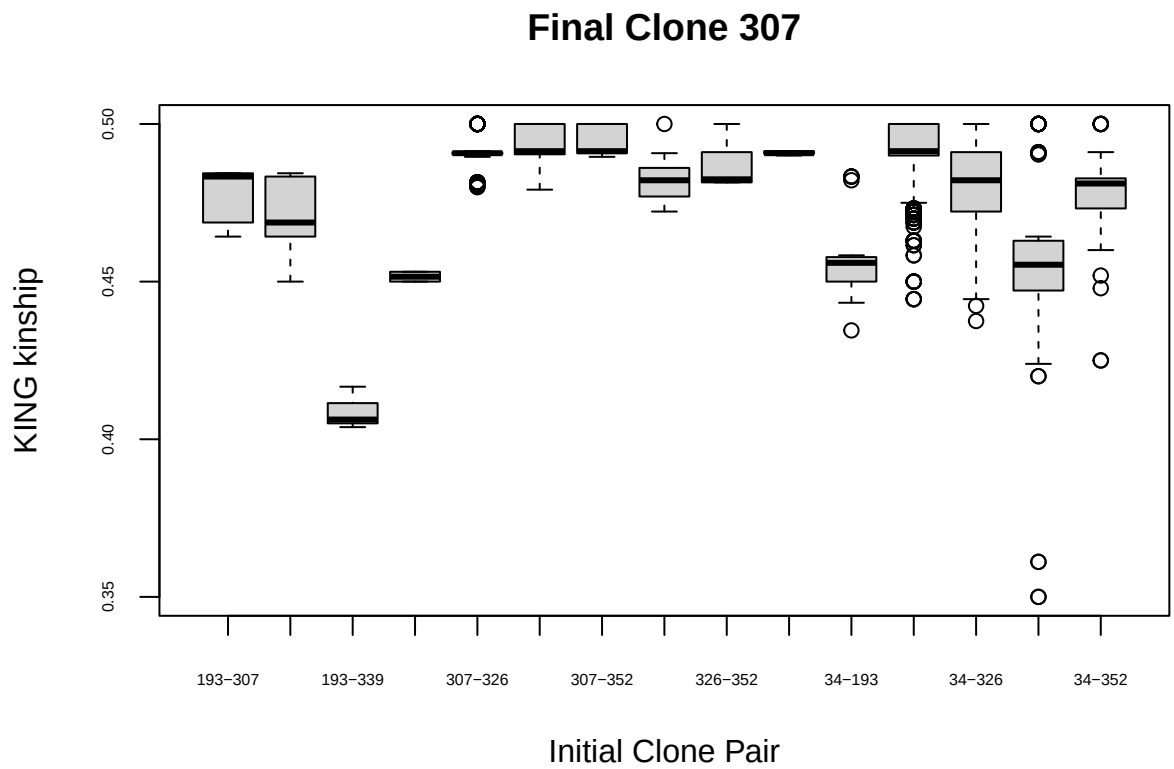

```
#Final Clone 307 - 5 other clones moved into clone 307
cloneInQ <- 307
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

```
##     initial init.size  prob final              possibles is.moved is.recip
## 34       34        27 0.337   307 193_256_307_326_339_352     TRUE    FALSE
## 193     193         1 0.276   307  34_256_307_326_339_352     TRUE    FALSE
## 307     307        28 0.337   307  326_34_339_352_193_256    FALSE     TRUE
## 326     326         5 0.337   307  307_34_339_352_193_256     TRUE    FALSE
## 339     339         4 0.337   307  34_326_352_307_193_256     TRUE    FALSE
## 352     352         2 0.337   307  307_339_326_34_193_256     TRUE    FALSE
```

```
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final              possibles is.moved is.recip
## 34       34        27 0.337   307 193_256_307_326_339_352     TRUE    FALSE
## 193     193         1 0.276   307  34_256_307_326_339_352     TRUE    FALSE
## 256     256         1 0.675   256  34_307_326_339_352_193    FALSE    FALSE
## 307     307        28 0.337   307  326_34_339_352_193_256    FALSE     TRUE
## 326     326         5 0.337   307  307_34_339_352_193_256     TRUE    FALSE
## 339     339         4 0.337   307  34_326_352_307_193_256     TRUE    FALSE
## 352     352         2 0.337   307  307_339_326_34_193_256     TRUE    FALSE
```

```r
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```




```r
#Final Clone 324 - 7 other clones moved into clone 324
cloneInQ <- 324
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

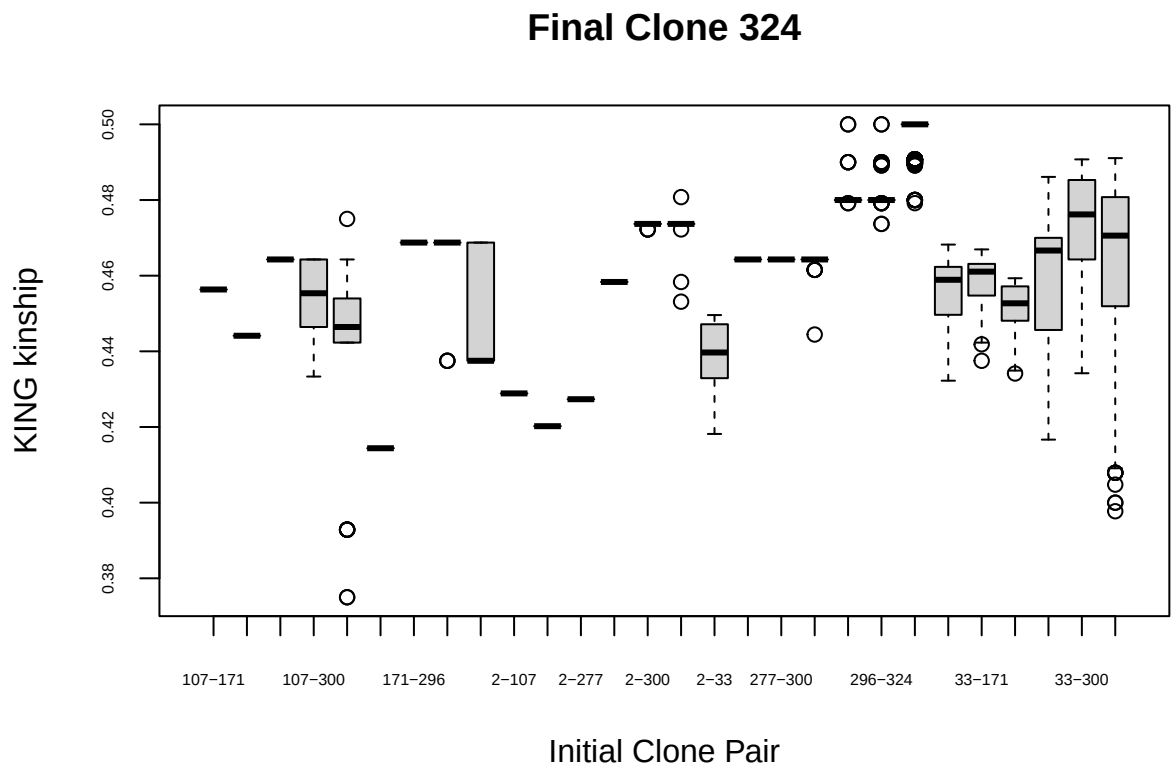
```
##     initial init.size  prob final                                possibles
## 2         2         1 0.221   324        33_107_171_234_260_277_296_300_324
## 33       33        34 0.346   324         107_171_234_260_277_296_300_324_2
## 107     107         1 0.289   324          33_171_234_260_277_296_300_324_2
## 171     171         1 0.332   324          33_234_260_277_296_300_324_2_107
## 277     277         1 0.451   324          33_296_300_324_2_107_171_234_260
## 296     296         2 0.346   324      300_324_2_33_107_171_234_260_269_277
## 300     300        18 0.346   324      324_296_2_33_107_171_234_260_269_277
## 324     324        36 0.346   324 300_296_2_33_107_171_234_258_260_269_277
```

```
##     is.moved is.recip
## 2       TRUE    FALSE
## 33      TRUE    FALSE
## 107     TRUE    FALSE
## 171     TRUE    FALSE
## 277     TRUE    FALSE
## 296     TRUE    FALSE
## 300     TRUE    FALSE
## 324    FALSE     TRUE
```

```
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final                                possibles
## 2         2         1 0.221   324       33_107_171_234_260_277_296_300_324
## 33       33        34 0.346   324        107_171_234_260_277_296_300_324_2
## 107     107         1 0.289   324         33_171_234_260_277_296_300_324_2
## 171     171         1 0.332   324         33_234_260_277_296_300_324_2_107
## 234     234         1 0.987   234             33_277_296_300_324_2_107_171
## 258     258         1 0.147   258           308_319_324_337_42_191_346_402
## 260     260         1 0.495   260             277_33_296_300_324_2_107_171
## 269     269         1 0.240   269                           292_296_300_324
## 277     277         1 0.451   324         33_296_300_324_2_107_171_234_260
## 296     296         2 0.346   324     300_324_2_33_107_171_234_260_269_277
## 300     300        18 0.346   324     324_296_2_33_107_171_234_260_269_277
## 324     324        36 0.346   324 300_296_2_33_107_171_234_258_260_269_277
##     is.moved is.recip
## 2       TRUE    FALSE
## 33      TRUE    FALSE
## 107     TRUE    FALSE
## 171     TRUE    FALSE
## 234    FALSE    FALSE
## 258    FALSE    FALSE
## 260    FALSE    FALSE
## 269    FALSE    FALSE
## 277     TRUE    FALSE
## 296     TRUE    FALSE
## 300     TRUE    FALSE
## 324    FALSE     TRUE
```

```
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```



Overall this looks pretty reasonble. There were some low-kinship outliers within some of these clones, though.

Another check, do we ever move samples into a clone that has already been removed from the population? In other words, clone B is identified as a recapture of clone A and moved accordingly. Subsequently, clone C is identified as a recapture of clone B (but not clone A) and moved into an "empty" clone. Does this happen?

```
#Any cases where a clone gets emptied out (moved to another, larger clone)
# but then another clone is assigned to the emptied clone?

#This should show up as rows where initial != final, but is.recip = TRUE
which(finalCloneDF$is.recip == TRUE &
        finalCloneDF$initial != finalCloneDF$final)
```

```
## integer(0)
```

```
# None... good!
```

Check for Odd Patterns in the FINAL clone assignments

Want to revisit the checks above, given our new clone assignments.

```
#Add COLclone to smd
smd$COLclone <- NA
smd$finalClone <- NA
```

```r
for(i in 1:nrow(smd)){
  thisclone <- col.out$CloneIndex[grep(smd$vcfID[i], col.out[,3])]

  if(length(thisclone) > 1) {
    this.this <- which(sapply(strsplit(col.out[thisclone,3],","), FUN=function(x) sum(x==smd$vcfID[i]))
    thisclone <- thisclone[this.this]
    rm(this.this)
  }

  if(length(thisclone) > 0) {
    smd$COLclone[i] <- thisclone
    smd$finalClone[i] <- finalCloneDF$final[finalCloneDF$initial == smd$COLclone[i]]
  }
  rm(thisclone)

  if(smd$FILT[i] == TRUE) {
    smd$finalClone[i] <- NA
  }
}
#Samples without a clone ID were those that were filtered from the dataset
sum(is.na(smd$finalClone))
```

```
## [1] 485
```

```r
sum(smd$FILT)
```

```
## [1] 485
```

```r
length(which(is.na(smd$finalClone) & smd$FILT))
```

```
## [1] 485
```

```r
nSex2 <- nSexGT2 <- nRegion2 <- c()
for(i in 1:max(smd$finalClone, na.rm = TRUE)) {
  thisclone <- which(smd$finalClone == i)
  if(length(thisclone) == 0) {
    nSex2[i] <- NA
    nSexGT2[i] <- NA
    nRegion2[i] <- NA
  } else {
    tmp <- smd[thisclone,]
    nSex2[i] <- length(unique(tmp$knownSEX[tmp$knownSEX != "" & !is.na(tmp$knownSEX)]))
    nSexGT2[i] <- length(unique(tmp$sexGT[tmp$sexGT != "undet" & !is.na(tmp$sexGT)]))
    nRegion2[i] <- length(unique(tmp$region))
    rm(tmp)
  }
}
```

Replicated samples

We don't want to have any replicates that are assigned to different final clones!

```r
repIDs <- unique(sub("R","",smd$ugaID[which(smd$is.rep == TRUE)]))
check.rep.clone <- data.frame(ugaID = repIDs,
                              clone1 = NA,
                              clone2 = NA)
for(i in 1:length(repIDs)) {
  tmp.match <- c(repIDs[i], paste0(repIDs[i],"R"))
  check.rep.clone[i,-1] <- smd$finalClone[which(smd$ugaID %in% tmp.match)]
  rm(tmp.match)
}
check.rep.clone
```

```
##      ugaID clone1 clone2
## 1   35-SGA     NA     NA
## 2   121-23     NA     NA
## 3     8-23     72     72
## 4   47-SGA     83     83
## 5     23-C      3      3
## 6   58-SGA    102    102
## 7   306-23     NA     NA
## 8   166-24    265    265
## 9   68-NGA     17     17
## 10   21-23     45     45
## 11  251-23     40     40
## 12  331-23    302    302
## 13  409-23     14     14
## 14    14-C    328     NA
## 15  241-23      1      1
## 16  358-24    307    307
## 17  280-23     NA     NA
## 18   50-23      4      4
## 19  358-23     38     38
## 20    5-24    324    324
## 21  134-24     NA    324
## 22  219-24     30     30
## 23  277-24    346    346
## 24  397-24     25     25
## 25  423-24    317    317
## 26  434-24     NA     NA
## 27     1-C     20     20
## 28    17-C    141    141
## 29  227-23     NA     NA
## 30  389-24    328    328
## 31  306-24    265    265
## 32  930-23    293    293
## 33  564-24     25     25
## 34  781-23     14     14
## 35  798-23    328    328
## 36  651-23     14     14
## 37  517-24     NA     NA
## 38  716-23     NA     NA
## 39  726-24     NA     NA
## 40  980-24    340    340
## 41  935-23    307    307
```

```
## 42 1306-24    353    353
## 43  567-24     25     25
## 44  881-24    323    323
## 45 2972-24    328    328
## 46 1384-24     NA     NA
## 47  933-24    144    144
## 48 1522-24    312    312
## 49 3058-24    353    353
## 50 2665-24    394    394
## 51 2869-24    265    265
## 52 2926-24     NA     NA
## 53  151-24    307    307
## 54  396-24    265     NA
## 55  424-23     NA     NA
## 56  191-24     NA     NA
## 57  466-23     NA     NA
## 58  286-23     NA     NA
## 59  500-24    353    353
## 60  311-23    321     NA
```

```
noNAs <- check.rep.clone[which(rowSums(is.na(check.rep.clone))==0),]
nrow(noNAs)
```

```
## [1] 41
```

```
sum(noNAs$clone1 == noNAs$clone2)
```

```
## [1] 41
```

Good! We don't assign any replicated samples to separate clones!

Known Sex

As above, we should not have multiple known sexes per clone.

```
nSex2
```

```
##   [1]  0 NA  1  1  1  0  0  0 NA NA  1 NA NA  0  1 NA  1  1  1  1  1  0 NA  0  0
##  [26] NA NA NA NA  0 NA NA NA NA  0  0 NA  0  0  0 NA  0  0  0  0  1  1 NA  1  1
##  [51]  0  1  0  1  1  1 NA  1  0  1  1  1 NA  1 NA  0 NA  1  0  1  0  1  1  1  1
##  [76]  0 NA  1  0  1 NA  1  1  1  1  1  1  0  0  1  0  1  1  1  1  1  1  1 NA  1
## [101]  1  1 NA  1  1 NA NA  1  1  0 NA  0  1  1  0  1  1  1  1  1  1  1  1  1  1
## [126] NA  1  1  1  1  1  1  1  1  1  1  1  1  1  0  1  1  1  1  0  0  0  1  1  1
## [151]  1  1  1  1  1  1  0  0  1  1  1  0  0  1  1  1  0  1  0  0 NA  1  1  0  0
## [176]  1  0  1  1  1  0  0  1  0  1  1  0 NA  0  0 NA  1 NA  0  1 NA  1  0  0  1
## [201]  1  1 NA  0 NA  1  0  0  0  1  0  0  0  0  0  0  0  0 NA  1  0 NA NA  0 NA
## [226]  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0
## [251]  0  0  1  0  0  0  0  0  0  0  0  0 NA  0  1  0  0  0  0  0  1  0  0  0  1
## [276]  1 NA  0  0  0  0 NA  0  0  0  0  0  0  0  0  0 NA  0 NA  0 NA NA  0  1 NA
## [301] NA  0  0 NA NA  1  0  0 NA  0 NA  0 NA NA  0  0  1  1  1 NA  0  0  0  1 NA
## [326] NA  0  1  0  0 NA NA  1  1  1  0  0  0 NA  0 NA  0  0  0  0  0  0  0  0  0
```

```
## [351] NA NA  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [376]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [401]  0  0  0  0  0  0  0
```

```
sum(!is.na(nSex2))
```

```
## [1] 342
```

```
max(nSex2, na.rm = TRUE)
```

```
## [1] 1
```

Genotype-inferred Sex

And again, we would hope that there were relatively few cases of multiple genotype-inferred sexes within clones.

```
nSexGT2
```

```
##   [1]  1 NA  1  1  1  1  1  2 NA NA  1 NA NA  2  1 NA  1  1  1  1  1  1 NA  1  1
##  [26] NA NA NA NA  1 NA NA NA NA  1  1 NA  1  1  1 NA  1  1  1  1  1  2 NA  1  1
##  [51]  1  1  1  1  1  1 NA  1  0  1  1  1 NA  1 NA  1 NA  1  1  1  1  1  1  1  1
##  [76]  1 NA  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1
## [101]  1  1 NA  1  1 NA NA  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [126] NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  0  1  1  1  1  1
## [151]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1
## [176]  1  0  1  1  1  1  1  1  1  1  1  1 NA  1  1 NA  1 NA  1  1 NA  1  1  1  1
## [201]  1  1 NA  1 NA  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1 NA NA  1 NA
## [226]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [251]  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1  1  1  0  1  1
## [276]  0 NA  1  1  1  1 NA  1  0  1  1  1  1  1  1  1 NA  1 NA  1 NA NA  1  1 NA
## [301] NA  1  1 NA NA  1  2  1 NA  1 NA  1 NA NA  1  1  1  1  1 NA  1  1  1  1 NA
## [326] NA  1  1  1  1 NA NA  1  2  1  1  1  1 NA  1 NA  1  1  1  1  1  1  1  1  1
## [351] NA NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [376]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  0  1
## [401]  1  1  1  1  1  1  1
```

```
sum(!is.na(nSexGT2))
```

```
## [1] 342
```

```
max(nSexGT2, na.rm = TRUE)
```

```
## [1] 2
```

```
which(nSexGT2 > 1)
```

```
## [1]   8  14  47 307 334
```

```r
#They do in a few cases...
#8, 14, 47, 307, 334

#All of these were noted previously...
#Clone 8 - 17 M 1 F, no direct captures
table(smd$sexGT[smd$finalClone == 8])
```

```
##
##  F  M
##  1 17
```

```r
smd[which(smd$finalClone == 8),]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 258                     UGA_173101_P017_WF06   30-23   TRUE  FALSE    6 2023
## 356                     UGA_173101_P018_WF08   61-24   TRUE  FALSE    7 2024
## 365                     UGA_173101_P018_WG05  139-24   TRUE  FALSE    7 2024
## 464                     UGA_173101_P020_WA02  135-24   TRUE  FALSE    7 2024
## 480                     UGA_173101_P020_WB06  481-24   TRUE  FALSE    7 2024
## 526                     UGA_173101_P020_WF04  482-24   TRUE  FALSE    7 2024
## 725                     UGA_173101_P022_WF11  257-24   TRUE  FALSE    7 2024
## 751                     UGA_173101_P023_WA01  285-24   TRUE  FALSE    4 2024
## 825                     UGA_173101_P023_WG03  326-24   TRUE  FALSE    4 2024
## 828      GTseek_024_384i5_001_P098.001_5-23    5-23   TRUE  FALSE    6 2023
## 1213  GTseek_025_384i5_003_P098.005_738-24  738-24   TRUE  FALSE    7 2024
## 1451 GTseek_025_384i5_241_P098.006_2962-24 2962-24   TRUE  FALSE    4 2024
## 1615 GTseek_026_384i5_025_P098.010_2748-24 2748-24   TRUE  FALSE    7 2024
## 1715 GTseek_026_384i5_163_P098.009_2320-24 2320-24   TRUE  FALSE    7 2024
## 1760 GTseek_026_384i5_253_P098.010_2864-24 2864-24   TRUE  FALSE    7 2024
## 1792 GTseek_026_384i5_317_P098.010_2894-24 2894-24   TRUE  FALSE    4 2024
## 1816 GTseek_026_384i5_365_P098.009_2729-24 2729-24   TRUE  FALSE    7 2024
## 2095 GTseek_028_384i5_270_P098.UGA3_260-24  260-24   TRUE  FALSE    4 2024
##       snare       utmSnare
## 258  HS 012 258412x3586648
## 356  HS 012 258412x3586648
## 365  HS 012 258412x3586648
## 464  HS 008 256165x3589588
## 480  HS 012 258412x3586648
## 526  HS 008 256165x3589588
## 725  HS 012 258412x3586648
## 751  HS 001 248187x3586614
## 825  HS 001 248187x3586614
## 828  HS 012 258412x3586648
## 1213 HS 001 248187x3586614
## 1451 HS 001 248187x3586614
## 1615 HS 008 256165x3589588
## 1715 HS 008 256165x3589588
## 1760 HS 012 258412x3586648
## 1792 HS 001 248187x3586614
## 1816 HS 012 258412x3586648
## 2095 HS 001 248187x3586614
##                                                                                  fqfile
## 258  RAPiD-Genomics_F534_UGA_173101_P017_WF06_i5-512_i7-162_S14995_L003_R1_001.fastq
```

```
## 356  RAPiD-Genomics_F534_UGA_173101_P018_WF08_i5-537_i7-164_S15095_L003_R1_001.fastq
## 365  RAPiD-Genomics_F534_UGA_173101_P018_WG05_i5-537_i7-173_S15104_L003_R1_001.fastq
## 464    RAPiD-Genomics_F566_UGA_173101_P020_WA02_i5-505_i7-27_S3892_L001_R1_001.fastq
## 480     RAPiD-Genomics_F566_UGA_173101_P020_WB06_i5-505_i7-6_S3908_L001_R1_001.fastq
## 526    RAPiD-Genomics_F566_UGA_173101_P020_WF04_i5-505_i7-45_S3954_L001_R1_001.fastq
## 725   RAPiD-Genomics_F566_UGA_173101_P022_WF11_i5-1098_i7-51_S7653_L003_R1_001.fastq
## 751   RAPiD-Genomics_F566_UGA_173101_P023_WA01_i5-1099_i7-59_S7679_L003_R1_001.fastq
## 825   RAPiD-Genomics_F566_UGA_173101_P023_WG03_i5-1099_i7-22_S7753_L003_R1_001.fastq
## 828                                                                            <NA>
## 1213                                                                           <NA>
## 1451                                                                           <NA>
## 1615                                                                           <NA>
## 1715                                                                           <NA>
## 1760                                                                           <NA>
## 1792                                                                           <NA>
## 1816                                                                           <NA>
## 2095                                                                           <NA>
##      knownSEX  sampSOURCE region nSNPgt               vcfID  FILT GTservice
## 258            Hair Snare    CGA   1234 UGA_173101_P017_WF06 FALSE       LGC
## 356            Hair Snare    CGA   1246 UGA_173101_P018_WF08 FALSE       LGC
## 365            Hair Snare    CGA   1280 UGA_173101_P018_WG05 FALSE       LGC
## 464            Hair Snare    CGA   1186 UGA_173101_P020_WA02 FALSE       LGC
## 480            Hair Snare    CGA   1268 UGA_173101_P020_WB06 FALSE       LGC
## 526            Hair Snare    CGA   1183 UGA_173101_P020_WF04 FALSE       LGC
## 725            Hair Snare    CGA   1245 UGA_173101_P022_WF11 FALSE       LGC
## 751            Hair Snare    CGA   1021 UGA_173101_P023_WA01 FALSE       LGC
## 825            Hair Snare    CGA    564 UGA_173101_P023_WG03 FALSE       LGC
## 828      <NA> Hair Sample    CGA    104             GTS_5-23 FALSE    GTSeek
## 1213     <NA> Hair Sample    CGA    119           GTS_738-24 FALSE    GTSeek
## 1451     <NA> Hair Sample    CGA    119          GTS_2962-24 FALSE    GTSeek
## 1615     <NA> Hair Sample    CGA    122          GTS_2748-24 FALSE    GTSeek
## 1715     <NA> Hair Sample    CGA    124          GTS_2320-24 FALSE    GTSeek
## 1760     <NA> Hair Sample    CGA    124          GTS_2864-24 FALSE    GTSeek
## 1792     <NA> Hair Sample    CGA    124          GTS_2894-24 FALSE    GTSeek
## 1816     <NA> Hair Sample    CGA    121          GTS_2729-24 FALSE    GTSeek
## 2095     <NA> Hair Sample    CGA    112           GTS_260-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 258      M        8          8
## 356      M        8          8
## 365      M        8          8
## 464      M        8          8
## 480      M        8          8
## 526      M        8          8
## 725      M        8          8
## 751      M        8          8
## 825      M        8          8
## 828      F      292          8
## 1213     M      292          8
## 1451     M      292          8
## 1615     M      292          8
## 1715     M      292          8
## 1760     M      292          8
## 1792     M      292          8
## 1816     M      292          8
```

```
## 2095     M      292          8
```

```r
unique(smd$COLclone[which(smd$finalClone == 8)])
```

```
## [1]   8 292
```

```r
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 8)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 8)] &
              compare.out$iscloneCOL==FALSE,]
```

```
##                 id1                  id2  kinship nsnp colclone1 colclone2
## 264636     GTS_5-23 UGA_173101_P020_WB06 0.434783   97       292         8
## 264697     GTS_5-23 UGA_173101_P018_WG05 0.437500  100       292         8
## 264701     GTS_5-23 UGA_173101_P018_WF08 0.427083   97       292         8
## 264892     GTS_5-23 UGA_173101_P017_WF06 0.423913   96       292         8
## 265001     GTS_5-23 UGA_173101_P020_WA02 0.443182   87       292         8
## 265066     GTS_5-23 UGA_173101_P023_WG03 0.473684   50       292         8
## 265091     GTS_5-23 UGA_173101_P023_WA01 0.416667   68       292         8
## 265109     GTS_5-23 UGA_173101_P022_WF11 0.420455   91       292         8
## 265255     GTS_5-23 UGA_173101_P020_WF04 0.431818   91       292         8
## 523784   GTS_738-24 UGA_173101_P020_WB06 0.491667  112       292         8
## 523845   GTS_738-24 UGA_173101_P018_WG05 0.491935  115       292         8
## 523849   GTS_738-24 UGA_173101_P018_WF08 0.483333  110       292         8
## 524040   GTS_738-24 UGA_173101_P017_WF06 0.483333  110       292         8
## 524149   GTS_738-24 UGA_173101_P020_WA02 0.491379  102       292         8
## 524214   GTS_738-24 UGA_173101_P023_WG03 0.467391   60       292         8
## 524239   GTS_738-24 UGA_173101_P023_WA01 0.477273   78       292         8
## 524257   GTS_738-24 UGA_173101_P022_WF11 0.482759  106       292         8
## 524403   GTS_738-24 UGA_173101_P020_WF04 0.482143  105       292         8
## 732663  GTS_2962-24 UGA_173101_P020_WB06 0.500000  112       292         8
## 732724  GTS_2962-24 UGA_173101_P018_WG05 0.500000  114       292         8
## 732728  GTS_2962-24 UGA_173101_P018_WF08 0.491071  110       292         8
## 732919  GTS_2962-24 UGA_173101_P017_WF06 0.491071  109       292         8
## 733028  GTS_2962-24 UGA_173101_P020_WA02 0.490385  101       292         8
## 733093  GTS_2962-24 UGA_173101_P023_WG03 0.465909   60       292         8
## 733118  GTS_2962-24 UGA_173101_P023_WA01 0.488095   78       292         8
## 733136  GTS_2962-24 UGA_173101_P022_WF11 0.490385  106       292         8
## 733282  GTS_2962-24 UGA_173101_P020_WF04 0.480769  105       292         8
## 901161  GTS_2748-24 UGA_173101_P020_WB06 0.467742  115       292         8
## 901222  GTS_2748-24 UGA_173101_P018_WG05 0.468750  117       292         8
## 901226  GTS_2748-24 UGA_173101_P018_WF08 0.458333  112       292         8
## 901417  GTS_2748-24 UGA_173101_P017_WF06 0.459677  112       292         8
## 901526  GTS_2748-24 UGA_173101_P020_WA02 0.458333  105       292         8
## 901591  GTS_2748-24 UGA_173101_P023_WG03 0.468750   61       292         8
## 901616  GTS_2748-24 UGA_173101_P023_WA01 0.443182   79       292         8
## 901634  GTS_2748-24 UGA_173101_P022_WF11 0.456897  109       292         8
## 901780  GTS_2748-24 UGA_173101_P020_WF04 0.448276  108       292         8
## 1010339 GTS_2320-24 UGA_173101_P020_WB06 0.468750  117       292         8
## 1010400 GTS_2320-24 UGA_173101_P018_WG05 0.469697  119       292         8
## 1010404 GTS_2320-24 UGA_173101_P018_WF08 0.459677  114       292         8
## 1010595 GTS_2320-24 UGA_173101_P017_WF06 0.460938  114       292         8
## 1010704 GTS_2320-24 UGA_173101_P020_WA02 0.458333  106       292         8
## 1010769 GTS_2320-24 UGA_173101_P023_WG03 0.468750   62       292         8
```

```
## 1010794 GTS_2320-24 UGA_173101_P023_WA01 0.445652   81        292          8
## 1010812 GTS_2320-24 UGA_173101_P022_WF11 0.458333  111        292          8
## 1010958 GTS_2320-24 UGA_173101_P020_WF04 0.448276  109        292          8
## 1060704 GTS_2864-24 UGA_173101_P020_WB06 0.492188  117        292          8
## 1060765 GTS_2864-24 UGA_173101_P018_WG05 0.492424  119        292          8
## 1060769 GTS_2864-24 UGA_173101_P018_WF08 0.484375  114        292          8
## 1060960 GTS_2864-24 UGA_173101_P017_WF06 0.484375  114        292          8
## 1061069 GTS_2864-24 UGA_173101_P020_WA02 0.483333  106        292          8
## 1061134 GTS_2864-24 UGA_173101_P023_WG03 0.460000   63        292          8
## 1061159 GTS_2864-24 UGA_173101_P023_WA01 0.479167   81        292          8
## 1061177 GTS_2864-24 UGA_173101_P022_WF11 0.483871  111        292          8
## 1061323 GTS_2864-24 UGA_173101_P020_WF04 0.474138  109        292          8
## 1098911 GTS_2894-24 UGA_173101_P020_WB06 0.492188  117        292          8
## 1098972 GTS_2894-24 UGA_173101_P018_WG05 0.492424  119        292          8
## 1098976 GTS_2894-24 UGA_173101_P018_WF08 0.483871  114        292          8
## 1099167 GTS_2894-24 UGA_173101_P017_WF06 0.484848  114        292          8
## 1099276 GTS_2894-24 UGA_173101_P020_WA02 0.483871  106        292          8
## 1099341 GTS_2894-24 UGA_173101_P023_WG03 0.468750   62        292          8
## 1099366 GTS_2894-24 UGA_173101_P023_WA01 0.478261   81        292          8
## 1099384 GTS_2894-24 UGA_173101_P022_WF11 0.483333  111        292          8
## 1099530 GTS_2894-24 UGA_173101_P020_WF04 0.474138  109        292          8
## 1125758 GTS_2729-24 UGA_173101_P020_WB06 0.483871  114        292          8
## 1125819 GTS_2729-24 UGA_173101_P018_WG05 0.484375  117        292          8
## 1125823 GTS_2729-24 UGA_173101_P018_WF08 0.475000  112        292          8
## 1126014 GTS_2729-24 UGA_173101_P017_WF06 0.475806  112        292          8
## 1126123 GTS_2729-24 UGA_173101_P020_WA02 0.475000  105        292          8
## 1126188 GTS_2729-24 UGA_173101_P023_WG03 0.458333   62        292          8
## 1126213 GTS_2729-24 UGA_173101_P023_WA01 0.465909   80        292          8
## 1126231 GTS_2729-24 UGA_173101_P022_WF11 0.474138  108        292          8
## 1126377 GTS_2729-24 UGA_173101_P020_WF04 0.465517  108        292          8
## 1380299  GTS_260-24 UGA_173101_P020_WB06 0.461538  106        292          8
## 1380360  GTS_260-24 UGA_173101_P018_WG05 0.462963  108        292          8
## 1380364  GTS_260-24 UGA_173101_P018_WF08 0.453704  104        292          8
## 1380555  GTS_260-24 UGA_173101_P017_WF06 0.451923  102        292          8
## 1380664  GTS_260-24 UGA_173101_P020_WA02 0.447917   96        292          8
## 1380729  GTS_260-24 UGA_173101_P023_WG03 0.462500   57        292          8
## 1380754  GTS_260-24 UGA_173101_P023_WA01 0.452381   73        292          8
## 1380772  GTS_260-24 UGA_173101_P022_WF11 0.450000   99        292          8
## 1380918  GTS_260-24 UGA_173101_P020_WF04 0.458333   99        292          8
##         isclonePLINK iscloneCOL
## 264636          TRUE      FALSE
## 264697          TRUE      FALSE
## 264701          TRUE      FALSE
## 264892          TRUE      FALSE
## 265001          TRUE      FALSE
## 265066          TRUE      FALSE
## 265091          TRUE      FALSE
## 265109          TRUE      FALSE
## 265255          TRUE      FALSE
## 523784          TRUE      FALSE
## 523845          TRUE      FALSE
## 523849          TRUE      FALSE
## 524040          TRUE      FALSE
## 524149          TRUE      FALSE
```

```
## 524214          TRUE      FALSE
## 524239          TRUE      FALSE
## 524257          TRUE      FALSE
## 524403          TRUE      FALSE
## 732663          TRUE      FALSE
## 732724          TRUE      FALSE
## 732728          TRUE      FALSE
## 732919          TRUE      FALSE
## 733028          TRUE      FALSE
## 733093          TRUE      FALSE
## 733118          TRUE      FALSE
## 733136          TRUE      FALSE
## 733282          TRUE      FALSE
## 901161          TRUE      FALSE
## 901222          TRUE      FALSE
## 901226          TRUE      FALSE
## 901417          TRUE      FALSE
## 901526          TRUE      FALSE
## 901591          TRUE      FALSE
## 901616          TRUE      FALSE
## 901634          TRUE      FALSE
## 901780          TRUE      FALSE
## 1010339         TRUE      FALSE
## 1010400         TRUE      FALSE
## 1010404         TRUE      FALSE
## 1010595         TRUE      FALSE
## 1010704         TRUE      FALSE
## 1010769         TRUE      FALSE
## 1010794         TRUE      FALSE
## 1010812         TRUE      FALSE
## 1010958         TRUE      FALSE
## 1060704         TRUE      FALSE
## 1060765         TRUE      FALSE
## 1060769         TRUE      FALSE
## 1060960         TRUE      FALSE
## 1061069         TRUE      FALSE
## 1061134         TRUE      FALSE
## 1061159         TRUE      FALSE
## 1061177         TRUE      FALSE
## 1061323         TRUE      FALSE
## 1098911         TRUE      FALSE
## 1098972         TRUE      FALSE
## 1098976         TRUE      FALSE
## 1099167         TRUE      FALSE
## 1099276         TRUE      FALSE
## 1099341         TRUE      FALSE
## 1099366         TRUE      FALSE
## 1099384         TRUE      FALSE
## 1099530         TRUE      FALSE
## 1125758         TRUE      FALSE
## 1125819         TRUE      FALSE
## 1125823         TRUE      FALSE
## 1126014         TRUE      FALSE
## 1126123         TRUE      FALSE
```

```
## 1126188          TRUE      FALSE
## 1126213          TRUE      FALSE
## 1126231          TRUE      FALSE
## 1126377          TRUE      FALSE
## 1380299          TRUE      FALSE
## 1380360          TRUE      FALSE
## 1380364          TRUE      FALSE
## 1380555          TRUE      FALSE
## 1380664          TRUE      FALSE
## 1380729          TRUE      FALSE
## 1380754          TRUE      FALSE
## 1380772          TRUE      FALSE
## 1380918          TRUE      FALSE
```

```r
#Clone 14 - 59 F, 2 M, 2 undet, no direct captures
table(smd$sexGT[smd$finalClone == 14])
```

```
##
##     F     M undet
##    59     2     2
```

```r
smd[which(smd$finalClone == 14),]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 83                    UGA_173101_P015_WG11   11-23   TRUE  FALSE    6 2023
## 91                    UGA_173101_P015_WH07  432-23   TRUE  FALSE    4 2023
## 125                   UGA_173101_P016_WC05   73-23   TRUE  FALSE    6 2023
## 138                   UGA_173101_P016_WD06  179-23   TRUE  FALSE    7 2023
## 186                   UGA_173101_P016_WH06  312-23   TRUE  FALSE    6 2023
## 188                   UGA_173101_P016_WH08  330-23   TRUE  FALSE    6 2023
## 194                   UGA_173101_P017_WA02  393-23   TRUE  FALSE    5 2023
## 233                   UGA_173101_P017_WD05  417-23   TRUE  FALSE    4 2023
## 256                   UGA_173101_P017_WF04  468-23   TRUE  FALSE    5 2023
## 260                   UGA_173101_P017_WF08  409-23   TRUE   TRUE    5 2023
## 285                   UGA_173101_P017_WH09  409-23   TRUE   TRUE    5 2023
## 338                   UGA_173101_P018_WE02   52-23   TRUE  FALSE    4 2023
## 345                   UGA_173101_P018_WE09  206-23   TRUE  FALSE    5 2023
## 351                   UGA_173101_P018_WF03   35-23   TRUE  FALSE    6 2023
## 392                   UGA_173101_P019_WC02  188-23   TRUE  FALSE    4 2023
## 396                   UGA_173101_P019_WC06  200-23   TRUE  FALSE    5 2023
## 406                   UGA_173101_P019_WD04  362-23   TRUE  FALSE    5 2023
## 467                   UGA_173101_P020_WA05  209-23   TRUE  FALSE    7 2023
## 469                   UGA_173101_P020_WA07   38-23   TRUE  FALSE    3 2023
## 471                   UGA_173101_P020_WA09  319-23   TRUE  FALSE    6 2023
## 540                   UGA_173101_P020_WG06  277-23   TRUE  FALSE    5 2023
## 541                   UGA_173101_P020_WG07  443-23   TRUE  FALSE    5 2023
## 549                   UGA_173101_P020_WH03  212-23   TRUE  FALSE    6 2023
## 550                   UGA_173101_P020_WH04  271-23   TRUE  FALSE    6 2023
## 557                   UGA_173101_P020_WH11  447-23   TRUE  FALSE    7 2023
## 583                   UGA_173101_P021_WC01   91-23   TRUE  FALSE    5 2023
## 666                   UGA_173101_P022_WA12  160-23   TRUE  FALSE    6 2023
## 735                   UGA_173101_P022_WG09  471-23   TRUE  FALSE    4 2023
## 766                   UGA_173101_P023_WB04   37-23   TRUE  FALSE    4 2023
```

```
## 768                   UGA_173101_P023_WB06  190-23   TRUE  FALSE    5 2023
## 773                   UGA_173101_P023_WB11    7-23   TRUE  FALSE    4 2023
## 826                   UGA_173101_P023_WG04  136-23   TRUE  FALSE    6 2023
## 861   GTseek_024_384i5_034_P098.003_938-23  938-23   TRUE  FALSE    5 2023
## 867   GTseek_024_384i5_040_P098.003_944-23  944-23   TRUE  FALSE    5 2023
## 902   GTseek_024_384i5_075_P098.001_631-23  631-23   TRUE  FALSE    5 2023
## 910   GTseek_024_384i5_083_P098.002_781-23  781-23   TRUE   TRUE    4 2023
## 922  GTseek_024_384i5_095_P098.002_781-23R 781-23R   TRUE   TRUE    4 2023
## 926   GTseek_024_384i5_099_P098.001_638-23  638-23   TRUE  FALSE    3 2023
## 935   GTseek_024_384i5_108_P098.003_972-23  972-23   TRUE  FALSE    6 2023
## 960   GTseek_024_384i5_133_P098.001_651-23  651-23   TRUE   TRUE    4 2023
## 961   GTseek_024_384i5_134_P098.003_982-23  982-23   TRUE  FALSE    4 2023
## 970  GTseek_024_384i5_143_P098.001_651-23R 651-23R   TRUE   TRUE    4 2023
## 977   GTseek_024_384i5_151_P098.002_803-23  803-23   TRUE  FALSE    6 2023
## 1005  GTseek_024_384i5_179_P098.002_809-23  809-23   TRUE  FALSE    5 2023
## 1026 GTseek_024_384i5_200_P098.003_1009-23 1009-23   TRUE  FALSE    6 2023
## 1032 GTseek_024_384i5_206_P098.003_1013-23 1013-23   TRUE  FALSE    5 2023
## 1113  GTseek_024_384i5_287_P098.002_871-23  871-23   TRUE  FALSE    3 2023
## 1119  GTseek_024_384i5_293_P098.001_713-23  713-23   TRUE  FALSE    6 2023
## 1155  GTseek_024_384i5_329_P098.001_732-23  732-23   TRUE  FALSE    7 2023
## 1201  GTseek_024_384i5_375_P098.002_919-23  919-23   TRUE  FALSE    4 2023
## 1224  GTseek_025_384i5_014_P098.007_671-23  671-23   TRUE  FALSE    4 2023
## 1256  GTseek_025_384i5_046_P098.007_706-23  706-23   TRUE  FALSE    5 2023
## 1286  GTseek_025_384i5_076_P098.007_776-23  776-23   TRUE  FALSE    6 2023
## 1312  GTseek_025_384i5_102_P098.007_847-23  847-23   TRUE  FALSE    5 2023
## 1340  GTseek_025_384i5_130_P098.007_881-23  881-23   TRUE  FALSE    5 2023
## 1344  GTseek_025_384i5_134_P098.007_890-23  890-23   TRUE  FALSE    5 2023
## 1406 GTseek_025_384i5_196_P098.007_1002-23 1002-23   TRUE  FALSE    5 2023
## 1914 GTseek_028_384i5_089_P098.UGA2_399-23  399-23   TRUE  FALSE    3 2023
## 1922 GTseek_028_384i5_097_P098.UGA1_104-23  104-23   TRUE  FALSE    5 2023
## 1956 GTseek_028_384i5_131_P098.UGA1_139-23  139-23   TRUE  FALSE    4 2023
## 1968 GTseek_028_384i5_143_P098.UGA1_150-23  150-23   TRUE  FALSE    5 2023
## 1998 GTseek_028_384i5_173_P098.UGA1_181-23  181-23   TRUE  FALSE    6 2023
## 2018 GTseek_028_384i5_193_P098.UGA1_187-23  187-23   TRUE  FALSE    5 2023
##       snare       utmSnare
## 83   HS 029 259736x3594236
## 91   HS 027 264464x3592495
## 125  HS 027 264464x3592495
## 138  HS 027 264464x3592495
## 186  HS 027 264464x3592495
## 188  HS 027 264464x3592495
## 194  HS 027 264464x3592495
## 233  HS 027 264464x3592495
## 256  HS 027 264464x3592495
## 260  HS 027 264464x3592495
## 285  HS 027 264464x3592495
## 338  HS 027 264464x3592495
## 345  HS 027 264464x3592495
## 351  HS 027 264464x3592495
## 392  HS 027 264464x3592495
## 396  HS 027 264464x3592495
## 406  HS 027 264464x3592495
## 467  HS 025 265276x3591787
## 469  HS 027 264464x3592495
```

```
## 471  HS 032 263672x3593292
## 540  HS 025 265276x3591787
## 541  HS 027 264464x3592495
## 549  HS 027 264464x3592495
## 550  HS 025 265276x3591787
## 557  HS 027 264464x3592495
## 583  HS 027 264464x3592495
## 666  HS 027 264464x3592495
## 735  HS 027 264464x3592495
## 766  HS 027 264464x3592495
## 768  HS 025 265276x3591787
## 773  HS 027 264464x3592495
## 826  HS 025 265276x3591787
## 861  HS 027 264464x3592495
## 867  HS 027 264464x3592495
## 902  HS 027 264464x3592495
## 910  HS 027 264464x3592495
## 922  HS 027 264464x3592495
## 926  HS 027 264464x3592495
## 935  HS 027 264464x3592495
## 960  HS 027 264464x3592495
## 961  HS 027 264464x3592495
## 970  HS 027 264464x3592495
## 977  HS 027 264464x3592495
## 1005 HS 027 264464x3592495
## 1026 HS 032 263672x3593292
## 1032 HS 027 264464x3592495
## 1113 HS 027 264464x3592495
## 1119 HS 025 265276x3591787
## 1155 HS 025 265276x3591787
## 1201 HS 027 264464x3592495
## 1224 HS 027 264464x3592495
## 1256 HS 027 264464x3592495
## 1286 HS 027 264464x3592495
## 1312 HS 027 264464x3592495
## 1340 HS 027 264464x3592495
## 1344 HS 027 264464x3592495
## 1406 HS 027 264464x3592495
## 1914 HS 027 264464x3592495
## 1922 HS 025 265276x3591787
## 1956 HS 027 264464x3592495
## 1968 HS 027 264464x3592495
## 1998 HS 032 263672x3593292
## 2018 HS 027 264464x3592495
##                                                                                    fqfile
## 83    RAPiD-Genomics_F534_UGA_173101_P015_WG11_i5-502_i7-96_S14821_L003_R1_001.fastq
## 91     RAPiD-Genomics_F534_UGA_173101_P015_WH07_i5-502_i7-1_S14829_L003_R1_001.fastq
## 125  RAPiD-Genomics_F534_UGA_173101_P016_WC05_i5-504_i7-125_S14863_L003_R1_001.fastq
## 138  RAPiD-Genomics_F534_UGA_173101_P016_WD06_i5-504_i7-138_S14876_L003_R1_001.fastq
## 186  RAPiD-Genomics_F534_UGA_173101_P016_WH06_i5-504_i7-186_S14923_L003_R1_001.fastq
## 188  RAPiD-Genomics_F534_UGA_173101_P016_WH08_i5-504_i7-188_S14925_L003_R1_001.fastq
## 194   RAPiD-Genomics_F534_UGA_173101_P017_WA02_i5-512_i7-98_S14931_L003_R1_001.fastq
## 233  RAPiD-Genomics_F534_UGA_173101_P017_WD05_i5-512_i7-137_S14970_L003_R1_001.fastq
## 256  RAPiD-Genomics_F534_UGA_173101_P017_WF04_i5-512_i7-160_S14993_L003_R1_001.fastq
```

```
## 260  RAPiD-Genomics_F534_UGA_173101_P017_WF08_i5-512_i7-164_S14997_L003_R1_001.fastq
## 285  RAPiD-Genomics_F534_UGA_173101_P017_WH09_i5-512_i7-189_S15022_L003_R1_001.fastq
## 338  RAPiD-Genomics_F534_UGA_173101_P018_WE02_i5-537_i7-146_S15075_L003_R1_001.fastq
## 345  RAPiD-Genomics_F534_UGA_173101_P018_WE09_i5-537_i7-153_S15084_L003_R1_001.fastq
## 351  RAPiD-Genomics_F534_UGA_173101_P018_WF03_i5-537_i7-159_S15090_L003_R1_001.fastq
## 392  RAPiD-Genomics_F534_UGA_173101_P019_WC02_i5-1102_i7-67_S15131_L003_R1_001.fastq
## 396   RAPiD-Genomics_F534_UGA_173101_P019_WC06_i5-1102_i7-3_S15135_L003_R1_001.fastq
## 406  RAPiD-Genomics_F534_UGA_173101_P019_WD04_i5-1102_i7-91_S15145_L003_R1_001.fastq
## 467    RAPiD-Genomics_F566_UGA_173101_P020_WA05_i5-505_i7-38_S3895_L001_R1_001.fastq
## 469    RAPiD-Genomics_F566_UGA_173101_P020_WA07_i5-505_i7-77_S3897_L001_R1_001.fastq
## 471    RAPiD-Genomics_F566_UGA_173101_P020_WA09_i5-505_i7-36_S3899_L001_R1_001.fastq
## 540    RAPiD-Genomics_F566_UGA_173101_P020_WG06_i5-505_i7-14_S3968_L001_R1_001.fastq
## 541    RAPiD-Genomics_F566_UGA_173101_P020_WG07_i5-505_i7-18_S3969_L001_R1_001.fastq
## 549    RAPiD-Genomics_F566_UGA_173101_P020_WH03_i5-505_i7-88_S3977_L001_R1_001.fastq
## 550    RAPiD-Genomics_F566_UGA_173101_P020_WH04_i5-505_i7-73_S3978_L001_R1_001.fastq
## 557    RAPiD-Genomics_F566_UGA_173101_P020_WH11_i5-505_i7-95_S3985_L001_R1_001.fastq
## 583      RAPiD-Genomics_F566_UGA_173101_P021_WC01_i5-501_i7-65_S25_L003_R1_001.fastq
## 666   RAPiD-Genomics_F566_UGA_173101_P022_WA12_i5-1098_i7-23_S7594_L003_R1_001.fastq
## 735   RAPiD-Genomics_F566_UGA_173101_P022_WG09_i5-1098_i7-41_S7663_L003_R1_001.fastq
## 766   RAPiD-Genomics_F566_UGA_173101_P023_WB04_i5-1099_i7-20_S7694_L003_R1_001.fastq
## 768    RAPiD-Genomics_F566_UGA_173101_P023_WB06_i5-1099_i7-6_S7696_L003_R1_001.fastq
## 773    RAPiD-Genomics_F566_UGA_173101_P023_WB11_i5-1099_i7-5_S7701_L003_R1_001.fastq
## 826   RAPiD-Genomics_F566_UGA_173101_P023_WG04_i5-1099_i7-10_S7754_L003_R1_001.fastq
## 861                                                                            <NA>
## 867                                                                            <NA>
## 902                                                                            <NA>
## 910                                                                            <NA>
## 922                                                                            <NA>
## 926                                                                            <NA>
## 935                                                                            <NA>
## 960                                                                            <NA>
## 961                                                                            <NA>
## 970                                                                            <NA>
## 977                                                                            <NA>
## 1005                                                                           <NA>
## 1026                                                                           <NA>
## 1032                                                                           <NA>
## 1113                                                                           <NA>
## 1119                                                                           <NA>
## 1155                                                                           <NA>
## 1201                                                                           <NA>
## 1224                                                                           <NA>
## 1256                                                                           <NA>
## 1286                                                                           <NA>
## 1312                                                                           <NA>
## 1340                                                                           <NA>
## 1344                                                                           <NA>
## 1406                                                                           <NA>
## 1914                                                                           <NA>
## 1922                                                                           <NA>
## 1956                                                                           <NA>
## 1968                                                                           <NA>
## 1998                                                                           <NA>
## 2018                                                                           <NA>
```

```
##      knownSEX  sampSOURCE region nSNPgt               vcfID  FILT GTservice
## 83             Hair Snare    CGA    333 UGA_173101_P015_WG11 FALSE       LGC
## 91             Hair Snare    CGA   1231 UGA_173101_P015_WH07 FALSE       LGC
## 125            Hair Snare    CGA    460 UGA_173101_P016_WC05 FALSE       LGC
## 138            Hair Snare    CGA   1018 UGA_173101_P016_WD06 FALSE       LGC
## 186            Hair Snare    CGA    940 UGA_173101_P016_WH06 FALSE       LGC
## 188            Hair Snare    CGA   1103 UGA_173101_P016_WH08 FALSE       LGC
## 194            Hair Snare    CGA   1274 UGA_173101_P017_WA02 FALSE       LGC
## 233            Hair Snare    CGA    486 UGA_173101_P017_WD05 FALSE       LGC
## 256            Hair Snare    CGA   1256 UGA_173101_P017_WF04 FALSE       LGC
## 260            Hair Snare    CGA    863 UGA_173101_P017_WF08 FALSE       LGC
## 285            Hair Snare    CGA    948 UGA_173101_P017_WH09 FALSE       LGC
## 338            Hair Snare    CGA   1272 UGA_173101_P018_WE02 FALSE       LGC
## 345            Hair Snare    CGA   1110 UGA_173101_P018_WE09 FALSE       LGC
## 351            Hair Snare    CGA   1174 UGA_173101_P018_WF03 FALSE       LGC
## 392            Hair Snare    CGA   1280 UGA_173101_P019_WC02 FALSE       LGC
## 396            Hair Snare    CGA   1282 UGA_173101_P019_WC06 FALSE       LGC
## 406            Hair Snare    CGA   1288 UGA_173101_P019_WD04 FALSE       LGC
## 467            Hair Snare    CGA   1252 UGA_173101_P020_WA05 FALSE       LGC
## 469            Hair Snare    CGA   1146 UGA_173101_P020_WA07 FALSE       LGC
## 471            Hair Snare    CGA   1059 UGA_173101_P020_WA09 FALSE       LGC
## 540            Hair Snare    CGA   1270 UGA_173101_P020_WG06 FALSE       LGC
## 541            Hair Snare    CGA    676 UGA_173101_P020_WG07 FALSE       LGC
## 549            Hair Snare    CGA   1301 UGA_173101_P020_WH03 FALSE       LGC
## 550            Hair Snare    CGA   1299 UGA_173101_P020_WH04 FALSE       LGC
## 557            Hair Snare    CGA   1303 UGA_173101_P020_WH11 FALSE       LGC
## 583            Hair Snare    CGA   1203 UGA_173101_P021_WC01 FALSE       LGC
## 666            Hair Snare    CGA   1103 UGA_173101_P022_WA12 FALSE       LGC
## 735            Hair Snare    CGA   1293 UGA_173101_P022_WG09 FALSE       LGC
## 766            Hair Snare    CGA   1262 UGA_173101_P023_WB04 FALSE       LGC
## 768            Hair Snare    CGA   1200 UGA_173101_P023_WB06 FALSE       LGC
## 773            Hair Snare    CGA   1230 UGA_173101_P023_WB11 FALSE       LGC
## 826            Hair Snare    CGA   1280 UGA_173101_P023_WG04 FALSE       LGC
## 861      <NA> Hair Sample    CGA    123           GTS_938-23 FALSE    GTSeek
## 867      <NA> Hair Sample    CGA    121           GTS_944-23 FALSE    GTSeek
## 902      <NA> Hair Sample    CGA    121           GTS_631-23 FALSE    GTSeek
## 910      <NA> Hair Sample    CGA    123           GTS_781-23 FALSE    GTSeek
## 922      <NA> Hair Sample    CGA    123          GTS_781-23R FALSE    GTSeek
## 926      <NA> Hair Sample    CGA    120           GTS_638-23 FALSE    GTSeek
## 935      <NA> Hair Sample    CGA    121           GTS_972-23 FALSE    GTSeek
## 960      <NA> Hair Sample    CGA    122           GTS_651-23 FALSE    GTSeek
## 961      <NA> Hair Sample    CGA    122           GTS_982-23 FALSE    GTSeek
## 970      <NA> Hair Sample    CGA    123          GTS_651-23R FALSE    GTSeek
## 977      <NA> Hair Sample    CGA    123           GTS_803-23 FALSE    GTSeek
## 1005     <NA> Hair Sample    CGA    123           GTS_809-23 FALSE    GTSeek
## 1026     <NA> Hair Sample    CGA    122          GTS_1009-23 FALSE    GTSeek
## 1032     <NA> Hair Sample    CGA    123          GTS_1013-23 FALSE    GTSeek
## 1113     <NA> Hair Sample    CGA    122           GTS_871-23 FALSE    GTSeek
## 1119     <NA> Hair Sample    CGA    121           GTS_713-23 FALSE    GTSeek
## 1155     <NA> Hair Sample    CGA    121           GTS_732-23 FALSE    GTSeek
## 1201     <NA> Hair Sample    CGA    123           GTS_919-23 FALSE    GTSeek
## 1224     <NA> Hair Sample    CGA    121           GTS_671-23 FALSE    GTSeek
## 1256     <NA> Hair Sample    CGA    122           GTS_706-23 FALSE    GTSeek
## 1286     <NA> Hair Sample    CGA    124           GTS_776-23 FALSE    GTSeek
```

```
## 1312     <NA> Hair Sample    CGA    124           GTS_847-23 FALSE    GTSeek
## 1340     <NA> Hair Sample    CGA    122           GTS_881-23 FALSE    GTSeek
## 1344     <NA> Hair Sample    CGA    124           GTS_890-23 FALSE    GTSeek
## 1406     <NA> Hair Sample    CGA    123          GTS_1002-23 FALSE    GTSeek
## 1914     <NA> Hair Sample    CGA    124           GTS_399-23 FALSE    GTSeek
## 1922     <NA> Hair Sample    CGA    109           GTS_104-23 FALSE    GTSeek
## 1956     <NA> Hair Sample    CGA    124           GTS_139-23 FALSE    GTSeek
## 1968     <NA> Hair Sample    CGA    114           GTS_150-23 FALSE    GTSeek
## 1998     <NA> Hair Sample    CGA    124           GTS_181-23 FALSE    GTSeek
## 2018     <NA> Hair Sample    CGA    120           GTS_187-23 FALSE    GTSeek
##      sexGT COLclone finalClone
## 83       F       14         14
## 91       F       14         14
## 125      F       14         14
## 138      F       14         14
## 186      F       14         14
## 188      F       14         14
## 194      F       14         14
## 233      F       14         14
## 256  undet       14         14
## 260      F       14         14
## 285      F       14         14
## 338      F       14         14
## 345      F       14         14
## 351      F       14         14
## 392      F       14         14
## 396      F       14         14
## 406      F       14         14
## 467      F       14         14
## 469      F       14         14
## 471      F       14         14
## 540      F       14         14
## 541      F       14         14
## 549      M       14         14
## 550      M       14         14
## 557      F       14         14
## 583      F       14         14
## 666  undet       14         14
## 735      F       14         14
## 766      F       14         14
## 768      F       14         14
## 773      F       14         14
## 826      F       14         14
## 861      F      301         14
## 867      F      305         14
## 902      F      301         14
## 910      F      305         14
## 922      F      305         14
## 926      F      301         14
## 935      F      301         14
## 960      F      301         14
## 961      F      301         14
## 970      F      301         14
## 977      F      305         14
```

```
## 1005     F      305         14
## 1026     F      301         14
## 1032     F      301         14
## 1113     F      305         14
## 1119     F      301         14
## 1155     F      301         14
## 1201     F      305         14
## 1224     F      301         14
## 1256     F      301         14
## 1286     F      301         14
## 1312     F      301         14
## 1340     F      301         14
## 1344     F      301         14
## 1406     F      301         14
## 1914     F      305         14
## 1922     F      305         14
## 1956     F      305         14
## 1968     F      305         14
## 1998     F      305         14
## 2018     F      305         14
```

```
unique(smd$COLclone[which(smd$finalClone == 14)])
```

```
## [1]  14 301 305
```

```
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 14)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 14)],]
```

```
##                          id1                  id2  kinship nsnp colclone1
## 1341    UGA_173101_P019_WC06 UGA_173101_P020_WA09 0.493939 1054        14
## 1500    UGA_173101_P019_WC02 UGA_173101_P020_WA09 0.494433 1054        14
## 1538    UGA_173101_P019_WC02 UGA_173101_P019_WC06 0.496287 1268        14
## 1845    UGA_173101_P019_WD04 UGA_173101_P020_WA09 0.492929 1057        14
## 1883    UGA_173101_P019_WD04 UGA_173101_P019_WC06 0.495058 1275        14
## 1886    UGA_173101_P019_WD04 UGA_173101_P019_WC02 0.495455 1272        14
## 2790    UGA_173101_P018_WF03 UGA_173101_P020_WA09 0.489741 1006        14
## 2828    UGA_173101_P018_WF03 UGA_173101_P019_WC06 0.490296 1171        14
## 2831    UGA_173101_P018_WF03 UGA_173101_P019_WC02 0.490758 1170        14
## 2837    UGA_173101_P018_WF03 UGA_173101_P019_WD04 0.488909 1173        14
## 3096    UGA_173101_P018_WE09 UGA_173101_P020_WA09 0.487832  964        14
## 3134    UGA_173101_P018_WE09 UGA_173101_P019_WC06 0.488942 1109        14
## 3137    UGA_173101_P018_WE09 UGA_173101_P019_WC02 0.488462 1109        14
## 3143    UGA_173101_P018_WE09 UGA_173101_P019_WD04 0.487958 1109        14
## 3157    UGA_173101_P018_WE09 UGA_173101_P018_WF03 0.482759 1055        14
## 5166    UGA_173101_P020_WA07 UGA_173101_P020_WA09 0.491342 1003        14
## 5204    UGA_173101_P020_WA07 UGA_173101_P019_WC06 0.493396 1141        14
## 5207    UGA_173101_P020_WA07 UGA_173101_P019_WC02 0.493396 1143        14
## 5213    UGA_173101_P020_WA07 UGA_173101_P019_WD04 0.491525 1145        14
## 5227    UGA_173101_P020_WA07 UGA_173101_P018_WF03 0.487500 1091        14
## 5231    UGA_173101_P020_WA07 UGA_173101_P018_WE09 0.483820 1026        14
## 5371    UGA_173101_P016_WD06 UGA_173101_P020_WA09 0.481459  907        14
## 5409    UGA_173101_P016_WD06 UGA_173101_P019_WC06 0.482979 1016        14
## 5412    UGA_173101_P016_WD06 UGA_173101_P019_WC02 0.485669 1018        14
```

```
## 5418    UGA_173101_P016_WD06 UGA_173101_P019_WD04 0.484574 1016        14
## 5432    UGA_173101_P016_WD06 UGA_173101_P018_WF03 0.480044  979        14
## 5436    UGA_173101_P016_WD06 UGA_173101_P018_WE09 0.481900  946        14
## 5459    UGA_173101_P016_WD06 UGA_173101_P020_WA07 0.483560  963        14
## 6918    UGA_173101_P016_WC05 UGA_173101_P020_WA09 0.478836  430        14
## 6956    UGA_173101_P016_WC05 UGA_173101_P019_WC06 0.473750  460        14
## 6959    UGA_173101_P016_WC05 UGA_173101_P019_WC02 0.476250  460        14
## 6965    UGA_173101_P016_WC05 UGA_173101_P019_WD04 0.474874  459        14
## 6979    UGA_173101_P016_WC05 UGA_173101_P018_WF03 0.472081  453        14
## 6983    UGA_173101_P016_WC05 UGA_173101_P018_WE09 0.471354  441        14
## 7006    UGA_173101_P016_WC05 UGA_173101_P020_WA07 0.474359  447        14
## 7008    UGA_173101_P016_WC05 UGA_173101_P016_WD06 0.472678  426        14
## 9606    UGA_173101_P017_WA02 UGA_173101_P020_WA09 0.491919 1056        14
## 9644    UGA_173101_P017_WA02 UGA_173101_P019_WC06 0.495025 1265        14
## 9647    UGA_173101_P017_WA02 UGA_173101_P019_WC02 0.496262 1262        14
## 9653    UGA_173101_P017_WA02 UGA_173101_P019_WD04 0.495439 1269        14
## 9667    UGA_173101_P017_WA02 UGA_173101_P018_WF03 0.488848 1164        14
## 9671    UGA_173101_P017_WA02 UGA_173101_P018_WE09 0.489884 1105        14
## 9694    UGA_173101_P017_WA02 UGA_173101_P020_WA07 0.492911 1142        14
## 9696    UGA_173101_P017_WA02 UGA_173101_P016_WD06 0.486170 1015        14
## 9710    UGA_173101_P017_WA02 UGA_173101_P016_WC05 0.476131  459        14
## 11643   UGA_173101_P015_WH07 UGA_173101_P020_WA09 0.488636 1034        14
## 11681   UGA_173101_P015_WH07 UGA_173101_P019_WC06 0.492268 1224        14
## 11684   UGA_173101_P015_WH07 UGA_173101_P019_WC02 0.493103 1223        14
## 11690   UGA_173101_P015_WH07 UGA_173101_P019_WD04 0.492308 1231        14
## 11704   UGA_173101_P015_WH07 UGA_173101_P018_WF03 0.485981 1150        14
## 11708   UGA_173101_P015_WH07 UGA_173101_P018_WE09 0.486381 1087        14
## 11731   UGA_173101_P015_WH07 UGA_173101_P020_WA07 0.488095 1122        14
## 11733   UGA_173101_P015_WH07 UGA_173101_P016_WD06 0.482833 1005        14
## 11747   UGA_173101_P015_WH07 UGA_173101_P016_WC05 0.473485  457        14
## 11768   UGA_173101_P015_WH07 UGA_173101_P017_WA02 0.492255 1218        14
## 16668   UGA_173101_P020_WA05 UGA_173101_P020_WA09 0.495455 1056        14
## 16706   UGA_173101_P020_WA05 UGA_173101_P019_WC06 0.496644 1241        14
## 16709   UGA_173101_P020_WA05 UGA_173101_P019_WC02 0.496644 1242        14
## 16715   UGA_173101_P020_WA05 UGA_173101_P019_WD04 0.496218 1246        14
## 16729   UGA_173101_P020_WA05 UGA_173101_P018_WF03 0.489700 1159        14
## 16733   UGA_173101_P020_WA05 UGA_173101_P018_WE09 0.489279 1092        14
## 16756   UGA_173101_P020_WA05 UGA_173101_P020_WA07 0.493868 1140        14
## 16758   UGA_173101_P020_WA05 UGA_173101_P016_WD06 0.484509 1012        14
## 16772   UGA_173101_P020_WA05 UGA_173101_P016_WC05 0.477500  459        14
## 16793   UGA_173101_P020_WA05 UGA_173101_P017_WA02 0.496627 1240        14
## 16807   UGA_173101_P020_WA05 UGA_173101_P015_WH07 0.492160 1205        14
## 30396   UGA_173101_P017_WD05 UGA_173101_P020_WA09 0.458791  444        14
## 30434   UGA_173101_P017_WD05 UGA_173101_P019_WC06 0.460396  485        14
## 30437   UGA_173101_P017_WD05 UGA_173101_P019_WC02 0.462871  484        14
## 30443   UGA_173101_P017_WD05 UGA_173101_P019_WD04 0.461634  486        14
## 30457   UGA_173101_P017_WD05 UGA_173101_P018_WF03 0.457500  478        14
## 30461   UGA_173101_P017_WD05 UGA_173101_P018_WE09 0.460459  469        14
## 30484   UGA_173101_P017_WD05 UGA_173101_P020_WA07 0.461735  472        14
## 30486   UGA_173101_P017_WD05 UGA_173101_P016_WD06 0.451117  446        14
## 30500   UGA_173101_P017_WD05 UGA_173101_P016_WC05 0.436170  244        14
## 30521   UGA_173101_P017_WD05 UGA_173101_P017_WA02 0.456683  486        14
## 30535   UGA_173101_P017_WD05 UGA_173101_P015_WH07 0.460396  483        14
## 30565   UGA_173101_P017_WD05 UGA_173101_P020_WA05 0.462871  484        14
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## 31390   UGA_173101_P018_WE02 UGA_173101_P020_WA09 0.493939 1057        14
## 31428   UGA_173101_P018_WE02 UGA_173101_P019_WC06 0.495432 1261        14
## 31431   UGA_173101_P018_WE02 UGA_173101_P019_WC02 0.496683 1262        14
## 31437   UGA_173101_P018_WE02 UGA_173101_P019_WD04 0.495833 1264        14
## 31451   UGA_173101_P018_WE02 UGA_173101_P018_WF03 0.489252 1165        14
## 31455   UGA_173101_P018_WE02 UGA_173101_P018_WE09 0.489341 1101        14
## 31478   UGA_173101_P018_WE02 UGA_173101_P020_WA07 0.492898 1142        14
## 31480   UGA_173101_P018_WE02 UGA_173101_P016_WD06 0.483511 1017        14
## 31494   UGA_173101_P018_WE02 UGA_173101_P016_WC05 0.475000  460        14
## 31515   UGA_173101_P018_WE02 UGA_173101_P017_WA02 0.495819 1258        14
## 31529   UGA_173101_P018_WE02 UGA_173101_P015_WH07 0.491304 1219        14
## 31559   UGA_173101_P018_WE02 UGA_173101_P020_WA05 0.497059 1240        14
## 31623   UGA_173101_P018_WE02 UGA_173101_P017_WD05 0.459158  486        14
## 34206   UGA_173101_P015_WG11 UGA_173101_P020_WA09 0.457364  310        14
## 34244   UGA_173101_P015_WG11 UGA_173101_P019_WC06 0.461957  333        14
## 34247   UGA_173101_P015_WG11 UGA_173101_P019_WC02 0.461957  331        14
## 34253   UGA_173101_P015_WG11 UGA_173101_P019_WD04 0.460145  333        14
## 34267   UGA_173101_P015_WG11 UGA_173101_P018_WF03 0.457721  327        14
## 34271   UGA_173101_P015_WG11 UGA_173101_P018_WE09 0.453358  324        14
## 34294   UGA_173101_P015_WG11 UGA_173101_P020_WA07 0.459559  327        14
## 34296   UGA_173101_P015_WG11 UGA_173101_P016_WD06 0.460227  317        14
## 34310   UGA_173101_P015_WG11 UGA_173101_P016_WC05 0.451493  173        14
## 34331   UGA_173101_P015_WG11 UGA_173101_P017_WA02 0.461957  333        14
## 34345   UGA_173101_P015_WG11 UGA_173101_P015_WH07 0.461679  331        14
## 34375   UGA_173101_P015_WG11 UGA_173101_P020_WA05 0.461957  332        14
## 34439   UGA_173101_P015_WG11 UGA_173101_P017_WD05 0.438356  185        14
## 34443   UGA_173101_P015_WG11 UGA_173101_P018_WE02 0.460145  333        14
## 49156   UGA_173101_P016_WH06 UGA_173101_P020_WA09 0.482868  839        14
## 49194   UGA_173101_P016_WH06 UGA_173101_P019_WC06 0.483634  938        14
## 49197   UGA_173101_P016_WH06 UGA_173101_P019_WC02 0.483708  939        14
## 49203   UGA_173101_P016_WH06 UGA_173101_P019_WD04 0.482022  940        14
## 49217   UGA_173101_P016_WH06 UGA_173101_P018_WF03 0.474419  907        14
## 49221   UGA_173101_P016_WH06 UGA_173101_P018_WE09 0.480120  878        14
## 49244   UGA_173101_P016_WH06 UGA_173101_P020_WA07 0.482656  892        14
## 49246   UGA_173101_P016_WH06 UGA_173101_P016_WD06 0.475911  825        14
## 49260   UGA_173101_P016_WH06 UGA_173101_P016_WC05 0.475410  407        14
## 49281   UGA_173101_P016_WH06 UGA_173101_P017_WA02 0.484831  940        14
## 49295   UGA_173101_P016_WH06 UGA_173101_P015_WH07 0.481818  928        14
## 49325   UGA_173101_P016_WH06 UGA_173101_P020_WA05 0.485294  933        14
## 49389   UGA_173101_P016_WH06 UGA_173101_P017_WD05 0.454670  429        14
## 49393   UGA_173101_P016_WH06 UGA_173101_P018_WE02 0.484763  936        14
## 49404   UGA_173101_P016_WH06 UGA_173101_P015_WG11 0.446000  301        14
## 50418   UGA_173101_P016_WH08 UGA_173101_P020_WA09 0.484763  953        14
## 50456   UGA_173101_P016_WH08 UGA_173101_P019_WC06 0.487044 1101        14
## 50459   UGA_173101_P016_WH08 UGA_173101_P019_WC02 0.488004 1099        14
## 50465   UGA_173101_P016_WH08 UGA_173101_P019_WD04 0.485660 1103        14
## 50479   UGA_173101_P016_WH08 UGA_173101_P018_WF03 0.481846 1045        14
## 50483   UGA_173101_P016_WH08 UGA_173101_P018_WE09 0.485169  995        14
## 50506   UGA_173101_P016_WH08 UGA_173101_P020_WA07 0.486897 1022        14
## 50508   UGA_173101_P016_WH08 UGA_173101_P016_WD06 0.476491  933        14
## 50522   UGA_173101_P016_WH08 UGA_173101_P016_WC05 0.466146  444        14
## 50543   UGA_173101_P016_WH08 UGA_173101_P017_WA02 0.486564 1098        14
## 50557   UGA_173101_P016_WH08 UGA_173101_P015_WH07 0.483559 1081        14
## 50587   UGA_173101_P016_WH08 UGA_173101_P020_WA05 0.487911 1088        14
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## 50651   UGA_173101_P016_WH08 UGA_173101_P017_WD05 0.457908  469        14
## 50655   UGA_173101_P016_WH08 UGA_173101_P018_WE02 0.487019 1098        14
## 50666   UGA_173101_P016_WH08 UGA_173101_P015_WG11 0.457090  321        14
## 50718   UGA_173101_P016_WH08 UGA_173101_P016_WH06 0.477163  874        14
## 58668   UGA_173101_P017_WF04 UGA_173101_P020_WA09 0.489754 1046        14
## 58706   UGA_173101_P017_WF04 UGA_173101_P019_WC06 0.491582 1247        14
## 58709   UGA_173101_P017_WF04 UGA_173101_P019_WC02 0.492437 1249        14
## 58715   UGA_173101_P017_WF04 UGA_173101_P019_WD04 0.491132 1250        14
## 58729   UGA_173101_P017_WF04 UGA_173101_P018_WF03 0.484991 1159        14
## 58733   UGA_173101_P017_WF04 UGA_173101_P018_WE09 0.485380 1098        14
## 58756   UGA_173101_P017_WF04 UGA_173101_P020_WA07 0.487143 1135        14
## 58758   UGA_173101_P017_WF04 UGA_173101_P016_WD06 0.479701 1012        14
## 58772   UGA_173101_P017_WF04 UGA_173101_P016_WC05 0.473485  457        14
## 58793   UGA_173101_P017_WF04 UGA_173101_P017_WA02 0.491117 1244        14
## 58807   UGA_173101_P017_WF04 UGA_173101_P015_WH07 0.488616 1210        14
## 58837   UGA_173101_P017_WF04 UGA_173101_P020_WA05 0.492747 1225        14
## 58901   UGA_173101_P017_WF04 UGA_173101_P017_WD05 0.460199  483        14
## 58905   UGA_173101_P017_WF04 UGA_173101_P018_WE02 0.491554 1241        14
## 58916   UGA_173101_P017_WF04 UGA_173101_P015_WG11 0.461679  332        14
## 58968   UGA_173101_P017_WF04 UGA_173101_P016_WH06 0.484127  933        14
## 58972   UGA_173101_P017_WF04 UGA_173101_P016_WH08 0.482977 1089        14
## 68280   UGA_173101_P017_WF08 UGA_173101_P020_WA09 0.480282  781        14
## 68318   UGA_173101_P017_WF08 UGA_173101_P019_WC06 0.479434  863        14
## 68321   UGA_173101_P017_WF08 UGA_173101_P019_WC02 0.480026  862        14
## 68327   UGA_173101_P017_WF08 UGA_173101_P019_WD04 0.479434  863        14
## 68341   UGA_173101_P017_WF08 UGA_173101_P018_WF03 0.474599  831        14
## 68345   UGA_173101_P017_WF08 UGA_173101_P018_WE09 0.472594  815        14
## 68368   UGA_173101_P017_WF08 UGA_173101_P020_WA07 0.479620  820        14
## 68370   UGA_173101_P017_WF08 UGA_173101_P016_WD06 0.469388  766        14
## 68384   UGA_173101_P017_WF08 UGA_173101_P016_WC05 0.466374  393        14
## 68405   UGA_173101_P017_WF08 UGA_173101_P017_WA02 0.480720  860        14
## 68419   UGA_173101_P017_WF08 UGA_173101_P015_WH07 0.476744  853        14
## 68449   UGA_173101_P017_WF08 UGA_173101_P020_WA05 0.479974  857        14
## 68513   UGA_173101_P017_WF08 UGA_173101_P017_WD05 0.456647  412        14
## 68517   UGA_173101_P017_WF08 UGA_173101_P018_WE02 0.480670  861        14
## 68528   UGA_173101_P017_WF08 UGA_173101_P015_WG11 0.441532  298        14
## 68580   UGA_173101_P017_WF08 UGA_173101_P016_WH06 0.472973  709        14
## 68584   UGA_173101_P017_WF08 UGA_173101_P016_WH08 0.470946  811        14
## 68609   UGA_173101_P017_WF08 UGA_173101_P017_WF04 0.476098  857        14
## 75870   UGA_173101_P017_WH09 UGA_173101_P020_WA09 0.476036  839        14
## 75908   UGA_173101_P017_WH09 UGA_173101_P019_WC06 0.478261  948        14
## 75911   UGA_173101_P017_WH09 UGA_173101_P019_WC02 0.477638  945        14
## 75917   UGA_173101_P017_WH09 UGA_173101_P019_WD04 0.479405  946        14
## 75931   UGA_173101_P017_WH09 UGA_173101_P018_WF03 0.474282  903        14
## 75935   UGA_173101_P017_WH09 UGA_173101_P018_WE09 0.475478  892        14
## 75958   UGA_173101_P017_WH09 UGA_173101_P020_WA07 0.475430  884        14
## 75960   UGA_173101_P017_WH09 UGA_173101_P016_WD06 0.466307  821        14
## 75974   UGA_173101_P017_WH09 UGA_173101_P016_WC05 0.465909  402        14
## 75995   UGA_173101_P017_WH09 UGA_173101_P017_WA02 0.478261  947        14
## 76009   UGA_173101_P017_WH09 UGA_173101_P015_WH07 0.478736  934        14
## 76039   UGA_173101_P017_WH09 UGA_173101_P020_WA05 0.478637  937        14
## 76103   UGA_173101_P017_WH09 UGA_173101_P017_WD05 0.450843  426        14
## 76107   UGA_173101_P017_WH09 UGA_173101_P018_WE02 0.476959  943        14
## 76118   UGA_173101_P017_WH09 UGA_173101_P015_WG11 0.449187  292        14
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## 76170   UGA_173101_P017_WH09 UGA_173101_P016_WH06 0.471311  769        14
## 76174   UGA_173101_P017_WH09 UGA_173101_P016_WH08 0.471130  869        14
## 76199   UGA_173101_P017_WH09 UGA_173101_P017_WF04 0.472801  940        14
## 76226   UGA_173101_P017_WH09 UGA_173101_P017_WF08 0.470896  725        14
## 94845   UGA_173101_P023_WG04 UGA_173101_P020_WA09 0.494399 1047        14
## 94883   UGA_173101_P023_WG04 UGA_173101_P019_WC06 0.495378 1247        14
## 94886   UGA_173101_P023_WG04 UGA_173101_P019_WC02 0.497054 1246        14
## 94892   UGA_173101_P023_WG04 UGA_173101_P019_WD04 0.495805 1253        14
## 94906   UGA_173101_P023_WG04 UGA_173101_P018_WF03 0.491071 1152        14
## 94910   UGA_173101_P023_WG04 UGA_173101_P018_WE09 0.489706 1090        14
## 94933   UGA_173101_P023_WG04 UGA_173101_P020_WA07 0.492366 1131        14
## 94935   UGA_173101_P023_WG04 UGA_173101_P016_WD06 0.485421 1003        14
## 94949   UGA_173101_P023_WG04 UGA_173101_P016_WC05 0.476131  455        14
## 94970   UGA_173101_P023_WG04 UGA_173101_P017_WA02 0.496186 1241        14
## 94984   UGA_173101_P023_WG04 UGA_173101_P015_WH07 0.493881 1206        14
## 95014   UGA_173101_P023_WG04 UGA_173101_P020_WA05 0.497889 1229        14
## 95078   UGA_173101_P023_WG04 UGA_173101_P017_WD05 0.461634  483        14
## 95082   UGA_173101_P023_WG04 UGA_173101_P018_WE02 0.496627 1241        14
## 95093   UGA_173101_P023_WG04 UGA_173101_P015_WG11 0.461397  329        14
## 95145   UGA_173101_P023_WG04 UGA_173101_P016_WH06 0.485126  922        14
## 95149   UGA_173101_P023_WG04 UGA_173101_P016_WH08 0.487379 1083        14
## 95174   UGA_173101_P023_WG04 UGA_173101_P017_WF04 0.493127 1223        14
## 95201   UGA_173101_P023_WG04 UGA_173101_P017_WF08 0.479818  850        14
## 95221   UGA_173101_P023_WG04 UGA_173101_P017_WH09 0.479021  932        14
## 102393  UGA_173101_P023_WB11 UGA_173101_P020_WA09 0.491201 1034        14
## 102431  UGA_173101_P023_WB11 UGA_173101_P019_WC06 0.492491 1198        14
## 102434  UGA_173101_P023_WB11 UGA_173101_P019_WC02 0.492105 1203        14
## 102440  UGA_173101_P023_WB11 UGA_173101_P019_WD04 0.491213 1206        14
## 102454  UGA_173101_P023_WB11 UGA_173101_P018_WF03 0.485294 1133        14
## 102458  UGA_173101_P023_WB11 UGA_173101_P018_WE09 0.486083 1066        14
## 102481  UGA_173101_P023_WB11 UGA_173101_P020_WA07 0.488878 1112        14
## 102483  UGA_173101_P023_WB11 UGA_173101_P016_WD06 0.478118  988        14
## 102497  UGA_173101_P023_WB11 UGA_173101_P016_WC05 0.476131  452        14
## 102518  UGA_173101_P023_WB11 UGA_173101_P017_WA02 0.490300 1199        14
## 102532  UGA_173101_P023_WB11 UGA_173101_P015_WH07 0.488309 1169        14
## 102562  UGA_173101_P023_WB11 UGA_173101_P020_WA05 0.492035 1193        14
## 102626  UGA_173101_P023_WB11 UGA_173101_P017_WD05 0.461634  480        14
## 102630  UGA_173101_P023_WB11 UGA_173101_P018_WE02 0.490691 1199        14
## 102641  UGA_173101_P023_WB11 UGA_173101_P015_WG11 0.457090  327        14
## 102693  UGA_173101_P023_WB11 UGA_173101_P016_WH06 0.478111  913        14
## 102697  UGA_173101_P023_WB11 UGA_173101_P016_WH08 0.484531 1055        14
## 102722  UGA_173101_P023_WB11 UGA_173101_P017_WF04 0.488799 1184        14
## 102749  UGA_173101_P023_WB11 UGA_173101_P017_WF08 0.475131  839        14
## 102769  UGA_173101_P023_WB11 UGA_173101_P017_WH09 0.473995  915        14
## 102815  UGA_173101_P023_WB11 UGA_173101_P023_WG04 0.491364 1223        14
## 103755  UGA_173101_P023_WB06 UGA_173101_P020_WA09 0.483589 1015        14
## 103793  UGA_173101_P023_WB06 UGA_173101_P019_WC06 0.482710 1170        14
## 103796  UGA_173101_P023_WB06 UGA_173101_P019_WC02 0.485158 1172        14
## 103802  UGA_173101_P023_WB06 UGA_173101_P019_WD04 0.483209 1176        14
## 103816  UGA_173101_P023_WB06 UGA_173101_P018_WF03 0.478500 1102        14
## 103820  UGA_173101_P023_WB06 UGA_173101_P018_WE09 0.478734 1042        14
## 103843  UGA_173101_P023_WB06 UGA_173101_P020_WA07 0.483266 1089        14
## 103845  UGA_173101_P023_WB06 UGA_173101_P016_WD06 0.473864  973        14
## 103859  UGA_173101_P023_WB06 UGA_173101_P016_WC05 0.466667  449        14
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## 103880  UGA_173101_P023_WB06 UGA_173101_P017_WA02 0.482775 1172        14
## 103894  UGA_173101_P023_WB06 UGA_173101_P015_WH07 0.481836 1140        14
## 103924  UGA_173101_P023_WB06 UGA_173101_P020_WA05 0.484522 1164        14
## 103988  UGA_173101_P023_WB06 UGA_173101_P017_WD05 0.456853  474        14
## 103992  UGA_173101_P023_WB06 UGA_173101_P018_WE02 0.483146 1172        14
## 104003  UGA_173101_P023_WB06 UGA_173101_P015_WG11 0.446296  326        14
## 104055  UGA_173101_P023_WB06 UGA_173101_P016_WH06 0.472749  894        14
## 104059  UGA_173101_P023_WB06 UGA_173101_P016_WH08 0.475480 1037        14
## 104084  UGA_173101_P023_WB06 UGA_173101_P017_WF04 0.480697 1161        14
## 104111  UGA_173101_P023_WB06 UGA_173101_P017_WF08 0.469595  823        14
## 104131  UGA_173101_P023_WB06 UGA_173101_P017_WH09 0.465770  893        14
## 104177  UGA_173101_P023_WB06 UGA_173101_P023_WG04 0.484973 1192        14
## 104194  UGA_173101_P023_WB06 UGA_173101_P023_WB11 0.480056 1167        14
## 106968  UGA_173101_P023_WB04 UGA_173101_P020_WA09 0.492284 1041        14
## 107006  UGA_173101_P023_WB04 UGA_173101_P019_WC06 0.493986 1230        14
## 107009  UGA_173101_P023_WB04 UGA_173101_P019_WC02 0.493162 1232        14
## 107015  UGA_173101_P023_WB04 UGA_173101_P019_WD04 0.492723 1235        14
## 107029  UGA_173101_P023_WB04 UGA_173101_P018_WF03 0.486347 1147        14
## 107033  UGA_173101_P023_WB04 UGA_173101_P018_WE09 0.485728 1086        14
## 107056  UGA_173101_P023_WB04 UGA_173101_P020_WA07 0.490883 1122        14
## 107058  UGA_173101_P023_WB04 UGA_173101_P016_WD06 0.484848 1000        14
## 107072  UGA_173101_P023_WB04 UGA_173101_P016_WC05 0.474619  453        14
## 107093  UGA_173101_P023_WB04 UGA_173101_P017_WA02 0.493127 1227        14
## 107107  UGA_173101_P023_WB04 UGA_173101_P015_WH07 0.489805 1189        14
## 107137  UGA_173101_P023_WB04 UGA_173101_P020_WA05 0.495250 1215        14
## 107201  UGA_173101_P023_WB04 UGA_173101_P017_WD05 0.460396  481        14
## 107205  UGA_173101_P023_WB04 UGA_173101_P018_WE02 0.493976 1223        14
## 107216  UGA_173101_P023_WB04 UGA_173101_P015_WG11 0.457721  329        14
## 107268  UGA_173101_P023_WB04 UGA_173101_P016_WH06 0.479931  918        14
## 107272  UGA_173101_P023_WB04 UGA_173101_P016_WH08 0.483301 1073        14
## 107297  UGA_173101_P023_WB04 UGA_173101_P017_WF04 0.489054 1211        14
## 107324  UGA_173101_P023_WB04 UGA_173101_P017_WF08 0.477749  844        14
## 107344  UGA_173101_P023_WB04 UGA_173101_P017_WH09 0.475236  922        14
## 107390  UGA_173101_P023_WB04 UGA_173101_P023_WG04 0.494098 1249        14
## 107407  UGA_173101_P023_WB04 UGA_173101_P023_WB11 0.488696 1214        14
## 107410  UGA_173101_P023_WB04 UGA_173101_P023_WB06 0.481193 1183        14
## 134436  UGA_173101_P020_WG06 UGA_173101_P020_WA09 0.493354 1044        14
## 134474  UGA_173101_P020_WG06 UGA_173101_P019_WC06 0.494519 1239        14
## 134477  UGA_173101_P020_WG06 UGA_173101_P019_WC02 0.495777 1239        14
## 134483  UGA_173101_P020_WG06 UGA_173101_P019_WD04 0.494924 1243        14
## 134497  UGA_173101_P020_WG06 UGA_173101_P018_WF03 0.488274 1152        14
## 134501  UGA_173101_P020_WG06 UGA_173101_P018_WE09 0.487721 1087        14
## 134524  UGA_173101_P020_WG06 UGA_173101_P020_WA07 0.492366 1130        14
## 134526  UGA_173101_P020_WG06 UGA_173101_P016_WD06 0.484881 1002        14
## 134540  UGA_173101_P020_WG06 UGA_173101_P016_WC05 0.476131  455        14
## 134561  UGA_173101_P020_WG06 UGA_173101_P017_WA02 0.494889 1234        14
## 134575  UGA_173101_P020_WG06 UGA_173101_P015_WH07 0.492091 1198        14
## 134605  UGA_173101_P020_WG06 UGA_173101_P020_WA05 0.497024 1226        14
## 134669  UGA_173101_P020_WG06 UGA_173101_P017_WD05 0.465174  479        14
## 134673  UGA_173101_P020_WG06 UGA_173101_P018_WE02 0.494907 1233        14
## 134684  UGA_173101_P020_WG06 UGA_173101_P015_WG11 0.459559  328        14
## 134736  UGA_173101_P020_WG06 UGA_173101_P016_WH06 0.481121  923        14
## 134740  UGA_173101_P020_WG06 UGA_173101_P016_WH08 0.484466 1080        14
## 134765  UGA_173101_P020_WG06 UGA_173101_P017_WF04 0.491409 1220        14
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## 134792  UGA_173101_P020_WG06 UGA_173101_P017_WF08 0.480418  844        14
## 134812  UGA_173101_P020_WG06 UGA_173101_P017_WH09 0.476636  929        14
## 134858  UGA_173101_P020_WG06 UGA_173101_P023_WG04 0.495895 1263        14
## 134875  UGA_173101_P020_WG06 UGA_173101_P023_WB11 0.490484 1221        14
## 134878  UGA_173101_P020_WG06 UGA_173101_P023_WB06 0.483547 1188        14
## 134885  UGA_173101_P020_WG06 UGA_173101_P023_WB04 0.492857 1248        14
## 153196  UGA_173101_P020_WH04 UGA_173101_P020_WA09 0.488313 1047        14
## 153234  UGA_173101_P020_WH04 UGA_173101_P019_WC06 0.490115 1263        14
## 153237  UGA_173101_P020_WH04 UGA_173101_P019_WC02 0.491322 1261        14
## 153243  UGA_173101_P020_WH04 UGA_173101_P019_WD04 0.490527 1269        14
## 153257  UGA_173101_P020_WH04 UGA_173101_P018_WF03 0.483675 1158        14
## 153261  UGA_173101_P020_WH04 UGA_173101_P018_WE09 0.483918 1094        14
## 153284  UGA_173101_P020_WH04 UGA_173101_P020_WA07 0.488095 1131        14
## 153286  UGA_173101_P020_WH04 UGA_173101_P016_WD06 0.478448 1006        14
## 153300  UGA_173101_P020_WH04 UGA_173101_P016_WC05 0.469849  455        14
## 153321  UGA_173101_P020_WH04 UGA_173101_P017_WA02 0.488750 1257        14
## 153335  UGA_173101_P020_WH04 UGA_173101_P015_WH07 0.488322 1214        14
## 153365  UGA_173101_P020_WH04 UGA_173101_P020_WA05 0.491597 1235        14
## 153429  UGA_173101_P020_WH04 UGA_173101_P017_WD05 0.457921  482        14
## 153433  UGA_173101_P020_WH04 UGA_173101_P018_WE02 0.490017 1253        14
## 153444  UGA_173101_P020_WH04 UGA_173101_P015_WG11 0.450368  329        14
## 153496  UGA_173101_P020_WH04 UGA_173101_P016_WH06 0.477169  925        14
## 153500  UGA_173101_P020_WH04 UGA_173101_P016_WH08 0.480658 1088        14
## 153525  UGA_173101_P020_WH04 UGA_173101_P017_WF04 0.488314 1238        14
## 153552  UGA_173101_P020_WH04 UGA_173101_P017_WF08 0.474609  849        14
## 153572  UGA_173101_P020_WH04 UGA_173101_P017_WH09 0.475000  935        14
## 153618  UGA_173101_P020_WH04 UGA_173101_P023_WG04 0.490244 1277        14
## 153635  UGA_173101_P020_WH04 UGA_173101_P023_WB11 0.486278 1228        14
## 153638  UGA_173101_P020_WH04 UGA_173101_P023_WB06 0.479204 1199        14
## 153645  UGA_173101_P020_WH04 UGA_173101_P023_WB04 0.488333 1258        14
## 153701  UGA_173101_P020_WH04 UGA_173101_P020_WG06 0.491013 1268        14
## 159910  UGA_173101_P020_WG07 UGA_173101_P020_WA09 0.474167  637        14
## 159948  UGA_173101_P020_WG07 UGA_173101_P019_WC06 0.473558  662        14
## 159951  UGA_173101_P020_WG07 UGA_173101_P019_WC02 0.474359  662        14
## 159957  UGA_173101_P020_WG07 UGA_173101_P019_WD04 0.475160  662        14
## 159971  UGA_173101_P020_WG07 UGA_173101_P018_WF03 0.470492  649        14
## 159975  UGA_173101_P020_WG07 UGA_173101_P018_WE09 0.471854  635        14
## 159998  UGA_173101_P020_WG07 UGA_173101_P020_WA07 0.472682  646        14
## 160000  UGA_173101_P020_WG07 UGA_173101_P016_WD06 0.471223  603        14
## 160014  UGA_173101_P020_WG07 UGA_173101_P016_WC05 0.467153  316        14
## 160035  UGA_173101_P020_WG07 UGA_173101_P017_WA02 0.474359  662        14
## 160049  UGA_173101_P020_WG07 UGA_173101_P015_WH07 0.474194  657        14
## 160079  UGA_173101_P020_WG07 UGA_173101_P020_WA05 0.475962  661        14
## 160143  UGA_173101_P020_WG07 UGA_173101_P017_WD05 0.451852  325        14
## 160147  UGA_173101_P020_WG07 UGA_173101_P018_WE02 0.474359  662        14
## 160158  UGA_173101_P020_WG07 UGA_173101_P015_WG11 0.449495  232        14
## 160210  UGA_173101_P020_WG07 UGA_173101_P016_WH06 0.462500  567        14
## 160214  UGA_173101_P020_WG07 UGA_173101_P016_WH08 0.461340  625        14
## 160239  UGA_173101_P020_WG07 UGA_173101_P017_WF04 0.475160  661        14
## 160266  UGA_173101_P020_WG07 UGA_173101_P017_WF08 0.461458  525        14
## 160286  UGA_173101_P020_WG07 UGA_173101_P017_WH09 0.459125  569        14
## 160332  UGA_173101_P020_WG07 UGA_173101_P023_WG04 0.475000  676        14
## 160349  UGA_173101_P020_WG07 UGA_173101_P023_WB11 0.469531  673        14
## 160352  UGA_173101_P020_WG07 UGA_173101_P023_WB06 0.466238  665        14
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## 160359  UGA_173101_P020_WG07 UGA_173101_P023_WB04 0.472656  676        14
## 160415  UGA_173101_P020_WG07 UGA_173101_P020_WG06 0.473438  676        14
## 160450  UGA_173101_P020_WG07 UGA_173101_P020_WH04 0.467969  676        14
## 162180  UGA_173101_P020_WH11 UGA_173101_P020_WA09 0.494919 1048        14
## 162218  UGA_173101_P020_WH11 UGA_173101_P019_WC06 0.496299 1266        14
## 162221  UGA_173101_P020_WH11 UGA_173101_P019_WC02 0.496700 1264        14
## 162227  UGA_173101_P020_WH11 UGA_173101_P019_WD04 0.495881 1272        14
## 162241  UGA_173101_P020_WH11 UGA_173101_P018_WF03 0.489272 1160        14
## 162245  UGA_173101_P020_WH11 UGA_173101_P018_WE09 0.489279 1096        14
## 162268  UGA_173101_P020_WH11 UGA_173101_P020_WA07 0.493333 1133        14
## 162270  UGA_173101_P020_WH11 UGA_173101_P016_WD06 0.484914 1006        14
## 162284  UGA_173101_P020_WH11 UGA_173101_P016_WC05 0.474874  455        14
## 162305  UGA_173101_P020_WH11 UGA_173101_P017_WA02 0.495417 1259        14
## 162319  UGA_173101_P020_WH11 UGA_173101_P015_WH07 0.493512 1215        14
## 162349  UGA_173101_P020_WH11 UGA_173101_P020_WA05 0.497899 1237        14
## 162413  UGA_173101_P020_WH11 UGA_173101_P017_WD05 0.461634  483        14
## 162417  UGA_173101_P020_WH11 UGA_173101_P018_WE02 0.496256 1255        14
## 162428  UGA_173101_P020_WH11 UGA_173101_P015_WG11 0.459559  329        14
## 162480  UGA_173101_P020_WH11 UGA_173101_P016_WH06 0.482306  926        14
## 162484  UGA_173101_P020_WH11 UGA_173101_P016_WH08 0.486460 1089        14
## 162509  UGA_173101_P020_WH11 UGA_173101_P017_WF04 0.493277 1239        14
## 162536  UGA_173101_P020_WH11 UGA_173101_P017_WF08 0.480469  850        14
## 162556  UGA_173101_P020_WH11 UGA_173101_P017_WH09 0.478488  935        14
## 162602  UGA_173101_P020_WH11 UGA_173101_P023_WG04 0.496748 1280        14
## 162619  UGA_173101_P020_WH11 UGA_173101_P023_WB11 0.492281 1230        14
## 162622  UGA_173101_P020_WH11 UGA_173101_P023_WB06 0.484177 1200        14
## 162629  UGA_173101_P020_WH11 UGA_173101_P023_WB04 0.493760 1262        14
## 162685  UGA_173101_P020_WH11 UGA_173101_P020_WG06 0.495922 1270        14
## 162720  UGA_173101_P020_WH11 UGA_173101_P020_WH04 0.491284 1298        14
## 162732  UGA_173101_P020_WH11 UGA_173101_P020_WG07 0.475781  676        14
## 179116  UGA_173101_P022_WG09 UGA_173101_P020_WA09 0.495408 1046        14
## 179154  UGA_173101_P022_WG09 UGA_173101_P019_WC06 0.495847 1258        14
## 179157  UGA_173101_P022_WG09 UGA_173101_P019_WC02 0.497088 1257        14
## 179163  UGA_173101_P022_WG09 UGA_173101_P019_WD04 0.496672 1261        14
## 179177  UGA_173101_P022_WG09 UGA_173101_P018_WF03 0.489700 1156        14
## 179181  UGA_173101_P022_WG09 UGA_173101_P018_WE09 0.489746 1094        14
## 179204  UGA_173101_P022_WG09 UGA_173101_P020_WA07 0.493308 1130        14
## 179206  UGA_173101_P022_WG09 UGA_173101_P016_WD06 0.484375 1006        14
## 179220  UGA_173101_P022_WG09 UGA_173101_P016_WC05 0.477387  454        14
## 179241  UGA_173101_P022_WG09 UGA_173101_P017_WA02 0.496225 1251        14
## 179255  UGA_173101_P022_WG09 UGA_173101_P015_WH07 0.493478 1210        14
## 179285  UGA_173101_P022_WG09 UGA_173101_P020_WA05 0.498735 1235        14
## 179349  UGA_173101_P022_WG09 UGA_173101_P017_WD05 0.462871  482        14
## 179353  UGA_173101_P022_WG09 UGA_173101_P018_WE02 0.496225 1249        14
## 179364  UGA_173101_P022_WG09 UGA_173101_P015_WG11 0.459559  329        14
## 179416  UGA_173101_P022_WG09 UGA_173101_P016_WH06 0.483982  924        14
## 179420  UGA_173101_P022_WG09 UGA_173101_P016_WH08 0.485977 1087        14
## 179445  UGA_173101_P022_WG09 UGA_173101_P017_WF04 0.492772 1233        14
## 179472  UGA_173101_P022_WG09 UGA_173101_P017_WF08 0.480469  849        14
## 179492  UGA_173101_P022_WG09 UGA_173101_P017_WH09 0.479651  932        14
## 179538  UGA_173101_P022_WG09 UGA_173101_P023_WG04 0.497954 1273        14
## 179555  UGA_173101_P022_WG09 UGA_173101_P023_WB11 0.490979 1228        14
## 179558  UGA_173101_P022_WG09 UGA_173101_P023_WB06 0.485000 1194        14
## 179565  UGA_173101_P022_WG09 UGA_173101_P023_WB04 0.494992 1258        14
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## 179621  UGA_173101_P022_WG09 UGA_173101_P020_WG06 0.495888 1266        14
## 179656  UGA_173101_P022_WG09 UGA_173101_P020_WH04 0.491559 1288        14
## 179668  UGA_173101_P022_WG09 UGA_173101_P020_WG07 0.474922  675        14
## 179672  UGA_173101_P022_WG09 UGA_173101_P020_WH11 0.497186 1292        14
## 185760  UGA_173101_P022_WA12 UGA_173101_P020_WA09 0.476351  967        14
## 185798  UGA_173101_P022_WA12 UGA_173101_P019_WC06 0.480263 1078        14
## 185801  UGA_173101_P022_WA12 UGA_173101_P019_WC02 0.481818 1076        14
## 185807  UGA_173101_P022_WA12 UGA_173101_P019_WD04 0.480769 1079        14
## 185821  UGA_173101_P022_WA12 UGA_173101_P018_WF03 0.473881 1032        14
## 185825  UGA_173101_P022_WA12 UGA_173101_P018_WE09 0.476217  982        14
## 185848  UGA_173101_P022_WA12 UGA_173101_P020_WA07 0.478448 1021        14
## 185850  UGA_173101_P022_WA12 UGA_173101_P016_WD06 0.469424  921        14
## 185864  UGA_173101_P022_WA12 UGA_173101_P016_WC05 0.463315  428        14
## 185885  UGA_173101_P022_WA12 UGA_173101_P017_WA02 0.480769 1077        14
## 185899  UGA_173101_P022_WA12 UGA_173101_P015_WH07 0.479466 1061        14
## 185929  UGA_173101_P022_WA12 UGA_173101_P020_WA05 0.482897 1080        14
## 185993  UGA_173101_P022_WA12 UGA_173101_P017_WD05 0.452128  457        14
## 185997  UGA_173101_P022_WA12 UGA_173101_P018_WE02 0.480769 1079        14
## 186008  UGA_173101_P022_WA12 UGA_173101_P015_WG11 0.453846  315        14
## 186060  UGA_173101_P022_WA12 UGA_173101_P016_WH06 0.469388  848        14
## 186064  UGA_173101_P022_WA12 UGA_173101_P016_WH08 0.474558  974        14
## 186089  UGA_173101_P022_WA12 UGA_173101_P017_WF04 0.477642 1074        14
## 186116  UGA_173101_P022_WA12 UGA_173101_P017_WF08 0.464697  778        14
## 186136  UGA_173101_P022_WA12 UGA_173101_P017_WH09 0.461886  847        14
## 186182  UGA_173101_P022_WA12 UGA_173101_P023_WG04 0.480769 1101        14
## 186199  UGA_173101_P022_WA12 UGA_173101_P023_WB11 0.476048 1084        14
## 186202  UGA_173101_P022_WA12 UGA_173101_P023_WB06 0.472973 1064        14
## 186209  UGA_173101_P022_WA12 UGA_173101_P023_WB04 0.481188 1098        14
## 186265  UGA_173101_P022_WA12 UGA_173101_P020_WG06 0.481791 1100        14
## 186300  UGA_173101_P022_WA12 UGA_173101_P020_WH04 0.478880 1102        14
## 186312  UGA_173101_P022_WA12 UGA_173101_P020_WG07 0.459967  650        14
## 186316  UGA_173101_P022_WA12 UGA_173101_P020_WH11 0.481791 1102        14
## 186345  UGA_173101_P022_WA12 UGA_173101_P022_WG09 0.482742 1101        14
## 201310  UGA_173101_P021_WC01 UGA_173101_P020_WA09 0.491183 1021        14
## 201348  UGA_173101_P021_WC01 UGA_173101_P019_WC06 0.491623 1174        14
## 201351  UGA_173101_P021_WC01 UGA_173101_P019_WC02 0.492504 1173        14
## 201357  UGA_173101_P021_WC01 UGA_173101_P019_WD04 0.491637 1177        14
## 201371  UGA_173101_P021_WC01 UGA_173101_P018_WF03 0.484951 1105        14
## 201375  UGA_173101_P021_WC01 UGA_173101_P018_WE09 0.483434 1051        14
## 201398  UGA_173101_P021_WC01 UGA_173101_P020_WA07 0.488235 1095        14
## 201400  UGA_173101_P021_WC01 UGA_173101_P016_WD06 0.480220  979        14
## 201414  UGA_173101_P021_WC01 UGA_173101_P016_WC05 0.471939  450        14
## 201435  UGA_173101_P021_WC01 UGA_173101_P017_WA02 0.491150 1174        14
## 201449  UGA_173101_P021_WC01 UGA_173101_P015_WH07 0.489091 1148        14
## 201479  UGA_173101_P021_WC01 UGA_173101_P020_WA05 0.493386 1171        14
## 201543  UGA_173101_P021_WC01 UGA_173101_P017_WD05 0.458333  475        14
## 201547  UGA_173101_P021_WC01 UGA_173101_P018_WE02 0.492077 1176        14
## 201558  UGA_173101_P021_WC01 UGA_173101_P015_WG11 0.453704  324        14
## 201610  UGA_173101_P021_WC01 UGA_173101_P016_WH06 0.478337  897        14
## 201614  UGA_173101_P021_WC01 UGA_173101_P016_WH08 0.483838 1038        14
## 201639  UGA_173101_P021_WC01 UGA_173101_P017_WF04 0.488309 1161        14
## 201666  UGA_173101_P021_WC01 UGA_173101_P017_WF08 0.475267  824        14
## 201686  UGA_173101_P021_WC01 UGA_173101_P017_WH09 0.471823  898        14
## 201732  UGA_173101_P021_WC01 UGA_173101_P023_WG04 0.493139 1199        14
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## 201749  UGA_173101_P021_WC01 UGA_173101_P023_WB11 0.486607 1172        14
## 201752  UGA_173101_P021_WC01 UGA_173101_P023_WB06 0.481707 1145        14
## 201759  UGA_173101_P021_WC01 UGA_173101_P023_WB04 0.490838 1188        14
## 201815  UGA_173101_P021_WC01 UGA_173101_P020_WG06 0.492215 1192        14
## 201850  UGA_173101_P021_WC01 UGA_173101_P020_WH04 0.486775 1202        14
## 201862  UGA_173101_P021_WC01 UGA_173101_P020_WG07 0.470820  671        14
## 201866  UGA_173101_P021_WC01 UGA_173101_P020_WH11 0.493162 1202        14
## 201895  UGA_173101_P021_WC01 UGA_173101_P022_WG09 0.493997 1200        14
## 201906  UGA_173101_P021_WC01 UGA_173101_P022_WA12 0.477545 1079        14
## 222793  UGA_173101_P021_WH06 UGA_173101_P020_WA09 0.485326 1007        14
## 222831  UGA_173101_P021_WH06 UGA_173101_P019_WC06 0.488383 1169        14
## 222834  UGA_173101_P021_WH06 UGA_173101_P019_WC02 0.487013 1169        14
## 222840  UGA_173101_P021_WH06 UGA_173101_P019_WD04 0.486965 1169        14
## 222854  UGA_173101_P021_WH06 UGA_173101_P018_WF03 0.480693 1098        14
## 222858  UGA_173101_P021_WH06 UGA_173101_P018_WE09 0.478615 1048        14
## 222881  UGA_173101_P021_WH06 UGA_173101_P020_WA07 0.483333 1086        14
## 222883  UGA_173101_P021_WH06 UGA_173101_P016_WD06 0.471088  972        14
## 222897  UGA_173101_P021_WH06 UGA_173101_P016_WC05 0.465385  443        14
## 222918  UGA_173101_P021_WH06 UGA_173101_P017_WA02 0.486059 1168        14
## 222932  UGA_173101_P021_WH06 UGA_173101_P015_WH07 0.484848 1137        14
## 222962  UGA_173101_P021_WH06 UGA_173101_P020_WA05 0.487312 1155        14
## 223026  UGA_173101_P021_WH06 UGA_173101_P017_WD05 0.456030  469        14
## 223030  UGA_173101_P021_WH06 UGA_173101_P018_WE02 0.486059 1167        14
## 223041  UGA_173101_P021_WH06 UGA_173101_P015_WG11 0.453704  322        14
## 223093  UGA_173101_P021_WH06 UGA_173101_P016_WH06 0.475768  889        14
## 223097  UGA_173101_P021_WH06 UGA_173101_P016_WH08 0.477510 1033        14
## 223122  UGA_173101_P021_WH06 UGA_173101_P017_WF04 0.482547 1155        14
## 223149  UGA_173101_P021_WH06 UGA_173101_P017_WF08 0.470080  824        14
## 223169  UGA_173101_P021_WH06 UGA_173101_P017_WH09 0.470339  900        14
## 223215  UGA_173101_P021_WH06 UGA_173101_P023_WG04 0.486289 1188        14
## 223232  UGA_173101_P021_WH06 UGA_173101_P023_WB11 0.484201 1160        14
## 223235  UGA_173101_P021_WH06 UGA_173101_P023_WB06 0.474038 1136        14
## 223242  UGA_173101_P021_WH06 UGA_173101_P023_WB04 0.484375 1177        14
## 223298  UGA_173101_P021_WH06 UGA_173101_P020_WG06 0.485806 1183        14
## 223333  UGA_173101_P021_WH06 UGA_173101_P020_WH04 0.481397 1192        14
## 223345  UGA_173101_P021_WH06 UGA_173101_P020_WG07 0.467742  664        14
## 223349  UGA_173101_P021_WH06 UGA_173101_P020_WH11 0.487296 1193        14
## 223378  UGA_173101_P021_WH06 UGA_173101_P022_WG09 0.487250 1190        14
## 223389  UGA_173101_P021_WH06 UGA_173101_P022_WA12 0.472050 1060        14
## 223414  UGA_173101_P021_WH06 UGA_173101_P021_WC01 0.483240 1143        14
## 247471  UGA_173101_P020_WH03 UGA_173101_P020_WA09 0.489329 1048        14
## 247509  UGA_173101_P020_WH03 UGA_173101_P019_WC06 0.491337 1263        14
## 247512  UGA_173101_P020_WH03 UGA_173101_P019_WC02 0.491736 1262        14
## 247518  UGA_173101_P020_WH03 UGA_173101_P019_WD04 0.489686 1270        14
## 247532  UGA_173101_P020_WH03 UGA_173101_P018_WF03 0.484579 1159        14
## 247536  UGA_173101_P020_WH03 UGA_173101_P018_WE09 0.484405 1096        14
## 247559  UGA_173101_P020_WH03 UGA_173101_P020_WA07 0.488095 1133        14
## 247561  UGA_173101_P020_WH03 UGA_173101_P016_WD06 0.478987 1006        14
## 247575  UGA_173101_P020_WH03 UGA_173101_P016_WC05 0.468593  455        14
## 247596  UGA_173101_P020_WH03 UGA_173101_P017_WA02 0.489583 1258        14
## 247610  UGA_173101_P020_WH03 UGA_173101_P015_WH07 0.487024 1214        14
## 247640  UGA_173101_P020_WH03 UGA_173101_P020_WA05 0.492017 1237        14
## 247704  UGA_173101_P020_WH03 UGA_173101_P017_WD05 0.454208  483        14
## 247708  UGA_173101_P020_WH03 UGA_173101_P018_WE02 0.490017 1254        14
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## 247719  UGA_173101_P020_WH03 UGA_173101_P015_WG11 0.452206  329        14
## 247771  UGA_173101_P020_WH03 UGA_173101_P016_WH06 0.478311  926        14
## 247775  UGA_173101_P020_WH03 UGA_173101_P016_WH08 0.483559 1089        14
## 247800  UGA_173101_P020_WH03 UGA_173101_P017_WF04 0.488769 1239        14
## 247827  UGA_173101_P020_WH03 UGA_173101_P017_WF08 0.473958  850        14
## 247847  UGA_173101_P020_WH03 UGA_173101_P017_WH09 0.473256  935        14
## 247893  UGA_173101_P020_WH03 UGA_173101_P023_WG04 0.491870 1279        14
## 247910  UGA_173101_P020_WH03 UGA_173101_P023_WB11 0.487136 1230        14
## 247913  UGA_173101_P020_WH03 UGA_173101_P023_WB06 0.478752 1200        14
## 247920  UGA_173101_P020_WH03 UGA_173101_P023_WB04 0.488333 1261        14
## 247976  UGA_173101_P020_WH03 UGA_173101_P020_WG06 0.490605 1269        14
## 248011  UGA_173101_P020_WH03 UGA_173101_P020_WH04 0.485737 1297        14
## 248023  UGA_173101_P020_WH03 UGA_173101_P020_WG07 0.470312  676        14
## 248027  UGA_173101_P020_WH03 UGA_173101_P020_WH11 0.492063 1300        14
## 248056  UGA_173101_P020_WH03 UGA_173101_P022_WG09 0.492363 1290        14
## 248067  UGA_173101_P020_WH03 UGA_173101_P022_WA12 0.476424 1103        14
## 248092  UGA_173101_P020_WH03 UGA_173101_P021_WC01 0.488481 1203        14
## 248125  UGA_173101_P020_WH03 UGA_173101_P021_WH06 0.483212 1193        14
## 281640            GTS_938-23 UGA_173101_P020_WA09 0.490741   94       301
## 281678            GTS_938-23 UGA_173101_P019_WC06 0.483871  119       301
## 281681            GTS_938-23 UGA_173101_P019_WC02 0.492188  119       301
## 281687            GTS_938-23 UGA_173101_P019_WD04 0.476562  122       301
## 281701            GTS_938-23 UGA_173101_P018_WF03 0.491667  110       301
## 281705            GTS_938-23 UGA_173101_P018_WE09 0.482143  105       301
## 281728            GTS_938-23 UGA_173101_P020_WA07 0.468750  104       301
## 281730            GTS_938-23 UGA_173101_P016_WD06 0.490385   99       301
## 281744            GTS_938-23 UGA_173101_P016_WC05 0.485294   59       301
## 281765            GTS_938-23 UGA_173101_P017_WA02 0.491935  117       301
## 281779            GTS_938-23 UGA_173101_P015_WH07 0.476562  113       301
## 281809            GTS_938-23 UGA_173101_P020_WA05 0.476562  114       301
## 281873            GTS_938-23 UGA_173101_P017_WD05 0.437500   58       301
## 281877            GTS_938-23 UGA_173101_P018_WE02 0.491935  115       301
## 281888            GTS_938-23 UGA_173101_P015_WG11 0.416667   45       301
## 281940            GTS_938-23 UGA_173101_P016_WH06 0.490385   90       301
## 281944            GTS_938-23 UGA_173101_P016_WH08 0.482759  102       301
## 281969            GTS_938-23 UGA_173101_P017_WF04 0.476562  117       301
## 281996            GTS_938-23 UGA_173101_P017_WF08 0.456522   94       301
## 282016            GTS_938-23 UGA_173101_P017_WH09 0.462963   96       301
## 282062            GTS_938-23 UGA_173101_P023_WG04 0.484375  115       301
## 282079            GTS_938-23 UGA_173101_P023_WB11 0.491935  106       301
## 282082            GTS_938-23 UGA_173101_P023_WB06 0.483333  107       301
## 282089            GTS_938-23 UGA_173101_P023_WB04 0.482759  108       301
## 282145            GTS_938-23 UGA_173101_P020_WG06 0.483871  113       301
## 282180            GTS_938-23 UGA_173101_P020_WH04 0.484375  117       301
## 282192            GTS_938-23 UGA_173101_P020_WG07 0.466667   57       301
## 282196            GTS_938-23 UGA_173101_P020_WH11 0.484375  118       301
## 282225            GTS_938-23 UGA_173101_P022_WG09 0.484375  114       301
## 282236            GTS_938-23 UGA_173101_P022_WA12 0.461538   94       301
## 282261            GTS_938-23 UGA_173101_P021_WC01 0.473214  100       301
## 282294            GTS_938-23 UGA_173101_P021_WH06 0.480769  102       301
## 282330            GTS_938-23 UGA_173101_P020_WH03 0.468750  118       301
## 284650            GTS_944-23 UGA_173101_P020_WA09 0.490741   93       305
## 284688            GTS_944-23 UGA_173101_P019_WC06 0.491935  117       305
## 284691            GTS_944-23 UGA_173101_P019_WC02 0.492188  118       305
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## 284697            GTS_944-23 UGA_173101_P019_WD04 0.484375  120        305
## 284711            GTS_944-23 UGA_173101_P018_WF03 0.491667  109        305
## 284715            GTS_944-23 UGA_173101_P018_WE09 0.482143  104        305
## 284738            GTS_944-23 UGA_173101_P020_WA07 0.468750  103        305
## 284740            GTS_944-23 UGA_173101_P016_WD06 0.490385   98        305
## 284754            GTS_944-23 UGA_173101_P016_WC05 0.485294   58        305
## 284775            GTS_944-23 UGA_173101_P017_WA02 0.491935  116        305
## 284789            GTS_944-23 UGA_173101_P015_WH07 0.484375  111        305
## 284819            GTS_944-23 UGA_173101_P020_WA05 0.484375  112        305
## 284883            GTS_944-23 UGA_173101_P017_WD05 0.437500   57        305
## 284887            GTS_944-23 UGA_173101_P018_WE02 0.491935  114        305
## 284898            GTS_944-23 UGA_173101_P015_WG11 0.416667   45        305
## 284950            GTS_944-23 UGA_173101_P016_WH06 0.490385   89        305
## 284954            GTS_944-23 UGA_173101_P016_WH08 0.482759  101        305
## 284979            GTS_944-23 UGA_173101_P017_WF04 0.476562  116        305
## 285006            GTS_944-23 UGA_173101_P017_WF08 0.456522   93        305
## 285026            GTS_944-23 UGA_173101_P017_WH09 0.462963   95        305
## 285072            GTS_944-23 UGA_173101_P023_WG04 0.492188  113        305
## 285089            GTS_944-23 UGA_173101_P023_WB11 0.491935  105        305
## 285092            GTS_944-23 UGA_173101_P023_WB06 0.483333  106        305
## 285099            GTS_944-23 UGA_173101_P023_WB04 0.482759  107        305
## 285155            GTS_944-23 UGA_173101_P020_WG06 0.491935  111        305
## 285190            GTS_944-23 UGA_173101_P020_WH04 0.492188  115        305
## 285202            GTS_944-23 UGA_173101_P020_WG07 0.466667   57        305
## 285206            GTS_944-23 UGA_173101_P020_WH11 0.492188  116        305
## 285235            GTS_944-23 UGA_173101_P022_WG09 0.492188  112        305
## 285246            GTS_944-23 UGA_173101_P022_WA12 0.471154   92        305
## 285271            GTS_944-23 UGA_173101_P021_WC01 0.482143   98        305
## 285304            GTS_944-23 UGA_173101_P021_WH06 0.480769  101        305
## 285340            GTS_944-23 UGA_173101_P020_WH03 0.476562  116        305
## 285387            GTS_944-23           GTS_938-23 0.500000  121        305
## 303825            GTS_631-23 UGA_173101_P020_WA09 0.451923   93        301
## 303863            GTS_631-23 UGA_173101_P019_WC06 0.450000  117        301
## 303866            GTS_631-23 UGA_173101_P019_WC02 0.459677  117        301
## 303872            GTS_631-23 UGA_173101_P019_WD04 0.443548  120        301
## 303886            GTS_631-23 UGA_173101_P018_WF03 0.456897  108        301
## 303890            GTS_631-23 UGA_173101_P018_WE09 0.444444  103        301
## 303913            GTS_631-23 UGA_173101_P020_WA07 0.423913  102        301
## 303915            GTS_631-23 UGA_173101_P016_WD06 0.450000   97        301
## 303929            GTS_631-23 UGA_173101_P016_WC05 0.426471   59        301
## 303950            GTS_631-23 UGA_173101_P017_WA02 0.458333  115        301
## 303964            GTS_631-23 UGA_173101_P015_WH07 0.443548  111        301
## 303994            GTS_631-23 UGA_173101_P020_WA05 0.443548  112        301
## 304058            GTS_631-23 UGA_173101_P017_WD05 0.354167   57        301
## 304062            GTS_631-23 UGA_173101_P018_WE02 0.458333  113        301
## 304073            GTS_631-23 UGA_173101_P015_WG11 0.333333   46        301
## 304125            GTS_631-23 UGA_173101_P016_WH06 0.450000   89        301
## 304129            GTS_631-23 UGA_173101_P016_WH08 0.446429  100        301
## 304154            GTS_631-23 UGA_173101_P017_WF04 0.443548  115        301
## 304181            GTS_631-23 UGA_173101_P017_WF08 0.409091   93        301
## 304201            GTS_631-23 UGA_173101_P017_WH09 0.423077   94        301
## 304247            GTS_631-23 UGA_173101_P023_WG04 0.451613  113        301
## 304264            GTS_631-23 UGA_173101_P023_WB11 0.458333  104        301
## 304267            GTS_631-23 UGA_173101_P023_WB06 0.448276  105        301
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## 304274          GTS_631-23 UGA_173101_P023_WB04 0.446429  106       301
## 304330          GTS_631-23 UGA_173101_P020_WG06 0.450000  111       301
## 304365          GTS_631-23 UGA_173101_P020_WH04 0.451613  115       301
## 304377          GTS_631-23 UGA_173101_P020_WG07 0.400000   58       301
## 304381          GTS_631-23 UGA_173101_P020_WH11 0.451613  116       301
## 304410          GTS_631-23 UGA_173101_P022_WG09 0.451613  112       301
## 304421          GTS_631-23 UGA_173101_P022_WA12 0.420000   92       301
## 304446          GTS_631-23 UGA_173101_P021_WC01 0.435185   98       301
## 304479          GTS_631-23 UGA_173101_P021_WH06 0.440000  100       301
## 304515          GTS_631-23 UGA_173101_P020_WH03 0.435484  116       301
## 304562          GTS_631-23           GTS_938-23 0.500000  120       301
## 304566          GTS_631-23           GTS_944-23 0.500000  119       301
## 309306          GTS_781-23 UGA_173101_P020_WA09 0.453704   94       305
## 309344          GTS_781-23 UGA_173101_P019_WC06 0.460938  119       305
## 309347          GTS_781-23 UGA_173101_P019_WC02 0.460938  119       305
## 309353          GTS_781-23 UGA_173101_P019_WD04 0.454545  122       305
## 309367          GTS_781-23 UGA_173101_P018_WF03 0.458333  110       305
## 309371          GTS_781-23 UGA_173101_P018_WE09 0.446429  105       305
## 309394          GTS_781-23 UGA_173101_P020_WA07 0.427083  104       305
## 309396          GTS_781-23 UGA_173101_P016_WD06 0.451923   99       305
## 309410          GTS_781-23 UGA_173101_P016_WC05 0.426471   59       305
## 309431          GTS_781-23 UGA_173101_P017_WA02 0.459677  117       305
## 309445          GTS_781-23 UGA_173101_P015_WH07 0.454545  113       305
## 309475          GTS_781-23 UGA_173101_P020_WA05 0.454545  114       305
## 309539          GTS_781-23 UGA_173101_P017_WD05 0.354167   58       305
## 309543          GTS_781-23 UGA_173101_P018_WE02 0.459677  115       305
## 309554          GTS_781-23 UGA_173101_P015_WG11 0.333333   46       305
## 309606          GTS_781-23 UGA_173101_P016_WH06 0.451923   90       305
## 309610          GTS_781-23 UGA_173101_P016_WH08 0.448276  102       305
## 309635          GTS_781-23 UGA_173101_P017_WF04 0.445312  117       305
## 309662          GTS_781-23 UGA_173101_P017_WF08 0.413043   94       305
## 309682          GTS_781-23 UGA_173101_P017_WH09 0.425926   96       305
## 309728          GTS_781-23 UGA_173101_P023_WG04 0.462121  115       305
## 309745          GTS_781-23 UGA_173101_P023_WB11 0.459677  106       305
## 309748          GTS_781-23 UGA_173101_P023_WB06 0.450000  107       305
## 309755          GTS_781-23 UGA_173101_P023_WB04 0.448276  108       305
## 309811          GTS_781-23 UGA_173101_P020_WG06 0.460938  113       305
## 309846          GTS_781-23 UGA_173101_P020_WH04 0.462121  117       305
## 309858          GTS_781-23 UGA_173101_P020_WG07 0.400000   58       305
## 309862          GTS_781-23 UGA_173101_P020_WH11 0.462121  118       305
## 309891          GTS_781-23 UGA_173101_P022_WG09 0.462121  114       305
## 309902          GTS_781-23 UGA_173101_P022_WA12 0.435185   94       305
## 309927          GTS_781-23 UGA_173101_P021_WC01 0.448276  100       305
## 309960          GTS_781-23 UGA_173101_P021_WH06 0.442308  102       305
## 309996          GTS_781-23 UGA_173101_P020_WH03 0.446970  118       305
## 310043          GTS_781-23           GTS_938-23 0.492188  122       305
## 310047          GTS_781-23           GTS_944-23 0.500000  121       305
## 310072          GTS_781-23           GTS_631-23 0.491935  121       305
## 316425         GTS_781-23R UGA_173101_P020_WA09 0.490741   93       305
## 316463         GTS_781-23R UGA_173101_P019_WC06 0.492188  118       305
## 316466         GTS_781-23R UGA_173101_P019_WC02 0.492188  118       305
## 316472         GTS_781-23R UGA_173101_P019_WD04 0.484848  121       305
## 316486         GTS_781-23R UGA_173101_P018_WF03 0.491667  109       305
## 316490         GTS_781-23R UGA_173101_P018_WE09 0.482143  104       305
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## 316513          GTS_781-23R UGA_173101_P020_WA07 0.468750  103       305
## 316515          GTS_781-23R UGA_173101_P016_WD06 0.490385   98       305
## 316529          GTS_781-23R UGA_173101_P016_WC05 0.485294   58       305
## 316550          GTS_781-23R UGA_173101_P017_WA02 0.491935  116       305
## 316564          GTS_781-23R UGA_173101_P015_WH07 0.484848  112       305
## 316594          GTS_781-23R UGA_173101_P020_WA05 0.484848  113       305
## 316658          GTS_781-23R UGA_173101_P017_WD05 0.437500   57       305
## 316662          GTS_781-23R UGA_173101_P018_WE02 0.491935  114       305
## 316673          GTS_781-23R UGA_173101_P015_WG11 0.416667   45       305
## 316725          GTS_781-23R UGA_173101_P016_WH06 0.490385   89       305
## 316729          GTS_781-23R UGA_173101_P016_WH08 0.482759  101       305
## 316754          GTS_781-23R UGA_173101_P017_WF04 0.476562  116       305
## 316781          GTS_781-23R UGA_173101_P017_WF08 0.456522   93       305
## 316801          GTS_781-23R UGA_173101_P017_WH09 0.462963   95       305
## 316847          GTS_781-23R UGA_173101_P023_WG04 0.492424  114       305
## 316864          GTS_781-23R UGA_173101_P023_WB11 0.491935  105       305
## 316867          GTS_781-23R UGA_173101_P023_WB06 0.483333  106       305
## 316874          GTS_781-23R UGA_173101_P023_WB04 0.482759  107       305
## 316930          GTS_781-23R UGA_173101_P020_WG06 0.492188  112       305
## 316965          GTS_781-23R UGA_173101_P020_WH04 0.492424  116       305
## 316977          GTS_781-23R UGA_173101_P020_WG07 0.466667   57       305
## 316981          GTS_781-23R UGA_173101_P020_WH11 0.492424  117       305
## 317010          GTS_781-23R UGA_173101_P022_WG09 0.492424  113       305
## 317021          GTS_781-23R UGA_173101_P022_WA12 0.472222   93       305
## 317046          GTS_781-23R UGA_173101_P021_WC01 0.482759   99       305
## 317079          GTS_781-23R UGA_173101_P021_WH06 0.480769  101       305
## 317115          GTS_781-23R UGA_173101_P020_WH03 0.477273  117       305
## 317162          GTS_781-23R           GTS_938-23 0.492188  122       305
## 317166          GTS_781-23R           GTS_944-23 0.500000  121       305
## 317191          GTS_781-23R           GTS_631-23 0.491935  120       305
## 317198          GTS_781-23R           GTS_781-23 0.500000  122       305
## 318816           GTS_638-23 UGA_173101_P020_WA09 0.480769   94       301
## 318854           GTS_638-23 UGA_173101_P019_WC06 0.483333  116       301
## 318857           GTS_638-23 UGA_173101_P019_WC02 0.483871  117       301
## 318863           GTS_638-23 UGA_173101_P019_WD04 0.475806  119       301
## 318877           GTS_638-23 UGA_173101_P018_WF03 0.482759  108       301
## 318881           GTS_638-23 UGA_173101_P018_WE09 0.472222  104       301
## 318904           GTS_638-23 UGA_173101_P020_WA07 0.458333  102       301
## 318906           GTS_638-23 UGA_173101_P016_WD06 0.480000   97       301
## 318920           GTS_638-23 UGA_173101_P016_WC05 0.485294   59       301
## 318941           GTS_638-23 UGA_173101_P017_WA02 0.483333  115       301
## 318955           GTS_638-23 UGA_173101_P015_WH07 0.475806  110       301
## 318985           GTS_638-23 UGA_173101_P020_WA05 0.475806  111       301
## 319049           GTS_638-23 UGA_173101_P017_WD05 0.437500   57       301
## 319053           GTS_638-23 UGA_173101_P018_WE02 0.483333  113       301
## 319064           GTS_638-23 UGA_173101_P015_WG11 0.416667   45       301
## 319116           GTS_638-23 UGA_173101_P016_WH06 0.480000   89       301
## 319120           GTS_638-23 UGA_173101_P016_WH08 0.473214  100       301
## 319145           GTS_638-23 UGA_173101_P017_WF04 0.467742  115       301
## 319172           GTS_638-23 UGA_173101_P017_WF08 0.445652   94       301
## 319192           GTS_638-23 UGA_173101_P017_WH09 0.451923   95       301
## 319238           GTS_638-23 UGA_173101_P023_WG04 0.483871  112       301
## 319255           GTS_638-23 UGA_173101_P023_WB11 0.483333  105       301
## 319258           GTS_638-23 UGA_173101_P023_WB06 0.474138  105       301
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## 319265            GTS_638-23 UGA_173101_P023_WB04 0.473214  107        301
## 319321            GTS_638-23 UGA_173101_P020_WG06 0.483333  110        301
## 319356            GTS_638-23 UGA_173101_P020_WH04 0.483871  114        301
## 319368            GTS_638-23 UGA_173101_P020_WG07 0.466667   57        301
## 319372            GTS_638-23 UGA_173101_P020_WH11 0.483871  115        301
## 319401            GTS_638-23 UGA_173101_P022_WG09 0.483871  111        301
## 319412            GTS_638-23 UGA_173101_P022_WA12 0.460000   92        301
## 319437            GTS_638-23 UGA_173101_P021_WC01 0.472222   98        301
## 319470            GTS_638-23 UGA_173101_P021_WH06 0.470000  100        301
## 319506            GTS_638-23 UGA_173101_P020_WH03 0.467742  115        301
## 319553            GTS_638-23           GTS_938-23 0.491935  120        301
## 319557            GTS_638-23           GTS_944-23 0.491935  119        301
## 319582            GTS_638-23           GTS_631-23 0.500000  118        301
## 319589            GTS_638-23           GTS_781-23 0.491935  119        301
## 319598            GTS_638-23          GTS_781-23R 0.491935  119        301
## 324430            GTS_972-23 UGA_173101_P020_WA09 0.490741   93        301
## 324468            GTS_972-23 UGA_173101_P019_WC06 0.491935  117        301
## 324471            GTS_972-23 UGA_173101_P019_WC02 0.492188  118        301
## 324477            GTS_972-23 UGA_173101_P019_WD04 0.484375  120        301
## 324491            GTS_972-23 UGA_173101_P018_WF03 0.491667  109        301
## 324495            GTS_972-23 UGA_173101_P018_WE09 0.482143  104        301
## 324518            GTS_972-23 UGA_173101_P020_WA07 0.468750  103        301
## 324520            GTS_972-23 UGA_173101_P016_WD06 0.490385   98        301
## 324534            GTS_972-23 UGA_173101_P016_WC05 0.485294   58        301
## 324555            GTS_972-23 UGA_173101_P017_WA02 0.491935  116        301
## 324569            GTS_972-23 UGA_173101_P015_WH07 0.484375  111        301
## 324599            GTS_972-23 UGA_173101_P020_WA05 0.484375  112        301
## 324663            GTS_972-23 UGA_173101_P017_WD05 0.437500   57        301
## 324667            GTS_972-23 UGA_173101_P018_WE02 0.491935  114        301
## 324678            GTS_972-23 UGA_173101_P015_WG11 0.416667   45        301
## 324730            GTS_972-23 UGA_173101_P016_WH06 0.490385   89        301
## 324734            GTS_972-23 UGA_173101_P016_WH08 0.482759  101        301
## 324759            GTS_972-23 UGA_173101_P017_WF04 0.476562  116        301
## 324786            GTS_972-23 UGA_173101_P017_WF08 0.456522   93        301
## 324806            GTS_972-23 UGA_173101_P017_WH09 0.462963   95        301
## 324852            GTS_972-23 UGA_173101_P023_WG04 0.492188  113        301
## 324869            GTS_972-23 UGA_173101_P023_WB11 0.491935  105        301
## 324872            GTS_972-23 UGA_173101_P023_WB06 0.483333  106        301
## 324879            GTS_972-23 UGA_173101_P023_WB04 0.482759  107        301
## 324935            GTS_972-23 UGA_173101_P020_WG06 0.491935  111        301
## 324970            GTS_972-23 UGA_173101_P020_WH04 0.492188  115        301
## 324982            GTS_972-23 UGA_173101_P020_WG07 0.466667   57        301
## 324986            GTS_972-23 UGA_173101_P020_WH11 0.492188  116        301
## 325015            GTS_972-23 UGA_173101_P022_WG09 0.492188  112        301
## 325026            GTS_972-23 UGA_173101_P022_WA12 0.471154   92        301
## 325051            GTS_972-23 UGA_173101_P021_WC01 0.482143   98        301
## 325084            GTS_972-23 UGA_173101_P021_WH06 0.480769  101        301
## 325120            GTS_972-23 UGA_173101_P020_WH03 0.476562  116        301
## 325167            GTS_972-23           GTS_938-23 0.500000  121        301
## 325171            GTS_972-23           GTS_944-23 0.500000  121        301
## 325196            GTS_972-23           GTS_631-23 0.500000  119        301
## 325203            GTS_972-23           GTS_781-23 0.500000  121        301
## 325212            GTS_972-23          GTS_781-23R 0.500000  121        301
## 325215            GTS_972-23           GTS_638-23 0.491935  119        301
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## 342393            GTS_651-23 UGA_173101_P020_WA09 0.490741   94        301
## 342431            GTS_651-23 UGA_173101_P019_WC06 0.491935  118        301
## 342434            GTS_651-23 UGA_173101_P019_WC02 0.492188  119        301
## 342440            GTS_651-23 UGA_173101_P019_WD04 0.484375  121        301
## 342454            GTS_651-23 UGA_173101_P018_WF03 0.491667  110        301
## 342458            GTS_651-23 UGA_173101_P018_WE09 0.482143  105        301
## 342481            GTS_651-23 UGA_173101_P020_WA07 0.468750  104        301
## 342483            GTS_651-23 UGA_173101_P016_WD06 0.490385   99        301
## 342497            GTS_651-23 UGA_173101_P016_WC05 0.485294   59        301
## 342518            GTS_651-23 UGA_173101_P017_WA02 0.491935  117        301
## 342532            GTS_651-23 UGA_173101_P015_WH07 0.484375  112        301
## 342562            GTS_651-23 UGA_173101_P020_WA05 0.484375  113        301
## 342626            GTS_651-23 UGA_173101_P017_WD05 0.437500   58        301
## 342630            GTS_651-23 UGA_173101_P018_WE02 0.491935  115        301
## 342641            GTS_651-23 UGA_173101_P015_WG11 0.416667   45        301
## 342693            GTS_651-23 UGA_173101_P016_WH06 0.490385   90        301
## 342697            GTS_651-23 UGA_173101_P016_WH08 0.482759  102        301
## 342722            GTS_651-23 UGA_173101_P017_WF04 0.476562  117        301
## 342749            GTS_651-23 UGA_173101_P017_WF08 0.456522   94        301
## 342769            GTS_651-23 UGA_173101_P017_WH09 0.462963   96        301
## 342815            GTS_651-23 UGA_173101_P023_WG04 0.492188  114        301
## 342832            GTS_651-23 UGA_173101_P023_WB11 0.491935  106        301
## 342835            GTS_651-23 UGA_173101_P023_WB06 0.483333  107        301
## 342842            GTS_651-23 UGA_173101_P023_WB04 0.482759  108        301
## 342898            GTS_651-23 UGA_173101_P020_WG06 0.491935  112        301
## 342933            GTS_651-23 UGA_173101_P020_WH04 0.492188  116        301
## 342945            GTS_651-23 UGA_173101_P020_WG07 0.466667   57        301
## 342949            GTS_651-23 UGA_173101_P020_WH11 0.492188  117        301
## 342978            GTS_651-23 UGA_173101_P022_WG09 0.492188  113        301
## 342989            GTS_651-23 UGA_173101_P022_WA12 0.471154   93        301
## 343014            GTS_651-23 UGA_173101_P021_WC01 0.482143   99        301
## 343047            GTS_651-23 UGA_173101_P021_WH06 0.480769  102        301
## 343083            GTS_651-23 UGA_173101_P020_WH03 0.476562  117        301
## 343130            GTS_651-23           GTS_938-23 0.500000  122        301
## 343134            GTS_651-23           GTS_944-23 0.500000  121        301
## 343159            GTS_651-23           GTS_631-23 0.500000  119        301
## 343166            GTS_651-23           GTS_781-23 0.500000  121        301
## 343175            GTS_651-23          GTS_781-23R 0.500000  121        301
## 343178            GTS_651-23           GTS_638-23 0.491935  120        301
## 343185            GTS_651-23           GTS_972-23 0.500000  121        301
## 343221            GTS_982-23 UGA_173101_P020_WA09 0.490741   94        301
## 343259            GTS_982-23 UGA_173101_P019_WC06 0.491935  118        301
## 343262            GTS_982-23 UGA_173101_P019_WC02 0.492188  119        301
## 343268            GTS_982-23 UGA_173101_P019_WD04 0.484375  121        301
## 343282            GTS_982-23 UGA_173101_P018_WF03 0.491667  110        301
## 343286            GTS_982-23 UGA_173101_P018_WE09 0.482143  105        301
## 343309            GTS_982-23 UGA_173101_P020_WA07 0.468750  104        301
## 343311            GTS_982-23 UGA_173101_P016_WD06 0.490385   99        301
## 343325            GTS_982-23 UGA_173101_P016_WC05 0.485294   59        301
## 343346            GTS_982-23 UGA_173101_P017_WA02 0.491935  117        301
## 343360            GTS_982-23 UGA_173101_P015_WH07 0.484375  112        301
## 343390            GTS_982-23 UGA_173101_P020_WA05 0.484375  113        301
## 343454            GTS_982-23 UGA_173101_P017_WD05 0.437500   58        301
## 343458            GTS_982-23 UGA_173101_P018_WE02 0.491935  115        301
```

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## 343469          GTS_982-23 UGA_173101_P015_WG11 0.416667   45       301
## 343521          GTS_982-23 UGA_173101_P016_WH06 0.490385   90       301
## 343525          GTS_982-23 UGA_173101_P016_WH08 0.482759  102       301
## 343550          GTS_982-23 UGA_173101_P017_WF04 0.476562  117       301
## 343577          GTS_982-23 UGA_173101_P017_WF08 0.456522   94       301
## 343597          GTS_982-23 UGA_173101_P017_WH09 0.462963   96       301
## 343643          GTS_982-23 UGA_173101_P023_WG04 0.492188  114       301
## 343660          GTS_982-23 UGA_173101_P023_WB11 0.491935  106       301
## 343663          GTS_982-23 UGA_173101_P023_WB06 0.483333  107       301
## 343670          GTS_982-23 UGA_173101_P023_WB04 0.482759  108       301
## 343726          GTS_982-23 UGA_173101_P020_WG06 0.491935  112       301
## 343761          GTS_982-23 UGA_173101_P020_WH04 0.492188  116       301
## 343773          GTS_982-23 UGA_173101_P020_WG07 0.466667   57       301
## 343777          GTS_982-23 UGA_173101_P020_WH11 0.492188  117       301
## 343806          GTS_982-23 UGA_173101_P022_WG09 0.492188  113       301
## 343817          GTS_982-23 UGA_173101_P022_WA12 0.471154   93       301
## 343842          GTS_982-23 UGA_173101_P021_WC01 0.482143   99       301
## 343875          GTS_982-23 UGA_173101_P021_WH06 0.480769  102       301
## 343911          GTS_982-23 UGA_173101_P020_WH03 0.476562  117       301
## 343958          GTS_982-23           GTS_938-23 0.500000  122       301
## 343962          GTS_982-23           GTS_944-23 0.500000  121       301
## 343987          GTS_982-23           GTS_631-23 0.500000  119       301
## 343994          GTS_982-23           GTS_781-23 0.500000  121       301
## 344003          GTS_982-23          GTS_781-23R 0.500000  121       301
## 344006          GTS_982-23           GTS_638-23 0.491935  120       301
## 344013          GTS_982-23           GTS_972-23 0.500000  121       301
## 344035          GTS_982-23           GTS_651-23 0.500000  122       301
## 350718         GTS_651-23R UGA_173101_P020_WA09 0.490741   94       301
## 350756         GTS_651-23R UGA_173101_P019_WC06 0.483871  119       301
## 350759         GTS_651-23R UGA_173101_P019_WC02 0.492188  119       301
## 350765         GTS_651-23R UGA_173101_P019_WD04 0.476562  122       301
## 350779         GTS_651-23R UGA_173101_P018_WF03 0.491667  110       301
## 350783         GTS_651-23R UGA_173101_P018_WE09 0.482143  105       301
## 350806         GTS_651-23R UGA_173101_P020_WA07 0.468750  104       301
## 350808         GTS_651-23R UGA_173101_P016_WD06 0.490385   99       301
## 350822         GTS_651-23R UGA_173101_P016_WC05 0.485294   59       301
## 350843         GTS_651-23R UGA_173101_P017_WA02 0.491935  117       301
## 350857         GTS_651-23R UGA_173101_P015_WH07 0.476562  113       301
## 350887         GTS_651-23R UGA_173101_P020_WA05 0.476562  114       301
## 350951         GTS_651-23R UGA_173101_P017_WD05 0.437500   58       301
## 350955         GTS_651-23R UGA_173101_P018_WE02 0.491935  115       301
## 350966         GTS_651-23R UGA_173101_P015_WG11 0.416667   45       301
## 351018         GTS_651-23R UGA_173101_P016_WH06 0.490385   90       301
## 351022         GTS_651-23R UGA_173101_P016_WH08 0.482759  102       301
## 351047         GTS_651-23R UGA_173101_P017_WF04 0.476562  117       301
## 351074         GTS_651-23R UGA_173101_P017_WF08 0.456522   94       301
## 351094         GTS_651-23R UGA_173101_P017_WH09 0.462963   96       301
## 351140         GTS_651-23R UGA_173101_P023_WG04 0.484375  115       301
## 351157         GTS_651-23R UGA_173101_P023_WB11 0.491935  106       301
## 351160         GTS_651-23R UGA_173101_P023_WB06 0.483333  107       301
## 351167         GTS_651-23R UGA_173101_P023_WB04 0.482759  108       301
## 351223         GTS_651-23R UGA_173101_P020_WG06 0.483871  113       301
## 351258         GTS_651-23R UGA_173101_P020_WH04 0.484375  117       301
## 351270         GTS_651-23R UGA_173101_P020_WG07 0.466667   57       301
```

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## 351274          GTS_651-23R UGA_173101_P020_WH11 0.484375  118       301
## 351303          GTS_651-23R UGA_173101_P022_WG09 0.484375  114       301
## 351314          GTS_651-23R UGA_173101_P022_WA12 0.461538   94       301
## 351339          GTS_651-23R UGA_173101_P021_WC01 0.473214  100       301
## 351372          GTS_651-23R UGA_173101_P021_WH06 0.480769  102       301
## 351408          GTS_651-23R UGA_173101_P020_WH03 0.468750  118       301
## 351455          GTS_651-23R           GTS_938-23 0.500000  123       301
## 351459          GTS_651-23R           GTS_944-23 0.500000  121       301
## 351484          GTS_651-23R           GTS_631-23 0.500000  120       301
## 351491          GTS_651-23R           GTS_781-23 0.492188  122       301
## 351500          GTS_651-23R          GTS_781-23R 0.492188  122       301
## 351503          GTS_651-23R           GTS_638-23 0.491935  120       301
## 351510          GTS_651-23R           GTS_972-23 0.500000  121       301
## 351532          GTS_651-23R           GTS_651-23 0.500000  122       301
## 351533          GTS_651-23R           GTS_982-23 0.500000  122       301
## 355761           GTS_803-23 UGA_173101_P020_WA09 0.490741   94       305
## 355799           GTS_803-23 UGA_173101_P019_WC06 0.492188  119       305
## 355802           GTS_803-23 UGA_173101_P019_WC02 0.492188  119       305
## 355808           GTS_803-23 UGA_173101_P019_WD04 0.484848  122       305
## 355822           GTS_803-23 UGA_173101_P018_WF03 0.491667  110       305
## 355826           GTS_803-23 UGA_173101_P018_WE09 0.482143  105       305
## 355849           GTS_803-23 UGA_173101_P020_WA07 0.468750  104       305
## 355851           GTS_803-23 UGA_173101_P016_WD06 0.490385   99       305
## 355865           GTS_803-23 UGA_173101_P016_WC05 0.485294   59       305
## 355886           GTS_803-23 UGA_173101_P017_WA02 0.491935  117       305
## 355900           GTS_803-23 UGA_173101_P015_WH07 0.484848  113       305
## 355930           GTS_803-23 UGA_173101_P020_WA05 0.484848  114       305
## 355994           GTS_803-23 UGA_173101_P017_WD05 0.437500   58       305
## 355998           GTS_803-23 UGA_173101_P018_WE02 0.491935  115       305
## 356009           GTS_803-23 UGA_173101_P015_WG11 0.416667   45       305
## 356061           GTS_803-23 UGA_173101_P016_WH06 0.490385   90       305
## 356065           GTS_803-23 UGA_173101_P016_WH08 0.482759  102       305
## 356090           GTS_803-23 UGA_173101_P017_WF04 0.476562  117       305
## 356117           GTS_803-23 UGA_173101_P017_WF08 0.456522   94       305
## 356137           GTS_803-23 UGA_173101_P017_WH09 0.462963   96       305
## 356183           GTS_803-23 UGA_173101_P023_WG04 0.492424  115       305
## 356200           GTS_803-23 UGA_173101_P023_WB11 0.491935  106       305
## 356203           GTS_803-23 UGA_173101_P023_WB06 0.483333  107       305
## 356210           GTS_803-23 UGA_173101_P023_WB04 0.482759  108       305
## 356266           GTS_803-23 UGA_173101_P020_WG06 0.492188  113       305
## 356301           GTS_803-23 UGA_173101_P020_WH04 0.492424  117       305
## 356313           GTS_803-23 UGA_173101_P020_WG07 0.466667   57       305
## 356317           GTS_803-23 UGA_173101_P020_WH11 0.492424  118       305
## 356346           GTS_803-23 UGA_173101_P022_WG09 0.492424  114       305
## 356357           GTS_803-23 UGA_173101_P022_WA12 0.472222   94       305
## 356382           GTS_803-23 UGA_173101_P021_WC01 0.482759  100       305
## 356415           GTS_803-23 UGA_173101_P021_WH06 0.480769  102       305
## 356451           GTS_803-23 UGA_173101_P020_WH03 0.477273  118       305
## 356498           GTS_803-23           GTS_938-23 0.492188  123       305
## 356502           GTS_803-23           GTS_944-23 0.500000  121       305
## 356527           GTS_803-23           GTS_631-23 0.491935  120       305
## 356534           GTS_803-23           GTS_781-23 0.500000  122       305
## 356543           GTS_803-23          GTS_781-23R 0.500000  122       305
## 356546           GTS_803-23           GTS_638-23 0.491935  120       305
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## 356553            GTS_803-23           GTS_972-23 0.500000  121        305
## 356575            GTS_803-23           GTS_651-23 0.500000  122        305
## 356576            GTS_803-23           GTS_982-23 0.500000  122        305
## 356585            GTS_803-23          GTS_651-23R 0.492188  123        305
## 369385            GTS_809-23 UGA_173101_P020_WA09 0.490741   94        305
## 369423            GTS_809-23 UGA_173101_P019_WC06 0.492188  119        305
## 369426            GTS_809-23 UGA_173101_P019_WC02 0.492188  119        305
## 369432            GTS_809-23 UGA_173101_P019_WD04 0.484848  122        305
## 369446            GTS_809-23 UGA_173101_P018_WF03 0.491667  110        305
## 369450            GTS_809-23 UGA_173101_P018_WE09 0.482143  105        305
## 369473            GTS_809-23 UGA_173101_P020_WA07 0.468750  104        305
## 369475            GTS_809-23 UGA_173101_P016_WD06 0.490385   99        305
## 369489            GTS_809-23 UGA_173101_P016_WC05 0.485294   59        305
## 369510            GTS_809-23 UGA_173101_P017_WA02 0.491935  117        305
## 369524            GTS_809-23 UGA_173101_P015_WH07 0.484848  113        305
## 369554            GTS_809-23 UGA_173101_P020_WA05 0.484848  114        305
## 369618            GTS_809-23 UGA_173101_P017_WD05 0.437500   58        305
## 369622            GTS_809-23 UGA_173101_P018_WE02 0.491935  115        305
## 369633            GTS_809-23 UGA_173101_P015_WG11 0.416667   45        305
## 369685            GTS_809-23 UGA_173101_P016_WH06 0.490385   90        305
## 369689            GTS_809-23 UGA_173101_P016_WH08 0.482759  102        305
## 369714            GTS_809-23 UGA_173101_P017_WF04 0.476562  117        305
## 369741            GTS_809-23 UGA_173101_P017_WF08 0.456522   94        305
## 369761            GTS_809-23 UGA_173101_P017_WH09 0.462963   96        305
## 369807            GTS_809-23 UGA_173101_P023_WG04 0.492424  115        305
## 369824            GTS_809-23 UGA_173101_P023_WB11 0.491935  106        305
## 369827            GTS_809-23 UGA_173101_P023_WB06 0.483333  107        305
## 369834            GTS_809-23 UGA_173101_P023_WB04 0.482759  108        305
## 369890            GTS_809-23 UGA_173101_P020_WG06 0.492188  113        305
## 369925            GTS_809-23 UGA_173101_P020_WH04 0.492424  117        305
## 369937            GTS_809-23 UGA_173101_P020_WG07 0.466667   57        305
## 369941            GTS_809-23 UGA_173101_P020_WH11 0.492424  118        305
## 369970            GTS_809-23 UGA_173101_P022_WG09 0.492424  114        305
## 369981            GTS_809-23 UGA_173101_P022_WA12 0.472222   94        305
## 370006            GTS_809-23 UGA_173101_P021_WC01 0.482759  100        305
## 370039            GTS_809-23 UGA_173101_P021_WH06 0.480769  102        305
## 370075            GTS_809-23 UGA_173101_P020_WH03 0.477273  118        305
## 370122            GTS_809-23           GTS_938-23 0.492188  123        305
## 370126            GTS_809-23           GTS_944-23 0.500000  121        305
## 370151            GTS_809-23           GTS_631-23 0.491935  120        305
## 370158            GTS_809-23           GTS_781-23 0.500000  122        305
## 370167            GTS_809-23          GTS_781-23R 0.500000  122        305
## 370170            GTS_809-23           GTS_638-23 0.491935  120        305
## 370177            GTS_809-23           GTS_972-23 0.500000  121        305
## 370199            GTS_809-23           GTS_651-23 0.500000  122        305
## 370200            GTS_809-23           GTS_982-23 0.500000  122        305
## 370209            GTS_809-23          GTS_651-23R 0.492188  123        305
## 370215            GTS_809-23           GTS_803-23 0.500000  123        305
## 385018           GTS_1009-23 UGA_173101_P020_WA09 0.490741   94        301
## 385056           GTS_1009-23 UGA_173101_P019_WC06 0.491935  118        301
## 385059           GTS_1009-23 UGA_173101_P019_WC02 0.492188  119        301
## 385065           GTS_1009-23 UGA_173101_P019_WD04 0.484375  121        301
## 385079           GTS_1009-23 UGA_173101_P018_WF03 0.491667  110        301
## 385083           GTS_1009-23 UGA_173101_P018_WE09 0.482143  105        301
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## 385106          GTS_1009-23 UGA_173101_P020_WA07 0.468750  104       301
## 385108          GTS_1009-23 UGA_173101_P016_WD06 0.490385   99       301
## 385122          GTS_1009-23 UGA_173101_P016_WC05 0.485294   59       301
## 385143          GTS_1009-23 UGA_173101_P017_WA02 0.491935  117       301
## 385157          GTS_1009-23 UGA_173101_P015_WH07 0.484375  112       301
## 385187          GTS_1009-23 UGA_173101_P020_WA05 0.484375  113       301
## 385251          GTS_1009-23 UGA_173101_P017_WD05 0.437500   58       301
## 385255          GTS_1009-23 UGA_173101_P018_WE02 0.491935  115       301
## 385266          GTS_1009-23 UGA_173101_P015_WG11 0.416667   45       301
## 385318          GTS_1009-23 UGA_173101_P016_WH06 0.490385   90       301
## 385322          GTS_1009-23 UGA_173101_P016_WH08 0.482759  102       301
## 385347          GTS_1009-23 UGA_173101_P017_WF04 0.476562  117       301
## 385374          GTS_1009-23 UGA_173101_P017_WF08 0.456522   94       301
## 385394          GTS_1009-23 UGA_173101_P017_WH09 0.462963   96       301
## 385440          GTS_1009-23 UGA_173101_P023_WG04 0.492188  114       301
## 385457          GTS_1009-23 UGA_173101_P023_WB11 0.491935  106       301
## 385460          GTS_1009-23 UGA_173101_P023_WB06 0.483333  107       301
## 385467          GTS_1009-23 UGA_173101_P023_WB04 0.482759  108       301
## 385523          GTS_1009-23 UGA_173101_P020_WG06 0.491935  112       301
## 385558          GTS_1009-23 UGA_173101_P020_WH04 0.492188  116       301
## 385570          GTS_1009-23 UGA_173101_P020_WG07 0.466667   57       301
## 385574          GTS_1009-23 UGA_173101_P020_WH11 0.492188  117       301
## 385603          GTS_1009-23 UGA_173101_P022_WG09 0.492188  113       301
## 385614          GTS_1009-23 UGA_173101_P022_WA12 0.471154   93       301
## 385639          GTS_1009-23 UGA_173101_P021_WC01 0.482143   99       301
## 385672          GTS_1009-23 UGA_173101_P021_WH06 0.480769  102       301
## 385708          GTS_1009-23 UGA_173101_P020_WH03 0.476562  117       301
## 385755          GTS_1009-23           GTS_938-23 0.500000  122       301
## 385759          GTS_1009-23           GTS_944-23 0.500000  121       301
## 385784          GTS_1009-23           GTS_631-23 0.500000  119       301
## 385791          GTS_1009-23           GTS_781-23 0.500000  121       301
## 385800          GTS_1009-23          GTS_781-23R 0.500000  121       301
## 385803          GTS_1009-23           GTS_638-23 0.491935  120       301
## 385810          GTS_1009-23           GTS_972-23 0.500000  121       301
## 385832          GTS_1009-23           GTS_651-23 0.500000  122       301
## 385833          GTS_1009-23           GTS_982-23 0.500000  122       301
## 385842          GTS_1009-23          GTS_651-23R 0.500000  122       301
## 385848          GTS_1009-23           GTS_803-23 0.500000  122       301
## 385864          GTS_1009-23           GTS_809-23 0.500000  122       301
## 389418          GTS_1013-23 UGA_173101_P020_WA09 0.490741   94       301
## 389456          GTS_1013-23 UGA_173101_P019_WC06 0.483871  119       301
## 389459          GTS_1013-23 UGA_173101_P019_WC02 0.492188  119       301
## 389465          GTS_1013-23 UGA_173101_P019_WD04 0.476562  122       301
## 389479          GTS_1013-23 UGA_173101_P018_WF03 0.491667  110       301
## 389483          GTS_1013-23 UGA_173101_P018_WE09 0.482143  105       301
## 389506          GTS_1013-23 UGA_173101_P020_WA07 0.468750  104       301
## 389508          GTS_1013-23 UGA_173101_P016_WD06 0.490385   99       301
## 389522          GTS_1013-23 UGA_173101_P016_WC05 0.485294   59       301
## 389543          GTS_1013-23 UGA_173101_P017_WA02 0.491935  117       301
## 389557          GTS_1013-23 UGA_173101_P015_WH07 0.476562  113       301
## 389587          GTS_1013-23 UGA_173101_P020_WA05 0.476562  114       301
## 389651          GTS_1013-23 UGA_173101_P017_WD05 0.437500   58       301
## 389655          GTS_1013-23 UGA_173101_P018_WE02 0.491935  115       301
## 389666          GTS_1013-23 UGA_173101_P015_WG11 0.416667   45       301
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## 389718          GTS_1013-23 UGA_173101_P016_WH06 0.490385   90       301
## 389722          GTS_1013-23 UGA_173101_P016_WH08 0.482759  102       301
## 389747          GTS_1013-23 UGA_173101_P017_WF04 0.476562  117       301
## 389774          GTS_1013-23 UGA_173101_P017_WF08 0.456522   94       301
## 389794          GTS_1013-23 UGA_173101_P017_WH09 0.462963   96       301
## 389840          GTS_1013-23 UGA_173101_P023_WG04 0.484375  115       301
## 389857          GTS_1013-23 UGA_173101_P023_WB11 0.491935  106       301
## 389860          GTS_1013-23 UGA_173101_P023_WB06 0.483333  107       301
## 389867          GTS_1013-23 UGA_173101_P023_WB04 0.482759  108       301
## 389923          GTS_1013-23 UGA_173101_P020_WG06 0.483871  113       301
## 389958          GTS_1013-23 UGA_173101_P020_WH04 0.484375  117       301
## 389970          GTS_1013-23 UGA_173101_P020_WG07 0.466667   57       301
## 389974          GTS_1013-23 UGA_173101_P020_WH11 0.484375  118       301
## 390003          GTS_1013-23 UGA_173101_P022_WG09 0.484375  114       301
## 390014          GTS_1013-23 UGA_173101_P022_WA12 0.461538   94       301
## 390039          GTS_1013-23 UGA_173101_P021_WC01 0.473214  100       301
## 390072          GTS_1013-23 UGA_173101_P021_WH06 0.480769  102       301
## 390108          GTS_1013-23 UGA_173101_P020_WH03 0.468750  118       301
## 390155          GTS_1013-23           GTS_938-23 0.500000  123       301
## 390159          GTS_1013-23           GTS_944-23 0.500000  121       301
## 390184          GTS_1013-23           GTS_631-23 0.500000  120       301
## 390191          GTS_1013-23           GTS_781-23 0.492188  122       301
## 390200          GTS_1013-23          GTS_781-23R 0.492188  122       301
## 390203          GTS_1013-23           GTS_638-23 0.491935  120       301
## 390210          GTS_1013-23           GTS_972-23 0.500000  121       301
## 390232          GTS_1013-23           GTS_651-23 0.500000  122       301
## 390233          GTS_1013-23           GTS_982-23 0.500000  122       301
## 390242          GTS_1013-23          GTS_651-23R 0.500000  123       301
## 390248          GTS_1013-23           GTS_803-23 0.492188  123       301
## 390264          GTS_1013-23           GTS_809-23 0.492188  123       301
## 390282          GTS_1013-23          GTS_1009-23 0.500000  122       301
## 450790           GTS_871-23 UGA_173101_P020_WA09 0.490741   93       305
## 450828           GTS_871-23 UGA_173101_P019_WC06 0.492188  118       305
## 450831           GTS_871-23 UGA_173101_P019_WC02 0.492188  118       305
## 450837           GTS_871-23 UGA_173101_P019_WD04 0.484848  121       305
## 450851           GTS_871-23 UGA_173101_P018_WF03 0.491667  109       305
## 450855           GTS_871-23 UGA_173101_P018_WE09 0.482143  104       305
## 450878           GTS_871-23 UGA_173101_P020_WA07 0.468750  103       305
## 450880           GTS_871-23 UGA_173101_P016_WD06 0.490385   98       305
## 450894           GTS_871-23 UGA_173101_P016_WC05 0.485294   58       305
## 450915           GTS_871-23 UGA_173101_P017_WA02 0.491935  116       305
## 450929           GTS_871-23 UGA_173101_P015_WH07 0.484848  112       305
## 450959           GTS_871-23 UGA_173101_P020_WA05 0.484848  113       305
## 451023           GTS_871-23 UGA_173101_P017_WD05 0.437500   57       305
## 451027           GTS_871-23 UGA_173101_P018_WE02 0.491935  114       305
## 451038           GTS_871-23 UGA_173101_P015_WG11 0.416667   45       305
## 451090           GTS_871-23 UGA_173101_P016_WH06 0.490385   89       305
## 451094           GTS_871-23 UGA_173101_P016_WH08 0.482759  101       305
## 451119           GTS_871-23 UGA_173101_P017_WF04 0.476562  116       305
## 451146           GTS_871-23 UGA_173101_P017_WF08 0.456522   93       305
## 451166           GTS_871-23 UGA_173101_P017_WH09 0.462963   95       305
## 451212           GTS_871-23 UGA_173101_P023_WG04 0.492424  114       305
## 451229           GTS_871-23 UGA_173101_P023_WB11 0.491935  105       305
## 451232           GTS_871-23 UGA_173101_P023_WB06 0.483333  106       305
```

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## 451239            GTS_871-23 UGA_173101_P023_WB04 0.482759  107        305
## 451295            GTS_871-23 UGA_173101_P020_WG06 0.492188  112        305
## 451330            GTS_871-23 UGA_173101_P020_WH04 0.492424  116        305
## 451342            GTS_871-23 UGA_173101_P020_WG07 0.466667   57        305
## 451346            GTS_871-23 UGA_173101_P020_WH11 0.492424  117        305
## 451375            GTS_871-23 UGA_173101_P022_WG09 0.492424  113        305
## 451386            GTS_871-23 UGA_173101_P022_WA12 0.472222   93        305
## 451411            GTS_871-23 UGA_173101_P021_WC01 0.482759   99        305
## 451444            GTS_871-23 UGA_173101_P021_WH06 0.480769  101        305
## 451480            GTS_871-23 UGA_173101_P020_WH03 0.477273  117        305
## 451527            GTS_871-23           GTS_938-23 0.492188  122        305
## 451531            GTS_871-23           GTS_944-23 0.500000  121        305
## 451556            GTS_871-23           GTS_631-23 0.491935  120        305
## 451563            GTS_871-23           GTS_781-23 0.500000  122        305
## 451572            GTS_871-23          GTS_781-23R 0.500000  122        305
## 451575            GTS_871-23           GTS_638-23 0.491935  119        305
## 451582            GTS_871-23           GTS_972-23 0.500000  121        305
## 451604            GTS_871-23           GTS_651-23 0.500000  121        305
## 451605            GTS_871-23           GTS_982-23 0.500000  121        305
## 451614            GTS_871-23          GTS_651-23R 0.492188  122        305
## 451620            GTS_871-23           GTS_803-23 0.500000  122        305
## 451636            GTS_871-23           GTS_809-23 0.500000  122        305
## 451654            GTS_871-23          GTS_1009-23 0.500000  121        305
## 451659            GTS_871-23          GTS_1013-23 0.492188  122        305
## 455550            GTS_713-23 UGA_173101_P020_WA09 0.482143   93        301
## 455588            GTS_713-23 UGA_173101_P019_WC06 0.484375  117        301
## 455591            GTS_713-23 UGA_173101_P019_WC02 0.484848  118        301
## 455597            GTS_713-23 UGA_173101_P019_WD04 0.477273  120        301
## 455611            GTS_713-23 UGA_173101_P018_WF03 0.483871  109        301
## 455615            GTS_713-23 UGA_173101_P018_WE09 0.473214  104        301
## 455638            GTS_713-23 UGA_173101_P020_WA07 0.458333  103        301
## 455640            GTS_713-23 UGA_173101_P016_WD06 0.481481   98        301
## 455654            GTS_713-23 UGA_173101_P016_WC05 0.472222   59        301
## 455675            GTS_713-23 UGA_173101_P017_WA02 0.484375  116        301
## 455689            GTS_713-23 UGA_173101_P015_WH07 0.477273  112        301
## 455719            GTS_713-23 UGA_173101_P020_WA05 0.476562  112        301
## 455783            GTS_713-23 UGA_173101_P017_WD05 0.437500   58        301
## 455787            GTS_713-23 UGA_173101_P018_WE02 0.484375  114        301
## 455798            GTS_713-23 UGA_173101_P015_WG11 0.416667   45        301
## 455850            GTS_713-23 UGA_173101_P016_WH06 0.490385   90        301
## 455854            GTS_713-23 UGA_173101_P016_WH08 0.474138  102        301
## 455879            GTS_713-23 UGA_173101_P017_WF04 0.469697  117        301
## 455906            GTS_713-23 UGA_173101_P017_WF08 0.445652   93        301
## 455926            GTS_713-23 UGA_173101_P017_WH09 0.455357   96        301
## 455972            GTS_713-23 UGA_173101_P023_WG04 0.484848  113        301
## 455989            GTS_713-23 UGA_173101_P023_WB11 0.484375  105        301
## 455992            GTS_713-23 UGA_173101_P023_WB06 0.475000  106        301
## 455999            GTS_713-23 UGA_173101_P023_WB04 0.475000  107        301
## 456055            GTS_713-23 UGA_173101_P020_WG06 0.484375  111        301
## 456090            GTS_713-23 UGA_173101_P020_WH04 0.484848  115        301
## 456102            GTS_713-23 UGA_173101_P020_WG07 0.466667   57        301
## 456106            GTS_713-23 UGA_173101_P020_WH11 0.484848  116        301
## 456135            GTS_713-23 UGA_173101_P022_WG09 0.484848  112        301
## 456146            GTS_713-23 UGA_173101_P022_WA12 0.471154   92        301
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## 456171            GTS_713-23 UGA_173101_P021_WC01 0.474138   98        301
## 456204            GTS_713-23 UGA_173101_P021_WH06 0.471154  102        301
## 456240            GTS_713-23 UGA_173101_P020_WH03 0.469697  116        301
## 456287            GTS_713-23           GTS_938-23 0.492188  121        301
## 456291            GTS_713-23           GTS_944-23 0.492188  120        301
## 456316            GTS_713-23           GTS_631-23 0.491935  118        301
## 456323            GTS_713-23           GTS_781-23 0.492188  120        301
## 456332            GTS_713-23          GTS_781-23R 0.492188  120        301
## 456335            GTS_713-23           GTS_638-23 0.483871  119        301
## 456342            GTS_713-23           GTS_972-23 0.492188  120        301
## 456364            GTS_713-23           GTS_651-23 0.492188  121        301
## 456365            GTS_713-23           GTS_982-23 0.492188  121        301
## 456374            GTS_713-23          GTS_651-23R 0.492188  121        301
## 456380            GTS_713-23           GTS_803-23 0.492188  121        301
## 456396            GTS_713-23           GTS_809-23 0.492188  121        301
## 456414            GTS_713-23          GTS_1009-23 0.492188  121        301
## 456419            GTS_713-23          GTS_1013-23 0.492188  121        301
## 456486            GTS_713-23           GTS_871-23 0.492188  120        301
## 478746            GTS_732-23 UGA_173101_P020_WA09 0.490385   93        301
## 478784            GTS_732-23 UGA_173101_P019_WC06 0.491667  117        301
## 478787            GTS_732-23 UGA_173101_P019_WC02 0.491935  118        301
## 478793            GTS_732-23 UGA_173101_P019_WD04 0.483871  120        301
## 478807            GTS_732-23 UGA_173101_P018_WF03 0.491379  109        301
## 478811            GTS_732-23 UGA_173101_P018_WE09 0.481481  104        301
## 478834            GTS_732-23 UGA_173101_P020_WA07 0.467391  103        301
## 478836            GTS_732-23 UGA_173101_P016_WD06 0.490000   98        301
## 478850            GTS_732-23 UGA_173101_P016_WC05 0.485294   59        301
## 478871            GTS_732-23 UGA_173101_P017_WA02 0.491667  116        301
## 478885            GTS_732-23 UGA_173101_P015_WH07 0.483871  111        301
## 478915            GTS_732-23 UGA_173101_P020_WA05 0.483871  112        301
## 478979            GTS_732-23 UGA_173101_P017_WD05 0.437500   58        301
## 478983            GTS_732-23 UGA_173101_P018_WE02 0.491667  114        301
## 478994            GTS_732-23 UGA_173101_P015_WG11 0.416667   45        301
## 479046            GTS_732-23 UGA_173101_P016_WH06 0.490000   89        301
## 479050            GTS_732-23 UGA_173101_P016_WH08 0.482143  101        301
## 479075            GTS_732-23 UGA_173101_P017_WF04 0.475806  116        301
## 479102            GTS_732-23 UGA_173101_P017_WF08 0.454545   93        301
## 479122            GTS_732-23 UGA_173101_P017_WH09 0.461538   95        301
## 479168            GTS_732-23 UGA_173101_P023_WG04 0.491935  113        301
## 479185            GTS_732-23 UGA_173101_P023_WB11 0.491667  105        301
## 479188            GTS_732-23 UGA_173101_P023_WB06 0.482759  106        301
## 479195            GTS_732-23 UGA_173101_P023_WB04 0.482143  107        301
## 479251            GTS_732-23 UGA_173101_P020_WG06 0.491667  111        301
## 479286            GTS_732-23 UGA_173101_P020_WH04 0.491935  115        301
## 479298            GTS_732-23 UGA_173101_P020_WG07 0.466667   57        301
## 479302            GTS_732-23 UGA_173101_P020_WH11 0.491935  116        301
## 479331            GTS_732-23 UGA_173101_P022_WG09 0.491935  112        301
## 479342            GTS_732-23 UGA_173101_P022_WA12 0.470000   92        301
## 479367            GTS_732-23 UGA_173101_P021_WC01 0.481481   98        301
## 479400            GTS_732-23 UGA_173101_P021_WH06 0.480000  101        301
## 479436            GTS_732-23 UGA_173101_P020_WH03 0.475806  116        301
## 479483            GTS_732-23           GTS_938-23 0.500000  121        301
## 479487            GTS_732-23           GTS_944-23 0.500000  120        301
## 479512            GTS_732-23           GTS_631-23 0.500000  119        301
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## 479519            GTS_732-23           GTS_781-23 0.500000  120        301
## 479528            GTS_732-23          GTS_781-23R 0.500000  120        301
## 479531            GTS_732-23           GTS_638-23 0.500000  119        301
## 479538            GTS_732-23           GTS_972-23 0.500000  120        301
## 479560            GTS_732-23           GTS_651-23 0.500000  121        301
## 479561            GTS_732-23           GTS_982-23 0.500000  121        301
## 479570            GTS_732-23          GTS_651-23R 0.500000  121        301
## 479576            GTS_732-23           GTS_803-23 0.500000  121        301
## 479592            GTS_732-23           GTS_809-23 0.500000  121        301
## 479610            GTS_732-23          GTS_1009-23 0.500000  121        301
## 479615            GTS_732-23          GTS_1013-23 0.500000  121        301
## 479682            GTS_732-23           GTS_871-23 0.500000  120        301
## 479687            GTS_732-23           GTS_713-23 0.491935  120        301
## 513606            GTS_919-23 UGA_173101_P020_WA09 0.490741   94        305
## 513644            GTS_919-23 UGA_173101_P019_WC06 0.492188  119        305
## 513647            GTS_919-23 UGA_173101_P019_WC02 0.492188  119        305
## 513653            GTS_919-23 UGA_173101_P019_WD04 0.484848  122        305
## 513667            GTS_919-23 UGA_173101_P018_WF03 0.491667  110        305
## 513671            GTS_919-23 UGA_173101_P018_WE09 0.482143  105        305
## 513694            GTS_919-23 UGA_173101_P020_WA07 0.468750  104        305
## 513696            GTS_919-23 UGA_173101_P016_WD06 0.490385   99        305
## 513710            GTS_919-23 UGA_173101_P016_WC05 0.485294   59        305
## 513731            GTS_919-23 UGA_173101_P017_WA02 0.491935  117        305
## 513745            GTS_919-23 UGA_173101_P015_WH07 0.484848  113        305
## 513775            GTS_919-23 UGA_173101_P020_WA05 0.484848  114        305
## 513839            GTS_919-23 UGA_173101_P017_WD05 0.437500   58        305
## 513843            GTS_919-23 UGA_173101_P018_WE02 0.491935  115        305
## 513854            GTS_919-23 UGA_173101_P015_WG11 0.416667   45        305
## 513906            GTS_919-23 UGA_173101_P016_WH06 0.490385   90        305
## 513910            GTS_919-23 UGA_173101_P016_WH08 0.482759  102        305
## 513935            GTS_919-23 UGA_173101_P017_WF04 0.476562  117        305
## 513962            GTS_919-23 UGA_173101_P017_WF08 0.456522   94        305
## 513982            GTS_919-23 UGA_173101_P017_WH09 0.462963   96        305
## 514028            GTS_919-23 UGA_173101_P023_WG04 0.492424  115        305
## 514045            GTS_919-23 UGA_173101_P023_WB11 0.491935  106        305
## 514048            GTS_919-23 UGA_173101_P023_WB06 0.483333  107        305
## 514055            GTS_919-23 UGA_173101_P023_WB04 0.482759  108        305
## 514111            GTS_919-23 UGA_173101_P020_WG06 0.492188  113        305
## 514146            GTS_919-23 UGA_173101_P020_WH04 0.492424  117        305
## 514158            GTS_919-23 UGA_173101_P020_WG07 0.466667   57        305
## 514162            GTS_919-23 UGA_173101_P020_WH11 0.492424  118        305
## 514191            GTS_919-23 UGA_173101_P022_WG09 0.492424  114        305
## 514202            GTS_919-23 UGA_173101_P022_WA12 0.472222   94        305
## 514227            GTS_919-23 UGA_173101_P021_WC01 0.482759  100        305
## 514260            GTS_919-23 UGA_173101_P021_WH06 0.480769  102        305
## 514296            GTS_919-23 UGA_173101_P020_WH03 0.477273  118        305
## 514343            GTS_919-23           GTS_938-23 0.492188  123        305
## 514347            GTS_919-23           GTS_944-23 0.500000  121        305
## 514372            GTS_919-23           GTS_631-23 0.491935  120        305
## 514379            GTS_919-23           GTS_781-23 0.500000  122        305
## 514388            GTS_919-23          GTS_781-23R 0.500000  122        305
## 514391            GTS_919-23           GTS_638-23 0.491935  120        305
## 514398            GTS_919-23           GTS_972-23 0.500000  121        305
## 514420            GTS_919-23           GTS_651-23 0.500000  122        305
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## 514421            GTS_919-23           GTS_982-23 0.500000  122       305
## 514430            GTS_919-23          GTS_651-23R 0.492188  123       305
## 514436            GTS_919-23           GTS_803-23 0.500000  123       305
## 514452            GTS_919-23           GTS_809-23 0.500000  123       305
## 514470            GTS_919-23          GTS_1009-23 0.500000  122       305
## 514475            GTS_919-23          GTS_1013-23 0.492188  123       305
## 514542            GTS_919-23           GTS_871-23 0.500000  122       305
## 514547            GTS_919-23           GTS_713-23 0.492188  121       305
## 514571            GTS_919-23           GTS_732-23 0.500000  121       305
## 534076            GTS_671-23 UGA_173101_P020_WA09 0.490385   93       301
## 534114            GTS_671-23 UGA_173101_P019_WC06 0.491667  116       301
## 534117            GTS_671-23 UGA_173101_P019_WC02 0.491935  117       301
## 534123            GTS_671-23 UGA_173101_P019_WD04 0.483871  119       301
## 534137            GTS_671-23 UGA_173101_P018_WF03 0.491379  108       301
## 534141            GTS_671-23 UGA_173101_P018_WE09 0.481481  104       301
## 534164            GTS_671-23 UGA_173101_P020_WA07 0.467391  102       301
## 534166            GTS_671-23 UGA_173101_P016_WD06 0.490000   97       301
## 534180            GTS_671-23 UGA_173101_P016_WC05 0.485294   59       301
## 534201            GTS_671-23 UGA_173101_P017_WA02 0.491667  115       301
## 534215            GTS_671-23 UGA_173101_P015_WH07 0.483871  110       301
## 534245            GTS_671-23 UGA_173101_P020_WA05 0.483871  111       301
## 534309            GTS_671-23 UGA_173101_P017_WD05 0.437500   58       301
## 534313            GTS_671-23 UGA_173101_P018_WE02 0.491667  113       301
## 534324            GTS_671-23 UGA_173101_P015_WG11 0.416667   45       301
## 534376            GTS_671-23 UGA_173101_P016_WH06 0.490000   88       301
## 534380            GTS_671-23 UGA_173101_P016_WH08 0.482143  100       301
## 534405            GTS_671-23 UGA_173101_P017_WF04 0.475806  115       301
## 534432            GTS_671-23 UGA_173101_P017_WF08 0.454545   93       301
## 534452            GTS_671-23 UGA_173101_P017_WH09 0.461538   95       301
## 534498            GTS_671-23 UGA_173101_P023_WG04 0.491935  112       301
## 534515            GTS_671-23 UGA_173101_P023_WB11 0.491667  105       301
## 534518            GTS_671-23 UGA_173101_P023_WB06 0.482759  105       301
## 534525            GTS_671-23 UGA_173101_P023_WB04 0.482143  107       301
## 534581            GTS_671-23 UGA_173101_P020_WG06 0.491667  110       301
## 534616            GTS_671-23 UGA_173101_P020_WH04 0.491935  114       301
## 534628            GTS_671-23 UGA_173101_P020_WG07 0.466667   57       301
## 534632            GTS_671-23 UGA_173101_P020_WH11 0.491935  115       301
## 534661            GTS_671-23 UGA_173101_P022_WG09 0.491935  111       301
## 534672            GTS_671-23 UGA_173101_P022_WA12 0.470000   92       301
## 534697            GTS_671-23 UGA_173101_P021_WC01 0.481481   98       301
## 534730            GTS_671-23 UGA_173101_P021_WH06 0.480000  100       301
## 534766            GTS_671-23 UGA_173101_P020_WH03 0.475806  115       301
## 534813            GTS_671-23           GTS_938-23 0.500000  120       301
## 534817            GTS_671-23           GTS_944-23 0.500000  119       301
## 534842            GTS_671-23           GTS_631-23 0.500000  118       301
## 534849            GTS_671-23           GTS_781-23 0.500000  119       301
## 534858            GTS_671-23          GTS_781-23R 0.500000  120       301
## 534861            GTS_671-23           GTS_638-23 0.500000  119       301
## 534868            GTS_671-23           GTS_972-23 0.500000  119       301
## 534890            GTS_671-23           GTS_651-23 0.500000  120       301
## 534891            GTS_671-23           GTS_982-23 0.500000  120       301
## 534900            GTS_671-23          GTS_651-23R 0.500000  120       301
## 534906            GTS_671-23           GTS_803-23 0.500000  120       301
## 534922            GTS_671-23           GTS_809-23 0.500000  120       301
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## 534940            GTS_671-23          GTS_1009-23 0.500000  120        301
## 534945            GTS_671-23          GTS_1013-23 0.500000  120        301
## 535012            GTS_671-23           GTS_871-23 0.500000  119        301
## 535017            GTS_671-23           GTS_713-23 0.491935  119        301
## 535041            GTS_671-23           GTS_732-23 0.500000  120        301
## 535076            GTS_671-23           GTS_919-23 0.500000  120        301
## 560226            GTS_706-23 UGA_173101_P020_WA09 0.490385   93        301
## 560264            GTS_706-23 UGA_173101_P019_WC06 0.491667  117        301
## 560267            GTS_706-23 UGA_173101_P019_WC02 0.491935  118        301
## 560273            GTS_706-23 UGA_173101_P019_WD04 0.483871  120        301
## 560287            GTS_706-23 UGA_173101_P018_WF03 0.491379  109        301
## 560291            GTS_706-23 UGA_173101_P018_WE09 0.481481  104        301
## 560314            GTS_706-23 UGA_173101_P020_WA07 0.467391  103        301
## 560316            GTS_706-23 UGA_173101_P016_WD06 0.490000   98        301
## 560330            GTS_706-23 UGA_173101_P016_WC05 0.485294   59        301
## 560351            GTS_706-23 UGA_173101_P017_WA02 0.491667  116        301
## 560365            GTS_706-23 UGA_173101_P015_WH07 0.483871  111        301
## 560395            GTS_706-23 UGA_173101_P020_WA05 0.483871  112        301
## 560459            GTS_706-23 UGA_173101_P017_WD05 0.437500   58        301
## 560463            GTS_706-23 UGA_173101_P018_WE02 0.491667  114        301
## 560474            GTS_706-23 UGA_173101_P015_WG11 0.416667   45        301
## 560526            GTS_706-23 UGA_173101_P016_WH06 0.490000   89        301
## 560530            GTS_706-23 UGA_173101_P016_WH08 0.482143  101        301
## 560555            GTS_706-23 UGA_173101_P017_WF04 0.475806  116        301
## 560582            GTS_706-23 UGA_173101_P017_WF08 0.454545   93        301
## 560602            GTS_706-23 UGA_173101_P017_WH09 0.461538   95        301
## 560648            GTS_706-23 UGA_173101_P023_WG04 0.491935  113        301
## 560665            GTS_706-23 UGA_173101_P023_WB11 0.491667  105        301
## 560668            GTS_706-23 UGA_173101_P023_WB06 0.482759  106        301
## 560675            GTS_706-23 UGA_173101_P023_WB04 0.482143  107        301
## 560731            GTS_706-23 UGA_173101_P020_WG06 0.491667  111        301
## 560766            GTS_706-23 UGA_173101_P020_WH04 0.491935  115        301
## 560778            GTS_706-23 UGA_173101_P020_WG07 0.466667   57        301
## 560782            GTS_706-23 UGA_173101_P020_WH11 0.491935  116        301
## 560811            GTS_706-23 UGA_173101_P022_WG09 0.491935  112        301
## 560822            GTS_706-23 UGA_173101_P022_WA12 0.470000   92        301
## 560847            GTS_706-23 UGA_173101_P021_WC01 0.481481   98        301
## 560880            GTS_706-23 UGA_173101_P021_WH06 0.480000  101        301
## 560916            GTS_706-23 UGA_173101_P020_WH03 0.475806  116        301
## 560963            GTS_706-23           GTS_938-23 0.500000  121        301
## 560967            GTS_706-23           GTS_944-23 0.500000  120        301
## 560992            GTS_706-23           GTS_631-23 0.500000  119        301
## 560999            GTS_706-23           GTS_781-23 0.500000  120        301
## 561008            GTS_706-23          GTS_781-23R 0.500000  121        301
## 561011            GTS_706-23           GTS_638-23 0.500000  119        301
## 561018            GTS_706-23           GTS_972-23 0.500000  120        301
## 561040            GTS_706-23           GTS_651-23 0.500000  121        301
## 561041            GTS_706-23           GTS_982-23 0.500000  121        301
## 561050            GTS_706-23          GTS_651-23R 0.500000  121        301
## 561056            GTS_706-23           GTS_803-23 0.500000  121        301
## 561072            GTS_706-23           GTS_809-23 0.500000  121        301
## 561090            GTS_706-23          GTS_1009-23 0.500000  121        301
## 561095            GTS_706-23          GTS_1013-23 0.500000  121        301
## 561162            GTS_706-23           GTS_871-23 0.500000  120        301
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## 561167            GTS_706-23           GTS_713-23 0.491935  120        301
## 561191            GTS_706-23           GTS_732-23 0.500000  121        301
## 561226            GTS_706-23           GTS_919-23 0.500000  121        301
## 561246            GTS_706-23           GTS_671-23 0.500000  121        301
## 583755            GTS_776-23 UGA_173101_P020_WA09 0.490741   94        301
## 583793            GTS_776-23 UGA_173101_P019_WC06 0.483871  119        301
## 583796            GTS_776-23 UGA_173101_P019_WC02 0.492188  119        301
## 583802            GTS_776-23 UGA_173101_P019_WD04 0.476562  122        301
## 583816            GTS_776-23 UGA_173101_P018_WF03 0.491667  110        301
## 583820            GTS_776-23 UGA_173101_P018_WE09 0.482143  105        301
## 583843            GTS_776-23 UGA_173101_P020_WA07 0.468750  104        301
## 583845            GTS_776-23 UGA_173101_P016_WD06 0.490385   99        301
## 583859            GTS_776-23 UGA_173101_P016_WC05 0.485294   59        301
## 583880            GTS_776-23 UGA_173101_P017_WA02 0.491935  117        301
## 583894            GTS_776-23 UGA_173101_P015_WH07 0.476562  113        301
## 583924            GTS_776-23 UGA_173101_P020_WA05 0.476562  114        301
## 583988            GTS_776-23 UGA_173101_P017_WD05 0.437500   58        301
## 583992            GTS_776-23 UGA_173101_P018_WE02 0.491935  115        301
## 584003            GTS_776-23 UGA_173101_P015_WG11 0.416667   45        301
## 584055            GTS_776-23 UGA_173101_P016_WH06 0.490385   90        301
## 584059            GTS_776-23 UGA_173101_P016_WH08 0.482759  102        301
## 584084            GTS_776-23 UGA_173101_P017_WF04 0.476562  117        301
## 584111            GTS_776-23 UGA_173101_P017_WF08 0.456522   94        301
## 584131            GTS_776-23 UGA_173101_P017_WH09 0.462963   96        301
## 584177            GTS_776-23 UGA_173101_P023_WG04 0.484375  115        301
## 584194            GTS_776-23 UGA_173101_P023_WB11 0.491935  106        301
## 584197            GTS_776-23 UGA_173101_P023_WB06 0.483333  107        301
## 584204            GTS_776-23 UGA_173101_P023_WB04 0.482759  108        301
## 584260            GTS_776-23 UGA_173101_P020_WG06 0.483871  113        301
## 584295            GTS_776-23 UGA_173101_P020_WH04 0.484375  117        301
## 584307            GTS_776-23 UGA_173101_P020_WG07 0.466667   57        301
## 584311            GTS_776-23 UGA_173101_P020_WH11 0.484375  118        301
## 584340            GTS_776-23 UGA_173101_P022_WG09 0.484375  114        301
## 584351            GTS_776-23 UGA_173101_P022_WA12 0.461538   94        301
## 584376            GTS_776-23 UGA_173101_P021_WC01 0.473214  100        301
## 584409            GTS_776-23 UGA_173101_P021_WH06 0.480769  102        301
## 584445            GTS_776-23 UGA_173101_P020_WH03 0.468750  118        301
## 584492            GTS_776-23           GTS_938-23 0.500000  123        301
## 584496            GTS_776-23           GTS_944-23 0.500000  121        301
## 584521            GTS_776-23           GTS_631-23 0.500000  120        301
## 584528            GTS_776-23           GTS_781-23 0.492188  122        301
## 584537            GTS_776-23          GTS_781-23R 0.492424  123        301
## 584540            GTS_776-23           GTS_638-23 0.491935  120        301
## 584547            GTS_776-23           GTS_972-23 0.500000  121        301
## 584569            GTS_776-23           GTS_651-23 0.500000  122        301
## 584570            GTS_776-23           GTS_982-23 0.500000  122        301
## 584579            GTS_776-23          GTS_651-23R 0.500000  123        301
## 584585            GTS_776-23           GTS_803-23 0.492188  123        301
## 584601            GTS_776-23           GTS_809-23 0.492188  123        301
## 584619            GTS_776-23          GTS_1009-23 0.500000  122        301
## 584624            GTS_776-23          GTS_1013-23 0.500000  123        301
## 584691            GTS_776-23           GTS_871-23 0.492188  122        301
## 584696            GTS_776-23           GTS_713-23 0.492188  121        301
## 584720            GTS_776-23           GTS_732-23 0.500000  121        301
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## 584755            GTS_776-23           GTS_919-23 0.492188  123        301
## 584775            GTS_776-23           GTS_671-23 0.500000  121        301
## 584800            GTS_776-23           GTS_706-23 0.500000  122        301
## 609975            GTS_847-23 UGA_173101_P020_WA09 0.453704   94        301
## 610013            GTS_847-23 UGA_173101_P019_WC06 0.451613  119        301
## 610016            GTS_847-23 UGA_173101_P019_WC02 0.460938  119        301
## 610022            GTS_847-23 UGA_173101_P019_WD04 0.445312  122        301
## 610036            GTS_847-23 UGA_173101_P018_WF03 0.458333  110        301
## 610040            GTS_847-23 UGA_173101_P018_WE09 0.446429  105        301
## 610063            GTS_847-23 UGA_173101_P020_WA07 0.427083  104        301
## 610065            GTS_847-23 UGA_173101_P016_WD06 0.451923   99        301
## 610079            GTS_847-23 UGA_173101_P016_WC05 0.426471   59        301
## 610100            GTS_847-23 UGA_173101_P017_WA02 0.459677  117        301
## 610114            GTS_847-23 UGA_173101_P015_WH07 0.445312  113        301
## 610144            GTS_847-23 UGA_173101_P020_WA05 0.445312  114        301
## 610208            GTS_847-23 UGA_173101_P017_WD05 0.354167   58        301
## 610212            GTS_847-23 UGA_173101_P018_WE02 0.459677  115        301
## 610223            GTS_847-23 UGA_173101_P015_WG11 0.333333   46        301
## 610275            GTS_847-23 UGA_173101_P016_WH06 0.451923   90        301
## 610279            GTS_847-23 UGA_173101_P016_WH08 0.448276  102        301
## 610304            GTS_847-23 UGA_173101_P017_WF04 0.445312  117        301
## 610331            GTS_847-23 UGA_173101_P017_WF08 0.413043   94        301
## 610351            GTS_847-23 UGA_173101_P017_WH09 0.425926   96        301
## 610397            GTS_847-23 UGA_173101_P023_WG04 0.453125  115        301
## 610414            GTS_847-23 UGA_173101_P023_WB11 0.459677  106        301
## 610417            GTS_847-23 UGA_173101_P023_WB06 0.450000  107        301
## 610424            GTS_847-23 UGA_173101_P023_WB04 0.448276  108        301
## 610480            GTS_847-23 UGA_173101_P020_WG06 0.451613  113        301
## 610515            GTS_847-23 UGA_173101_P020_WH04 0.453125  117        301
## 610527            GTS_847-23 UGA_173101_P020_WG07 0.400000   58        301
## 610531            GTS_847-23 UGA_173101_P020_WH11 0.453125  118        301
## 610560            GTS_847-23 UGA_173101_P022_WG09 0.453125  114        301
## 610571            GTS_847-23 UGA_173101_P022_WA12 0.423077   94        301
## 610596            GTS_847-23 UGA_173101_P021_WC01 0.437500  100        301
## 610629            GTS_847-23 UGA_173101_P021_WH06 0.442308  102        301
## 610665            GTS_847-23 UGA_173101_P020_WH03 0.437500  118        301
## 610712            GTS_847-23           GTS_938-23 0.500000  122        301
## 610716            GTS_847-23           GTS_944-23 0.500000  121        301
## 610741            GTS_847-23           GTS_631-23 0.500000  121        301
## 610748            GTS_847-23           GTS_781-23 0.492188  123        301
## 610757            GTS_847-23          GTS_781-23R 0.492424  123        301
## 610760            GTS_847-23           GTS_638-23 0.491935  119        301
## 610767            GTS_847-23           GTS_972-23 0.500000  121        301
## 610789            GTS_847-23           GTS_651-23 0.500000  121        301
## 610790            GTS_847-23           GTS_982-23 0.500000  121        301
## 610799            GTS_847-23          GTS_651-23R 0.500000  122        301
## 610805            GTS_847-23           GTS_803-23 0.492188  122        301
## 610821            GTS_847-23           GTS_809-23 0.492188  122        301
## 610839            GTS_847-23          GTS_1009-23 0.500000  121        301
## 610844            GTS_847-23          GTS_1013-23 0.500000  122        301
## 610911            GTS_847-23           GTS_871-23 0.492188  122        301
## 610916            GTS_847-23           GTS_713-23 0.492188  120        301
## 610940            GTS_847-23           GTS_732-23 0.500000  120        301
## 610975            GTS_847-23           GTS_919-23 0.492188  122        301
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## 610995            GTS_847-23           GTS_671-23 0.500000  120        301
## 611020            GTS_847-23           GTS_706-23 0.500000  121        301
## 611042            GTS_847-23           GTS_776-23 0.500000  123        301
## 636771            GTS_881-23 UGA_173101_P020_WA09 0.490385   92        301
## 636809            GTS_881-23 UGA_173101_P019_WC06 0.483333  117        301
## 636812            GTS_881-23 UGA_173101_P019_WC02 0.491935  117        301
## 636818            GTS_881-23 UGA_173101_P019_WD04 0.475806  120        301
## 636832            GTS_881-23 UGA_173101_P018_WF03 0.491379  108        301
## 636836            GTS_881-23 UGA_173101_P018_WE09 0.481481  103        301
## 636859            GTS_881-23 UGA_173101_P020_WA07 0.467391  102        301
## 636861            GTS_881-23 UGA_173101_P016_WD06 0.490000   97        301
## 636875            GTS_881-23 UGA_173101_P016_WC05 0.485294   58        301
## 636896            GTS_881-23 UGA_173101_P017_WA02 0.491667  115        301
## 636910            GTS_881-23 UGA_173101_P015_WH07 0.475806  111        301
## 636940            GTS_881-23 UGA_173101_P020_WA05 0.475806  112        301
## 637004            GTS_881-23 UGA_173101_P017_WD05 0.437500   57        301
## 637008            GTS_881-23 UGA_173101_P018_WE02 0.491667  113        301
## 637019            GTS_881-23 UGA_173101_P015_WG11 0.416667   45        301
## 637071            GTS_881-23 UGA_173101_P016_WH06 0.490000   88        301
## 637075            GTS_881-23 UGA_173101_P016_WH08 0.482143  100        301
## 637100            GTS_881-23 UGA_173101_P017_WF04 0.475806  115        301
## 637127            GTS_881-23 UGA_173101_P017_WF08 0.454545   92        301
## 637147            GTS_881-23 UGA_173101_P017_WH09 0.461538   94        301
## 637193            GTS_881-23 UGA_173101_P023_WG04 0.483871  113        301
## 637210            GTS_881-23 UGA_173101_P023_WB11 0.491667  104        301
## 637213            GTS_881-23 UGA_173101_P023_WB06 0.482759  105        301
## 637220            GTS_881-23 UGA_173101_P023_WB04 0.482143  106        301
## 637276            GTS_881-23 UGA_173101_P020_WG06 0.483333  111        301
## 637311            GTS_881-23 UGA_173101_P020_WH04 0.483871  115        301
## 637323            GTS_881-23 UGA_173101_P020_WG07 0.466667   57        301
## 637327            GTS_881-23 UGA_173101_P020_WH11 0.483871  116        301
## 637356            GTS_881-23 UGA_173101_P022_WG09 0.483871  112        301
## 637367            GTS_881-23 UGA_173101_P022_WA12 0.460000   92        301
## 637392            GTS_881-23 UGA_173101_P021_WC01 0.472222   98        301
## 637425            GTS_881-23 UGA_173101_P021_WH06 0.480000  100        301
## 637461            GTS_881-23 UGA_173101_P020_WH03 0.467742  116        301
## 637508            GTS_881-23           GTS_938-23 0.500000  121        301
## 637512            GTS_881-23           GTS_944-23 0.500000  120        301
## 637537            GTS_881-23           GTS_631-23 0.500000  120        301
## 637544            GTS_881-23           GTS_781-23 0.491935  121        301
## 637553            GTS_881-23          GTS_781-23R 0.492188  122        301
## 637556            GTS_881-23           GTS_638-23 0.500000  118        301
## 637563            GTS_881-23           GTS_972-23 0.500000  120        301
## 637585            GTS_881-23           GTS_651-23 0.500000  120        301
## 637586            GTS_881-23           GTS_982-23 0.500000  120        301
## 637595            GTS_881-23          GTS_651-23R 0.500000  121        301
## 637601            GTS_881-23           GTS_803-23 0.491935  121        301
## 637617            GTS_881-23           GTS_809-23 0.491935  121        301
## 637635            GTS_881-23          GTS_1009-23 0.500000  120        301
## 637640            GTS_881-23          GTS_1013-23 0.500000  121        301
## 637707            GTS_881-23           GTS_871-23 0.491935  121        301
## 637712            GTS_881-23           GTS_713-23 0.491935  119        301
## 637736            GTS_881-23           GTS_732-23 0.500000  120        301
## 637771            GTS_881-23           GTS_919-23 0.491935  121        301
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## 637791            GTS_881-23           GTS_671-23 0.500000  120        301
## 637816            GTS_881-23           GTS_706-23 0.500000  121        301
## 637838            GTS_881-23           GTS_776-23 0.500000  122        301
## 637862            GTS_881-23           GTS_847-23 0.500000  122        301
## 641293            GTS_890-23 UGA_173101_P020_WA09 0.453704   94        301
## 641331            GTS_890-23 UGA_173101_P019_WC06 0.451613  119        301
## 641334            GTS_890-23 UGA_173101_P019_WC02 0.460938  119        301
## 641340            GTS_890-23 UGA_173101_P019_WD04 0.445312  122        301
## 641354            GTS_890-23 UGA_173101_P018_WF03 0.458333  110        301
## 641358            GTS_890-23 UGA_173101_P018_WE09 0.446429  105        301
## 641381            GTS_890-23 UGA_173101_P020_WA07 0.427083  104        301
## 641383            GTS_890-23 UGA_173101_P016_WD06 0.451923   99        301
## 641397            GTS_890-23 UGA_173101_P016_WC05 0.426471   59        301
## 641418            GTS_890-23 UGA_173101_P017_WA02 0.459677  117        301
## 641432            GTS_890-23 UGA_173101_P015_WH07 0.445312  113        301
## 641462            GTS_890-23 UGA_173101_P020_WA05 0.445312  114        301
## 641526            GTS_890-23 UGA_173101_P017_WD05 0.354167   58        301
## 641530            GTS_890-23 UGA_173101_P018_WE02 0.459677  115        301
## 641541            GTS_890-23 UGA_173101_P015_WG11 0.333333   46        301
## 641593            GTS_890-23 UGA_173101_P016_WH06 0.451923   90        301
## 641597            GTS_890-23 UGA_173101_P016_WH08 0.448276  102        301
## 641622            GTS_890-23 UGA_173101_P017_WF04 0.445312  117        301
## 641649            GTS_890-23 UGA_173101_P017_WF08 0.413043   94        301
## 641669            GTS_890-23 UGA_173101_P017_WH09 0.425926   96        301
## 641715            GTS_890-23 UGA_173101_P023_WG04 0.453125  115        301
## 641732            GTS_890-23 UGA_173101_P023_WB11 0.459677  106        301
## 641735            GTS_890-23 UGA_173101_P023_WB06 0.450000  107        301
## 641742            GTS_890-23 UGA_173101_P023_WB04 0.448276  108        301
## 641798            GTS_890-23 UGA_173101_P020_WG06 0.451613  113        301
## 641833            GTS_890-23 UGA_173101_P020_WH04 0.453125  117        301
## 641845            GTS_890-23 UGA_173101_P020_WG07 0.400000   58        301
## 641849            GTS_890-23 UGA_173101_P020_WH11 0.453125  118        301
## 641878            GTS_890-23 UGA_173101_P022_WG09 0.453125  114        301
## 641889            GTS_890-23 UGA_173101_P022_WA12 0.423077   94        301
## 641914            GTS_890-23 UGA_173101_P021_WC01 0.437500  100        301
## 641947            GTS_890-23 UGA_173101_P021_WH06 0.442308  102        301
## 641983            GTS_890-23 UGA_173101_P020_WH03 0.437500  118        301
## 642030            GTS_890-23           GTS_938-23 0.500000  122        301
## 642034            GTS_890-23           GTS_944-23 0.500000  121        301
## 642059            GTS_890-23           GTS_631-23 0.500000  121        301
## 642066            GTS_890-23           GTS_781-23 0.492188  123        301
## 642075            GTS_890-23          GTS_781-23R 0.492424  123        301
## 642078            GTS_890-23           GTS_638-23 0.491935  119        301
## 642085            GTS_890-23           GTS_972-23 0.500000  121        301
## 642107            GTS_890-23           GTS_651-23 0.500000  121        301
## 642108            GTS_890-23           GTS_982-23 0.500000  121        301
## 642117            GTS_890-23          GTS_651-23R 0.500000  122        301
## 642123            GTS_890-23           GTS_803-23 0.492188  122        301
## 642139            GTS_890-23           GTS_809-23 0.492188  122        301
## 642157            GTS_890-23          GTS_1009-23 0.500000  121        301
## 642162            GTS_890-23          GTS_1013-23 0.500000  122        301
## 642229            GTS_890-23           GTS_871-23 0.492188  122        301
## 642234            GTS_890-23           GTS_713-23 0.492188  120        301
## 642258            GTS_890-23           GTS_732-23 0.500000  120        301
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## 642293          GTS_890-23          GTS_919-23 0.492188  122       301
## 642313          GTS_890-23          GTS_671-23 0.500000  120       301
## 642338          GTS_890-23          GTS_706-23 0.500000  121       301
## 642360          GTS_890-23          GTS_776-23 0.500000  123       301
## 642384          GTS_890-23          GTS_847-23 0.500000  124       301
## 642408          GTS_890-23          GTS_881-23 0.500000  122       301
## 692091         GTS_1002-23 UGA_173101_P020_WA09 0.490741   93       301
## 692129         GTS_1002-23 UGA_173101_P019_WC06 0.483871  118       301
## 692132         GTS_1002-23 UGA_173101_P019_WC02 0.492188  118       301
## 692138         GTS_1002-23 UGA_173101_P019_WD04 0.476562  121       301
## 692152         GTS_1002-23 UGA_173101_P018_WF03 0.491667  109       301
## 692156         GTS_1002-23 UGA_173101_P018_WE09 0.482143  104       301
## 692179         GTS_1002-23 UGA_173101_P020_WA07 0.468750  103       301
## 692181         GTS_1002-23 UGA_173101_P016_WD06 0.490385   98       301
## 692195         GTS_1002-23 UGA_173101_P016_WC05 0.485294   58       301
## 692216         GTS_1002-23 UGA_173101_P017_WA02 0.491935  116       301
## 692230         GTS_1002-23 UGA_173101_P015_WH07 0.476562  112       301
## 692260         GTS_1002-23 UGA_173101_P020_WA05 0.476562  113       301
## 692324         GTS_1002-23 UGA_173101_P017_WD05 0.437500   57       301
## 692328         GTS_1002-23 UGA_173101_P018_WE02 0.491935  114       301
## 692339         GTS_1002-23 UGA_173101_P015_WG11 0.416667   45       301
## 692391         GTS_1002-23 UGA_173101_P016_WH06 0.490385   89       301
## 692395         GTS_1002-23 UGA_173101_P016_WH08 0.482759  101       301
## 692420         GTS_1002-23 UGA_173101_P017_WF04 0.476562  116       301
## 692447         GTS_1002-23 UGA_173101_P017_WF08 0.456522   93       301
## 692467         GTS_1002-23 UGA_173101_P017_WH09 0.462963   95       301
## 692513         GTS_1002-23 UGA_173101_P023_WG04 0.484375  114       301
## 692530         GTS_1002-23 UGA_173101_P023_WB11 0.491935  105       301
## 692533         GTS_1002-23 UGA_173101_P023_WB06 0.483333  106       301
## 692540         GTS_1002-23 UGA_173101_P023_WB04 0.482759  107       301
## 692596         GTS_1002-23 UGA_173101_P020_WG06 0.483871  112       301
## 692631         GTS_1002-23 UGA_173101_P020_WH04 0.484375  116       301
## 692643         GTS_1002-23 UGA_173101_P020_WG07 0.466667   57       301
## 692647         GTS_1002-23 UGA_173101_P020_WH11 0.484375  117       301
## 692676         GTS_1002-23 UGA_173101_P022_WG09 0.484375  113       301
## 692687         GTS_1002-23 UGA_173101_P022_WA12 0.461538   93       301
## 692712         GTS_1002-23 UGA_173101_P021_WC01 0.473214   99       301
## 692745         GTS_1002-23 UGA_173101_P021_WH06 0.480769  101       301
## 692781         GTS_1002-23 UGA_173101_P020_WH03 0.468750  117       301
## 692828         GTS_1002-23          GTS_938-23 0.500000  122       301
## 692832         GTS_1002-23          GTS_944-23 0.500000  121       301
## 692857         GTS_1002-23          GTS_631-23 0.500000  120       301
## 692864         GTS_1002-23          GTS_781-23 0.492188  122       301
## 692873         GTS_1002-23         GTS_781-23R 0.492424  123       301
## 692876         GTS_1002-23          GTS_638-23 0.491935  119       301
## 692883         GTS_1002-23          GTS_972-23 0.500000  121       301
## 692905         GTS_1002-23          GTS_651-23 0.500000  121       301
## 692906         GTS_1002-23          GTS_982-23 0.500000  121       301
## 692915         GTS_1002-23         GTS_651-23R 0.500000  122       301
## 692921         GTS_1002-23          GTS_803-23 0.492188  122       301
## 692937         GTS_1002-23          GTS_809-23 0.492188  122       301
## 692955         GTS_1002-23         GTS_1009-23 0.500000  121       301
## 692960         GTS_1002-23         GTS_1013-23 0.500000  122       301
## 693027         GTS_1002-23          GTS_871-23 0.492188  122       301
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## 693032          GTS_1002-23           GTS_713-23 0.492188  120       301
## 693056          GTS_1002-23           GTS_732-23 0.500000  120       301
## 693091          GTS_1002-23           GTS_919-23 0.492188  122       301
## 693111          GTS_1002-23           GTS_671-23 0.500000  120       301
## 693136          GTS_1002-23           GTS_706-23 0.500000  121       301
## 693158          GTS_1002-23           GTS_776-23 0.500000  123       301
## 693182          GTS_1002-23           GTS_847-23 0.500000  123       301
## 693206          GTS_1002-23           GTS_881-23 0.500000  122       301
## 693210          GTS_1002-23           GTS_890-23 0.500000  123       301
## 1208250          GTS_399-23 UGA_173101_P020_WA09 0.490741   93       305
## 1208288          GTS_399-23 UGA_173101_P019_WC06 0.492188  119       305
## 1208291          GTS_399-23 UGA_173101_P019_WC02 0.492188  119       305
## 1208297          GTS_399-23 UGA_173101_P019_WD04 0.484848  122       305
## 1208311          GTS_399-23 UGA_173101_P018_WF03 0.491667  110       305
## 1208315          GTS_399-23 UGA_173101_P018_WE09 0.482143  104       305
## 1208338          GTS_399-23 UGA_173101_P020_WA07 0.468750  103       305
## 1208340          GTS_399-23 UGA_173101_P016_WD06 0.490385   98       305
## 1208354          GTS_399-23 UGA_173101_P016_WC05 0.485294   58       305
## 1208375          GTS_399-23 UGA_173101_P017_WA02 0.491935  116       305
## 1208389          GTS_399-23 UGA_173101_P015_WH07 0.484848  112       305
## 1208419          GTS_399-23 UGA_173101_P020_WA05 0.484848  113       305
## 1208483          GTS_399-23 UGA_173101_P017_WD05 0.437500   57       305
## 1208487          GTS_399-23 UGA_173101_P018_WE02 0.491935  115       305
## 1208498          GTS_399-23 UGA_173101_P015_WG11 0.416667   45       305
## 1208550          GTS_399-23 UGA_173101_P016_WH06 0.490385   89       305
## 1208554          GTS_399-23 UGA_173101_P016_WH08 0.482759  101       305
## 1208579          GTS_399-23 UGA_173101_P017_WF04 0.476562  117       305
## 1208606          GTS_399-23 UGA_173101_P017_WF08 0.456522   93       305
## 1208626          GTS_399-23 UGA_173101_P017_WH09 0.462963   95       305
## 1208672          GTS_399-23 UGA_173101_P023_WG04 0.492424  114       305
## 1208689          GTS_399-23 UGA_173101_P023_WB11 0.491935  106       305
## 1208692          GTS_399-23 UGA_173101_P023_WB06 0.483333  107       305
## 1208699          GTS_399-23 UGA_173101_P023_WB04 0.482759  108       305
## 1208755          GTS_399-23 UGA_173101_P020_WG06 0.492188  112       305
## 1208790          GTS_399-23 UGA_173101_P020_WH04 0.492424  117       305
## 1208802          GTS_399-23 UGA_173101_P020_WG07 0.466667   57       305
## 1208806          GTS_399-23 UGA_173101_P020_WH11 0.492424  118       305
## 1208835          GTS_399-23 UGA_173101_P022_WG09 0.492424  114       305
## 1208846          GTS_399-23 UGA_173101_P022_WA12 0.472222   93       305
## 1208871          GTS_399-23 UGA_173101_P021_WC01 0.482759   99       305
## 1208904          GTS_399-23 UGA_173101_P021_WH06 0.480769  101       305
## 1208940          GTS_399-23 UGA_173101_P020_WH03 0.477273  118       305
## 1208987          GTS_399-23           GTS_938-23 0.492188  122       305
## 1208991          GTS_399-23           GTS_944-23 0.500000  121       305
## 1209016          GTS_399-23           GTS_631-23 0.491935  120       305
## 1209023          GTS_399-23           GTS_781-23 0.500000  122       305
## 1209032          GTS_399-23          GTS_781-23R 0.500000  123       305
## 1209035          GTS_399-23           GTS_638-23 0.491935  119       305
## 1209042          GTS_399-23           GTS_972-23 0.500000  121       305
## 1209064          GTS_399-23           GTS_651-23 0.500000  121       305
## 1209065          GTS_399-23           GTS_982-23 0.500000  121       305
## 1209074          GTS_399-23          GTS_651-23R 0.492188  122       305
## 1209080          GTS_399-23           GTS_803-23 0.500000  122       305
## 1209096          GTS_399-23           GTS_809-23 0.500000  122       305
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## 1209114          GTS_399-23          GTS_1009-23 0.500000  121       305
## 1209119          GTS_399-23          GTS_1013-23 0.492188  122       305
## 1209186          GTS_399-23           GTS_871-23 0.500000  122       305
## 1209191          GTS_399-23           GTS_713-23 0.492188  120       305
## 1209215          GTS_399-23           GTS_732-23 0.500000  120       305
## 1209250          GTS_399-23           GTS_919-23 0.500000  122       305
## 1209270          GTS_399-23           GTS_671-23 0.500000  120       305
## 1209295          GTS_399-23           GTS_706-23 0.500000  121       305
## 1209317          GTS_399-23           GTS_776-23 0.492424  123       305
## 1209341          GTS_399-23           GTS_847-23 0.492424  123       305
## 1209365          GTS_399-23           GTS_881-23 0.492188  122       305
## 1209369          GTS_399-23           GTS_890-23 0.492424  123       305
## 1209413          GTS_399-23          GTS_1002-23 0.492424  123       305
## 1219156          GTS_104-23 UGA_173101_P020_WA09 0.487500   83       305
## 1219194          GTS_104-23 UGA_173101_P019_WC06 0.489583  107       305
## 1219197          GTS_104-23 UGA_173101_P019_WC02 0.489130  106       305
## 1219203          GTS_104-23 UGA_173101_P019_WD04 0.479167  109       305
## 1219217          GTS_104-23 UGA_173101_P018_WF03 0.488636   98       305
## 1219221          GTS_104-23 UGA_173101_P018_WE09 0.487500   92       305
## 1219244          GTS_104-23 UGA_173101_P020_WA07 0.472222   92       305
## 1219246          GTS_104-23 UGA_173101_P016_WD06 0.487500   88       305
## 1219260          GTS_104-23 UGA_173101_P016_WC05 0.477273   50       305
## 1219281          GTS_104-23 UGA_173101_P017_WA02 0.489130  104       305
## 1219295          GTS_104-23 UGA_173101_P015_WH07 0.479167  100       305
## 1219325          GTS_104-23 UGA_173101_P020_WA05 0.479167  100       305
## 1219389          GTS_104-23 UGA_173101_P017_WD05 0.406250   52       305
## 1219393          GTS_104-23 UGA_173101_P018_WE02 0.489130  103       305
## 1219404          GTS_104-23 UGA_173101_P015_WG11 0.428571   39       305
## 1219456          GTS_104-23 UGA_173101_P016_WH06 0.486111   80       305
## 1219460          GTS_104-23 UGA_173101_P016_WH08 0.476190   90       305
## 1219485          GTS_104-23 UGA_173101_P017_WF04 0.467391  105       305
## 1219512          GTS_104-23 UGA_173101_P017_WF08 0.441176   84       305
## 1219532          GTS_104-23 UGA_173101_P017_WH09 0.450000   85       305
## 1219578          GTS_104-23 UGA_173101_P023_WG04 0.489583  101       305
## 1219595          GTS_104-23 UGA_173101_P023_WB11 0.488636   93       305
## 1219598          GTS_104-23 UGA_173101_P023_WB06 0.488095   94       305
## 1219605          GTS_104-23 UGA_173101_P023_WB04 0.475000   95       305
## 1219661          GTS_104-23 UGA_173101_P020_WG06 0.489130   99       305
## 1219696          GTS_104-23 UGA_173101_P020_WH04 0.489583  104       305
## 1219708          GTS_104-23 UGA_173101_P020_WG07 0.450000   50       305
## 1219712          GTS_104-23 UGA_173101_P020_WH11 0.489583  105       305
## 1219741          GTS_104-23 UGA_173101_P022_WG09 0.489583  101       305
## 1219752          GTS_104-23 UGA_173101_P022_WA12 0.460526   81       305
## 1219777          GTS_104-23 UGA_173101_P021_WC01 0.476190   87       305
## 1219810          GTS_104-23 UGA_173101_P021_WH06 0.472222   90       305
## 1219846          GTS_104-23 UGA_173101_P020_WH03 0.468750  105       305
## 1219893          GTS_104-23           GTS_938-23 0.489130  108       305
## 1219897          GTS_104-23           GTS_944-23 0.500000  107       305
## 1219922          GTS_104-23           GTS_631-23 0.488636  107       305
## 1219929          GTS_104-23           GTS_781-23 0.500000  108       305
## 1219938          GTS_104-23          GTS_781-23R 0.500000  108       305
## 1219941          GTS_104-23           GTS_638-23 0.488636  106       305
## 1219948          GTS_104-23           GTS_972-23 0.500000  107       305
## 1219970          GTS_104-23           GTS_651-23 0.500000  107       305
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## 1219971          GTS_104-23           GTS_982-23 0.500000  107       305
## 1219980          GTS_104-23          GTS_651-23R 0.489130  108       305
## 1219986          GTS_104-23           GTS_803-23 0.500000  108       305
## 1220002          GTS_104-23           GTS_809-23 0.500000  108       305
## 1220020          GTS_104-23          GTS_1009-23 0.500000  107       305
## 1220025          GTS_104-23          GTS_1013-23 0.489130  108       305
## 1220092          GTS_104-23           GTS_871-23 0.500000  108       305
## 1220097          GTS_104-23           GTS_713-23 0.489130  107       305
## 1220121          GTS_104-23           GTS_732-23 0.500000  106       305
## 1220156          GTS_104-23           GTS_919-23 0.500000  108       305
## 1220176          GTS_104-23           GTS_671-23 0.500000  105       305
## 1220201          GTS_104-23           GTS_706-23 0.500000  106       305
## 1220223          GTS_104-23           GTS_776-23 0.489130  108       305
## 1220247          GTS_104-23           GTS_847-23 0.489130  108       305
## 1220271          GTS_104-23           GTS_881-23 0.488636  107       305
## 1220275          GTS_104-23           GTS_890-23 0.489130  108       305
## 1220319          GTS_104-23          GTS_1002-23 0.489130  108       305
## 1220697          GTS_104-23           GTS_399-23 0.500000  109       305
## 1250586          GTS_139-23 UGA_173101_P020_WA09 0.490741   93       305
## 1250624          GTS_139-23 UGA_173101_P019_WC06 0.492188  119       305
## 1250627          GTS_139-23 UGA_173101_P019_WC02 0.492188  119       305
## 1250633          GTS_139-23 UGA_173101_P019_WD04 0.484848  122       305
## 1250647          GTS_139-23 UGA_173101_P018_WF03 0.491667  110       305
## 1250651          GTS_139-23 UGA_173101_P018_WE09 0.482143  104       305
## 1250674          GTS_139-23 UGA_173101_P020_WA07 0.468750  103       305
## 1250676          GTS_139-23 UGA_173101_P016_WD06 0.490385   98       305
## 1250690          GTS_139-23 UGA_173101_P016_WC05 0.485294   58       305
## 1250711          GTS_139-23 UGA_173101_P017_WA02 0.491935  116       305
## 1250725          GTS_139-23 UGA_173101_P015_WH07 0.484848  112       305
## 1250755          GTS_139-23 UGA_173101_P020_WA05 0.484848  113       305
## 1250819          GTS_139-23 UGA_173101_P017_WD05 0.437500   57       305
## 1250823          GTS_139-23 UGA_173101_P018_WE02 0.491935  115       305
## 1250834          GTS_139-23 UGA_173101_P015_WG11 0.416667   45       305
## 1250886          GTS_139-23 UGA_173101_P016_WH06 0.490385   89       305
## 1250890          GTS_139-23 UGA_173101_P016_WH08 0.482759  101       305
## 1250915          GTS_139-23 UGA_173101_P017_WF04 0.476562  117       305
## 1250942          GTS_139-23 UGA_173101_P017_WF08 0.456522   93       305
## 1250962          GTS_139-23 UGA_173101_P017_WH09 0.462963   95       305
## 1251008          GTS_139-23 UGA_173101_P023_WG04 0.492424  114       305
## 1251025          GTS_139-23 UGA_173101_P023_WB11 0.491935  106       305
## 1251028          GTS_139-23 UGA_173101_P023_WB06 0.483333  107       305
## 1251035          GTS_139-23 UGA_173101_P023_WB04 0.482759  108       305
## 1251091          GTS_139-23 UGA_173101_P020_WG06 0.492188  112       305
## 1251126          GTS_139-23 UGA_173101_P020_WH04 0.492424  117       305
## 1251138          GTS_139-23 UGA_173101_P020_WG07 0.466667   57       305
## 1251142          GTS_139-23 UGA_173101_P020_WH11 0.492424  118       305
## 1251171          GTS_139-23 UGA_173101_P022_WG09 0.492424  114       305
## 1251182          GTS_139-23 UGA_173101_P022_WA12 0.472222   93       305
## 1251207          GTS_139-23 UGA_173101_P021_WC01 0.482759   99       305
## 1251240          GTS_139-23 UGA_173101_P021_WH06 0.480769  101       305
## 1251276          GTS_139-23 UGA_173101_P020_WH03 0.477273  118       305
## 1251323          GTS_139-23           GTS_938-23 0.492188  122       305
## 1251327          GTS_139-23           GTS_944-23 0.500000  121       305
## 1251352          GTS_139-23           GTS_631-23 0.491935  120       305
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## 1251359          GTS_139-23           GTS_781-23 0.500000  122       305
## 1251368          GTS_139-23          GTS_781-23R 0.500000  123       305
## 1251371          GTS_139-23           GTS_638-23 0.491935  119       305
## 1251378          GTS_139-23           GTS_972-23 0.500000  121       305
## 1251400          GTS_139-23           GTS_651-23 0.500000  121       305
## 1251401          GTS_139-23           GTS_982-23 0.500000  121       305
## 1251410          GTS_139-23          GTS_651-23R 0.492188  122       305
## 1251416          GTS_139-23           GTS_803-23 0.500000  122       305
## 1251432          GTS_139-23           GTS_809-23 0.500000  122       305
## 1251450          GTS_139-23          GTS_1009-23 0.500000  121       305
## 1251455          GTS_139-23          GTS_1013-23 0.492188  122       305
## 1251522          GTS_139-23           GTS_871-23 0.500000  122       305
## 1251527          GTS_139-23           GTS_713-23 0.492188  120       305
## 1251551          GTS_139-23           GTS_732-23 0.500000  120       305
## 1251586          GTS_139-23           GTS_919-23 0.500000  122       305
## 1251606          GTS_139-23           GTS_671-23 0.500000  120       305
## 1251631          GTS_139-23           GTS_706-23 0.500000  121       305
## 1251653          GTS_139-23           GTS_776-23 0.492424  123       305
## 1251677          GTS_139-23           GTS_847-23 0.492424  123       305
## 1251701          GTS_139-23           GTS_881-23 0.492188  122       305
## 1251705          GTS_139-23           GTS_890-23 0.492424  123       305
## 1251749          GTS_139-23          GTS_1002-23 0.492424  123       305
## 1252127          GTS_139-23           GTS_399-23 0.500000  124       305
## 1252134          GTS_139-23           GTS_104-23 0.500000  109       305
## 1261681          GTS_150-23 UGA_173101_P020_WA09 0.488095   86       305
## 1261719          GTS_150-23 UGA_173101_P019_WC06 0.490385  112       305
## 1261722          GTS_150-23 UGA_173101_P019_WC02 0.490000  111       305
## 1261728          GTS_150-23 UGA_173101_P019_WD04 0.480769  114       305
## 1261742          GTS_150-23 UGA_173101_P018_WF03 0.489583  103       305
## 1261746          GTS_150-23 UGA_173101_P018_WE09 0.488636   97       305
## 1261769          GTS_150-23 UGA_173101_P020_WA07 0.473684   96       305
## 1261771          GTS_150-23 UGA_173101_P016_WD06 0.488636   93       305
## 1261785          GTS_150-23 UGA_173101_P016_WC05 0.479167   53       305
## 1261806          GTS_150-23 UGA_173101_P017_WA02 0.490000  109       305
## 1261820          GTS_150-23 UGA_173101_P015_WH07 0.480769  105       305
## 1261850          GTS_150-23 UGA_173101_P020_WA05 0.480769  105       305
## 1261914          GTS_150-23 UGA_173101_P017_WD05 0.416667   54       305
## 1261918          GTS_150-23 UGA_173101_P018_WE02 0.490000  108       305
## 1261929          GTS_150-23 UGA_173101_P015_WG11 0.425000   43       305
## 1261981          GTS_150-23 UGA_173101_P016_WH06 0.487500   83       305
## 1261985          GTS_150-23 UGA_173101_P016_WH08 0.478261   95       305
## 1262010          GTS_150-23 UGA_173101_P017_WF04 0.470000  110       305
## 1262037          GTS_150-23 UGA_173101_P017_WF08 0.460526   87       305
## 1262057          GTS_150-23 UGA_173101_P017_WH09 0.454545   90       305
## 1262103          GTS_150-23 UGA_173101_P023_WG04 0.490385  106       305
## 1262120          GTS_150-23 UGA_173101_P023_WB11 0.489583   98       305
## 1262123          GTS_150-23 UGA_173101_P023_WB06 0.478261   99       305
## 1262130          GTS_150-23 UGA_173101_P023_WB04 0.477273  100       305
## 1262186          GTS_150-23 UGA_173101_P020_WG06 0.490000  104       305
## 1262221          GTS_150-23 UGA_173101_P020_WH04 0.490385  109       305
## 1262233          GTS_150-23 UGA_173101_P020_WG07 0.458333   54       305
## 1262237          GTS_150-23 UGA_173101_P020_WH11 0.490385  110       305
## 1262266          GTS_150-23 UGA_173101_P022_WG09 0.490385  106       305
## 1262277          GTS_150-23 UGA_173101_P022_WA12 0.464286   86       305
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## 1262302          GTS_150-23 UGA_173101_P021_WC01 0.478261   92       305
## 1262335          GTS_150-23 UGA_173101_P021_WH06 0.475000   95       305
## 1262371          GTS_150-23 UGA_173101_P020_WH03 0.471154  110       305
## 1262418          GTS_150-23           GTS_938-23 0.490000  113       305
## 1262422          GTS_150-23           GTS_944-23 0.500000  112       305
## 1262447          GTS_150-23           GTS_631-23 0.489583  111       305
## 1262454          GTS_150-23           GTS_781-23 0.500000  113       305
## 1262463          GTS_150-23          GTS_781-23R 0.500000  113       305
## 1262466          GTS_150-23           GTS_638-23 0.489583  110       305
## 1262473          GTS_150-23           GTS_972-23 0.500000  112       305
## 1262495          GTS_150-23           GTS_651-23 0.500000  112       305
## 1262496          GTS_150-23           GTS_982-23 0.500000  112       305
## 1262505          GTS_150-23          GTS_651-23R 0.490000  113       305
## 1262511          GTS_150-23           GTS_803-23 0.500000  113       305
## 1262527          GTS_150-23           GTS_809-23 0.500000  113       305
## 1262545          GTS_150-23          GTS_1009-23 0.500000  112       305
## 1262550          GTS_150-23          GTS_1013-23 0.490000  113       305
## 1262617          GTS_150-23           GTS_871-23 0.500000  113       305
## 1262622          GTS_150-23           GTS_713-23 0.490000  112       305
## 1262646          GTS_150-23           GTS_732-23 0.500000  111       305
## 1262681          GTS_150-23           GTS_919-23 0.500000  113       305
## 1262701          GTS_150-23           GTS_671-23 0.500000  110       305
## 1262726          GTS_150-23           GTS_706-23 0.500000  111       305
## 1262748          GTS_150-23           GTS_776-23 0.490000  113       305
## 1262772          GTS_150-23           GTS_847-23 0.490000  113       305
## 1262796          GTS_150-23           GTS_881-23 0.489583  112       305
## 1262800          GTS_150-23           GTS_890-23 0.490000  113       305
## 1262844          GTS_150-23          GTS_1002-23 0.490000  113       305
## 1263222          GTS_150-23           GTS_399-23 0.500000  114       305
## 1263229          GTS_150-23           GTS_104-23 0.500000  107       305
## 1263249          GTS_150-23           GTS_139-23 0.500000  114       305
## 1290436          GTS_181-23 UGA_173101_P020_WA09 0.490741   93       305
## 1290474          GTS_181-23 UGA_173101_P019_WC06 0.492188  119       305
## 1290477          GTS_181-23 UGA_173101_P019_WC02 0.492188  119       305
## 1290483          GTS_181-23 UGA_173101_P019_WD04 0.484848  122       305
## 1290497          GTS_181-23 UGA_173101_P018_WF03 0.491667  110       305
## 1290501          GTS_181-23 UGA_173101_P018_WE09 0.482143  104       305
## 1290524          GTS_181-23 UGA_173101_P020_WA07 0.468750  103       305
## 1290526          GTS_181-23 UGA_173101_P016_WD06 0.490385   98       305
## 1290540          GTS_181-23 UGA_173101_P016_WC05 0.485294   58       305
## 1290561          GTS_181-23 UGA_173101_P017_WA02 0.491935  116       305
## 1290575          GTS_181-23 UGA_173101_P015_WH07 0.484848  112       305
## 1290605          GTS_181-23 UGA_173101_P020_WA05 0.484848  113       305
## 1290669          GTS_181-23 UGA_173101_P017_WD05 0.437500   57       305
## 1290673          GTS_181-23 UGA_173101_P018_WE02 0.491935  115       305
## 1290684          GTS_181-23 UGA_173101_P015_WG11 0.416667   45       305
## 1290736          GTS_181-23 UGA_173101_P016_WH06 0.490385   89       305
## 1290740          GTS_181-23 UGA_173101_P016_WH08 0.482759  101       305
## 1290765          GTS_181-23 UGA_173101_P017_WF04 0.476562  117       305
## 1290792          GTS_181-23 UGA_173101_P017_WF08 0.456522   93       305
## 1290812          GTS_181-23 UGA_173101_P017_WH09 0.462963   95       305
## 1290858          GTS_181-23 UGA_173101_P023_WG04 0.492424  114       305
## 1290875          GTS_181-23 UGA_173101_P023_WB11 0.491935  106       305
## 1290878          GTS_181-23 UGA_173101_P023_WB06 0.483333  107       305
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## 1290885          GTS_181-23 UGA_173101_P023_WB04 0.482759  108       305
## 1290941          GTS_181-23 UGA_173101_P020_WG06 0.492188  112       305
## 1290976          GTS_181-23 UGA_173101_P020_WH04 0.492424  117       305
## 1290988          GTS_181-23 UGA_173101_P020_WG07 0.466667   57       305
## 1290992          GTS_181-23 UGA_173101_P020_WH11 0.492424  118       305
## 1291021          GTS_181-23 UGA_173101_P022_WG09 0.492424  114       305
## 1291032          GTS_181-23 UGA_173101_P022_WA12 0.472222   93       305
## 1291057          GTS_181-23 UGA_173101_P021_WC01 0.482759   99       305
## 1291090          GTS_181-23 UGA_173101_P021_WH06 0.480769  101       305
## 1291126          GTS_181-23 UGA_173101_P020_WH03 0.477273  118       305
## 1291173          GTS_181-23           GTS_938-23 0.492188  122       305
## 1291177          GTS_181-23           GTS_944-23 0.500000  121       305
## 1291202          GTS_181-23           GTS_631-23 0.491935  120       305
## 1291209          GTS_181-23           GTS_781-23 0.500000  122       305
## 1291218          GTS_181-23          GTS_781-23R 0.500000  123       305
## 1291221          GTS_181-23           GTS_638-23 0.491935  119       305
## 1291228          GTS_181-23           GTS_972-23 0.500000  121       305
## 1291250          GTS_181-23           GTS_651-23 0.500000  121       305
## 1291251          GTS_181-23           GTS_982-23 0.500000  121       305
## 1291260          GTS_181-23          GTS_651-23R 0.492188  122       305
## 1291266          GTS_181-23           GTS_803-23 0.500000  122       305
## 1291282          GTS_181-23           GTS_809-23 0.500000  122       305
## 1291300          GTS_181-23          GTS_1009-23 0.500000  121       305
## 1291305          GTS_181-23          GTS_1013-23 0.492188  122       305
## 1291372          GTS_181-23           GTS_871-23 0.500000  122       305
## 1291377          GTS_181-23           GTS_713-23 0.492188  120       305
## 1291401          GTS_181-23           GTS_732-23 0.500000  120       305
## 1291436          GTS_181-23           GTS_919-23 0.500000  122       305
## 1291456          GTS_181-23           GTS_671-23 0.500000  120       305
## 1291481          GTS_181-23           GTS_706-23 0.500000  121       305
## 1291503          GTS_181-23           GTS_776-23 0.492424  123       305
## 1291527          GTS_181-23           GTS_847-23 0.492424  123       305
## 1291551          GTS_181-23           GTS_881-23 0.492188  122       305
## 1291555          GTS_181-23           GTS_890-23 0.492424  123       305
## 1291599          GTS_181-23          GTS_1002-23 0.492424  123       305
## 1291977          GTS_181-23           GTS_399-23 0.500000  124       305
## 1291984          GTS_181-23           GTS_104-23 0.500000  109       305
## 1292004          GTS_181-23           GTS_139-23 0.500000  124       305
## 1292011          GTS_181-23           GTS_150-23 0.500000  114       305
## 1308168          GTS_187-23 UGA_173101_P020_WA09 0.490000   91       305
## 1308206          GTS_187-23 UGA_173101_P019_WC06 0.491379  116       305
## 1308209          GTS_187-23 UGA_173101_P019_WC02 0.491379  116       305
## 1308215          GTS_187-23 UGA_173101_P019_WD04 0.483333  119       305
## 1308229          GTS_187-23 UGA_173101_P018_WF03 0.490741  107       305
## 1308233          GTS_187-23 UGA_173101_P018_WE09 0.490385  102       305
## 1308256          GTS_187-23 UGA_173101_P020_WA07 0.467391  102       305
## 1308258          GTS_187-23 UGA_173101_P016_WD06 0.489583   96       305
## 1308272          GTS_187-23 UGA_173101_P016_WC05 0.484375   57       305
## 1308293          GTS_187-23 UGA_173101_P017_WA02 0.491071  113       305
## 1308307          GTS_187-23 UGA_173101_P015_WH07 0.483333  109       305
## 1308337          GTS_187-23 UGA_173101_P020_WA05 0.483333  110       305
## 1308401          GTS_187-23 UGA_173101_P017_WD05 0.437500   57       305
## 1308405          GTS_187-23 UGA_173101_P018_WE02 0.491071  112       305
## 1308416          GTS_187-23 UGA_173101_P015_WG11 0.416667   45       305
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## 1308468          GTS_187-23 UGA_173101_P016_WH06 0.489583   87       305
## 1308472          GTS_187-23 UGA_173101_P016_WH08 0.490385   98       305
## 1308497          GTS_187-23 UGA_173101_P017_WF04 0.474138  114       305
## 1308524          GTS_187-23 UGA_173101_P017_WF08 0.452381   91       305
## 1308544          GTS_187-23 UGA_173101_P017_WH09 0.468750   92       305
## 1308590          GTS_187-23 UGA_173101_P023_WG04 0.491667  111       305
## 1308607          GTS_187-23 UGA_173101_P023_WB11 0.491379  104       305
## 1308610          GTS_187-23 UGA_173101_P023_WB06 0.481481  104       305
## 1308617          GTS_187-23 UGA_173101_P023_WB04 0.481481  106       305
## 1308673          GTS_187-23 UGA_173101_P020_WG06 0.491379  109       305
## 1308708          GTS_187-23 UGA_173101_P020_WH04 0.491667  114       305
## 1308720          GTS_187-23 UGA_173101_P020_WG07 0.482143   55       305
## 1308724          GTS_187-23 UGA_173101_P020_WH11 0.491667  115       305
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## 1308764          GTS_187-23 UGA_173101_P022_WA12 0.470000   91       305
## 1308789          GTS_187-23 UGA_173101_P021_WC01 0.481481   97       305
## 1308822          GTS_187-23 UGA_173101_P021_WH06 0.489583   99       305
## 1308858          GTS_187-23 UGA_173101_P020_WH03 0.475000  115       305
## 1308905          GTS_187-23           GTS_938-23 0.491379  119       305
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## 1308524        14         TRUE      FALSE
## 1308544        14         TRUE      FALSE
## 1308590        14         TRUE      FALSE
## 1308607        14         TRUE      FALSE
## 1308610        14         TRUE      FALSE
## 1308617        14         TRUE      FALSE
## 1308673        14         TRUE      FALSE
## 1308708        14         TRUE      FALSE
## 1308720        14         TRUE      FALSE
## 1308724        14         TRUE      FALSE
## 1308753        14         TRUE      FALSE
## 1308764        14         TRUE      FALSE
## 1308789        14         TRUE      FALSE
## 1308822        14         TRUE      FALSE
## 1308858        14         TRUE      FALSE
## 1308905       301         TRUE      FALSE
## 1308909       305         TRUE       TRUE
## 1308934       301         TRUE      FALSE
## 1308941       305         TRUE       TRUE
## 1308950       305         TRUE       TRUE
## 1308953       301         TRUE      FALSE
## 1308960       301         TRUE      FALSE
## 1308982       301         TRUE      FALSE
## 1308983       301         TRUE      FALSE
## 1308992       301         TRUE      FALSE
## 1308998       305         TRUE       TRUE
## 1309014       305         TRUE       TRUE
## 1309032       301         TRUE      FALSE
## 1309037       301         TRUE      FALSE
## 1309104       305         TRUE       TRUE
## 1309109       301         TRUE      FALSE
## 1309133       301         TRUE      FALSE
```

```
## 1309168       305         TRUE       TRUE
## 1309188       301         TRUE      FALSE
## 1309213       301         TRUE      FALSE
## 1309235       301         TRUE      FALSE
## 1309259       301         TRUE      FALSE
## 1309283       301         TRUE      FALSE
## 1309287       301         TRUE      FALSE
## 1309331       301         TRUE      FALSE
## 1309709       305         TRUE       TRUE
## 1309716       305         TRUE       TRUE
## 1309736       305         TRUE       TRUE
## 1309743       305         TRUE       TRUE
## 1309761       305         TRUE       TRUE
```

```r
#Clone 47 - 5 F, 1 M, direct capture confirms majority
table(smd$sexGT[smd$finalClone == 47])
```

```
##
## F M
## 5 1
```

```r
smd[which(smd$finalClone == 47),]
```

```
##                                     extID   ugaID is.SCR is.rep week year
## 49                    UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                   UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                   UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>           <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                        fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                           <NA>
## 1598                                                                           <NA>
## 1819                                                                           <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F     Capture    CGA   1126 UGA_173101_P015_WE01 FALSE       LGC
## 492            Hair Snare    CGA   1259 UGA_173101_P020_WC06 FALSE       LGC
## 629            Hair Snare    CGA   1257 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120           GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123          GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124          GTS_2920-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 49       F       47         47
```

```
## 492      F       47         47
## 629      M       47         47
## 1048     F       47         47
## 1598     F       47         47
## 1819     F       47         47
```

```
unique(smd$COLclone[which(smd$finalClone == 47)])
```

```
## [1] 47
```

```
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 47)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 47)],]
```

```
##                          id1                  id2  kinship nsnp colclone1
## 23083   UGA_173101_P015_WE01 UGA_173101_P020_WC06 0.490157 1110        47
## 214918  UGA_173101_P021_WF11 UGA_173101_P020_WC06 0.466549 1215        47
## 215056  UGA_173101_P021_WF11 UGA_173101_P015_WE01 0.460813 1102        47
## 399249            GTS_664-24 UGA_173101_P020_WC06 0.483333  113        47
## 399387            GTS_664-24 UGA_173101_P015_WE01 0.481481  101        47
## 399828            GTS_664-24 UGA_173101_P021_WF11 0.474138  109        47
## 882534           GTS_3057-24 UGA_173101_P020_WC06 0.484375  115        47
## 882672           GTS_3057-24 UGA_173101_P015_WE01 0.482143  102        47
## 883113           GTS_3057-24 UGA_173101_P021_WF11 0.475806  112        47
## 883351           GTS_3057-24           GTS_664-24 0.500000  120        47
## 1130334          GTS_2920-24 UGA_173101_P020_WC06 0.453125  116        47
## 1130472          GTS_2920-24 UGA_173101_P015_WE01 0.446429  103        47
## 1130913          GTS_2920-24 UGA_173101_P021_WF11 0.443548  113        47
## 1131151          GTS_2920-24           GTS_664-24 0.500000  120        47
## 1131586          GTS_2920-24          GTS_3057-24 0.500000  123        47
##         colclone2 isclonePLINK iscloneCOL
## 23083          47         TRUE       TRUE
## 214918         47         TRUE       TRUE
## 215056         47         TRUE       TRUE
## 399249         47         TRUE       TRUE
## 399387         47         TRUE       TRUE
## 399828         47         TRUE       TRUE
## 882534         47         TRUE       TRUE
## 882672         47         TRUE       TRUE
## 883113         47         TRUE       TRUE
## 883351         47         TRUE       TRUE
## 1130334        47         TRUE       TRUE
## 1130472        47         TRUE       TRUE
## 1130913        47         TRUE       TRUE
## 1131151        47         TRUE       TRUE
## 1131586        47         TRUE       TRUE
```

```
#Clone 307 - 61 M, 1 F, 5 undet, no direct captures
table(smd$sexGT[smd$finalClone == 307])
```

```
##
##     F     M undet
##     1    61     5
```

```r
smd[which(smd$finalClone == 307),]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 132                     UGA_173101_P016_WC12  452-23   TRUE  FALSE    7 2023
## 152                     UGA_173101_P016_WE08  194-24   TRUE  FALSE    2 2024
## 163                     UGA_173101_P016_WF07  496-24   TRUE  FALSE    3 2024
## 189                     UGA_173101_P016_WH09  480-23   TRUE  FALSE    6 2023
## 218                     UGA_173101_P017_WC02  216-23   TRUE  FALSE    7 2023
## 228                     UGA_173101_P017_WC12  101-24   TRUE  FALSE    3 2024
## 237                     UGA_173101_P017_WD09  318-23   TRUE  FALSE    7 2023
## 304                     UGA_173101_P018_WB04  186-24   TRUE  FALSE    3 2024
## 306                     UGA_173101_P018_WB06  490-23   TRUE  FALSE    3 2023
## 323                     UGA_173101_P018_WC11  358-24   TRUE   TRUE    3 2024
## 334                     UGA_173101_P018_WD10  334-23   TRUE  FALSE    6 2023
## 367                     UGA_173101_P019_WA01   25-23   TRUE  FALSE    6 2023
## 368                     UGA_173101_P019_WA02   32-23   TRUE  FALSE    7 2023
## 370                     UGA_173101_P019_WA04   48-23   TRUE  FALSE    6 2023
## 419                     UGA_173101_P019_WE05   48-24   TRUE  FALSE    3 2024
## 441                     UGA_173101_P019_WG03  358-24   TRUE   TRUE    3 2024
## 619                     UGA_173101_P021_WF01  274-23   TRUE  FALSE    7 2023
## 663                     UGA_173101_P022_WA09  272-23   TRUE  FALSE    6 2023
## 692                     UGA_173101_P022_WD02   63-24   TRUE  FALSE    3 2024
## 754                     UGA_173101_P023_WA04  189-23   TRUE  FALSE    3 2023
## 757                     UGA_173101_P023_WA07  307-23   TRUE  FALSE    7 2023
## 774                     UGA_173101_P023_WB12  374-23   TRUE  FALSE    6 2023
## 815                     UGA_173101_P023_WF05  333-24   TRUE  FALSE    3 2024
## 873    GTseek_024_384i5_046_P098.003_949-23  949-23   TRUE  FALSE    6 2023
## 936    GTseek_024_384i5_109_P098.001_646-23  646-23   TRUE  FALSE    7 2023
## 951    GTseek_024_384i5_124_P098.004_616-24  616-24   TRUE  FALSE    2 2024
## 962    GTseek_024_384i5_135_P098.001_655-23  655-23   TRUE  FALSE    6 2023
## 963    GTseek_024_384i5_136_P098.003_983-23  983-23   TRUE  FALSE    3 2023
## 974    GTseek_024_384i5_148_P098.004_630-24  630-24   TRUE  FALSE    3 2024
## 993    GTseek_024_384i5_167_P098.001_665-23  665-23   TRUE  FALSE    7 2023
## 998    GTseek_024_384i5_172_P098.003_997-23  997-23   TRUE  FALSE    7 2023
## 1006   GTseek_024_384i5_180_P098.004_644-24  644-24   TRUE  FALSE    3 2024
## 1013   GTseek_024_384i5_187_P098.002_818-23  818-23   TRUE  FALSE    6 2023
## 1029   GTseek_024_384i5_203_P098.001_677-23  677-23   TRUE  FALSE    3 2023
## 1030  GTseek_024_384i5_204_P098.003_1012-23 1012-23   TRUE  FALSE    7 2023
## 1038   GTseek_024_384i5_212_P098.004_657-24  657-24   TRUE  FALSE    3 2024
## 1061   GTseek_024_384i5_235_P098.001_691-23  691-23   TRUE  FALSE    6 2023
## 1151   GTseek_024_384i5_325_P098.001_725-23  725-23   TRUE  FALSE    3 2023
## 1161   GTseek_024_384i5_335_P098.001_736-23  736-23   TRUE  FALSE    6 2023
## 1294  GTseek_025_384i5_084_P098.008_1202-24 1202-24   TRUE  FALSE    3 2024
## 1314   GTseek_025_384i5_104_P098.007_850-23  850-23   TRUE  FALSE    6 2023
## 1353   GTseek_025_384i5_143_P098.005_825-24  825-24   TRUE  FALSE    3 2024
## 1374   GTseek_025_384i5_164_P098.007_935-23  935-23   TRUE   TRUE    7 2023
## 1386  GTseek_025_384i5_176_P098.007_935-23R 935-23R   TRUE   TRUE    7 2023
## 1411   GTseek_025_384i5_201_P098.005_855-24  855-24   TRUE  FALSE    3 2024
## 1426  GTseek_025_384i5_216_P098.008_1344-24 1344-24   TRUE  FALSE    3 2024
## 1473   GTseek_025_384i5_263_P098.005_886-24  886-24   TRUE  FALSE    3 2024
## 1490  GTseek_025_384i5_280_P098.008_1403-24 1403-24   TRUE  FALSE    3 2024
## 1513   GTseek_025_384i5_303_P098.005_931-24  931-24   TRUE  FALSE    3 2024
## 1518  GTseek_025_384i5_308_P098.008_1428-24 1428-24   TRUE  FALSE    2 2024
```

```
## 1527  GTseek_025_384i5_317_P098.006_3012-24 3012-24   TRUE  FALSE    6 2024
## 1548  GTseek_025_384i5_338_P098.008_1451-24 1451-24   TRUE  FALSE    3 2024
## 1563   GTseek_025_384i5_353_P098.005_955-24  955-24   TRUE  FALSE    2 2024
## 1595  GTseek_026_384i5_001_P098.009_1516-24 1516-24   TRUE  FALSE    3 2024
## 1641  GTseek_026_384i5_061_P098.010_2777-24 2777-24   TRUE  FALSE    3 2024
## 1677  GTseek_026_384i5_107_P098.009_1633-24 1633-24   TRUE  FALSE    3 2024
## 1691  GTseek_026_384i5_129_P098.009_1672-24 1672-24   TRUE  FALSE    3 2024
## 1717  GTseek_026_384i5_167_P098.009_2376-24 2376-24   TRUE  FALSE    6 2024
## 1747  GTseek_026_384i5_227_P098.009_2669-24 2669-24   TRUE  FALSE    6 2024
## 1756  GTseek_026_384i5_245_P098.010_2857-24 2857-24   TRUE  FALSE    3 2024
## 1782  GTseek_026_384i5_297_P098.009_2697-24 2697-24   TRUE  FALSE    2 2024
## 1805  GTseek_026_384i5_343_P098.010_2906-24 2906-24   TRUE  FALSE    6 2024
## 1903  GTseek_028_384i5_078_P098.UGA3_151-24  151-24   TRUE   TRUE    3 2024
## 1920  GTseek_028_384i5_095_P098.UGA2_426-23  426-23   TRUE  FALSE    6 2023
## 1937 GTseek_028_384i5_112_P098.UGA3_151-24R 151-24R   TRUE   TRUE    3 2024
## 1964  GTseek_028_384i5_139_P098.UGA1_146-23  146-23   TRUE  FALSE    3 2023
## 2185  GTseek_028_384i5_360_P098.UGA3_311-24  311-24   TRUE  FALSE    3 2024
##       snare       utmSnare
## 132  HS 009 259209x3585507
## 152  HS 014 261409x3586557
## 163  HS 017 260417x3588089
## 189  HS 009 259209x3585507
## 218  HS 009 259209x3585507
## 228  HS 012 258412x3586648
## 237  HS 002 255814x3587061
## 304  HS 011 257416x3586308
## 306  HS 012 258412x3586648
## 323  HS 017 260417x3588089
## 334  HS 012 258412x3586648
## 367  HS 009 259209x3585507
## 368  HS 009 259209x3585507
## 370  HS 012 258412x3586648
## 419  HS 017 260417x3588089
## 441  HS 017 260417x3588089
## 619  HS 007 255057x3589730
## 663  HS 009 259209x3585507
## 692  HS 011 257416x3586308
## 754  HS 012 258412x3586648
## 757  HS 009 259209x3585507
## 774  HS 009 259209x3585507
## 815  HS 017 260417x3588089
## 873  HS 009 259209x3585507
## 936  HS 009 259209x3585507
## 951  HS 014 261409x3586557
## 962  HS 012 258412x3586648
## 963  HS 012 258412x3586648
## 974  HS 012 258412x3586648
## 993  HS 002 255814x3587061
## 998  HS 009 259209x3585507
## 1006 HS 012 258412x3586648
## 1013 HS 009 259209x3585507
## 1029 HS 012 258412x3586648
## 1030 HS 002 255814x3587061
## 1038 HS 012 258412x3586648
```

```
## 1061 HS 012 258412x3586648
## 1151 HS 012 258412x3586648
## 1161 HS 012 258412x3586648
## 1294 HS 012 258412x3586648
## 1314 HS 012 258412x3586648
## 1353 HS 017 260417x3588089
## 1374 HS 002 255814x3587061
## 1386 HS 002 255814x3587061
## 1411 HS 011 257416x3586308
## 1426 HS 011 257416x3586308
## 1473 HS 017 260417x3588089
## 1490 HS 012 258412x3586648
## 1513 HS 011 257416x3586308
## 1518 HS 014 261409x3586557
## 1527 HS 009 259092x3585434
## 1548 HS 012 258412x3586648
## 1563 HS 014 261409x3586557
## 1595 HS 017 260417x3588089
## 1641 HS 011 257416x3586308
## 1677 HS 011 257416x3586308
## 1691 HS 011 257416x3586308
## 1717 HS 009 259092x3585434
## 1747 HS 009 259092x3585434
## 1756 HS 011 257416x3586308
## 1782 HS 014 261409x3586557
## 1805 HS 009 259092x3585434
## 1903 HS 012 258412x3586648
## 1920 HS 012 258412x3586648
## 1937 HS 012 258412x3586648
## 1964 HS 012 258412x3586648
## 2185 HS 012 258412x3586648
##                                                                                  fqfile
## 132  RAPiD-Genomics_F534_UGA_173101_P016_WC12_i5-504_i7-132_S14870_L003_R1_001.fastq
## 152  RAPiD-Genomics_F534_UGA_173101_P016_WE08_i5-504_i7-152_S14889_L003_R1_001.fastq
## 163  RAPiD-Genomics_F534_UGA_173101_P016_WF07_i5-504_i7-163_S14900_L003_R1_001.fastq
## 189  RAPiD-Genomics_F534_UGA_173101_P016_WH09_i5-504_i7-189_S14926_L003_R1_001.fastq
## 218  RAPiD-Genomics_F534_UGA_173101_P017_WC02_i5-512_i7-122_S14955_L003_R1_001.fastq
## 228  RAPiD-Genomics_F534_UGA_173101_P017_WC12_i5-512_i7-132_S14965_L003_R1_001.fastq
## 237  RAPiD-Genomics_F534_UGA_173101_P017_WD09_i5-512_i7-141_S14974_L003_R1_001.fastq
## 304  RAPiD-Genomics_F534_UGA_173101_P018_WB04_i5-537_i7-112_S15041_L003_R1_001.fastq
## 306  RAPiD-Genomics_F534_UGA_173101_P018_WB06_i5-537_i7-114_S15043_L003_R1_001.fastq
## 323  RAPiD-Genomics_F534_UGA_173101_P018_WC11_i5-537_i7-131_S15060_L003_R1_001.fastq
## 334  RAPiD-Genomics_F534_UGA_173101_P018_WD10_i5-537_i7-142_S15071_L003_R1_001.fastq
## 367  RAPiD-Genomics_F534_UGA_173101_P019_WA01_i5-1102_i7-59_S15106_L003_R1_001.fastq
## 368  RAPiD-Genomics_F534_UGA_173101_P019_WA02_i5-1102_i7-27_S15107_L003_R1_001.fastq
## 370   RAPiD-Genomics_F534_UGA_173101_P019_WA04_i5-1102_i7-7_S15109_L003_R1_001.fastq
## 419  RAPiD-Genomics_F534_UGA_173101_P019_WE05_i5-1102_i7-94_S15158_L003_R1_001.fastq
## 441  RAPiD-Genomics_F534_UGA_173101_P019_WG03_i5-1102_i7-22_S15277_L003_R1_001.fastq
## 619      RAPiD-Genomics_F566_UGA_173101_P021_WF01_i5-501_i7-28_S61_L003_R1_001.fastq
## 663   RAPiD-Genomics_F566_UGA_173101_P022_WA09_i5-1098_i7-36_S7591_L003_R1_001.fastq
## 692   RAPiD-Genomics_F566_UGA_173101_P022_WD02_i5-1098_i7-72_S7620_L003_R1_001.fastq
## 754    RAPiD-Genomics_F566_UGA_173101_P023_WA04_i5-1099_i7-7_S7682_L003_R1_001.fastq
## 757   RAPiD-Genomics_F566_UGA_173101_P023_WA07_i5-1099_i7-77_S7685_L003_R1_001.fastq
## 774   RAPiD-Genomics_F566_UGA_173101_P023_WB12_i5-1099_i7-75_S7702_L003_R1_001.fastq
```

```
## 815   RAPiD-Genomics_F566_UGA_173101_P023_WF05_i5-1099_i7-92_S7743_L003_R1_001.fastq
## 873                                                                                <NA>
## 936                                                                                <NA>
## 951                                                                                <NA>
## 962                                                                                <NA>
## 963                                                                                <NA>
## 974                                                                                <NA>
## 993                                                                                <NA>
## 998                                                                                <NA>
## 1006                                                                               <NA>
## 1013                                                                               <NA>
## 1029                                                                               <NA>
## 1030                                                                               <NA>
## 1038                                                                               <NA>
## 1061                                                                               <NA>
## 1151                                                                               <NA>
## 1161                                                                               <NA>
## 1294                                                                               <NA>
## 1314                                                                               <NA>
## 1353                                                                               <NA>
## 1374                                                                               <NA>
## 1386                                                                               <NA>
## 1411                                                                               <NA>
## 1426                                                                               <NA>
## 1473                                                                               <NA>
## 1490                                                                               <NA>
## 1513                                                                               <NA>
## 1518                                                                               <NA>
## 1527                                                                               <NA>
## 1548                                                                               <NA>
## 1563                                                                               <NA>
## 1595                                                                               <NA>
## 1641                                                                               <NA>
## 1677                                                                               <NA>
## 1691                                                                               <NA>
## 1717                                                                               <NA>
## 1747                                                                               <NA>
## 1756                                                                               <NA>
## 1782                                                                               <NA>
## 1805                                                                               <NA>
## 1903                                                                               <NA>
## 1920                                                                               <NA>
## 1937                                                                               <NA>
## 1964                                                                               <NA>
## 2185                                                                               <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 132            Hair Snare    CGA   1172 UGA_173101_P016_WC12 FALSE       LGC
## 152            Hair Snare    CGA   1266 UGA_173101_P016_WE08 FALSE       LGC
## 163            Hair Snare    CGA   1084 UGA_173101_P016_WF07 FALSE       LGC
## 189            Hair Snare    CGA   1214 UGA_173101_P016_WH09 FALSE       LGC
## 218            Hair Snare    CGA   1258 UGA_173101_P017_WC02 FALSE       LGC
## 228            Hair Snare    CGA   1269 UGA_173101_P017_WC12 FALSE       LGC
## 237            Hair Snare    CGA   1204 UGA_173101_P017_WD09 FALSE       LGC
## 304            Hair Snare    CGA   1249 UGA_173101_P018_WB04 FALSE       LGC
```

```
## 306            Hair Snare    CGA    518 UGA_173101_P018_WB06 FALSE       LGC
## 323            Hair Snare    CGA   1297 UGA_173101_P018_WC11 FALSE       LGC
## 334            Hair Snare    CGA   1113 UGA_173101_P018_WD10 FALSE       LGC
## 367            Hair Snare    CGA   1288 UGA_173101_P019_WA01 FALSE       LGC
## 368            Hair Snare    CGA   1272 UGA_173101_P019_WA02 FALSE       LGC
## 370            Hair Snare    CGA    900 UGA_173101_P019_WA04 FALSE       LGC
## 419            Hair Snare    CGA   1197 UGA_173101_P019_WE05 FALSE       LGC
## 441            Hair Snare    CGA   1292 UGA_173101_P019_WG03 FALSE       LGC
## 619            Hair Snare    CGA   1227 UGA_173101_P021_WF01 FALSE       LGC
## 663            Hair Snare    CGA   1209 UGA_173101_P022_WA09 FALSE       LGC
## 692            Hair Snare    CGA    508 UGA_173101_P022_WD02 FALSE       LGC
## 754            Hair Snare    CGA   1155 UGA_173101_P023_WA04 FALSE       LGC
## 757            Hair Snare    CGA   1227 UGA_173101_P023_WA07 FALSE       LGC
## 774            Hair Snare    CGA   1266 UGA_173101_P023_WB12 FALSE       LGC
## 815            Hair Snare    CGA   1082 UGA_173101_P023_WF05 FALSE       LGC
## 873      <NA> Hair Sample    CGA    124           GTS_949-23 FALSE    GTSeek
## 936      <NA> Hair Sample    CGA    120           GTS_646-23 FALSE    GTSeek
## 951      <NA> Hair Sample    CGA    122           GTS_616-24 FALSE    GTSeek
## 962      <NA> Hair Sample    CGA    116           GTS_655-23 FALSE    GTSeek
## 963      <NA> Hair Sample    CGA    121           GTS_983-23 FALSE    GTSeek
## 974      <NA> Hair Sample    CGA    119           GTS_630-24 FALSE    GTSeek
## 993      <NA> Hair Sample    CGA    122           GTS_665-23 FALSE    GTSeek
## 998      <NA> Hair Sample    CGA    123           GTS_997-23 FALSE    GTSeek
## 1006     <NA> Hair Sample    CGA    119           GTS_644-24 FALSE    GTSeek
## 1013     <NA> Hair Sample    CGA    123           GTS_818-23 FALSE    GTSeek
## 1029     <NA> Hair Sample    CGA    119           GTS_677-23 FALSE    GTSeek
## 1030     <NA> Hair Sample    CGA    121          GTS_1012-23 FALSE    GTSeek
## 1038     <NA> Hair Sample    CGA    120           GTS_657-24 FALSE    GTSeek
## 1061     <NA> Hair Sample    CGA    119           GTS_691-23 FALSE    GTSeek
## 1151     <NA> Hair Sample    CGA    119           GTS_725-23 FALSE    GTSeek
## 1161     <NA> Hair Sample    CGA    120           GTS_736-23 FALSE    GTSeek
## 1294     <NA> Hair Sample    CGA    122          GTS_1202-24 FALSE    GTSeek
## 1314     <NA> Hair Sample    CGA    124           GTS_850-23 FALSE    GTSeek
## 1353     <NA> Hair Sample    CGA    120           GTS_825-24 FALSE    GTSeek
## 1374     <NA> Hair Sample    CGA    123           GTS_935-23 FALSE    GTSeek
## 1386     <NA> Hair Sample    CGA    123          GTS_935-23R FALSE    GTSeek
## 1411     <NA> Hair Sample    CGA    123           GTS_855-24 FALSE    GTSeek
## 1426     <NA> Hair Sample    CGA    124          GTS_1344-24 FALSE    GTSeek
## 1473     <NA> Hair Sample    CGA    124           GTS_886-24 FALSE    GTSeek
## 1490     <NA> Hair Sample    CGA    122          GTS_1403-24 FALSE    GTSeek
## 1513     <NA> Hair Sample    CGA    119           GTS_931-24 FALSE    GTSeek
## 1518     <NA> Hair Sample    CGA    123          GTS_1428-24 FALSE    GTSeek
## 1527     <NA> Hair Sample    CGA    122          GTS_3012-24 FALSE    GTSeek
## 1548     <NA> Hair Sample    CGA    123          GTS_1451-24 FALSE    GTSeek
## 1563     <NA> Hair Sample    CGA    122           GTS_955-24 FALSE    GTSeek
## 1595     <NA> Hair Sample    CGA    120          GTS_1516-24 FALSE    GTSeek
## 1641     <NA> Hair Sample    CGA    121          GTS_2777-24 FALSE    GTSeek
## 1677     <NA> Hair Sample    CGA    124          GTS_1633-24 FALSE    GTSeek
## 1691     <NA> Hair Sample    CGA    121          GTS_1672-24 FALSE    GTSeek
## 1717     <NA> Hair Sample    CGA    123          GTS_2376-24 FALSE    GTSeek
## 1747     <NA> Hair Sample    CGA    123          GTS_2669-24 FALSE    GTSeek
## 1756     <NA> Hair Sample    CGA    124          GTS_2857-24 FALSE    GTSeek
## 1782     <NA> Hair Sample    CGA    122          GTS_2697-24 FALSE    GTSeek
## 1805     <NA> Hair Sample    CGA    124          GTS_2906-24 FALSE    GTSeek
```

```
## 1903     <NA> Hair Sample    CGA    122           GTS_151-24 FALSE    GTSeek
## 1920     <NA> Hair Sample    CGA    119           GTS_426-23 FALSE    GTSeek
## 1937     <NA> Hair Sample    CGA    118          GTS_151-24R FALSE    GTSeek
## 1964     <NA> Hair Sample    CGA    122           GTS_146-23 FALSE    GTSeek
## 2185     <NA> Hair Sample    CGA    121           GTS_311-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 132      M       34        307
## 152      M       34        307
## 163      M       34        307
## 189      M       34        307
## 218      M       34        307
## 228      M       34        307
## 237      M       34        307
## 304      M       34        307
## 306      M      193        307
## 323      M       34        307
## 334      M       34        307
## 367      M       34        307
## 368      M       34        307
## 370      M       34        307
## 419      M       34        307
## 441      M       34        307
## 619      M       34        307
## 663      M       34        307
## 692      M       34        307
## 754      M       34        307
## 757      M       34        307
## 774      M       34        307
## 815      M       34        307
## 873      M      307        307
## 936  undet      307        307
## 951      M      326        307
## 962      F      307        307
## 963      M       34        307
## 974      M      307        307
## 993  undet      307        307
## 998      M      307        307
## 1006     M      326        307
## 1013     M      307        307
## 1029     M      307        307
## 1030     M      307        307
## 1038     M      326        307
## 1061     M      339        307
## 1151 undet       34        307
## 1161 undet      326        307
## 1294     M      339        307
## 1314     M      352        307
## 1353     M      352        307
## 1374     M      307        307
## 1386     M      339        307
## 1411     M      307        307
## 1426     M      307        307
## 1473     M      307        307
## 1490     M      307        307
```

```
## 1513     M      307        307
## 1518     M      307        307
## 1527     M      307        307
## 1548     M      307        307
## 1563     M      307        307
## 1595     M      326        307
## 1641     M      307        307
## 1677     M      339        307
## 1691     M      307        307
## 1717     M      307        307
## 1747     M      307        307
## 1756     M      307        307
## 1782     M      307        307
## 1805     M      307        307
## 1903     M      307        307
## 1920     M      307        307
## 1937     M       34        307
## 1964     M       34        307
## 2185 undet       34        307
```

```
unique(smd$COLclone[which(smd$finalClone == 307)])
```

```
## [1]  34 193 307 326 339 352
```

```
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 307)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 307)],]
```

```
##                          id1                   id2  kinship nsnp colclone1
## 2186    UGA_173101_P019_WA04 UGA_173101_P019_WE05 0.476515  876        34
## 5712    UGA_173101_P016_WC12 UGA_173101_P019_WE05 0.493664 1122        34
## 5738    UGA_173101_P016_WC12 UGA_173101_P019_WA04 0.478593  866        34
## 5927    UGA_173101_P017_WC02 UGA_173101_P019_WE05 0.492257 1185        34
## 5953    UGA_173101_P017_WC02 UGA_173101_P019_WA04 0.477477  893        34
## 5994    UGA_173101_P017_WC02 UGA_173101_P016_WC12 0.492778 1159        34
## 6711    UGA_173101_P018_WB04 UGA_173101_P019_WE05 0.492778 1180        34
## 6737    UGA_173101_P018_WB04 UGA_173101_P019_WA04 0.478163  887        34
## 6778    UGA_173101_P018_WB04 UGA_173101_P016_WC12 0.494910 1151        34
## 6780    UGA_173101_P018_WB04 UGA_173101_P017_WC02 0.493085 1227        34
## 7301    UGA_173101_P016_WF07 UGA_173101_P019_WE05 0.485802 1051        34
## 7327    UGA_173101_P016_WF07 UGA_173101_P019_WA04 0.468850  830        34
## 7368    UGA_173101_P016_WF07 UGA_173101_P016_WC12 0.485661 1029        34
## 7370    UGA_173101_P016_WF07 UGA_173101_P017_WC02 0.483929 1079        34
## 7377    UGA_173101_P016_WF07 UGA_173101_P018_WB04 0.487440 1076        34
## 14406   UGA_173101_P019_WG03 UGA_173101_P019_WE05 0.494505 1197        34
## 14432   UGA_173101_P019_WG03 UGA_173101_P019_WA04 0.481399  900        34
## 14473   UGA_173101_P019_WG03 UGA_173101_P016_WC12 0.496145 1170        34
## 14475   UGA_173101_P019_WG03 UGA_173101_P017_WC02 0.495322 1255        34
## 14482   UGA_173101_P019_WG03 UGA_173101_P018_WB04 0.496855 1248        34
## 14487   UGA_173101_P019_WG03 UGA_173101_P016_WF07 0.487530 1084        34
## 15441   UGA_173101_P016_WE08 UGA_173101_P019_WE05 0.493943 1190        34
## 15467   UGA_173101_P016_WE08 UGA_173101_P019_WA04 0.479167  897        34
## 15508   UGA_173101_P016_WE08 UGA_173101_P016_WC12 0.495565 1161        34
## 15510   UGA_173101_P016_WE08 UGA_173101_P017_WC02 0.494198 1237        34
```

```
## 15517   UGA_173101_P016_WE08 UGA_173101_P018_WB04 0.495763 1232        34
## 15522   UGA_173101_P016_WE08 UGA_173101_P016_WF07 0.486905 1080        34
## 15571   UGA_173101_P016_WE08 UGA_173101_P019_WG03 0.497938 1264        34
## 21362   UGA_173101_P017_WC12 UGA_173101_P019_WE05 0.491685 1189        34
## 21388   UGA_173101_P017_WC12 UGA_173101_P019_WA04 0.478979  896        34
## 21429   UGA_173101_P017_WC12 UGA_173101_P016_WC12 0.493392 1165        34
## 21431   UGA_173101_P017_WC12 UGA_173101_P017_WC02 0.492693 1244        34
## 21438   UGA_173101_P017_WC12 UGA_173101_P018_WB04 0.493658 1236        34
## 21443   UGA_173101_P017_WC12 UGA_173101_P016_WF07 0.485714 1081        34
## 21492   UGA_173101_P017_WC12 UGA_173101_P019_WG03 0.496933 1265        34
## 21498   UGA_173101_P017_WC12 UGA_173101_P016_WE08 0.494792 1250        34
## 25919   UGA_173101_P019_WA02 UGA_173101_P019_WE05 0.492257 1188        34
## 25945   UGA_173101_P019_WA02 UGA_173101_P019_WA04 0.479042  896        34
## 25986   UGA_173101_P019_WA02 UGA_173101_P016_WC12 0.494469 1165        34
## 25988   UGA_173101_P019_WA02 UGA_173101_P017_WC02 0.494223 1244        34
## 25995   UGA_173101_P019_WA02 UGA_173101_P018_WB04 0.494703 1235        34
## 26000   UGA_173101_P019_WA02 UGA_173101_P016_WF07 0.486310 1080        34
## 26049   UGA_173101_P019_WA02 UGA_173101_P019_WG03 0.497433 1268        34
## 26055   UGA_173101_P019_WA02 UGA_173101_P016_WE08 0.495842 1253        34
## 26086   UGA_173101_P019_WA02 UGA_173101_P017_WC12 0.496369 1253        34
## 32937   UGA_173101_P018_WD10 UGA_173101_P019_WE05 0.488095 1071        34
## 32963   UGA_173101_P018_WD10 UGA_173101_P019_WA04 0.475000  842        34
## 33004   UGA_173101_P018_WD10 UGA_173101_P016_WC12 0.487440 1057        34
## 33006   UGA_173101_P018_WD10 UGA_173101_P017_WC02 0.490385 1103        34
## 33013   UGA_173101_P018_WD10 UGA_173101_P018_WB04 0.490847 1103        34
## 33018   UGA_173101_P018_WD10 UGA_173101_P016_WF07 0.479167  998        34
## 33067   UGA_173101_P018_WD10 UGA_173101_P019_WG03 0.492117 1112        34
## 33073   UGA_173101_P018_WD10 UGA_173101_P016_WE08 0.490297 1103        34
## 33104   UGA_173101_P018_WD10 UGA_173101_P017_WC12 0.489796 1108        34
## 33125   UGA_173101_P018_WD10 UGA_173101_P019_WA02 0.492045 1107        34
## 50762   UGA_173101_P016_WH09 UGA_173101_P019_WE05 0.491592 1159        34
## 50788   UGA_173101_P016_WH09 UGA_173101_P019_WA04 0.476586  882        34
## 50829   UGA_173101_P016_WH09 UGA_173101_P016_WC12 0.491573 1134        34
## 50831   UGA_173101_P016_WH09 UGA_173101_P017_WC02 0.489807 1201        34
## 50838   UGA_173101_P016_WH09 UGA_173101_P018_WB04 0.491830 1190        34
## 50843   UGA_173101_P016_WH09 UGA_173101_P016_WF07 0.482446 1055        34
## 50892   UGA_173101_P016_WH09 UGA_173101_P019_WG03 0.493644 1212        34
## 50898   UGA_173101_P016_WH09 UGA_173101_P016_WE08 0.491935 1201        34
## 50929   UGA_173101_P016_WH09 UGA_173101_P017_WC12 0.490405 1204        34
## 50950   UGA_173101_P016_WH09 UGA_173101_P019_WA02 0.491398 1205        34
## 50979   UGA_173101_P016_WH09 UGA_173101_P018_WD10 0.490162 1086        34
## 62876   UGA_173101_P017_WD09 UGA_173101_P019_WE05 0.487613 1150        34
## 62902   UGA_173101_P017_WD09 UGA_173101_P019_WA04 0.475758  882        34
## 62943   UGA_173101_P017_WD09 UGA_173101_P016_WC12 0.490363 1133        34
## 62945   UGA_173101_P017_WD09 UGA_173101_P017_WC02 0.487445 1187        34
## 62952   UGA_173101_P017_WD09 UGA_173101_P018_WB04 0.489606 1180        34
## 62957   UGA_173101_P017_WD09 UGA_173101_P016_WF07 0.481235 1055        34
## 63006   UGA_173101_P017_WD09 UGA_173101_P019_WG03 0.490841 1203        34
## 63012   UGA_173101_P017_WD09 UGA_173101_P016_WE08 0.491342 1193        34
## 63043   UGA_173101_P017_WD09 UGA_173101_P017_WC12 0.488069 1195        34
## 63064   UGA_173101_P017_WD09 UGA_173101_P019_WA02 0.489674 1194        34
## 63093   UGA_173101_P017_WD09 UGA_173101_P018_WD10 0.483645 1080        34
## 63155   UGA_173101_P017_WD09 UGA_173101_P016_WH09 0.485000 1158        34
## 70166   UGA_173101_P019_WA01 UGA_173101_P019_WE05 0.493956 1197        34
```

```
## 70192   UGA_173101_P019_WA01 UGA_173101_P019_WA04 0.478358  899        34
## 70233   UGA_173101_P019_WA01 UGA_173101_P016_WC12 0.494457 1169        34
## 70235   UGA_173101_P019_WA01 UGA_173101_P017_WC02 0.494235 1253        34
## 70242   UGA_173101_P019_WA01 UGA_173101_P018_WB04 0.494186 1246        34
## 70247   UGA_173101_P019_WA01 UGA_173101_P016_WF07 0.483254 1083        34
## 70296   UGA_173101_P019_WA01 UGA_173101_P019_WG03 0.495885 1282        34
## 70302   UGA_173101_P019_WA01 UGA_173101_P016_WE08 0.494792 1262        34
## 70333   UGA_173101_P019_WA01 UGA_173101_P017_WC12 0.493229 1263        34
## 70354   UGA_173101_P019_WA01 UGA_173101_P019_WA02 0.494802 1266        34
## 70383   UGA_173101_P019_WA01 UGA_173101_P018_WD10 0.490888 1111        34
## 70445   UGA_173101_P019_WA01 UGA_173101_P016_WH09 0.491416 1210        34
## 70481   UGA_173101_P019_WA01 UGA_173101_P017_WD09 0.487554 1202        34
## 80642   UGA_173101_P018_WB06 UGA_173101_P019_WE05 0.458333  508       193
## 80668   UGA_173101_P018_WB06 UGA_173101_P019_WA04 0.444099  432       193
## 80709   UGA_173101_P018_WB06 UGA_173101_P016_WC12 0.456349  506       193
## 80711   UGA_173101_P018_WB06 UGA_173101_P017_WC02 0.455959  517       193
## 80718   UGA_173101_P018_WB06 UGA_173101_P018_WB04 0.458333  512       193
## 80723   UGA_173101_P018_WB06 UGA_173101_P016_WF07 0.445652  493       193
## 80772   UGA_173101_P018_WB06 UGA_173101_P019_WG03 0.454663  518       193
## 80778   UGA_173101_P018_WB06 UGA_173101_P016_WE08 0.455959  516       193
## 80809   UGA_173101_P018_WB06 UGA_173101_P017_WC12 0.454663  517       193
## 80830   UGA_173101_P018_WB06 UGA_173101_P019_WA02 0.454663  517       193
## 80859   UGA_173101_P018_WB06 UGA_173101_P018_WD10 0.445767  494       193
## 80921   UGA_173101_P018_WB06 UGA_173101_P016_WH09 0.449482  512       193
## 80957   UGA_173101_P018_WB06 UGA_173101_P017_WD09 0.450521  513       193
## 80977   UGA_173101_P018_WB06 UGA_173101_P019_WA01 0.455959  518       193
## 90566   UGA_173101_P018_WC11 UGA_173101_P019_WE05 0.495055 1197        34
## 90592   UGA_173101_P018_WC11 UGA_173101_P019_WA04 0.480655  900        34
## 90633   UGA_173101_P018_WC11 UGA_173101_P016_WC12 0.496703 1172        34
## 90635   UGA_173101_P018_WC11 UGA_173101_P017_WC02 0.495859 1258        34
## 90642   UGA_173101_P018_WC11 UGA_173101_P018_WB04 0.496855 1249        34
## 90647   UGA_173101_P018_WC11 UGA_173101_P016_WF07 0.488124 1084        34
## 90696   UGA_173101_P018_WC11 UGA_173101_P019_WG03 0.498988 1290        34
## 90702   UGA_173101_P018_WC11 UGA_173101_P016_WE08 0.497938 1266        34
## 90733   UGA_173101_P018_WC11 UGA_173101_P017_WC12 0.496429 1268        34
## 90754   UGA_173101_P018_WC11 UGA_173101_P019_WA02 0.496914 1270        34
## 90783   UGA_173101_P018_WC11 UGA_173101_P018_WD10 0.491535 1113        34
## 90845   UGA_173101_P018_WC11 UGA_173101_P016_WH09 0.492585 1213        34
## 90881   UGA_173101_P018_WC11 UGA_173101_P017_WD09 0.490841 1204        34
## 90901   UGA_173101_P018_WC11 UGA_173101_P019_WA01 0.495893 1287        34
## 90928   UGA_173101_P018_WC11 UGA_173101_P018_WB06 0.455959  518        34
## 96182   UGA_173101_P023_WF05 UGA_173101_P019_WE05 0.475658 1023        34
## 96208   UGA_173101_P023_WF05 UGA_173101_P019_WA04 0.459866  802        34
## 96249   UGA_173101_P023_WF05 UGA_173101_P016_WC12 0.478117 1014        34
## 96251   UGA_173101_P023_WF05 UGA_173101_P017_WC02 0.475578 1056        34
## 96258   UGA_173101_P023_WF05 UGA_173101_P018_WB04 0.474871 1049        34
## 96263   UGA_173101_P023_WF05 UGA_173101_P016_WF07 0.464489  952        34
## 96312   UGA_173101_P023_WF05 UGA_173101_P019_WG03 0.479167 1064        34
## 96318   UGA_173101_P023_WF05 UGA_173101_P016_WE08 0.477215 1059        34
## 96349   UGA_173101_P023_WF05 UGA_173101_P017_WC12 0.477157 1059        34
## 96370   UGA_173101_P023_WF05 UGA_173101_P019_WA02 0.478426 1061        34
## 96399   UGA_173101_P023_WF05 UGA_173101_P018_WD10 0.472603  966        34
## 96461   UGA_173101_P023_WF05 UGA_173101_P016_WH09 0.473958 1035        34
## 96497   UGA_173101_P023_WF05 UGA_173101_P017_WD09 0.473890 1029        34
```

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## 96517   UGA_173101_P023_WF05 UGA_173101_P019_WA01 0.478535 1063        34
## 96544   UGA_173101_P023_WF05 UGA_173101_P018_WB06 0.434524  485        34
## 96568   UGA_173101_P023_WF05 UGA_173101_P018_WC11 0.477904 1065        34
## 101967  UGA_173101_P023_WB12 UGA_173101_P019_WE05 0.493258 1169        34
## 101993  UGA_173101_P023_WB12 UGA_173101_P019_WA04 0.481003  881        34
## 102034  UGA_173101_P023_WB12 UGA_173101_P016_WC12 0.494407 1147        34
## 102036  UGA_173101_P023_WB12 UGA_173101_P017_WC02 0.493631 1222        34
## 102043  UGA_173101_P023_WB12 UGA_173101_P018_WB04 0.495699 1215        34
## 102048  UGA_173101_P023_WB12 UGA_173101_P016_WF07 0.488554 1063        34
## 102097  UGA_173101_P023_WB12 UGA_173101_P019_WG03 0.497912 1245        34
## 102103  UGA_173101_P023_WB12 UGA_173101_P016_WE08 0.496809 1230        34
## 102134  UGA_173101_P023_WB12 UGA_173101_P017_WC12 0.495273 1230        34
## 102155  UGA_173101_P023_WB12 UGA_173101_P019_WA02 0.496285 1228        34
## 102184  UGA_173101_P023_WB12 UGA_173101_P018_WD10 0.490230 1091        34
## 102246  UGA_173101_P023_WB12 UGA_173101_P016_WH09 0.491361 1187        34
## 102282  UGA_173101_P023_WB12 UGA_173101_P017_WD09 0.489537 1176        34
## 102302  UGA_173101_P023_WB12 UGA_173101_P019_WA01 0.494186 1242        34
## 102329  UGA_173101_P023_WB12 UGA_173101_P018_WB06 0.456349  506        34
## 102353  UGA_173101_P023_WB12 UGA_173101_P018_WC11 0.496868 1247        34
## 102366  UGA_173101_P023_WB12 UGA_173101_P023_WF05 0.478342 1078        34
## 106071  UGA_173101_P023_WA04 UGA_173101_P019_WE05 0.484375 1096        34
## 106097  UGA_173101_P023_WA04 UGA_173101_P019_WA04 0.472741  855        34
## 106138  UGA_173101_P023_WA04 UGA_173101_P016_WC12 0.487195 1073        34
## 106140  UGA_173101_P023_WA04 UGA_173101_P017_WC02 0.485363 1130        34
## 106147  UGA_173101_P023_WA04 UGA_173101_P018_WB04 0.487028 1120        34
## 106152  UGA_173101_P023_WA04 UGA_173101_P016_WF07 0.482955 1008        34
## 106201  UGA_173101_P023_WA04 UGA_173101_P019_WG03 0.489510 1138        34
## 106207  UGA_173101_P023_WA04 UGA_173101_P016_WE08 0.486565 1133        34
## 106238  UGA_173101_P023_WA04 UGA_173101_P017_WC12 0.487705 1135        34
## 106259  UGA_173101_P023_WA04 UGA_173101_P019_WA02 0.488263 1131        34
## 106288  UGA_173101_P023_WA04 UGA_173101_P018_WD10 0.482587 1027        34
## 106350  UGA_173101_P023_WA04 UGA_173101_P016_WH09 0.484524 1103        34
## 106386  UGA_173101_P023_WA04 UGA_173101_P017_WD09 0.482270 1099        34
## 106406  UGA_173101_P023_WA04 UGA_173101_P019_WA01 0.484813 1137        34
## 106433  UGA_173101_P023_WA04 UGA_173101_P018_WB06 0.455163  491        34
## 106457  UGA_173101_P023_WA04 UGA_173101_P018_WC11 0.489510 1138        34
## 106470  UGA_173101_P023_WA04 UGA_173101_P023_WF05 0.471204 1025        34
## 106483  UGA_173101_P023_WA04 UGA_173101_P023_WB12 0.488584 1154        34
## 132911  UGA_173101_P023_WA07 UGA_173101_P019_WE05 0.488532 1138        34
## 132937  UGA_173101_P023_WA07 UGA_173101_P019_WA04 0.477064  873        34
## 132978  UGA_173101_P023_WA07 UGA_173101_P016_WC12 0.490252 1117        34
## 132980  UGA_173101_P023_WA07 UGA_173101_P017_WC02 0.490721 1192        34
## 132987  UGA_173101_P023_WA07 UGA_173101_P018_WB04 0.490022 1181        34
## 132992  UGA_173101_P023_WA07 UGA_173101_P016_WF07 0.483415 1044        34
## 133041  UGA_173101_P023_WA07 UGA_173101_P019_WG03 0.492950 1203        34
## 133047  UGA_173101_P023_WA07 UGA_173101_P016_WE08 0.491189 1190        34
## 133078  UGA_173101_P023_WA07 UGA_173101_P017_WC12 0.490217 1199        34
## 133099  UGA_173101_P023_WA07 UGA_173101_P019_WA02 0.491758 1193        34
## 133128  UGA_173101_P023_WA07 UGA_173101_P018_WD10 0.487089 1067        34
## 133190  UGA_173101_P023_WA07 UGA_173101_P016_WH09 0.486018 1159        34
## 133226  UGA_173101_P023_WA07 UGA_173101_P017_WD09 0.484270 1147        34
## 133246  UGA_173101_P023_WA07 UGA_173101_P019_WA01 0.489674 1202        34
## 133273  UGA_173101_P023_WA07 UGA_173101_P018_WB06 0.452703  498        34
## 133297  UGA_173101_P023_WA07 UGA_173101_P018_WC11 0.492981 1206        34
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## 133310  UGA_173101_P023_WA07 UGA_173101_P023_WF05 0.472222 1058        34
## 133323  UGA_173101_P023_WA07 UGA_173101_P023_WB12 0.492505 1214        34
## 133332  UGA_173101_P023_WA07 UGA_173101_P023_WA04 0.483683 1133        34
## 145032  UGA_173101_P022_WD02 UGA_173101_P019_WE05 0.465084  496        34
## 145058  UGA_173101_P022_WD02 UGA_173101_P019_WA04 0.449346  408        34
## 145099  UGA_173101_P022_WD02 UGA_173101_P016_WC12 0.466667  495        34
## 145101  UGA_173101_P022_WD02 UGA_173101_P017_WC02 0.466667  499        34
## 145108  UGA_173101_P022_WD02 UGA_173101_P018_WB04 0.465278  499        34
## 145113  UGA_173101_P022_WD02 UGA_173101_P016_WF07 0.460756  472        34
## 145162  UGA_173101_P022_WD02 UGA_173101_P019_WG03 0.468056  500        34
## 145168  UGA_173101_P022_WD02 UGA_173101_P016_WE08 0.468056  500        34
## 145199  UGA_173101_P022_WD02 UGA_173101_P017_WC12 0.466667  500        34
## 145220  UGA_173101_P022_WD02 UGA_173101_P019_WA02 0.468056  500        34
## 145249  UGA_173101_P022_WD02 UGA_173101_P018_WD10 0.461207  472        34
## 145311  UGA_173101_P022_WD02 UGA_173101_P016_WH09 0.463687  497        34
## 145347  UGA_173101_P022_WD02 UGA_173101_P017_WD09 0.462500  495        34
## 145367  UGA_173101_P022_WD02 UGA_173101_P019_WA01 0.468056  500        34
## 145394  UGA_173101_P022_WD02 UGA_173101_P018_WB06 0.443299  265        34
## 145418  UGA_173101_P022_WD02 UGA_173101_P018_WC11 0.468056  500        34
## 145431  UGA_173101_P022_WD02 UGA_173101_P023_WF05 0.452514  487        34
## 145444  UGA_173101_P022_WD02 UGA_173101_P023_WB12 0.470430  507        34
## 145453  UGA_173101_P022_WD02 UGA_173101_P023_WA04 0.460383  497        34
## 145508  UGA_173101_P022_WD02 UGA_173101_P023_WA07 0.467742  508        34
## 230222  UGA_173101_P021_WF01 UGA_173101_P019_WE05 0.481132 1139        34
## 230248  UGA_173101_P021_WF01 UGA_173101_P019_WA04 0.465278  869        34
## 230289  UGA_173101_P021_WF01 UGA_173101_P016_WC12 0.483529 1125        34
## 230291  UGA_173101_P021_WF01 UGA_173101_P017_WC02 0.482022 1187        34
## 230298  UGA_173101_P021_WF01 UGA_173101_P018_WB04 0.483485 1180        34
## 230303  UGA_173101_P021_WF01 UGA_173101_P016_WF07 0.475379 1040        34
## 230352  UGA_173101_P021_WF01 UGA_173101_P019_WG03 0.486696 1206        34
## 230358  UGA_173101_P021_WF01 UGA_173101_P016_WE08 0.485426 1192        34
## 230389  UGA_173101_P021_WF01 UGA_173101_P017_WC12 0.484410 1198        34
## 230410  UGA_173101_P021_WF01 UGA_173101_P019_WA02 0.484899 1194        34
## 230439  UGA_173101_P021_WF01 UGA_173101_P018_WD10 0.478554 1066        34
## 230501  UGA_173101_P021_WF01 UGA_173101_P016_WH09 0.478161 1159        34
## 230537  UGA_173101_P021_WF01 UGA_173101_P017_WD09 0.478784 1149        34
## 230557  UGA_173101_P021_WF01 UGA_173101_P019_WA01 0.483370 1204        34
## 230584  UGA_173101_P021_WF01 UGA_173101_P018_WB06 0.449176  501        34
## 230608  UGA_173101_P021_WF01 UGA_173101_P018_WC11 0.486726 1207        34
## 230621  UGA_173101_P021_WF01 UGA_173101_P023_WF05 0.466625 1061        34
## 230634  UGA_173101_P021_WF01 UGA_173101_P023_WB12 0.485808 1213        34
## 230643  UGA_173101_P021_WF01 UGA_173101_P023_WA04 0.479117 1123        34
## 230698  UGA_173101_P021_WF01 UGA_173101_P023_WA07 0.479866 1183        34
## 230721  UGA_173101_P021_WF01 UGA_173101_P022_WD02 0.448370  504        34
## 260322  UGA_173101_P022_WA09 UGA_173101_P019_WE05 0.489655 1132        34
## 260348  UGA_173101_P022_WA09 UGA_173101_P019_WA04 0.474615  866        34
## 260389  UGA_173101_P022_WA09 UGA_173101_P016_WC12 0.490762 1111        34
## 260391  UGA_173101_P022_WA09 UGA_173101_P017_WC02 0.491685 1174        34
## 260398  UGA_173101_P022_WA09 UGA_173101_P018_WB04 0.492205 1169        34
## 260403  UGA_173101_P022_WA09 UGA_173101_P016_WF07 0.484606 1034        34
## 260452  UGA_173101_P022_WA09 UGA_173101_P019_WG03 0.493982 1189        34
## 260458  UGA_173101_P022_WA09 UGA_173101_P016_WE08 0.493929 1177        34
## 260489  UGA_173101_P022_WA09 UGA_173101_P017_WC12 0.493407 1181        34
## 260510  UGA_173101_P022_WA09 UGA_173101_P019_WA02 0.493392 1181        34
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## 260539  UGA_173101_P022_WA09 UGA_173101_P018_WD10 0.487530 1057        34
## 260601  UGA_173101_P022_WA09 UGA_173101_P016_WH09 0.487668 1152        34
## 260637  UGA_173101_P022_WA09 UGA_173101_P017_WD09 0.485923 1139        34
## 260657  UGA_173101_P022_WA09 UGA_173101_P019_WA01 0.492888 1189        34
## 260684  UGA_173101_P022_WA09 UGA_173101_P018_WB06 0.457219  499        34
## 260708  UGA_173101_P022_WA09 UGA_173101_P018_WC11 0.493982 1190        34
## 260721  UGA_173101_P022_WA09 UGA_173101_P023_WF05 0.477330 1061        34
## 260734  UGA_173101_P022_WA09 UGA_173101_P023_WB12 0.493548 1206        34
## 260743  UGA_173101_P022_WA09 UGA_173101_P023_WA04 0.483834 1127        34
## 260798  UGA_173101_P022_WA09 UGA_173101_P023_WA07 0.487390 1179        34
## 260821  UGA_173101_P022_WA09 UGA_173101_P022_WD02 0.469086  508        34
## 260961  UGA_173101_P022_WA09 UGA_173101_P021_WF01 0.481461 1170        34
## 286944            GTS_949-23 UGA_173101_P019_WE05 0.490741  115       307
## 286970            GTS_949-23 UGA_173101_P019_WA04 0.489130   92       307
## 287011            GTS_949-23 UGA_173101_P016_WC12 0.491379  110       307
## 287013            GTS_949-23 UGA_173101_P017_WC02 0.483333  117       307
## 287020            GTS_949-23 UGA_173101_P018_WB04 0.491071  118       307
## 287025            GTS_949-23 UGA_173101_P016_WF07 0.462963  103       307
## 287074            GTS_949-23 UGA_173101_P019_WG03 0.491379  122       307
## 287080            GTS_949-23 UGA_173101_P016_WE08 0.491379  117       307
## 287111            GTS_949-23 UGA_173101_P017_WC12 0.482143  117       307
## 287132            GTS_949-23 UGA_173101_P019_WA02 0.491379  119       307
## 287161            GTS_949-23 UGA_173101_P018_WD10 0.491071  111       307
## 287223            GTS_949-23 UGA_173101_P016_WH09 0.491071  115       307
## 287259            GTS_949-23 UGA_173101_P017_WD09 0.490741  109       307
## 287279            GTS_949-23 UGA_173101_P019_WA01 0.491379  121       307
## 287306            GTS_949-23 UGA_173101_P018_WB06 0.468750   56       307
## 287330            GTS_949-23 UGA_173101_P018_WC11 0.491379  122       307
## 287343            GTS_949-23 UGA_173101_P023_WF05 0.480769   94       307
## 287356            GTS_949-23 UGA_173101_P023_WB12 0.473214  111       307
## 287365            GTS_949-23 UGA_173101_P023_WA04 0.489130   93       307
## 287420            GTS_949-23 UGA_173101_P023_WA07 0.481481  108       307
## 287443            GTS_949-23 UGA_173101_P022_WD02 0.450000   52       307
## 287583            GTS_949-23 UGA_173101_P021_WF01 0.481481  104       307
## 287626            GTS_949-23 UGA_173101_P022_WA09 0.490000  102       307
## 325262            GTS_646-23 UGA_173101_P019_WE05 0.500000  112       307
## 325288            GTS_646-23 UGA_173101_P019_WA04 0.500000   90       307
## 325329            GTS_646-23 UGA_173101_P016_WC12 0.500000  107       307
## 325331            GTS_646-23 UGA_173101_P017_WC02 0.490741  114       307
## 325338            GTS_646-23 UGA_173101_P018_WB04 0.500000  115       307
## 325343            GTS_646-23 UGA_173101_P016_WF07 0.468750  100       307
## 325392            GTS_646-23 UGA_173101_P019_WG03 0.500000  119       307
## 325398            GTS_646-23 UGA_173101_P016_WE08 0.500000  114       307
## 325429            GTS_646-23 UGA_173101_P017_WC12 0.490385  114       307
## 325450            GTS_646-23 UGA_173101_P019_WA02 0.500000  116       307
## 325479            GTS_646-23 UGA_173101_P018_WD10 0.500000  108       307
## 325541            GTS_646-23 UGA_173101_P016_WH09 0.500000  112       307
## 325577            GTS_646-23 UGA_173101_P017_WD09 0.500000  106       307
## 325597            GTS_646-23 UGA_173101_P019_WA01 0.500000  118       307
## 325624            GTS_646-23 UGA_173101_P018_WB06 0.484375   55       307
## 325648            GTS_646-23 UGA_173101_P018_WC11 0.500000  119       307
## 325661            GTS_646-23 UGA_173101_P023_WF05 0.489583   92       307
## 325674            GTS_646-23 UGA_173101_P023_WB12 0.480000  108       307
## 325683            GTS_646-23 UGA_173101_P023_WA04 0.500000   90       307
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## 325738            GTS_646-23 UGA_173101_P023_WA07 0.490000  105        307
## 325761            GTS_646-23 UGA_173101_P022_WD02 0.475000   51        307
## 325901            GTS_646-23 UGA_173101_P021_WF01 0.489583  101        307
## 325944            GTS_646-23 UGA_173101_P022_WA09 0.500000   99        307
## 325980            GTS_646-23           GTS_949-23 0.500000  120        307
## 336651            GTS_616-24 UGA_173101_P019_WE05 0.490385  114        326
## 336677            GTS_616-24 UGA_173101_P019_WA04 0.488636   91        326
## 336718            GTS_616-24 UGA_173101_P016_WC12 0.491071  109        326
## 336720            GTS_616-24 UGA_173101_P017_WC02 0.482759  116        326
## 336727            GTS_616-24 UGA_173101_P018_WB04 0.490741  117        326
## 336732            GTS_616-24 UGA_173101_P016_WF07 0.461538  102        326
## 336781            GTS_616-24 UGA_173101_P019_WG03 0.491071  120        326
## 336787            GTS_616-24 UGA_173101_P016_WE08 0.491071  116        326
## 336818            GTS_616-24 UGA_173101_P017_WC12 0.481481  116        326
## 336839            GTS_616-24 UGA_173101_P019_WA02 0.491071  118        326
## 336868            GTS_616-24 UGA_173101_P018_WD10 0.490741  109        326
## 336930            GTS_616-24 UGA_173101_P016_WH09 0.490741  114        326
## 336966            GTS_616-24 UGA_173101_P017_WD09 0.490385  108        326
## 336986            GTS_616-24 UGA_173101_P019_WA01 0.491071  119        326
## 337013            GTS_616-24 UGA_173101_P018_WB06 0.468750   56        326
## 337037            GTS_616-24 UGA_173101_P018_WC11 0.491071  120        326
## 337050            GTS_616-24 UGA_173101_P023_WF05 0.480000   93        326
## 337063            GTS_616-24 UGA_173101_P023_WB12 0.472222  109        326
## 337072            GTS_616-24 UGA_173101_P023_WA04 0.488636   92        326
## 337127            GTS_616-24 UGA_173101_P023_WA07 0.480769  107        326
## 337150            GTS_616-24 UGA_173101_P022_WD02 0.450000   52        326
## 337290            GTS_616-24 UGA_173101_P021_WF01 0.480769  102        326
## 337333            GTS_616-24 UGA_173101_P022_WA09 0.489583  100        326
## 337369            GTS_616-24           GTS_949-23 0.491379  122        326
## 337418            GTS_616-24           GTS_646-23 0.500000  119        326
## 344076            GTS_655-23 UGA_173101_P019_WE05 0.489130  109        307
## 344102            GTS_655-23 UGA_173101_P019_WA04 0.488095   89        307
## 344143            GTS_655-23 UGA_173101_P016_WC12 0.490000  104        307
## 344145            GTS_655-23 UGA_173101_P017_WC02 0.480769  111        307
## 344152            GTS_655-23 UGA_173101_P018_WB04 0.489583  112        307
## 344157            GTS_655-23 UGA_173101_P016_WF07 0.458333   98        307
## 344206            GTS_655-23 UGA_173101_P019_WG03 0.490000  115        307
## 344212            GTS_655-23 UGA_173101_P016_WE08 0.490000  111        307
## 344243            GTS_655-23 UGA_173101_P017_WC12 0.490000  111        307
## 344264            GTS_655-23 UGA_173101_P019_WA02 0.490000  113        307
## 344293            GTS_655-23 UGA_173101_P018_WD10 0.489583  105        307
## 344355            GTS_655-23 UGA_173101_P016_WH09 0.489583  109        307
## 344391            GTS_655-23 UGA_173101_P017_WD09 0.489130  104        307
## 344411            GTS_655-23 UGA_173101_P019_WA01 0.490000  114        307
## 344438            GTS_655-23 UGA_173101_P018_WB06 0.466667   54        307
## 344462            GTS_655-23 UGA_173101_P018_WC11 0.490000  115        307
## 344475            GTS_655-23 UGA_173101_P023_WF05 0.479167   92        307
## 344488            GTS_655-23 UGA_173101_P023_WB12 0.470000  105        307
## 344497            GTS_655-23 UGA_173101_P023_WA04 0.487500   90        307
## 344552            GTS_655-23 UGA_173101_P023_WA07 0.478261  102        307
## 344575            GTS_655-23 UGA_173101_P022_WD02 0.444444   50        307
## 344715            GTS_655-23 UGA_173101_P021_WF01 0.479167   98        307
## 344758            GTS_655-23 UGA_173101_P022_WA09 0.488636   96        307
## 344794            GTS_655-23           GTS_949-23 0.500000  116        307
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## 344843            GTS_655-23           GTS_646-23 0.500000  115        307
## 344857            GTS_655-23           GTS_616-24 0.500000  116        307
## 344906            GTS_983-23 UGA_173101_P019_WE05 0.500000  113         34
## 344932            GTS_983-23 UGA_173101_P019_WA04 0.500000   90         34
## 344973            GTS_983-23 UGA_173101_P016_WC12 0.500000  108         34
## 344975            GTS_983-23 UGA_173101_P017_WC02 0.491071  115         34
## 344982            GTS_983-23 UGA_173101_P018_WB04 0.500000  116         34
## 344987            GTS_983-23 UGA_173101_P016_WF07 0.470000  101         34
## 345036            GTS_983-23 UGA_173101_P019_WG03 0.500000  120         34
## 345042            GTS_983-23 UGA_173101_P016_WE08 0.500000  115         34
## 345073            GTS_983-23 UGA_173101_P017_WC12 0.490741  115         34
## 345094            GTS_983-23 UGA_173101_P019_WA02 0.500000  117         34
## 345123            GTS_983-23 UGA_173101_P018_WD10 0.500000  109         34
## 345185            GTS_983-23 UGA_173101_P016_WH09 0.500000  113         34
## 345221            GTS_983-23 UGA_173101_P017_WD09 0.500000  107         34
## 345241            GTS_983-23 UGA_173101_P019_WA01 0.500000  119         34
## 345268            GTS_983-23 UGA_173101_P018_WB06 0.483333   54         34
## 345292            GTS_983-23 UGA_173101_P018_WC11 0.500000  120         34
## 345305            GTS_983-23 UGA_173101_P023_WF05 0.489583   92         34
## 345318            GTS_983-23 UGA_173101_P023_WB12 0.480769  109         34
## 345327            GTS_983-23 UGA_173101_P023_WA04 0.500000   91         34
## 345382            GTS_983-23 UGA_173101_P023_WA07 0.490385  106         34
## 345405            GTS_983-23 UGA_173101_P022_WD02 0.472222   50         34
## 345545            GTS_983-23 UGA_173101_P021_WF01 0.490000  102         34
## 345588            GTS_983-23 UGA_173101_P022_WA09 0.500000  100         34
## 345624            GTS_983-23           GTS_949-23 0.500000  121         34
## 345673            GTS_983-23           GTS_646-23 0.500000  119         34
## 345687            GTS_983-23           GTS_616-24 0.500000  119         34
## 345696            GTS_983-23           GTS_655-23 0.500000  114         34
## 354102            GTS_630-24 UGA_173101_P019_WE05 0.500000  112        307
## 354128            GTS_630-24 UGA_173101_P019_WA04 0.500000   89        307
## 354169            GTS_630-24 UGA_173101_P016_WC12 0.500000  107        307
## 354171            GTS_630-24 UGA_173101_P017_WC02 0.490741  114        307
## 354178            GTS_630-24 UGA_173101_P018_WB04 0.500000  115        307
## 354183            GTS_630-24 UGA_173101_P016_WF07 0.468750  100        307
## 354232            GTS_630-24 UGA_173101_P019_WG03 0.500000  118        307
## 354238            GTS_630-24 UGA_173101_P016_WE08 0.500000  114        307
## 354269            GTS_630-24 UGA_173101_P017_WC12 0.490385  114        307
## 354290            GTS_630-24 UGA_173101_P019_WA02 0.500000  116        307
## 354319            GTS_630-24 UGA_173101_P018_WD10 0.500000  107        307
## 354381            GTS_630-24 UGA_173101_P016_WH09 0.500000  112        307
## 354417            GTS_630-24 UGA_173101_P017_WD09 0.500000  106        307
## 354437            GTS_630-24 UGA_173101_P019_WA01 0.500000  117        307
## 354464            GTS_630-24 UGA_173101_P018_WB06 0.483333   54        307
## 354488            GTS_630-24 UGA_173101_P018_WC11 0.500000  118        307
## 354501            GTS_630-24 UGA_173101_P023_WF05 0.489130   91        307
## 354514            GTS_630-24 UGA_173101_P023_WB12 0.480000  107        307
## 354523            GTS_630-24 UGA_173101_P023_WA04 0.500000   90        307
## 354578            GTS_630-24 UGA_173101_P023_WA07 0.500000  105        307
## 354601            GTS_630-24 UGA_173101_P022_WD02 0.475000   51        307
## 354741            GTS_630-24 UGA_173101_P021_WF01 0.490000  101        307
## 354784            GTS_630-24 UGA_173101_P022_WA09 0.500000   98        307
## 354820            GTS_630-24           GTS_949-23 0.500000  119        307
## 354869            GTS_630-24           GTS_646-23 0.500000  118        307
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## 354883            GTS_630-24           GTS_616-24 0.500000  119        307
## 354892            GTS_630-24           GTS_655-23 0.500000  114        307
## 354893            GTS_630-24           GTS_983-23 0.500000  118        307
## 362567            GTS_665-23 UGA_173101_P019_WE05 0.490741  115        307
## 362593            GTS_665-23 UGA_173101_P019_WA04 0.489130   92        307
## 362634            GTS_665-23 UGA_173101_P016_WC12 0.491379  110        307
## 362636            GTS_665-23 UGA_173101_P017_WC02 0.483333  117        307
## 362643            GTS_665-23 UGA_173101_P018_WB04 0.491071  118        307
## 362648            GTS_665-23 UGA_173101_P016_WF07 0.462963  103        307
## 362697            GTS_665-23 UGA_173101_P019_WG03 0.491379  121        307
## 362703            GTS_665-23 UGA_173101_P016_WE08 0.491379  117        307
## 362734            GTS_665-23 UGA_173101_P017_WC12 0.482143  117        307
## 362755            GTS_665-23 UGA_173101_P019_WA02 0.491379  119        307
## 362784            GTS_665-23 UGA_173101_P018_WD10 0.491071  110        307
## 362846            GTS_665-23 UGA_173101_P016_WH09 0.491071  115        307
## 362882            GTS_665-23 UGA_173101_P017_WD09 0.490741  109        307
## 362902            GTS_665-23 UGA_173101_P019_WA01 0.491379  120        307
## 362929            GTS_665-23 UGA_173101_P018_WB06 0.468750   56        307
## 362953            GTS_665-23 UGA_173101_P018_WC11 0.491379  121        307
## 362966            GTS_665-23 UGA_173101_P023_WF05 0.480769   94        307
## 362979            GTS_665-23 UGA_173101_P023_WB12 0.473214  110        307
## 362988            GTS_665-23 UGA_173101_P023_WA04 0.489130   93        307
## 363043            GTS_665-23 UGA_173101_P023_WA07 0.481481  108        307
## 363066            GTS_665-23 UGA_173101_P022_WD02 0.450000   52        307
## 363206            GTS_665-23 UGA_173101_P021_WF01 0.481481  103        307
## 363249            GTS_665-23 UGA_173101_P022_WA09 0.490000  101        307
## 363285            GTS_665-23           GTS_949-23 0.500000  122        307
## 363334            GTS_665-23           GTS_646-23 0.500000  119        307
## 363348            GTS_665-23           GTS_616-24 0.500000  121        307
## 363357            GTS_665-23           GTS_655-23 0.500000  116        307
## 363358            GTS_665-23           GTS_983-23 0.500000  120        307
## 363369            GTS_665-23           GTS_630-24 0.500000  119        307
## 365126            GTS_997-23 UGA_173101_P019_WE05 0.490741  115        307
## 365152            GTS_997-23 UGA_173101_P019_WA04 0.489130   92        307
## 365193            GTS_997-23 UGA_173101_P016_WC12 0.491379  110        307
## 365195            GTS_997-23 UGA_173101_P017_WC02 0.483333  117        307
## 365202            GTS_997-23 UGA_173101_P018_WB04 0.491071  118        307
## 365207            GTS_997-23 UGA_173101_P016_WF07 0.462963  103        307
## 365256            GTS_997-23 UGA_173101_P019_WG03 0.491379  122        307
## 365262            GTS_997-23 UGA_173101_P016_WE08 0.491379  117        307
## 365293            GTS_997-23 UGA_173101_P017_WC12 0.482143  117        307
## 365314            GTS_997-23 UGA_173101_P019_WA02 0.491379  119        307
## 365343            GTS_997-23 UGA_173101_P018_WD10 0.491071  111        307
## 365405            GTS_997-23 UGA_173101_P016_WH09 0.491071  115        307
## 365441            GTS_997-23 UGA_173101_P017_WD09 0.490741  109        307
## 365461            GTS_997-23 UGA_173101_P019_WA01 0.491379  121        307
## 365488            GTS_997-23 UGA_173101_P018_WB06 0.468750   56        307
## 365512            GTS_997-23 UGA_173101_P018_WC11 0.491379  122        307
## 365525            GTS_997-23 UGA_173101_P023_WF05 0.480769   94        307
## 365538            GTS_997-23 UGA_173101_P023_WB12 0.473214  111        307
## 365547            GTS_997-23 UGA_173101_P023_WA04 0.489130   93        307
## 365602            GTS_997-23 UGA_173101_P023_WA07 0.481481  108        307
## 365625            GTS_997-23 UGA_173101_P022_WD02 0.450000   52        307
## 365765            GTS_997-23 UGA_173101_P021_WF01 0.481481  104        307
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## 365808            GTS_997-23 UGA_173101_P022_WA09 0.490000  102        307
## 365844            GTS_997-23           GTS_949-23 0.500000  123        307
## 365893            GTS_997-23           GTS_646-23 0.500000  120        307
## 365907            GTS_997-23           GTS_616-24 0.500000  121        307
## 365916            GTS_997-23           GTS_655-23 0.500000  116        307
## 365917            GTS_997-23           GTS_983-23 0.500000  121        307
## 365928            GTS_997-23           GTS_630-24 0.500000  119        307
## 365938            GTS_997-23           GTS_665-23 0.500000  122        307
## 370271            GTS_644-24 UGA_173101_P019_WE05 0.500000  111        326
## 370297            GTS_644-24 UGA_173101_P019_WA04 0.500000   89        326
## 370338            GTS_644-24 UGA_173101_P016_WC12 0.500000  106        326
## 370340            GTS_644-24 UGA_173101_P017_WC02 0.490741  113        326
## 370347            GTS_644-24 UGA_173101_P018_WB04 0.500000  114        326
## 370352            GTS_644-24 UGA_173101_P016_WF07 0.468750   99        326
## 370401            GTS_644-24 UGA_173101_P019_WG03 0.500000  117        326
## 370407            GTS_644-24 UGA_173101_P016_WE08 0.500000  113        326
## 370438            GTS_644-24 UGA_173101_P017_WC12 0.490385  113        326
## 370459            GTS_644-24 UGA_173101_P019_WA02 0.500000  115        326
## 370488            GTS_644-24 UGA_173101_P018_WD10 0.500000  106        326
## 370550            GTS_644-24 UGA_173101_P016_WH09 0.500000  111        326
## 370586            GTS_644-24 UGA_173101_P017_WD09 0.500000  105        326
## 370606            GTS_644-24 UGA_173101_P019_WA01 0.500000  116        326
## 370633            GTS_644-24 UGA_173101_P018_WB06 0.483333   54        326
## 370657            GTS_644-24 UGA_173101_P018_WC11 0.500000  117        326
## 370670            GTS_644-24 UGA_173101_P023_WF05 0.489130   91        326
## 370683            GTS_644-24 UGA_173101_P023_WB12 0.480000  106        326
## 370692            GTS_644-24 UGA_173101_P023_WA04 0.500000   90        326
## 370747            GTS_644-24 UGA_173101_P023_WA07 0.500000  104        326
## 370770            GTS_644-24 UGA_173101_P022_WD02 0.475000   51        326
## 370910            GTS_644-24 UGA_173101_P021_WF01 0.490000  101        326
## 370953            GTS_644-24 UGA_173101_P022_WA09 0.500000   97        326
## 370989            GTS_644-24           GTS_949-23 0.490741  119        326
## 371038            GTS_644-24           GTS_646-23 0.500000  117        326
## 371052            GTS_644-24           GTS_616-24 0.500000  119        326
## 371061            GTS_644-24           GTS_655-23 0.500000  113        326
## 371062            GTS_644-24           GTS_983-23 0.500000  117        326
## 371073            GTS_644-24           GTS_630-24 0.500000  118        326
## 371083            GTS_644-24           GTS_665-23 0.500000  118        326
## 371086            GTS_644-24           GTS_997-23 0.500000  118        326
## 375452            GTS_818-23 UGA_173101_P019_WE05 0.490741  115        307
## 375478            GTS_818-23 UGA_173101_P019_WA04 0.489130   92        307
## 375519            GTS_818-23 UGA_173101_P016_WC12 0.491379  110        307
## 375521            GTS_818-23 UGA_173101_P017_WC02 0.483333  117        307
## 375528            GTS_818-23 UGA_173101_P018_WB04 0.491071  118        307
## 375533            GTS_818-23 UGA_173101_P016_WF07 0.462963  103        307
## 375582            GTS_818-23 UGA_173101_P019_WG03 0.491379  122        307
## 375588            GTS_818-23 UGA_173101_P016_WE08 0.491379  117        307
## 375619            GTS_818-23 UGA_173101_P017_WC12 0.482143  117        307
## 375640            GTS_818-23 UGA_173101_P019_WA02 0.491379  119        307
## 375669            GTS_818-23 UGA_173101_P018_WD10 0.491071  111        307
## 375731            GTS_818-23 UGA_173101_P016_WH09 0.491071  115        307
## 375767            GTS_818-23 UGA_173101_P017_WD09 0.490741  109        307
## 375787            GTS_818-23 UGA_173101_P019_WA01 0.491379  121        307
## 375814            GTS_818-23 UGA_173101_P018_WB06 0.468750   56        307
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## 375838            GTS_818-23 UGA_173101_P018_WC11 0.491379  122        307
## 375851            GTS_818-23 UGA_173101_P023_WF05 0.480769   94        307
## 375864            GTS_818-23 UGA_173101_P023_WB12 0.473214  111        307
## 375873            GTS_818-23 UGA_173101_P023_WA04 0.489130   93        307
## 375928            GTS_818-23 UGA_173101_P023_WA07 0.481481  108        307
## 375951            GTS_818-23 UGA_173101_P022_WD02 0.450000   52        307
## 376091            GTS_818-23 UGA_173101_P021_WF01 0.481481  104        307
## 376134            GTS_818-23 UGA_173101_P022_WA09 0.490000  102        307
## 376170            GTS_818-23           GTS_949-23 0.500000  123        307
## 376219            GTS_818-23           GTS_646-23 0.500000  120        307
## 376233            GTS_818-23           GTS_616-24 0.500000  121        307
## 376242            GTS_818-23           GTS_655-23 0.500000  116        307
## 376243            GTS_818-23           GTS_983-23 0.500000  121        307
## 376254            GTS_818-23           GTS_630-24 0.500000  119        307
## 376264            GTS_818-23           GTS_665-23 0.500000  122        307
## 376267            GTS_818-23           GTS_997-23 0.500000  123        307
## 376273            GTS_818-23           GTS_644-24 0.500000  118        307
## 386801            GTS_677-23 UGA_173101_P019_WE05 0.489583  112        307
## 386827            GTS_677-23 UGA_173101_P019_WA04 0.487500   89        307
## 386868            GTS_677-23 UGA_173101_P016_WC12 0.490385  107        307
## 386870            GTS_677-23 UGA_173101_P017_WC02 0.481481  114        307
## 386877            GTS_677-23 UGA_173101_P018_WB04 0.490000  115        307
## 386882            GTS_677-23 UGA_173101_P016_WF07 0.458333  100        307
## 386931            GTS_677-23 UGA_173101_P019_WG03 0.490385  118        307
## 386937            GTS_677-23 UGA_173101_P016_WE08 0.490385  114        307
## 386968            GTS_677-23 UGA_173101_P017_WC12 0.480000  114        307
## 386989            GTS_677-23 UGA_173101_P019_WA02 0.490385  116        307
## 387018            GTS_677-23 UGA_173101_P018_WD10 0.490000  107        307
## 387080            GTS_677-23 UGA_173101_P016_WH09 0.490000  112        307
## 387116            GTS_677-23 UGA_173101_P017_WD09 0.489583  106        307
## 387136            GTS_677-23 UGA_173101_P019_WA01 0.490385  117        307
## 387163            GTS_677-23 UGA_173101_P018_WB06 0.464286   54        307
## 387187            GTS_677-23 UGA_173101_P018_WC11 0.490385  118        307
## 387200            GTS_677-23 UGA_173101_P023_WF05 0.478261   91        307
## 387213            GTS_677-23 UGA_173101_P023_WB12 0.470000  107        307
## 387222            GTS_677-23 UGA_173101_P023_WA04 0.487500   90        307
## 387277            GTS_677-23 UGA_173101_P023_WA07 0.490000  105        307
## 387300            GTS_677-23 UGA_173101_P022_WD02 0.444444   51        307
## 387440            GTS_677-23 UGA_173101_P021_WF01 0.480000  101        307
## 387483            GTS_677-23 UGA_173101_P022_WA09 0.488636   98        307
## 387519            GTS_677-23           GTS_949-23 0.500000  119        307
## 387568            GTS_677-23           GTS_646-23 0.500000  117        307
## 387582            GTS_677-23           GTS_616-24 0.500000  119        307
## 387591            GTS_677-23           GTS_655-23 0.500000  114        307
## 387592            GTS_677-23           GTS_983-23 0.500000  118        307
## 387603            GTS_677-23           GTS_630-24 0.500000  118        307
## 387613            GTS_677-23           GTS_665-23 0.500000  119        307
## 387616            GTS_677-23           GTS_997-23 0.500000  119        307
## 387622            GTS_677-23           GTS_644-24 0.500000  117        307
## 387628            GTS_677-23           GTS_818-23 0.500000  119        307
## 387681           GTS_1012-23 UGA_173101_P019_WE05 0.500000  113        307
## 387707           GTS_1012-23 UGA_173101_P019_WA04 0.500000   90        307
## 387748           GTS_1012-23 UGA_173101_P016_WC12 0.500000  108        307
## 387750           GTS_1012-23 UGA_173101_P017_WC02 0.491071  115        307
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## 387757          GTS_1012-23 UGA_173101_P018_WB04 0.500000  116       307
## 387762          GTS_1012-23 UGA_173101_P016_WF07 0.470000  101       307
## 387811          GTS_1012-23 UGA_173101_P019_WG03 0.500000  120       307
## 387817          GTS_1012-23 UGA_173101_P016_WE08 0.500000  115       307
## 387848          GTS_1012-23 UGA_173101_P017_WC12 0.490741  115       307
## 387869          GTS_1012-23 UGA_173101_P019_WA02 0.500000  117       307
## 387898          GTS_1012-23 UGA_173101_P018_WD10 0.500000  109       307
## 387960          GTS_1012-23 UGA_173101_P016_WH09 0.500000  113       307
## 387996          GTS_1012-23 UGA_173101_P017_WD09 0.500000  107       307
## 388016          GTS_1012-23 UGA_173101_P019_WA01 0.500000  119       307
## 388043          GTS_1012-23 UGA_173101_P018_WB06 0.483333   54       307
## 388067          GTS_1012-23 UGA_173101_P018_WC11 0.500000  120       307
## 388080          GTS_1012-23 UGA_173101_P023_WF05 0.489583   92       307
## 388093          GTS_1012-23 UGA_173101_P023_WB12 0.480769  109       307
## 388102          GTS_1012-23 UGA_173101_P023_WA04 0.500000   91       307
## 388157          GTS_1012-23 UGA_173101_P023_WA07 0.490385  106       307
## 388180          GTS_1012-23 UGA_173101_P022_WD02 0.472222   50       307
## 388320          GTS_1012-23 UGA_173101_P021_WF01 0.490000  102       307
## 388363          GTS_1012-23 UGA_173101_P022_WA09 0.500000  100       307
## 388399          GTS_1012-23           GTS_949-23 0.500000  121       307
## 388448          GTS_1012-23           GTS_646-23 0.500000  119       307
## 388462          GTS_1012-23           GTS_616-24 0.500000  119       307
## 388471          GTS_1012-23           GTS_655-23 0.500000  114       307
## 388472          GTS_1012-23           GTS_983-23 0.500000  121       307
## 388483          GTS_1012-23           GTS_630-24 0.500000  118       307
## 388493          GTS_1012-23           GTS_665-23 0.500000  120       307
## 388496          GTS_1012-23           GTS_997-23 0.500000  121       307
## 388502          GTS_1012-23           GTS_644-24 0.500000  117       307
## 388508          GTS_1012-23           GTS_818-23 0.500000  121       307
## 388521          GTS_1012-23           GTS_677-23 0.500000  118       307
## 392982           GTS_657-24 UGA_173101_P019_WE05 0.490385  113       326
## 393008           GTS_657-24 UGA_173101_P019_WA04 0.488636   91       326
## 393049           GTS_657-24 UGA_173101_P016_WC12 0.491071  108       326
## 393051           GTS_657-24 UGA_173101_P017_WC02 0.482143  115       326
## 393058           GTS_657-24 UGA_173101_P018_WB04 0.490741  116       326
## 393063           GTS_657-24 UGA_173101_P016_WF07 0.460000  101       326
## 393112           GTS_657-24 UGA_173101_P019_WG03 0.491071  119       326
## 393118           GTS_657-24 UGA_173101_P016_WE08 0.491071  115       326
## 393149           GTS_657-24 UGA_173101_P017_WC12 0.482143  115       326
## 393170           GTS_657-24 UGA_173101_P019_WA02 0.491071  117       326
## 393199           GTS_657-24 UGA_173101_P018_WD10 0.490741  108       326
## 393261           GTS_657-24 UGA_173101_P016_WH09 0.490741  113       326
## 393297           GTS_657-24 UGA_173101_P017_WD09 0.490385  107       326
## 393317           GTS_657-24 UGA_173101_P019_WA01 0.491071  118       326
## 393344           GTS_657-24 UGA_173101_P018_WB06 0.484375   55       326
## 393368           GTS_657-24 UGA_173101_P018_WC11 0.491071  119       326
## 393381           GTS_657-24 UGA_173101_P023_WF05 0.479167   93       326
## 393394           GTS_657-24 UGA_173101_P023_WB12 0.471154  108       326
## 393403           GTS_657-24 UGA_173101_P023_WA04 0.488636   92       326
## 393458           GTS_657-24 UGA_173101_P023_WA07 0.481481  106       326
## 393481           GTS_657-24 UGA_173101_P022_WD02 0.475000   51       326
## 393621           GTS_657-24 UGA_173101_P021_WF01 0.480000  102       326
## 393664           GTS_657-24 UGA_173101_P022_WA09 0.489583   99       326
## 393700           GTS_657-24           GTS_949-23 0.491071  120       326
```

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## 393749            GTS_657-24           GTS_646-23 0.500000  118        326
## 393763            GTS_657-24           GTS_616-24 0.500000  119        326
## 393772            GTS_657-24           GTS_655-23 0.500000  114        326
## 393773            GTS_657-24           GTS_983-23 0.490741  119        326
## 393784            GTS_657-24           GTS_630-24 0.500000  118        326
## 393794            GTS_657-24           GTS_665-23 0.491071  120        326
## 393797            GTS_657-24           GTS_997-23 0.491071  120        326
## 393803            GTS_657-24           GTS_644-24 0.500000  118        326
## 393809            GTS_657-24           GTS_818-23 0.491071  120        326
## 393822            GTS_657-24           GTS_677-23 0.500000  117        326
## 393823            GTS_657-24          GTS_1012-23 0.490741  119        326
## 410006            GTS_691-23 UGA_173101_P019_WE05 0.458333  112        339
## 410032            GTS_691-23 UGA_173101_P019_WA04 0.450000   89        339
## 410073            GTS_691-23 UGA_173101_P016_WC12 0.461538  107        339
## 410075            GTS_691-23 UGA_173101_P017_WC02 0.451923  114        339
## 410082            GTS_691-23 UGA_173101_P018_WB04 0.460000  115        339
## 410087            GTS_691-23 UGA_173101_P016_WF07 0.423913  100        339
## 410136            GTS_691-23 UGA_173101_P019_WG03 0.461538  118        339
## 410142            GTS_691-23 UGA_173101_P016_WE08 0.461538  114        339
## 410173            GTS_691-23 UGA_173101_P017_WC12 0.450000  114        339
## 410194            GTS_691-23 UGA_173101_P019_WA02 0.461538  116        339
## 410223            GTS_691-23 UGA_173101_P018_WD10 0.460000  107        339
## 410285            GTS_691-23 UGA_173101_P016_WH09 0.460000  112        339
## 410321            GTS_691-23 UGA_173101_P017_WD09 0.458333  106        339
## 410341            GTS_691-23 UGA_173101_P019_WA01 0.461538  117        339
## 410368            GTS_691-23 UGA_173101_P018_WB06 0.403846   53        339
## 410392            GTS_691-23 UGA_173101_P018_WC11 0.461538  118        339
## 410405            GTS_691-23 UGA_173101_P023_WF05 0.443182   91        339
## 410418            GTS_691-23 UGA_173101_P023_WB12 0.437500  107        339
## 410427            GTS_691-23 UGA_173101_P023_WA04 0.450000   90        339
## 410482            GTS_691-23 UGA_173101_P023_WA07 0.460000  105        339
## 410505            GTS_691-23 UGA_173101_P022_WD02 0.361111   51        339
## 410645            GTS_691-23 UGA_173101_P021_WF01 0.447917  101        339
## 410688            GTS_691-23 UGA_173101_P022_WA09 0.454545   98        339
## 410724            GTS_691-23           GTS_949-23 0.490385  119        339
## 410773            GTS_691-23           GTS_646-23 0.500000  116        339
## 410787            GTS_691-23           GTS_616-24 0.490000  118        339
## 410796            GTS_691-23           GTS_655-23 0.488636  113        339
## 410797            GTS_691-23           GTS_983-23 0.500000  118        339
## 410808            GTS_691-23           GTS_630-24 0.500000  117        339
## 410818            GTS_691-23           GTS_665-23 0.490385  119        339
## 410821            GTS_691-23           GTS_997-23 0.490385  119        339
## 410827            GTS_691-23           GTS_644-24 0.500000  116        339
## 410833            GTS_691-23           GTS_818-23 0.490385  119        339
## 410846            GTS_691-23           GTS_677-23 0.490000  118        339
## 410847            GTS_691-23          GTS_1012-23 0.500000  118        339
## 410853            GTS_691-23           GTS_657-24 0.490000  117        339
## 474866            GTS_725-23 UGA_173101_P019_WE05 0.500000  112         34
## 474892            GTS_725-23 UGA_173101_P019_WA04 0.500000   89         34
## 474933            GTS_725-23 UGA_173101_P016_WC12 0.500000  107         34
## 474935            GTS_725-23 UGA_173101_P017_WC02 0.490741  114         34
## 474942            GTS_725-23 UGA_173101_P018_WB04 0.500000  115         34
## 474947            GTS_725-23 UGA_173101_P016_WF07 0.468750  100         34
## 474996            GTS_725-23 UGA_173101_P019_WG03 0.500000  118         34
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## 475002            GTS_725-23 UGA_173101_P016_WE08 0.500000  114        34
## 475033            GTS_725-23 UGA_173101_P017_WC12 0.490385  114        34
## 475054            GTS_725-23 UGA_173101_P019_WA02 0.500000  116        34
## 475083            GTS_725-23 UGA_173101_P018_WD10 0.500000  107        34
## 475145            GTS_725-23 UGA_173101_P016_WH09 0.500000  112        34
## 475181            GTS_725-23 UGA_173101_P017_WD09 0.500000  106        34
## 475201            GTS_725-23 UGA_173101_P019_WA01 0.500000  117        34
## 475228            GTS_725-23 UGA_173101_P018_WB06 0.483333   54        34
## 475252            GTS_725-23 UGA_173101_P018_WC11 0.500000  118        34
## 475265            GTS_725-23 UGA_173101_P023_WF05 0.489130   91        34
## 475278            GTS_725-23 UGA_173101_P023_WB12 0.480000  107        34
## 475287            GTS_725-23 UGA_173101_P023_WA04 0.500000   90        34
## 475342            GTS_725-23 UGA_173101_P023_WA07 0.500000  105        34
## 475365            GTS_725-23 UGA_173101_P022_WD02 0.475000   51        34
## 475505            GTS_725-23 UGA_173101_P021_WF01 0.490000  101        34
## 475548            GTS_725-23 UGA_173101_P022_WA09 0.500000   98        34
## 475584            GTS_725-23           GTS_949-23 0.500000  119        34
## 475633            GTS_725-23           GTS_646-23 0.500000  118        34
## 475647            GTS_725-23           GTS_616-24 0.500000  119        34
## 475656            GTS_725-23           GTS_655-23 0.500000  114        34
## 475657            GTS_725-23           GTS_983-23 0.500000  118        34
## 475668            GTS_725-23           GTS_630-24 0.500000  119        34
## 475678            GTS_725-23           GTS_665-23 0.500000  119        34
## 475681            GTS_725-23           GTS_997-23 0.500000  119        34
## 475687            GTS_725-23           GTS_644-24 0.500000  118        34
## 475693            GTS_725-23           GTS_818-23 0.500000  119        34
## 475706            GTS_725-23           GTS_677-23 0.500000  118        34
## 475707            GTS_725-23          GTS_1012-23 0.500000  118        34
## 475713            GTS_725-23           GTS_657-24 0.500000  118        34
## 475732            GTS_725-23           GTS_691-23 0.500000  117        34
## 483677            GTS_736-23 UGA_173101_P019_WE05 0.470000  113       326
## 483703            GTS_736-23 UGA_173101_P019_WA04 0.464286   91       326
## 483744            GTS_736-23 UGA_173101_P016_WC12 0.472222  108       326
## 483746            GTS_736-23 UGA_173101_P017_WC02 0.462963  115       326
## 483753            GTS_736-23 UGA_173101_P018_WB04 0.471154  116       326
## 483758            GTS_736-23 UGA_173101_P016_WF07 0.437500  101       326
## 483807            GTS_736-23 UGA_173101_P019_WG03 0.472222  119       326
## 483813            GTS_736-23 UGA_173101_P016_WE08 0.472222  115       326
## 483844            GTS_736-23 UGA_173101_P017_WC12 0.462963  115       326
## 483865            GTS_736-23 UGA_173101_P019_WA02 0.472222  117       326
## 483894            GTS_736-23 UGA_173101_P018_WD10 0.471154  108       326
## 483956            GTS_736-23 UGA_173101_P016_WH09 0.471154  113       326
## 483992            GTS_736-23 UGA_173101_P017_WD09 0.470000  107       326
## 484012            GTS_736-23 UGA_173101_P019_WA01 0.472222  118       326
## 484039            GTS_736-23 UGA_173101_P018_WB06 0.450000   55       326
## 484063            GTS_736-23 UGA_173101_P018_WC11 0.472222  119       326
## 484076            GTS_736-23 UGA_173101_P023_WF05 0.456522   93       326
## 484089            GTS_736-23 UGA_173101_P023_WB12 0.450000  108       326
## 484098            GTS_736-23 UGA_173101_P023_WA04 0.464286   92       326
## 484153            GTS_736-23 UGA_173101_P023_WA07 0.442308  106       326
## 484176            GTS_736-23 UGA_173101_P022_WD02 0.444444   51       326
## 484316            GTS_736-23 UGA_173101_P021_WF01 0.470000  102       326
## 484359            GTS_736-23 UGA_173101_P022_WA09 0.467391   99       326
## 484395            GTS_736-23           GTS_949-23 0.481481  120       326
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## 484444          GTS_736-23          GTS_646-23 0.490000  117       326
## 484458          GTS_736-23          GTS_616-24 0.490741  119       326
## 484467          GTS_736-23          GTS_655-23 0.489583  114       326
## 484468          GTS_736-23          GTS_983-23 0.480769  119       326
## 484479          GTS_736-23          GTS_630-24 0.500000  117       326
## 484489          GTS_736-23          GTS_665-23 0.481481  120       326
## 484492          GTS_736-23          GTS_997-23 0.481481  120       326
## 484498          GTS_736-23          GTS_644-24 0.500000  117       326
## 484504          GTS_736-23          GTS_818-23 0.481481  120       326
## 484517          GTS_736-23          GTS_677-23 0.500000  118       326
## 484518          GTS_736-23         GTS_1012-23 0.480769  119       326
## 484524          GTS_736-23          GTS_657-24 0.490385  119       326
## 484543          GTS_736-23          GTS_691-23 0.480769  118       326
## 484612          GTS_736-23          GTS_725-23 0.500000  117       326
## 591369         GTS_1202-24 UGA_173101_P019_WE05 0.451923  113       339
## 591395         GTS_1202-24 UGA_173101_P019_WA04 0.443182   92       339
## 591436         GTS_1202-24 UGA_173101_P016_WC12 0.455357  109       339
## 591438         GTS_1202-24 UGA_173101_P017_WC02 0.446429  116       339
## 591445         GTS_1202-24 UGA_173101_P018_WB04 0.453704  117       339
## 591450         GTS_1202-24 UGA_173101_P016_WF07 0.420000  102       339
## 591499         GTS_1202-24 UGA_173101_P019_WG03 0.455357  120       339
## 591505         GTS_1202-24 UGA_173101_P016_WE08 0.455357  116       339
## 591536         GTS_1202-24 UGA_173101_P017_WC12 0.446429  116       339
## 591557         GTS_1202-24 UGA_173101_P019_WA02 0.455357  118       339
## 591586         GTS_1202-24 UGA_173101_P018_WD10 0.453704  110       339
## 591648         GTS_1202-24 UGA_173101_P016_WH09 0.453704  114       339
## 591684         GTS_1202-24 UGA_173101_P017_WD09 0.451923  108       339
## 591704         GTS_1202-24 UGA_173101_P019_WA01 0.455357  119       339
## 591731         GTS_1202-24 UGA_173101_P018_WB06 0.406250   56       339
## 591755         GTS_1202-24 UGA_173101_P018_WC11 0.455357  120       339
## 591768         GTS_1202-24 UGA_173101_P023_WF05 0.437500   93       339
## 591781         GTS_1202-24 UGA_173101_P023_WB12 0.432692  110       339
## 591790         GTS_1202-24 UGA_173101_P023_WA04 0.443182   92       339
## 591845         GTS_1202-24 UGA_173101_P023_WA07 0.444444  107       339
## 591868         GTS_1202-24 UGA_173101_P022_WD02 0.350000   51       339
## 592008         GTS_1202-24 UGA_173101_P021_WF01 0.440000  103       339
## 592051         GTS_1202-24 UGA_173101_P022_WA09 0.447917  101       339
## 592087         GTS_1202-24          GTS_949-23 0.482759  122       339
## 592136         GTS_1202-24          GTS_646-23 0.490385  118       339
## 592150         GTS_1202-24          GTS_616-24 0.473214  120       339
## 592159         GTS_1202-24          GTS_655-23 0.479167  114       339
## 592160         GTS_1202-24          GTS_983-23 0.500000  119       339
## 592171         GTS_1202-24          GTS_630-24 0.490385  117       339
## 592181         GTS_1202-24          GTS_665-23 0.482143  120       339
## 592184         GTS_1202-24          GTS_997-23 0.482143  121       339
## 592190         GTS_1202-24          GTS_644-24 0.481481  117       339
## 592196         GTS_1202-24          GTS_818-23 0.482143  121       339
## 592209         GTS_1202-24          GTS_677-23 0.490385  117       339
## 592210         GTS_1202-24         GTS_1012-23 0.500000  119       339
## 592216         GTS_1202-24          GTS_657-24 0.482143  118       339
## 592235         GTS_1202-24          GTS_691-23 0.500000  117       339
## 592304         GTS_1202-24          GTS_725-23 0.490385  117       339
## 592313         GTS_1202-24          GTS_736-23 0.472222  118       339
## 611106          GTS_850-23 UGA_173101_P019_WE05 0.481481  115       352
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## 611132            GTS_850-23 UGA_173101_P019_WA04 0.478261   92        352
## 611173            GTS_850-23 UGA_173101_P016_WC12 0.482759  110        352
## 611175            GTS_850-23 UGA_173101_P017_WC02 0.474138  117        352
## 611182            GTS_850-23 UGA_173101_P018_WB04 0.482143  118        352
## 611187            GTS_850-23 UGA_173101_P016_WF07 0.451923  103        352
## 611236            GTS_850-23 UGA_173101_P019_WG03 0.482759  122        352
## 611242            GTS_850-23 UGA_173101_P016_WE08 0.482759  117        352
## 611273            GTS_850-23 UGA_173101_P017_WC12 0.473214  117        352
## 611294            GTS_850-23 UGA_173101_P019_WA02 0.482759  119        352
## 611323            GTS_850-23 UGA_173101_P018_WD10 0.482143  111        352
## 611385            GTS_850-23 UGA_173101_P016_WH09 0.482143  115        352
## 611421            GTS_850-23 UGA_173101_P017_WD09 0.481481  109        352
## 611441            GTS_850-23 UGA_173101_P019_WA01 0.482759  121        352
## 611468            GTS_850-23 UGA_173101_P018_WB06 0.453125   56        352
## 611492            GTS_850-23 UGA_173101_P018_WC11 0.482759  122        352
## 611505            GTS_850-23 UGA_173101_P023_WF05 0.470000   94        352
## 611518            GTS_850-23 UGA_173101_P023_WB12 0.462963  111        352
## 611527            GTS_850-23 UGA_173101_P023_WA04 0.478261   93        352
## 611582            GTS_850-23 UGA_173101_P023_WA07 0.472222  108        352
## 611605            GTS_850-23 UGA_173101_P022_WD02 0.425000   52        352
## 611745            GTS_850-23 UGA_173101_P021_WF01 0.471154  104        352
## 611788            GTS_850-23 UGA_173101_P022_WA09 0.480000  102        352
## 611824            GTS_850-23           GTS_949-23 0.491667  124        352
## 611873            GTS_850-23           GTS_646-23 0.490385  120        352
## 611887            GTS_850-23           GTS_616-24 0.482759  122        352
## 611896            GTS_850-23           GTS_655-23 0.490000  116        352
## 611897            GTS_850-23           GTS_983-23 0.500000  121        352
## 611908            GTS_850-23           GTS_630-24 0.490385  119        352
## 611918            GTS_850-23           GTS_665-23 0.491379  122        352
## 611921            GTS_850-23           GTS_997-23 0.491379  123        352
## 611927            GTS_850-23           GTS_644-24 0.481481  119        352
## 611933            GTS_850-23           GTS_818-23 0.491379  123        352
## 611946            GTS_850-23           GTS_677-23 0.500000  119        352
## 611947            GTS_850-23          GTS_1012-23 0.500000  121        352
## 611953            GTS_850-23           GTS_657-24 0.482143  120        352
## 611972            GTS_850-23           GTS_691-23 0.490385  119        352
## 612041            GTS_850-23           GTS_725-23 0.490385  119        352
## 612050            GTS_850-23           GTS_736-23 0.481481  120        352
## 612154            GTS_850-23          GTS_1202-24 0.491379  122        352
## 649271            GTS_825-24 UGA_173101_P019_WE05 0.480000  112        352
## 649297            GTS_825-24 UGA_173101_P019_WA04 0.476190   90        352
## 649338            GTS_825-24 UGA_173101_P016_WC12 0.481481  107        352
## 649340            GTS_825-24 UGA_173101_P017_WC02 0.472222  114        352
## 649347            GTS_825-24 UGA_173101_P018_WB04 0.480769  115        352
## 649352            GTS_825-24 UGA_173101_P016_WF07 0.447917  100        352
## 649401            GTS_825-24 UGA_173101_P019_WG03 0.481481  118        352
## 649407            GTS_825-24 UGA_173101_P016_WE08 0.481481  114        352
## 649438            GTS_825-24 UGA_173101_P017_WC12 0.471154  114        352
## 649459            GTS_825-24 UGA_173101_P019_WA02 0.481481  116        352
## 649488            GTS_825-24 UGA_173101_P018_WD10 0.480769  107        352
## 649550            GTS_825-24 UGA_173101_P016_WH09 0.480769  112        352
## 649586            GTS_825-24 UGA_173101_P017_WD09 0.480000  106        352
## 649606            GTS_825-24 UGA_173101_P019_WA01 0.481481  117        352
## 649633            GTS_825-24 UGA_173101_P018_WB06 0.450000   55        352
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## 649657            GTS_825-24 UGA_173101_P018_WC11 0.481481  118        352
## 649670            GTS_825-24 UGA_173101_P023_WF05 0.467391   92        352
## 649683            GTS_825-24 UGA_173101_P023_WB12 0.460000  107        352
## 649692            GTS_825-24 UGA_173101_P023_WA04 0.476190   91        352
## 649747            GTS_825-24 UGA_173101_P023_WA07 0.480769  105        352
## 649770            GTS_825-24 UGA_173101_P022_WD02 0.425000   52        352
## 649910            GTS_825-24 UGA_173101_P021_WF01 0.470000  102        352
## 649953            GTS_825-24 UGA_173101_P022_WA09 0.478261   98        352
## 649989            GTS_825-24           GTS_949-23 0.491071  120        352
## 650038            GTS_825-24           GTS_646-23 0.490000  117        352
## 650052            GTS_825-24           GTS_616-24 0.482143  120        352
## 650061            GTS_825-24           GTS_655-23 0.489583  114        352
## 650062            GTS_825-24           GTS_983-23 0.500000  117        352
## 650073            GTS_825-24           GTS_630-24 0.490385  118        352
## 650083            GTS_825-24           GTS_665-23 0.490741  119        352
## 650086            GTS_825-24           GTS_997-23 0.490741  119        352
## 650092            GTS_825-24           GTS_644-24 0.481481  119        352
## 650098            GTS_825-24           GTS_818-23 0.490741  119        352
## 650111            GTS_825-24           GTS_677-23 0.500000  118        352
## 650112            GTS_825-24          GTS_1012-23 0.500000  117        352
## 650118            GTS_825-24           GTS_657-24 0.490385  118        352
## 650137            GTS_825-24           GTS_691-23 0.490000  117        352
## 650206            GTS_825-24           GTS_725-23 0.490385  118        352
## 650215            GTS_825-24           GTS_736-23 0.500000  118        352
## 650319            GTS_825-24          GTS_1202-24 0.490741  118        352
## 650337            GTS_825-24           GTS_850-23 0.500000  120        352
## 664169            GTS_935-23 UGA_173101_P019_WE05 0.500000  114        307
## 664195            GTS_935-23 UGA_173101_P019_WA04 0.500000   91        307
## 664236            GTS_935-23 UGA_173101_P016_WC12 0.500000  109        307
## 664238            GTS_935-23 UGA_173101_P017_WC02 0.491379  116        307
## 664245            GTS_935-23 UGA_173101_P018_WB04 0.500000  117        307
## 664250            GTS_935-23 UGA_173101_P016_WF07 0.471154  102        307
## 664299            GTS_935-23 UGA_173101_P019_WG03 0.500000  121        307
## 664305            GTS_935-23 UGA_173101_P016_WE08 0.500000  116        307
## 664336            GTS_935-23 UGA_173101_P017_WC12 0.491071  116        307
## 664357            GTS_935-23 UGA_173101_P019_WA02 0.500000  118        307
## 664386            GTS_935-23 UGA_173101_P018_WD10 0.500000  110        307
## 664448            GTS_935-23 UGA_173101_P016_WH09 0.500000  114        307
## 664484            GTS_935-23 UGA_173101_P017_WD09 0.500000  108        307
## 664504            GTS_935-23 UGA_173101_P019_WA01 0.500000  120        307
## 664531            GTS_935-23 UGA_173101_P018_WB06 0.484375   55        307
## 664555            GTS_935-23 UGA_173101_P018_WC11 0.500000  121        307
## 664568            GTS_935-23 UGA_173101_P023_WF05 0.490000   93        307
## 664581            GTS_935-23 UGA_173101_P023_WB12 0.481481  110        307
## 664590            GTS_935-23 UGA_173101_P023_WA04 0.500000   92        307
## 664645            GTS_935-23 UGA_173101_P023_WA07 0.490741  107        307
## 664668            GTS_935-23 UGA_173101_P022_WD02 0.475000   51        307
## 664808            GTS_935-23 UGA_173101_P021_WF01 0.490385  103        307
## 664851            GTS_935-23 UGA_173101_P022_WA09 0.500000  101        307
## 664887            GTS_935-23           GTS_949-23 0.500000  123        307
## 664936            GTS_935-23           GTS_646-23 0.500000  120        307
## 664950            GTS_935-23           GTS_616-24 0.491071  121        307
## 664959            GTS_935-23           GTS_655-23 0.500000  115        307
## 664960            GTS_935-23           GTS_983-23 0.500000  121        307
```

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## 664971            GTS_935-23           GTS_630-24 0.500000  119        307
## 664981            GTS_935-23           GTS_665-23 0.500000  121        307
## 664984            GTS_935-23           GTS_997-23 0.500000  122        307
## 664990            GTS_935-23           GTS_644-24 0.490741  119        307
## 664996            GTS_935-23           GTS_818-23 0.500000  122        307
## 665009            GTS_935-23           GTS_677-23 0.500000  118        307
## 665010            GTS_935-23          GTS_1012-23 0.500000  121        307
## 665016            GTS_935-23           GTS_657-24 0.491071  120        307
## 665035            GTS_935-23           GTS_691-23 0.500000  118        307
## 665104            GTS_935-23           GTS_725-23 0.500000  119        307
## 665113            GTS_935-23           GTS_736-23 0.480769  119        307
## 665217            GTS_935-23          GTS_1202-24 0.491379  121        307
## 665235            GTS_935-23           GTS_850-23 0.491379  123        307
## 665269            GTS_935-23           GTS_825-24 0.490741  119        307
## 675744           GTS_935-23R UGA_173101_P019_WE05 0.461538  114        339
## 675770           GTS_935-23R UGA_173101_P019_WA04 0.454545   91        339
## 675811           GTS_935-23R UGA_173101_P016_WC12 0.464286  109        339
## 675813           GTS_935-23R UGA_173101_P017_WC02 0.455357  116        339
## 675820           GTS_935-23R UGA_173101_P018_WB04 0.462963  117        339
## 675825           GTS_935-23R UGA_173101_P016_WF07 0.430000  102        339
## 675874           GTS_935-23R UGA_173101_P019_WG03 0.464286  121        339
## 675880           GTS_935-23R UGA_173101_P016_WE08 0.464286  116        339
## 675911           GTS_935-23R UGA_173101_P017_WC12 0.453704  116        339
## 675932           GTS_935-23R UGA_173101_P019_WA02 0.464286  118        339
## 675961           GTS_935-23R UGA_173101_P018_WD10 0.462963  110        339
## 676023           GTS_935-23R UGA_173101_P016_WH09 0.462963  114        339
## 676059           GTS_935-23R UGA_173101_P017_WD09 0.461538  108        339
## 676079           GTS_935-23R UGA_173101_P019_WA01 0.464286  120        339
## 676106           GTS_935-23R UGA_173101_P018_WB06 0.416667   55        339
## 676130           GTS_935-23R UGA_173101_P018_WC11 0.464286  121        339
## 676143           GTS_935-23R UGA_173101_P023_WF05 0.447917   93        339
## 676156           GTS_935-23R UGA_173101_P023_WB12 0.442308  110        339
## 676165           GTS_935-23R UGA_173101_P023_WA04 0.454545   92        339
## 676220           GTS_935-23R UGA_173101_P023_WA07 0.451923  107        339
## 676243           GTS_935-23R UGA_173101_P022_WD02 0.361111   51        339
## 676383           GTS_935-23R UGA_173101_P021_WF01 0.450000  103        339
## 676426           GTS_935-23R UGA_173101_P022_WA09 0.458333  101        339
## 676462           GTS_935-23R           GTS_949-23 0.491379  123        339
## 676511           GTS_935-23R           GTS_646-23 0.500000  119        339
## 676525           GTS_935-23R           GTS_616-24 0.482143  121        339
## 676534           GTS_935-23R           GTS_655-23 0.489583  115        339
## 676535           GTS_935-23R           GTS_983-23 0.500000  121        339
## 676546           GTS_935-23R           GTS_630-24 0.500000  118        339
## 676556           GTS_935-23R           GTS_665-23 0.491071  121        339
## 676559           GTS_935-23R           GTS_997-23 0.491071  122        339
## 676565           GTS_935-23R           GTS_644-24 0.490385  118        339
## 676571           GTS_935-23R           GTS_818-23 0.491071  122        339
## 676584           GTS_935-23R           GTS_677-23 0.490385  119        339
## 676585           GTS_935-23R          GTS_1012-23 0.500000  121        339
## 676591           GTS_935-23R           GTS_657-24 0.490741  119        339
## 676610           GTS_935-23R           GTS_691-23 0.500000  119        339
## 676679           GTS_935-23R           GTS_725-23 0.500000  118        339
## 676688           GTS_935-23R           GTS_736-23 0.472222  120        339
## 676792           GTS_935-23R          GTS_1202-24 0.500000  121        339
```

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## 676810          GTS_935-23R           GTS_850-23 0.491379  123       339
## 676844          GTS_935-23R           GTS_825-24 0.490741  119       339
## 676857          GTS_935-23R           GTS_935-23 0.500000  122       339
## 696831           GTS_855-24 UGA_173101_P019_WE05 0.500000  114       307
## 696857           GTS_855-24 UGA_173101_P019_WA04 0.500000   91       307
## 696898           GTS_855-24 UGA_173101_P016_WC12 0.500000  109       307
## 696900           GTS_855-24 UGA_173101_P017_WC02 0.491379  116       307
## 696907           GTS_855-24 UGA_173101_P018_WB04 0.500000  117       307
## 696912           GTS_855-24 UGA_173101_P016_WF07 0.471154  102       307
## 696961           GTS_855-24 UGA_173101_P019_WG03 0.500000  121       307
## 696967           GTS_855-24 UGA_173101_P016_WE08 0.500000  116       307
## 696998           GTS_855-24 UGA_173101_P017_WC12 0.491071  116       307
## 697019           GTS_855-24 UGA_173101_P019_WA02 0.500000  118       307
## 697048           GTS_855-24 UGA_173101_P018_WD10 0.500000  110       307
## 697110           GTS_855-24 UGA_173101_P016_WH09 0.500000  114       307
## 697146           GTS_855-24 UGA_173101_P017_WD09 0.500000  108       307
## 697166           GTS_855-24 UGA_173101_P019_WA01 0.500000  120       307
## 697193           GTS_855-24 UGA_173101_P018_WB06 0.484375   55       307
## 697217           GTS_855-24 UGA_173101_P018_WC11 0.500000  121       307
## 697230           GTS_855-24 UGA_173101_P023_WF05 0.490000   93       307
## 697243           GTS_855-24 UGA_173101_P023_WB12 0.481481  110       307
## 697252           GTS_855-24 UGA_173101_P023_WA04 0.500000   92       307
## 697307           GTS_855-24 UGA_173101_P023_WA07 0.490741  107       307
## 697330           GTS_855-24 UGA_173101_P022_WD02 0.475000   51       307
## 697470           GTS_855-24 UGA_173101_P021_WF01 0.490385  103       307
## 697513           GTS_855-24 UGA_173101_P022_WA09 0.500000  101       307
## 697549           GTS_855-24           GTS_949-23 0.500000  123       307
## 697598           GTS_855-24           GTS_646-23 0.500000  120       307
## 697612           GTS_855-24           GTS_616-24 0.491071  121       307
## 697621           GTS_855-24           GTS_655-23 0.500000  115       307
## 697622           GTS_855-24           GTS_983-23 0.500000  121       307
## 697633           GTS_855-24           GTS_630-24 0.500000  119       307
## 697643           GTS_855-24           GTS_665-23 0.500000  121       307
## 697646           GTS_855-24           GTS_997-23 0.500000  122       307
## 697652           GTS_855-24           GTS_644-24 0.490741  119       307
## 697658           GTS_855-24           GTS_818-23 0.500000  122       307
## 697671           GTS_855-24           GTS_677-23 0.500000  118       307
## 697672           GTS_855-24          GTS_1012-23 0.500000  121       307
## 697678           GTS_855-24           GTS_657-24 0.491071  120       307
## 697697           GTS_855-24           GTS_691-23 0.500000  118       307
## 697766           GTS_855-24           GTS_725-23 0.500000  119       307
## 697775           GTS_855-24           GTS_736-23 0.480769  119       307
## 697879           GTS_855-24          GTS_1202-24 0.491379  121       307
## 697897           GTS_855-24           GTS_850-23 0.491379  123       307
## 697931           GTS_855-24           GTS_825-24 0.490741  119       307
## 697944           GTS_855-24           GTS_935-23 0.500000  123       307
## 697954           GTS_855-24          GTS_935-23R 0.500000  122       307
## 711069          GTS_1344-24 UGA_173101_P019_WE05 0.500000  115       307
## 711095          GTS_1344-24 UGA_173101_P019_WA04 0.500000   92       307
## 711136          GTS_1344-24 UGA_173101_P016_WC12 0.500000  110       307
## 711138          GTS_1344-24 UGA_173101_P017_WC02 0.491379  117       307
## 711145          GTS_1344-24 UGA_173101_P018_WB04 0.500000  118       307
## 711150          GTS_1344-24 UGA_173101_P016_WF07 0.471154  103       307
## 711199          GTS_1344-24 UGA_173101_P019_WG03 0.500000  122       307
```

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## 711205          GTS_1344-24 UGA_173101_P016_WE08 0.500000  117       307
## 711236          GTS_1344-24 UGA_173101_P017_WC12 0.491071  117       307
## 711257          GTS_1344-24 UGA_173101_P019_WA02 0.500000  119       307
## 711286          GTS_1344-24 UGA_173101_P018_WD10 0.500000  111       307
## 711348          GTS_1344-24 UGA_173101_P016_WH09 0.500000  115       307
## 711384          GTS_1344-24 UGA_173101_P017_WD09 0.500000  109       307
## 711404          GTS_1344-24 UGA_173101_P019_WA01 0.500000  121       307
## 711431          GTS_1344-24 UGA_173101_P018_WB06 0.484375   55       307
## 711455          GTS_1344-24 UGA_173101_P018_WC11 0.500000  122       307
## 711468          GTS_1344-24 UGA_173101_P023_WF05 0.490000   94       307
## 711481          GTS_1344-24 UGA_173101_P023_WB12 0.481481  111       307
## 711490          GTS_1344-24 UGA_173101_P023_WA04 0.500000   93       307
## 711545          GTS_1344-24 UGA_173101_P023_WA07 0.490741  108       307
## 711568          GTS_1344-24 UGA_173101_P022_WD02 0.475000   51       307
## 711708          GTS_1344-24 UGA_173101_P021_WF01 0.490385  104       307
## 711751          GTS_1344-24 UGA_173101_P022_WA09 0.500000  102       307
## 711787          GTS_1344-24           GTS_949-23 0.500000  123       307
## 711836          GTS_1344-24           GTS_646-23 0.500000  120       307
## 711850          GTS_1344-24           GTS_616-24 0.491071  121       307
## 711859          GTS_1344-24           GTS_655-23 0.500000  115       307
## 711860          GTS_1344-24           GTS_983-23 0.500000  121       307
## 711871          GTS_1344-24           GTS_630-24 0.500000  119       307
## 711881          GTS_1344-24           GTS_665-23 0.500000  121       307
## 711884          GTS_1344-24           GTS_997-23 0.500000  122       307
## 711890          GTS_1344-24           GTS_644-24 0.490741  119       307
## 711896          GTS_1344-24           GTS_818-23 0.500000  122       307
## 711909          GTS_1344-24           GTS_677-23 0.500000  118       307
## 711910          GTS_1344-24          GTS_1012-23 0.500000  121       307
## 711916          GTS_1344-24           GTS_657-24 0.491071  120       307
## 711935          GTS_1344-24           GTS_691-23 0.500000  118       307
## 712004          GTS_1344-24           GTS_725-23 0.500000  119       307
## 712013          GTS_1344-24           GTS_736-23 0.480769  119       307
## 712117          GTS_1344-24          GTS_1202-24 0.491379  121       307
## 712135          GTS_1344-24           GTS_850-23 0.491379  123       307
## 712169          GTS_1344-24           GTS_825-24 0.490741  119       307
## 712182          GTS_1344-24           GTS_935-23 0.500000  123       307
## 712192          GTS_1344-24          GTS_935-23R 0.500000  122       307
## 712210          GTS_1344-24           GTS_855-24 0.500000  123       307
## 755876           GTS_886-24 UGA_173101_P019_WE05 0.490741  115       307
## 755902           GTS_886-24 UGA_173101_P019_WA04 0.489130   92       307
## 755943           GTS_886-24 UGA_173101_P016_WC12 0.491379  110       307
## 755945           GTS_886-24 UGA_173101_P017_WC02 0.483333  117       307
## 755952           GTS_886-24 UGA_173101_P018_WB04 0.491071  118       307
## 755957           GTS_886-24 UGA_173101_P016_WF07 0.462963  103       307
## 756006           GTS_886-24 UGA_173101_P019_WG03 0.491379  122       307
## 756012           GTS_886-24 UGA_173101_P016_WE08 0.491379  117       307
## 756043           GTS_886-24 UGA_173101_P017_WC12 0.482143  117       307
## 756064           GTS_886-24 UGA_173101_P019_WA02 0.491379  119       307
## 756093           GTS_886-24 UGA_173101_P018_WD10 0.491071  111       307
## 756155           GTS_886-24 UGA_173101_P016_WH09 0.491071  115       307
## 756191           GTS_886-24 UGA_173101_P017_WD09 0.490741  109       307
## 756211           GTS_886-24 UGA_173101_P019_WA01 0.491379  121       307
## 756238           GTS_886-24 UGA_173101_P018_WB06 0.468750   56       307
## 756262           GTS_886-24 UGA_173101_P018_WC11 0.491379  122       307
```

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## 756275            GTS_886-24 UGA_173101_P023_WF05 0.480769   94        307
## 756288            GTS_886-24 UGA_173101_P023_WB12 0.473214  111        307
## 756297            GTS_886-24 UGA_173101_P023_WA04 0.489130   93        307
## 756352            GTS_886-24 UGA_173101_P023_WA07 0.481481  108        307
## 756375            GTS_886-24 UGA_173101_P022_WD02 0.450000   52        307
## 756515            GTS_886-24 UGA_173101_P021_WF01 0.481481  104        307
## 756558            GTS_886-24 UGA_173101_P022_WA09 0.490000  102        307
## 756594            GTS_886-24           GTS_949-23 0.500000  124        307
## 756643            GTS_886-24           GTS_646-23 0.500000  120        307
## 756657            GTS_886-24           GTS_616-24 0.491379  122        307
## 756666            GTS_886-24           GTS_655-23 0.500000  116        307
## 756667            GTS_886-24           GTS_983-23 0.500000  121        307
## 756678            GTS_886-24           GTS_630-24 0.500000  119        307
## 756688            GTS_886-24           GTS_665-23 0.500000  122        307
## 756691            GTS_886-24           GTS_997-23 0.500000  123        307
## 756697            GTS_886-24           GTS_644-24 0.490741  119        307
## 756703            GTS_886-24           GTS_818-23 0.500000  123        307
## 756716            GTS_886-24           GTS_677-23 0.500000  119        307
## 756717            GTS_886-24          GTS_1012-23 0.500000  121        307
## 756723            GTS_886-24           GTS_657-24 0.491071  120        307
## 756742            GTS_886-24           GTS_691-23 0.490385  119        307
## 756811            GTS_886-24           GTS_725-23 0.500000  119        307
## 756820            GTS_886-24           GTS_736-23 0.481481  120        307
## 756924            GTS_886-24          GTS_1202-24 0.482759  122        307
## 756942            GTS_886-24           GTS_850-23 0.491667  124        307
## 756976            GTS_886-24           GTS_825-24 0.491071  120        307
## 756989            GTS_886-24           GTS_935-23 0.500000  123        307
## 756999            GTS_886-24          GTS_935-23R 0.491379  123        307
## 757017            GTS_886-24           GTS_855-24 0.500000  123        307
## 757029            GTS_886-24          GTS_1344-24 0.500000  123        307
## 773187           GTS_1403-24 UGA_173101_P019_WE05 0.500000  113        307
## 773213           GTS_1403-24 UGA_173101_P019_WA04 0.500000   90        307
## 773254           GTS_1403-24 UGA_173101_P016_WC12 0.500000  108        307
## 773256           GTS_1403-24 UGA_173101_P017_WC02 0.491071  115        307
## 773263           GTS_1403-24 UGA_173101_P018_WB04 0.500000  116        307
## 773268           GTS_1403-24 UGA_173101_P016_WF07 0.470000  101        307
## 773317           GTS_1403-24 UGA_173101_P019_WG03 0.500000  120        307
## 773323           GTS_1403-24 UGA_173101_P016_WE08 0.500000  115        307
## 773354           GTS_1403-24 UGA_173101_P017_WC12 0.490741  115        307
## 773375           GTS_1403-24 UGA_173101_P019_WA02 0.500000  117        307
## 773404           GTS_1403-24 UGA_173101_P018_WD10 0.500000  109        307
## 773466           GTS_1403-24 UGA_173101_P016_WH09 0.500000  113        307
## 773502           GTS_1403-24 UGA_173101_P017_WD09 0.500000  107        307
## 773522           GTS_1403-24 UGA_173101_P019_WA01 0.500000  119        307
## 773549           GTS_1403-24 UGA_173101_P018_WB06 0.483333   54        307
## 773573           GTS_1403-24 UGA_173101_P018_WC11 0.500000  120        307
## 773586           GTS_1403-24 UGA_173101_P023_WF05 0.489583   92        307
## 773599           GTS_1403-24 UGA_173101_P023_WB12 0.480769  109        307
## 773608           GTS_1403-24 UGA_173101_P023_WA04 0.500000   91        307
## 773663           GTS_1403-24 UGA_173101_P023_WA07 0.490385  106        307
## 773686           GTS_1403-24 UGA_173101_P022_WD02 0.472222   50        307
## 773826           GTS_1403-24 UGA_173101_P021_WF01 0.490000  102        307
## 773869           GTS_1403-24 UGA_173101_P022_WA09 0.500000  100        307
## 773905           GTS_1403-24           GTS_949-23 0.500000  122        307
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## 773954          GTS_1403-24           GTS_646-23 0.500000  119       307
## 773968          GTS_1403-24           GTS_616-24 0.490741  120       307
## 773977          GTS_1403-24           GTS_655-23 0.500000  114       307
## 773978          GTS_1403-24           GTS_983-23 0.500000  121       307
## 773989          GTS_1403-24           GTS_630-24 0.500000  118       307
## 773999          GTS_1403-24           GTS_665-23 0.500000  120       307
## 774002          GTS_1403-24           GTS_997-23 0.500000  121       307
## 774008          GTS_1403-24           GTS_644-24 0.490385  118       307
## 774014          GTS_1403-24           GTS_818-23 0.500000  121       307
## 774027          GTS_1403-24           GTS_677-23 0.500000  118       307
## 774028          GTS_1403-24          GTS_1012-23 0.500000  121       307
## 774034          GTS_1403-24           GTS_657-24 0.490741  119       307
## 774053          GTS_1403-24           GTS_691-23 0.500000  118       307
## 774122          GTS_1403-24           GTS_725-23 0.500000  118       307
## 774131          GTS_1403-24           GTS_736-23 0.480769  119       307
## 774235          GTS_1403-24          GTS_1202-24 0.500000  120       307
## 774253          GTS_1403-24           GTS_850-23 0.500000  122       307
## 774287          GTS_1403-24           GTS_825-24 0.500000  118       307
## 774300          GTS_1403-24           GTS_935-23 0.500000  122       307
## 774310          GTS_1403-24          GTS_935-23R 0.500000  122       307
## 774328          GTS_1403-24           GTS_855-24 0.500000  122       307
## 774340          GTS_1403-24          GTS_1344-24 0.500000  122       307
## 774377          GTS_1403-24           GTS_886-24 0.500000  122       307
## 794471           GTS_931-24 UGA_173101_P019_WE05 0.500000  111       307
## 794497           GTS_931-24 UGA_173101_P019_WA04 0.500000   89       307
## 794538           GTS_931-24 UGA_173101_P016_WC12 0.500000  106       307
## 794540           GTS_931-24 UGA_173101_P017_WC02 0.490741  113       307
## 794547           GTS_931-24 UGA_173101_P018_WB04 0.500000  114       307
## 794552           GTS_931-24 UGA_173101_P016_WF07 0.468750   99       307
## 794601           GTS_931-24 UGA_173101_P019_WG03 0.500000  117       307
## 794607           GTS_931-24 UGA_173101_P016_WE08 0.500000  113       307
## 794638           GTS_931-24 UGA_173101_P017_WC12 0.490385  113       307
## 794659           GTS_931-24 UGA_173101_P019_WA02 0.500000  115       307
## 794688           GTS_931-24 UGA_173101_P018_WD10 0.500000  106       307
## 794750           GTS_931-24 UGA_173101_P016_WH09 0.500000  111       307
## 794786           GTS_931-24 UGA_173101_P017_WD09 0.500000  105       307
## 794806           GTS_931-24 UGA_173101_P019_WA01 0.500000  116       307
## 794833           GTS_931-24 UGA_173101_P018_WB06 0.483333   54       307
## 794857           GTS_931-24 UGA_173101_P018_WC11 0.500000  117       307
## 794870           GTS_931-24 UGA_173101_P023_WF05 0.489130   91       307
## 794883           GTS_931-24 UGA_173101_P023_WB12 0.480000  106       307
## 794892           GTS_931-24 UGA_173101_P023_WA04 0.500000   90       307
## 794947           GTS_931-24 UGA_173101_P023_WA07 0.490000  104       307
## 794970           GTS_931-24 UGA_173101_P022_WD02 0.472222   50       307
## 795110           GTS_931-24 UGA_173101_P021_WF01 0.489583  100       307
## 795153           GTS_931-24 UGA_173101_P022_WA09 0.500000   97       307
## 795189           GTS_931-24           GTS_949-23 0.500000  119       307
## 795238           GTS_931-24           GTS_646-23 0.500000  117       307
## 795252           GTS_931-24           GTS_616-24 0.490741  119       307
## 795261           GTS_931-24           GTS_655-23 0.500000  113       307
## 795262           GTS_931-24           GTS_983-23 0.500000  118       307
## 795273           GTS_931-24           GTS_630-24 0.500000  117       307
## 795283           GTS_931-24           GTS_665-23 0.500000  118       307
## 795286           GTS_931-24           GTS_997-23 0.500000  118       307
```

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## 795292          GTS_931-24          GTS_644-24 0.490385  118       307
## 795298          GTS_931-24          GTS_818-23 0.500000  118       307
## 795311          GTS_931-24          GTS_677-23 0.500000  117       307
## 795312          GTS_931-24         GTS_1012-23 0.500000  118       307
## 795318          GTS_931-24          GTS_657-24 0.500000  118       307
## 795337          GTS_931-24          GTS_691-23 0.500000  116       307
## 795406          GTS_931-24          GTS_725-23 0.500000  117       307
## 795415          GTS_931-24          GTS_736-23 0.490385  118       307
## 795519          GTS_931-24         GTS_1202-24 0.500000  117       307
## 795537          GTS_931-24          GTS_850-23 0.500000  119       307
## 795571          GTS_931-24          GTS_825-24 0.500000  118       307
## 795584          GTS_931-24          GTS_935-23 0.500000  119       307
## 795594          GTS_931-24         GTS_935-23R 0.500000  119       307
## 795612          GTS_931-24          GTS_855-24 0.500000  119       307
## 795624          GTS_931-24         GTS_1344-24 0.500000  119       307
## 795661          GTS_931-24          GTS_886-24 0.500000  119       307
## 795675          GTS_931-24         GTS_1403-24 0.500000  119       307
## 799521         GTS_1428-24 UGA_173101_P019_WE05 0.500000  114       307
## 799547         GTS_1428-24 UGA_173101_P019_WA04 0.500000   91       307
## 799588         GTS_1428-24 UGA_173101_P016_WC12 0.500000  109       307
## 799590         GTS_1428-24 UGA_173101_P017_WC02 0.491379  116       307
## 799597         GTS_1428-24 UGA_173101_P018_WB04 0.500000  117       307
## 799602         GTS_1428-24 UGA_173101_P016_WF07 0.471154  102       307
## 799651         GTS_1428-24 UGA_173101_P019_WG03 0.500000  121       307
## 799657         GTS_1428-24 UGA_173101_P016_WE08 0.500000  116       307
## 799688         GTS_1428-24 UGA_173101_P017_WC12 0.491071  116       307
## 799709         GTS_1428-24 UGA_173101_P019_WA02 0.500000  118       307
## 799738         GTS_1428-24 UGA_173101_P018_WD10 0.500000  110       307
## 799800         GTS_1428-24 UGA_173101_P016_WH09 0.500000  114       307
## 799836         GTS_1428-24 UGA_173101_P017_WD09 0.500000  108       307
## 799856         GTS_1428-24 UGA_173101_P019_WA01 0.500000  120       307
## 799883         GTS_1428-24 UGA_173101_P018_WB06 0.484375   55       307
## 799907         GTS_1428-24 UGA_173101_P018_WC11 0.500000  121       307
## 799920         GTS_1428-24 UGA_173101_P023_WF05 0.490000   93       307
## 799933         GTS_1428-24 UGA_173101_P023_WB12 0.481481  110       307
## 799942         GTS_1428-24 UGA_173101_P023_WA04 0.500000   92       307
## 799997         GTS_1428-24 UGA_173101_P023_WA07 0.490741  107       307
## 800020         GTS_1428-24 UGA_173101_P022_WD02 0.475000   51       307
## 800160         GTS_1428-24 UGA_173101_P021_WF01 0.490385  103       307
## 800203         GTS_1428-24 UGA_173101_P022_WA09 0.500000  101       307
## 800239         GTS_1428-24          GTS_949-23 0.500000  123       307
## 800288         GTS_1428-24          GTS_646-23 0.500000  120       307
## 800302         GTS_1428-24          GTS_616-24 0.491071  121       307
## 800311         GTS_1428-24          GTS_655-23 0.500000  115       307
## 800312         GTS_1428-24          GTS_983-23 0.500000  121       307
## 800323         GTS_1428-24          GTS_630-24 0.500000  119       307
## 800333         GTS_1428-24          GTS_665-23 0.500000  121       307
## 800336         GTS_1428-24          GTS_997-23 0.500000  122       307
## 800342         GTS_1428-24          GTS_644-24 0.490741  119       307
## 800348         GTS_1428-24          GTS_818-23 0.500000  122       307
## 800361         GTS_1428-24          GTS_677-23 0.500000  118       307
## 800362         GTS_1428-24         GTS_1012-23 0.500000  121       307
## 800368         GTS_1428-24          GTS_657-24 0.491071  120       307
## 800387         GTS_1428-24          GTS_691-23 0.500000  118       307
```

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## 800456          GTS_1428-24           GTS_725-23 0.500000  119       307
## 800465          GTS_1428-24           GTS_736-23 0.480769  119       307
## 800569          GTS_1428-24          GTS_1202-24 0.491379  121       307
## 800587          GTS_1428-24           GTS_850-23 0.491379  123       307
## 800621          GTS_1428-24           GTS_825-24 0.490741  119       307
## 800634          GTS_1428-24           GTS_935-23 0.500000  123       307
## 800644          GTS_1428-24          GTS_935-23R 0.500000  122       307
## 800662          GTS_1428-24           GTS_855-24 0.500000  123       307
## 800674          GTS_1428-24          GTS_1344-24 0.500000  123       307
## 800711          GTS_1428-24           GTS_886-24 0.500000  123       307
## 800725          GTS_1428-24          GTS_1403-24 0.500000  122       307
## 800742          GTS_1428-24           GTS_931-24 0.500000  119       307
## 810942          GTS_3012-24 UGA_173101_P019_WE05 0.500000  114       307
## 810968          GTS_3012-24 UGA_173101_P019_WA04 0.500000   91       307
## 811009          GTS_3012-24 UGA_173101_P016_WC12 0.500000  108       307
## 811011          GTS_3012-24 UGA_173101_P017_WC02 0.491071  115       307
## 811018          GTS_3012-24 UGA_173101_P018_WB04 0.500000  116       307
## 811023          GTS_3012-24 UGA_173101_P016_WF07 0.471154  102       307
## 811072          GTS_3012-24 UGA_173101_P019_WG03 0.500000  120       307
## 811078          GTS_3012-24 UGA_173101_P016_WE08 0.500000  115       307
## 811109          GTS_3012-24 UGA_173101_P017_WC12 0.490741  115       307
## 811130          GTS_3012-24 UGA_173101_P019_WA02 0.500000  117       307
## 811159          GTS_3012-24 UGA_173101_P018_WD10 0.500000  110       307
## 811221          GTS_3012-24 UGA_173101_P016_WH09 0.500000  114       307
## 811257          GTS_3012-24 UGA_173101_P017_WD09 0.500000  108       307
## 811277          GTS_3012-24 UGA_173101_P019_WA01 0.500000  119       307
## 811304          GTS_3012-24 UGA_173101_P018_WB06 0.484375   55       307
## 811328          GTS_3012-24 UGA_173101_P018_WC11 0.500000  120       307
## 811341          GTS_3012-24 UGA_173101_P023_WF05 0.490000   93       307
## 811354          GTS_3012-24 UGA_173101_P023_WB12 0.480769  109       307
## 811363          GTS_3012-24 UGA_173101_P023_WA04 0.500000   92       307
## 811418          GTS_3012-24 UGA_173101_P023_WA07 0.490385  106       307
## 811441          GTS_3012-24 UGA_173101_P022_WD02 0.475000   51       307
## 811581          GTS_3012-24 UGA_173101_P021_WF01 0.490000  102       307
## 811624          GTS_3012-24 UGA_173101_P022_WA09 0.500000  101       307
## 811660          GTS_3012-24           GTS_949-23 0.500000  122       307
## 811709          GTS_3012-24           GTS_646-23 0.500000  119       307
## 811723          GTS_3012-24           GTS_616-24 0.490741  120       307
## 811732          GTS_3012-24           GTS_655-23 0.500000  114       307
## 811733          GTS_3012-24           GTS_983-23 0.500000  120       307
## 811744          GTS_3012-24           GTS_630-24 0.500000  118       307
## 811754          GTS_3012-24           GTS_665-23 0.500000  120       307
## 811757          GTS_3012-24           GTS_997-23 0.500000  121       307
## 811763          GTS_3012-24           GTS_644-24 0.490385  118       307
## 811769          GTS_3012-24           GTS_818-23 0.500000  121       307
## 811782          GTS_3012-24           GTS_677-23 0.500000  117       307
## 811783          GTS_3012-24          GTS_1012-23 0.500000  120       307
## 811789          GTS_3012-24           GTS_657-24 0.490741  119       307
## 811808          GTS_3012-24           GTS_691-23 0.500000  117       307
## 811877          GTS_3012-24           GTS_725-23 0.500000  118       307
## 811886          GTS_3012-24           GTS_736-23 0.480000  118       307
## 811990          GTS_3012-24          GTS_1202-24 0.491071  120       307
## 812008          GTS_3012-24           GTS_850-23 0.491071  122       307
## 812042          GTS_3012-24           GTS_825-24 0.490385  118       307
```

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## 812055          GTS_3012-24           GTS_935-23 0.500000  122       307
## 812065          GTS_3012-24          GTS_935-23R 0.500000  121       307
## 812083          GTS_3012-24           GTS_855-24 0.500000  122       307
## 812095          GTS_3012-24          GTS_1344-24 0.500000  122       307
## 812132          GTS_3012-24           GTS_886-24 0.500000  122       307
## 812146          GTS_3012-24          GTS_1403-24 0.500000  121       307
## 812163          GTS_3012-24           GTS_931-24 0.500000  118       307
## 812167          GTS_3012-24          GTS_1428-24 0.500000  122       307
## 831446          GTS_1451-24 UGA_173101_P019_WE05 0.500000  114       307
## 831472          GTS_1451-24 UGA_173101_P019_WA04 0.500000   91       307
## 831513          GTS_1451-24 UGA_173101_P016_WC12 0.500000  109       307
## 831515          GTS_1451-24 UGA_173101_P017_WC02 0.491379  116       307
## 831522          GTS_1451-24 UGA_173101_P018_WB04 0.500000  117       307
## 831527          GTS_1451-24 UGA_173101_P016_WF07 0.471154  102       307
## 831576          GTS_1451-24 UGA_173101_P019_WG03 0.500000  121       307
## 831582          GTS_1451-24 UGA_173101_P016_WE08 0.500000  116       307
## 831613          GTS_1451-24 UGA_173101_P017_WC12 0.491071  116       307
## 831634          GTS_1451-24 UGA_173101_P019_WA02 0.500000  118       307
## 831663          GTS_1451-24 UGA_173101_P018_WD10 0.500000  110       307
## 831725          GTS_1451-24 UGA_173101_P016_WH09 0.500000  114       307
## 831761          GTS_1451-24 UGA_173101_P017_WD09 0.500000  108       307
## 831781          GTS_1451-24 UGA_173101_P019_WA01 0.500000  120       307
## 831808          GTS_1451-24 UGA_173101_P018_WB06 0.484375   55       307
## 831832          GTS_1451-24 UGA_173101_P018_WC11 0.500000  121       307
## 831845          GTS_1451-24 UGA_173101_P023_WF05 0.490000   93       307
## 831858          GTS_1451-24 UGA_173101_P023_WB12 0.481481  110       307
## 831867          GTS_1451-24 UGA_173101_P023_WA04 0.500000   92       307
## 831922          GTS_1451-24 UGA_173101_P023_WA07 0.490741  107       307
## 831945          GTS_1451-24 UGA_173101_P022_WD02 0.475000   51       307
## 832085          GTS_1451-24 UGA_173101_P021_WF01 0.490385  103       307
## 832128          GTS_1451-24 UGA_173101_P022_WA09 0.500000  101       307
## 832164          GTS_1451-24           GTS_949-23 0.500000  123       307
## 832213          GTS_1451-24           GTS_646-23 0.500000  120       307
## 832227          GTS_1451-24           GTS_616-24 0.491071  121       307
## 832236          GTS_1451-24           GTS_655-23 0.500000  115       307
## 832237          GTS_1451-24           GTS_983-23 0.500000  121       307
## 832248          GTS_1451-24           GTS_630-24 0.500000  119       307
## 832258          GTS_1451-24           GTS_665-23 0.500000  121       307
## 832261          GTS_1451-24           GTS_997-23 0.500000  122       307
## 832267          GTS_1451-24           GTS_644-24 0.490741  119       307
## 832273          GTS_1451-24           GTS_818-23 0.500000  122       307
## 832286          GTS_1451-24           GTS_677-23 0.500000  118       307
## 832287          GTS_1451-24          GTS_1012-23 0.500000  121       307
## 832293          GTS_1451-24           GTS_657-24 0.491071  120       307
## 832312          GTS_1451-24           GTS_691-23 0.500000  118       307
## 832381          GTS_1451-24           GTS_725-23 0.500000  119       307
## 832390          GTS_1451-24           GTS_736-23 0.480769  119       307
## 832494          GTS_1451-24          GTS_1202-24 0.491379  121       307
## 832512          GTS_1451-24           GTS_850-23 0.491379  123       307
## 832546          GTS_1451-24           GTS_825-24 0.490741  119       307
## 832559          GTS_1451-24           GTS_935-23 0.500000  123       307
## 832569          GTS_1451-24          GTS_935-23R 0.500000  122       307
## 832587          GTS_1451-24           GTS_855-24 0.500000  123       307
## 832599          GTS_1451-24          GTS_1344-24 0.500000  123       307
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## 832636          GTS_1451-24           GTS_886-24 0.500000  123       307
## 832650          GTS_1451-24          GTS_1403-24 0.500000  122       307
## 832667          GTS_1451-24           GTS_931-24 0.500000  119       307
## 832671          GTS_1451-24          GTS_1428-24 0.500000  123       307
## 832680          GTS_1451-24          GTS_3012-24 0.500000  122       307
## 843092           GTS_955-24 UGA_173101_P019_WE05 0.500000  113       307
## 843118           GTS_955-24 UGA_173101_P019_WA04 0.500000   90       307
## 843159           GTS_955-24 UGA_173101_P016_WC12 0.500000  108       307
## 843161           GTS_955-24 UGA_173101_P017_WC02 0.491071  115       307
## 843168           GTS_955-24 UGA_173101_P018_WB04 0.500000  116       307
## 843173           GTS_955-24 UGA_173101_P016_WF07 0.470000  101       307
## 843222           GTS_955-24 UGA_173101_P019_WG03 0.500000  120       307
## 843228           GTS_955-24 UGA_173101_P016_WE08 0.500000  115       307
## 843259           GTS_955-24 UGA_173101_P017_WC12 0.490741  115       307
## 843280           GTS_955-24 UGA_173101_P019_WA02 0.500000  117       307
## 843309           GTS_955-24 UGA_173101_P018_WD10 0.500000  109       307
## 843371           GTS_955-24 UGA_173101_P016_WH09 0.500000  113       307
## 843407           GTS_955-24 UGA_173101_P017_WD09 0.500000  107       307
## 843427           GTS_955-24 UGA_173101_P019_WA01 0.500000  119       307
## 843454           GTS_955-24 UGA_173101_P018_WB06 0.483333   54       307
## 843478           GTS_955-24 UGA_173101_P018_WC11 0.500000  120       307
## 843491           GTS_955-24 UGA_173101_P023_WF05 0.489583   92       307
## 843504           GTS_955-24 UGA_173101_P023_WB12 0.480769  109       307
## 843513           GTS_955-24 UGA_173101_P023_WA04 0.500000   91       307
## 843568           GTS_955-24 UGA_173101_P023_WA07 0.490385  106       307
## 843591           GTS_955-24 UGA_173101_P022_WD02 0.472222   50       307
## 843731           GTS_955-24 UGA_173101_P021_WF01 0.490000  102       307
## 843774           GTS_955-24 UGA_173101_P022_WA09 0.500000  100       307
## 843810           GTS_955-24           GTS_949-23 0.500000  122       307
## 843859           GTS_955-24           GTS_646-23 0.500000  119       307
## 843873           GTS_955-24           GTS_616-24 0.490741  120       307
## 843882           GTS_955-24           GTS_655-23 0.500000  114       307
## 843883           GTS_955-24           GTS_983-23 0.500000  121       307
## 843894           GTS_955-24           GTS_630-24 0.500000  118       307
## 843904           GTS_955-24           GTS_665-23 0.500000  120       307
## 843907           GTS_955-24           GTS_997-23 0.500000  121       307
## 843913           GTS_955-24           GTS_644-24 0.490385  118       307
## 843919           GTS_955-24           GTS_818-23 0.500000  121       307
## 843932           GTS_955-24           GTS_677-23 0.500000  118       307
## 843933           GTS_955-24          GTS_1012-23 0.500000  121       307
## 843939           GTS_955-24           GTS_657-24 0.490741  119       307
## 843958           GTS_955-24           GTS_691-23 0.500000  118       307
## 844027           GTS_955-24           GTS_725-23 0.500000  118       307
## 844036           GTS_955-24           GTS_736-23 0.480769  119       307
## 844140           GTS_955-24          GTS_1202-24 0.500000  120       307
## 844158           GTS_955-24           GTS_850-23 0.500000  122       307
## 844192           GTS_955-24           GTS_825-24 0.500000  118       307
## 844205           GTS_955-24           GTS_935-23 0.500000  122       307
## 844215           GTS_955-24          GTS_935-23R 0.500000  122       307
## 844233           GTS_955-24           GTS_855-24 0.500000  122       307
## 844245           GTS_955-24          GTS_1344-24 0.500000  122       307
## 844282           GTS_955-24           GTS_886-24 0.500000  122       307
## 844296           GTS_955-24          GTS_1403-24 0.500000  122       307
## 844313           GTS_955-24           GTS_931-24 0.500000  119       307
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## 844317            GTS_955-24          GTS_1428-24 0.500000  122        307
## 844326            GTS_955-24          GTS_3012-24 0.500000  121        307
## 844342            GTS_955-24          GTS_1451-24 0.500000  122        307
## 878516           GTS_1516-24 UGA_173101_P019_WE05 0.480000  112        326
## 878542           GTS_1516-24 UGA_173101_P019_WA04 0.476190   90        326
## 878583           GTS_1516-24 UGA_173101_P016_WC12 0.481481  107        326
## 878585           GTS_1516-24 UGA_173101_P017_WC02 0.472222  114        326
## 878592           GTS_1516-24 UGA_173101_P018_WB04 0.480769  115        326
## 878597           GTS_1516-24 UGA_173101_P016_WF07 0.447917  100        326
## 878646           GTS_1516-24 UGA_173101_P019_WG03 0.481481  118        326
## 878652           GTS_1516-24 UGA_173101_P016_WE08 0.481481  114        326
## 878683           GTS_1516-24 UGA_173101_P017_WC12 0.471154  114        326
## 878704           GTS_1516-24 UGA_173101_P019_WA02 0.481481  116        326
## 878733           GTS_1516-24 UGA_173101_P018_WD10 0.480769  107        326
## 878795           GTS_1516-24 UGA_173101_P016_WH09 0.480769  112        326
## 878831           GTS_1516-24 UGA_173101_P017_WD09 0.480000  106        326
## 878851           GTS_1516-24 UGA_173101_P019_WA01 0.481481  117        326
## 878878           GTS_1516-24 UGA_173101_P018_WB06 0.464286   54        326
## 878902           GTS_1516-24 UGA_173101_P018_WC11 0.481481  118        326
## 878915           GTS_1516-24 UGA_173101_P023_WF05 0.467391   92        326
## 878928           GTS_1516-24 UGA_173101_P023_WB12 0.460000  107        326
## 878937           GTS_1516-24 UGA_173101_P023_WA04 0.476190   91        326
## 878992           GTS_1516-24 UGA_173101_P023_WA07 0.480769  105        326
## 879015           GTS_1516-24 UGA_173101_P022_WD02 0.444444   51        326
## 879155           GTS_1516-24 UGA_173101_P021_WF01 0.470000  102        326
## 879198           GTS_1516-24 UGA_173101_P022_WA09 0.478261   98        326
## 879234           GTS_1516-24           GTS_949-23 0.491071  120        326
## 879283           GTS_1516-24           GTS_646-23 0.500000  116        326
## 879297           GTS_1516-24           GTS_616-24 0.490741  119        326
## 879306           GTS_1516-24           GTS_655-23 0.500000  113        326
## 879307           GTS_1516-24           GTS_983-23 0.490385  118        326
## 879318           GTS_1516-24           GTS_630-24 0.500000  117        326
## 879328           GTS_1516-24           GTS_665-23 0.490741  119        326
## 879331           GTS_1516-24           GTS_997-23 0.490741  119        326
## 879337           GTS_1516-24           GTS_644-24 0.490385  118        326
## 879343           GTS_1516-24           GTS_818-23 0.490741  119        326
## 879356           GTS_1516-24           GTS_677-23 0.500000  118        326
## 879357           GTS_1516-24          GTS_1012-23 0.490385  118        326
## 879363           GTS_1516-24           GTS_657-24 0.500000  118        326
## 879382           GTS_1516-24           GTS_691-23 0.480769  118        326
## 879451           GTS_1516-24           GTS_725-23 0.500000  117        326
## 879460           GTS_1516-24           GTS_736-23 0.500000  119        326
## 879564           GTS_1516-24          GTS_1202-24 0.482143  118        326
## 879582           GTS_1516-24           GTS_850-23 0.491071  120        326
## 879616           GTS_1516-24           GTS_825-24 0.500000  119        326
## 879629           GTS_1516-24           GTS_935-23 0.490741  119        326
## 879639           GTS_1516-24          GTS_935-23R 0.482143  120        326
## 879657           GTS_1516-24           GTS_855-24 0.490741  119        326
## 879669           GTS_1516-24          GTS_1344-24 0.490741  119        326
## 879706           GTS_1516-24           GTS_886-24 0.491071  120        326
## 879720           GTS_1516-24          GTS_1403-24 0.490741  119        326
## 879737           GTS_1516-24           GTS_931-24 0.500000  118        326
## 879741           GTS_1516-24          GTS_1428-24 0.490741  119        326
## 879750           GTS_1516-24          GTS_3012-24 0.490385  118        326
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## 879766          GTS_1516-24          GTS_1451-24 0.490741  119       326
## 879775          GTS_1516-24           GTS_955-24 0.490741  119       326
## 928244          GTS_2777-24 UGA_173101_P019_WE05 0.500000  114       307
## 928270          GTS_2777-24 UGA_173101_P019_WA04 0.500000   91       307
## 928311          GTS_2777-24 UGA_173101_P016_WC12 0.500000  109       307
## 928313          GTS_2777-24 UGA_173101_P017_WC02 0.491379  116       307
## 928320          GTS_2777-24 UGA_173101_P018_WB04 0.500000  117       307
## 928325          GTS_2777-24 UGA_173101_P016_WF07 0.471154  102       307
## 928374          GTS_2777-24 UGA_173101_P019_WG03 0.500000  120       307
## 928380          GTS_2777-24 UGA_173101_P016_WE08 0.500000  116       307
## 928411          GTS_2777-24 UGA_173101_P017_WC12 0.491071  116       307
## 928432          GTS_2777-24 UGA_173101_P019_WA02 0.500000  118       307
## 928461          GTS_2777-24 UGA_173101_P018_WD10 0.500000  109       307
## 928523          GTS_2777-24 UGA_173101_P016_WH09 0.500000  114       307
## 928559          GTS_2777-24 UGA_173101_P017_WD09 0.500000  108       307
## 928579          GTS_2777-24 UGA_173101_P019_WA01 0.500000  119       307
## 928606          GTS_2777-24 UGA_173101_P018_WB06 0.484375   55       307
## 928630          GTS_2777-24 UGA_173101_P018_WC11 0.500000  120       307
## 928643          GTS_2777-24 UGA_173101_P023_WF05 0.490000   93       307
## 928656          GTS_2777-24 UGA_173101_P023_WB12 0.481481  109       307
## 928665          GTS_2777-24 UGA_173101_P023_WA04 0.500000   92       307
## 928720          GTS_2777-24 UGA_173101_P023_WA07 0.490741  107       307
## 928743          GTS_2777-24 UGA_173101_P022_WD02 0.475000   51       307
## 928883          GTS_2777-24 UGA_173101_P021_WF01 0.490385  102       307
## 928926          GTS_2777-24 UGA_173101_P022_WA09 0.500000  100       307
## 928962          GTS_2777-24           GTS_949-23 0.500000  121       307
## 929011          GTS_2777-24           GTS_646-23 0.500000  119       307
## 929025          GTS_2777-24           GTS_616-24 0.500000  120       307
## 929034          GTS_2777-24           GTS_655-23 0.500000  115       307
## 929035          GTS_2777-24           GTS_983-23 0.500000  120       307
## 929046          GTS_2777-24           GTS_630-24 0.500000  119       307
## 929056          GTS_2777-24           GTS_665-23 0.500000  121       307
## 929059          GTS_2777-24           GTS_997-23 0.500000  121       307
## 929065          GTS_2777-24           GTS_644-24 0.500000  118       307
## 929071          GTS_2777-24           GTS_818-23 0.500000  121       307
## 929084          GTS_2777-24           GTS_677-23 0.500000  118       307
## 929085          GTS_2777-24          GTS_1012-23 0.500000  120       307
## 929091          GTS_2777-24           GTS_657-24 0.491071  120       307
## 929110          GTS_2777-24           GTS_691-23 0.500000  118       307
## 929179          GTS_2777-24           GTS_725-23 0.500000  119       307
## 929188          GTS_2777-24           GTS_736-23 0.480769  119       307
## 929292          GTS_2777-24          GTS_1202-24 0.491071  119       307
## 929310          GTS_2777-24           GTS_850-23 0.491071  121       307
## 929344          GTS_2777-24           GTS_825-24 0.490385  118       307
## 929357          GTS_2777-24           GTS_935-23 0.500000  121       307
## 929367          GTS_2777-24          GTS_935-23R 0.500000  120       307
## 929385          GTS_2777-24           GTS_855-24 0.500000  121       307
## 929397          GTS_2777-24          GTS_1344-24 0.500000  121       307
## 929434          GTS_2777-24           GTS_886-24 0.500000  121       307
## 929448          GTS_2777-24          GTS_1403-24 0.500000  120       307
## 929465          GTS_2777-24           GTS_931-24 0.500000  118       307
## 929469          GTS_2777-24          GTS_1428-24 0.500000  121       307
## 929478          GTS_2777-24          GTS_3012-24 0.500000  120       307
## 929494          GTS_2777-24          GTS_1451-24 0.500000  121       307
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## 929503          GTS_2777-24           GTS_955-24 0.500000  120        307
## 929530          GTS_2777-24          GTS_1516-24 0.490385  118        307
## 968177          GTS_1633-24 UGA_173101_P019_WE05 0.451923  115        339
## 968203          GTS_1633-24 UGA_173101_P019_WA04 0.443182   92        339
## 968244          GTS_1633-24 UGA_173101_P016_WC12 0.455357  110        339
## 968246          GTS_1633-24 UGA_173101_P017_WC02 0.446429  117        339
## 968253          GTS_1633-24 UGA_173101_P018_WB04 0.453704  118        339
## 968258          GTS_1633-24 UGA_173101_P016_WF07 0.420000  103        339
## 968307          GTS_1633-24 UGA_173101_P019_WG03 0.455357  122        339
## 968313          GTS_1633-24 UGA_173101_P016_WE08 0.455357  117        339
## 968344          GTS_1633-24 UGA_173101_P017_WC12 0.446429  117        339
## 968365          GTS_1633-24 UGA_173101_P019_WA02 0.455357  119        339
## 968394          GTS_1633-24 UGA_173101_P018_WD10 0.453704  111        339
## 968456          GTS_1633-24 UGA_173101_P016_WH09 0.453704  115        339
## 968492          GTS_1633-24 UGA_173101_P017_WD09 0.451923  109        339
## 968512          GTS_1633-24 UGA_173101_P019_WA01 0.455357  121        339
## 968539          GTS_1633-24 UGA_173101_P018_WB06 0.406250   56        339
## 968563          GTS_1633-24 UGA_173101_P018_WC11 0.455357  122        339
## 968576          GTS_1633-24 UGA_173101_P023_WF05 0.437500   94        339
## 968589          GTS_1633-24 UGA_173101_P023_WB12 0.432692  111        339
## 968598          GTS_1633-24 UGA_173101_P023_WA04 0.443182   93        339
## 968653          GTS_1633-24 UGA_173101_P023_WA07 0.444444  108        339
## 968676          GTS_1633-24 UGA_173101_P022_WD02 0.350000   52        339
## 968816          GTS_1633-24 UGA_173101_P021_WF01 0.440000  104        339
## 968859          GTS_1633-24 UGA_173101_P022_WA09 0.447917  102        339
## 968895          GTS_1633-24           GTS_949-23 0.482759  124        339
## 968944          GTS_1633-24           GTS_646-23 0.490385  120        339
## 968958          GTS_1633-24           GTS_616-24 0.473214  122        339
## 968967          GTS_1633-24           GTS_655-23 0.479167  116        339
## 968968          GTS_1633-24           GTS_983-23 0.500000  121        339
## 968979          GTS_1633-24           GTS_630-24 0.490385  119        339
## 968989          GTS_1633-24           GTS_665-23 0.482143  122        339
## 968992          GTS_1633-24           GTS_997-23 0.482143  123        339
## 968998          GTS_1633-24           GTS_644-24 0.481481  119        339
## 969004          GTS_1633-24           GTS_818-23 0.482143  123        339
## 969017          GTS_1633-24           GTS_677-23 0.490385  119        339
## 969018          GTS_1633-24          GTS_1012-23 0.500000  121        339
## 969024          GTS_1633-24           GTS_657-24 0.482143  120        339
## 969043          GTS_1633-24           GTS_691-23 0.500000  119        339
## 969112          GTS_1633-24           GTS_725-23 0.490385  119        339
## 969121          GTS_1633-24           GTS_736-23 0.472222  120        339
## 969225          GTS_1633-24          GTS_1202-24 0.500000  122        339
## 969243          GTS_1633-24           GTS_850-23 0.491379  124        339
## 969277          GTS_1633-24           GTS_825-24 0.490741  120        339
## 969290          GTS_1633-24           GTS_935-23 0.491379  123        339
## 969300          GTS_1633-24          GTS_935-23R 0.500000  123        339
## 969318          GTS_1633-24           GTS_855-24 0.491379  123        339
## 969330          GTS_1633-24          GTS_1344-24 0.491379  123        339
## 969367          GTS_1633-24           GTS_886-24 0.482759  124        339
## 969381          GTS_1633-24          GTS_1403-24 0.500000  122        339
## 969398          GTS_1633-24           GTS_931-24 0.500000  119        339
## 969402          GTS_1633-24          GTS_1428-24 0.491379  123        339
## 969411          GTS_1633-24          GTS_3012-24 0.491071  122        339
## 969427          GTS_1633-24          GTS_1451-24 0.491379  123        339
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## 969436          GTS_1633-24           GTS_955-24 0.500000  122       339
## 969463          GTS_1633-24          GTS_1516-24 0.482143  120       339
## 969500          GTS_1633-24          GTS_2777-24 0.491071  121       339
## 980741          GTS_1672-24 UGA_173101_P019_WE05 0.500000  113       307
## 980767          GTS_1672-24 UGA_173101_P019_WA04 0.500000   90       307
## 980808          GTS_1672-24 UGA_173101_P016_WC12 0.500000  108       307
## 980810          GTS_1672-24 UGA_173101_P017_WC02 0.491071  115       307
## 980817          GTS_1672-24 UGA_173101_P018_WB04 0.500000  116       307
## 980822          GTS_1672-24 UGA_173101_P016_WF07 0.470000  101       307
## 980871          GTS_1672-24 UGA_173101_P019_WG03 0.500000  119       307
## 980877          GTS_1672-24 UGA_173101_P016_WE08 0.500000  115       307
## 980908          GTS_1672-24 UGA_173101_P017_WC12 0.490741  115       307
## 980929          GTS_1672-24 UGA_173101_P019_WA02 0.500000  117       307
## 980958          GTS_1672-24 UGA_173101_P018_WD10 0.500000  108       307
## 981020          GTS_1672-24 UGA_173101_P016_WH09 0.500000  113       307
## 981056          GTS_1672-24 UGA_173101_P017_WD09 0.500000  107       307
## 981076          GTS_1672-24 UGA_173101_P019_WA01 0.500000  118       307
## 981103          GTS_1672-24 UGA_173101_P018_WB06 0.483333   54       307
## 981127          GTS_1672-24 UGA_173101_P018_WC11 0.500000  119       307
## 981140          GTS_1672-24 UGA_173101_P023_WF05 0.489583   92       307
## 981153          GTS_1672-24 UGA_173101_P023_WB12 0.480769  108       307
## 981162          GTS_1672-24 UGA_173101_P023_WA04 0.500000   91       307
## 981217          GTS_1672-24 UGA_173101_P023_WA07 0.490385  106       307
## 981240          GTS_1672-24 UGA_173101_P022_WD02 0.472222   50       307
## 981380          GTS_1672-24 UGA_173101_P021_WF01 0.490000  101       307
## 981423          GTS_1672-24 UGA_173101_P022_WA09 0.500000   99       307
## 981459          GTS_1672-24           GTS_949-23 0.500000  121       307
## 981508          GTS_1672-24           GTS_646-23 0.500000  118       307
## 981522          GTS_1672-24           GTS_616-24 0.490741  120       307
## 981531          GTS_1672-24           GTS_655-23 0.500000  114       307
## 981532          GTS_1672-24           GTS_983-23 0.500000  120       307
## 981543          GTS_1672-24           GTS_630-24 0.500000  118       307
## 981553          GTS_1672-24           GTS_665-23 0.500000  120       307
## 981556          GTS_1672-24           GTS_997-23 0.500000  120       307
## 981562          GTS_1672-24           GTS_644-24 0.490385  118       307
## 981568          GTS_1672-24           GTS_818-23 0.500000  120       307
## 981581          GTS_1672-24           GTS_677-23 0.500000  118       307
## 981582          GTS_1672-24          GTS_1012-23 0.500000  120       307
## 981588          GTS_1672-24           GTS_657-24 0.490741  119       307
## 981607          GTS_1672-24           GTS_691-23 0.500000  118       307
## 981676          GTS_1672-24           GTS_725-23 0.500000  118       307
## 981685          GTS_1672-24           GTS_736-23 0.480769  119       307
## 981789          GTS_1672-24          GTS_1202-24 0.500000  119       307
## 981807          GTS_1672-24           GTS_850-23 0.500000  121       307
## 981841          GTS_1672-24           GTS_825-24 0.500000  118       307
## 981854          GTS_1672-24           GTS_935-23 0.500000  121       307
## 981864          GTS_1672-24          GTS_935-23R 0.500000  121       307
## 981882          GTS_1672-24           GTS_855-24 0.500000  121       307
## 981894          GTS_1672-24          GTS_1344-24 0.500000  121       307
## 981931          GTS_1672-24           GTS_886-24 0.500000  121       307
## 981945          GTS_1672-24          GTS_1403-24 0.500000  121       307
## 981962          GTS_1672-24           GTS_931-24 0.500000  119       307
## 981966          GTS_1672-24          GTS_1428-24 0.500000  121       307
## 981975          GTS_1672-24          GTS_3012-24 0.500000  120       307
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## 981991          GTS_1672-24          GTS_1451-24 0.500000  121       307
## 982000          GTS_1672-24           GTS_955-24 0.500000  121       307
## 982027          GTS_1672-24          GTS_1516-24 0.490741  119       307
## 982064          GTS_1672-24          GTS_2777-24 0.500000  120       307
## 982093          GTS_1672-24          GTS_1633-24 0.500000  121       307
## 1013217         GTS_2376-24 UGA_173101_P019_WE05 0.490385  114       307
## 1013243         GTS_2376-24 UGA_173101_P019_WA04 0.488636   91       307
## 1013284         GTS_2376-24 UGA_173101_P016_WC12 0.491071  109       307
## 1013286         GTS_2376-24 UGA_173101_P017_WC02 0.482759  116       307
## 1013293         GTS_2376-24 UGA_173101_P018_WB04 0.490741  117       307
## 1013298         GTS_2376-24 UGA_173101_P016_WF07 0.461538  102       307
## 1013347         GTS_2376-24 UGA_173101_P019_WG03 0.491071  121       307
## 1013353         GTS_2376-24 UGA_173101_P016_WE08 0.491071  116       307
## 1013384         GTS_2376-24 UGA_173101_P017_WC12 0.481481  116       307
## 1013405         GTS_2376-24 UGA_173101_P019_WA02 0.491071  118       307
## 1013434         GTS_2376-24 UGA_173101_P018_WD10 0.490741  110       307
## 1013496         GTS_2376-24 UGA_173101_P016_WH09 0.490741  114       307
## 1013532         GTS_2376-24 UGA_173101_P017_WD09 0.490385  108       307
## 1013552         GTS_2376-24 UGA_173101_P019_WA01 0.491071  120       307
## 1013579         GTS_2376-24 UGA_173101_P018_WB06 0.466667   55       307
## 1013603         GTS_2376-24 UGA_173101_P018_WC11 0.491071  121       307
## 1013616         GTS_2376-24 UGA_173101_P023_WF05 0.480000   93       307
## 1013629         GTS_2376-24 UGA_173101_P023_WB12 0.472222  110       307
## 1013638         GTS_2376-24 UGA_173101_P023_WA04 0.488636   92       307
## 1013693         GTS_2376-24 UGA_173101_P023_WA07 0.480769  107       307
## 1013716         GTS_2376-24 UGA_173101_P022_WD02 0.444444   51       307
## 1013856         GTS_2376-24 UGA_173101_P021_WF01 0.480769  103       307
## 1013899         GTS_2376-24 UGA_173101_P022_WA09 0.489583  101       307
## 1013935         GTS_2376-24           GTS_949-23 0.500000  123       307
## 1013984         GTS_2376-24           GTS_646-23 0.500000  119       307
## 1013998         GTS_2376-24           GTS_616-24 0.491071  121       307
## 1014007         GTS_2376-24           GTS_655-23 0.500000  115       307
## 1014008         GTS_2376-24           GTS_983-23 0.500000  121       307
## 1014019         GTS_2376-24           GTS_630-24 0.500000  118       307
## 1014029         GTS_2376-24           GTS_665-23 0.500000  121       307
## 1014032         GTS_2376-24           GTS_997-23 0.500000  122       307
## 1014038         GTS_2376-24           GTS_644-24 0.490385  118       307
## 1014044         GTS_2376-24           GTS_818-23 0.500000  122       307
## 1014057         GTS_2376-24           GTS_677-23 0.500000  119       307
## 1014058         GTS_2376-24          GTS_1012-23 0.500000  121       307
## 1014064         GTS_2376-24           GTS_657-24 0.490741  119       307
## 1014083         GTS_2376-24           GTS_691-23 0.490385  119       307
## 1014152         GTS_2376-24           GTS_725-23 0.500000  118       307
## 1014161         GTS_2376-24           GTS_736-23 0.481481  120       307
## 1014265         GTS_2376-24          GTS_1202-24 0.491379  121       307
## 1014283         GTS_2376-24           GTS_850-23 0.500000  123       307
## 1014317         GTS_2376-24           GTS_825-24 0.500000  119       307
## 1014330         GTS_2376-24           GTS_935-23 0.500000  122       307
## 1014340         GTS_2376-24          GTS_935-23R 0.491379  123       307
## 1014358         GTS_2376-24           GTS_855-24 0.500000  122       307
## 1014370         GTS_2376-24          GTS_1344-24 0.500000  122       307
## 1014407         GTS_2376-24           GTS_886-24 0.500000  123       307
## 1014421         GTS_2376-24          GTS_1403-24 0.500000  122       307
## 1014438         GTS_2376-24           GTS_931-24 0.500000  119       307
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## 1014442          GTS_2376-24          GTS_1428-24 0.500000  122        307
## 1014451          GTS_2376-24          GTS_3012-24 0.500000  121        307
## 1014467          GTS_2376-24          GTS_1451-24 0.500000  122        307
## 1014476          GTS_2376-24           GTS_955-24 0.500000  122        307
## 1014503          GTS_2376-24          GTS_1516-24 0.491071  120        307
## 1014540          GTS_2376-24          GTS_2777-24 0.500000  120        307
## 1014569          GTS_2376-24          GTS_1633-24 0.491379  123        307
## 1014578          GTS_2376-24          GTS_1672-24 0.500000  121        307
## 1044776          GTS_2669-24 UGA_173101_P019_WE05 0.500000  114        307
## 1044802          GTS_2669-24 UGA_173101_P019_WA04 0.500000   91        307
## 1044843          GTS_2669-24 UGA_173101_P016_WC12 0.500000  109        307
## 1044845          GTS_2669-24 UGA_173101_P017_WC02 0.491379  116        307
## 1044852          GTS_2669-24 UGA_173101_P018_WB04 0.500000  117        307
## 1044857          GTS_2669-24 UGA_173101_P016_WF07 0.471154  102        307
## 1044906          GTS_2669-24 UGA_173101_P019_WG03 0.500000  121        307
## 1044912          GTS_2669-24 UGA_173101_P016_WE08 0.500000  116        307
## 1044943          GTS_2669-24 UGA_173101_P017_WC12 0.491071  116        307
## 1044964          GTS_2669-24 UGA_173101_P019_WA02 0.500000  118        307
## 1044993          GTS_2669-24 UGA_173101_P018_WD10 0.500000  110        307
## 1045055          GTS_2669-24 UGA_173101_P016_WH09 0.500000  114        307
## 1045091          GTS_2669-24 UGA_173101_P017_WD09 0.500000  108        307
## 1045111          GTS_2669-24 UGA_173101_P019_WA01 0.500000  120        307
## 1045138          GTS_2669-24 UGA_173101_P018_WB06 0.484375   55        307
## 1045162          GTS_2669-24 UGA_173101_P018_WC11 0.500000  121        307
## 1045175          GTS_2669-24 UGA_173101_P023_WF05 0.490000   93        307
## 1045188          GTS_2669-24 UGA_173101_P023_WB12 0.481481  110        307
## 1045197          GTS_2669-24 UGA_173101_P023_WA04 0.500000   92        307
## 1045252          GTS_2669-24 UGA_173101_P023_WA07 0.490741  107        307
## 1045275          GTS_2669-24 UGA_173101_P022_WD02 0.475000   51        307
## 1045415          GTS_2669-24 UGA_173101_P021_WF01 0.490385  103        307
## 1045458          GTS_2669-24 UGA_173101_P022_WA09 0.500000  101        307
## 1045494          GTS_2669-24           GTS_949-23 0.500000  123        307
## 1045543          GTS_2669-24           GTS_646-23 0.500000  120        307
## 1045557          GTS_2669-24           GTS_616-24 0.491071  121        307
## 1045566          GTS_2669-24           GTS_655-23 0.500000  115        307
## 1045567          GTS_2669-24           GTS_983-23 0.500000  121        307
## 1045578          GTS_2669-24           GTS_630-24 0.500000  119        307
## 1045588          GTS_2669-24           GTS_665-23 0.500000  121        307
## 1045591          GTS_2669-24           GTS_997-23 0.500000  122        307
## 1045597          GTS_2669-24           GTS_644-24 0.490741  119        307
## 1045603          GTS_2669-24           GTS_818-23 0.500000  122        307
## 1045616          GTS_2669-24           GTS_677-23 0.500000  118        307
## 1045617          GTS_2669-24          GTS_1012-23 0.500000  121        307
## 1045623          GTS_2669-24           GTS_657-24 0.491071  120        307
## 1045642          GTS_2669-24           GTS_691-23 0.500000  118        307
## 1045711          GTS_2669-24           GTS_725-23 0.500000  119        307
## 1045720          GTS_2669-24           GTS_736-23 0.480769  119        307
## 1045824          GTS_2669-24          GTS_1202-24 0.491379  121        307
## 1045842          GTS_2669-24           GTS_850-23 0.491379  123        307
## 1045876          GTS_2669-24           GTS_825-24 0.490741  119        307
## 1045889          GTS_2669-24           GTS_935-23 0.500000  123        307
## 1045899          GTS_2669-24          GTS_935-23R 0.500000  122        307
## 1045917          GTS_2669-24           GTS_855-24 0.500000  123        307
## 1045929          GTS_2669-24          GTS_1344-24 0.500000  123        307
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## 1045966          GTS_2669-24           GTS_886-24 0.500000  123        307
## 1045980          GTS_2669-24          GTS_1403-24 0.500000  122        307
## 1045997          GTS_2669-24           GTS_931-24 0.500000  119        307
## 1046001          GTS_2669-24          GTS_1428-24 0.500000  123        307
## 1046010          GTS_2669-24          GTS_3012-24 0.500000  122        307
## 1046026          GTS_2669-24          GTS_1451-24 0.500000  123        307
## 1046035          GTS_2669-24           GTS_955-24 0.500000  122        307
## 1046062          GTS_2669-24          GTS_1516-24 0.490741  119        307
## 1046099          GTS_2669-24          GTS_2777-24 0.500000  121        307
## 1046128          GTS_2669-24          GTS_1633-24 0.491379  123        307
## 1046137          GTS_2669-24          GTS_1672-24 0.500000  121        307
## 1046160          GTS_2669-24          GTS_2376-24 0.500000  122        307
## 1056372          GTS_2857-24 UGA_173101_P019_WE05 0.490741  115        307
## 1056398          GTS_2857-24 UGA_173101_P019_WA04 0.489130   92        307
## 1056439          GTS_2857-24 UGA_173101_P016_WC12 0.491379  110        307
## 1056441          GTS_2857-24 UGA_173101_P017_WC02 0.483333  117        307
## 1056448          GTS_2857-24 UGA_173101_P018_WB04 0.491071  118        307
## 1056453          GTS_2857-24 UGA_173101_P016_WF07 0.462963  103        307
## 1056502          GTS_2857-24 UGA_173101_P019_WG03 0.491379  122        307
## 1056508          GTS_2857-24 UGA_173101_P016_WE08 0.491379  117        307
## 1056539          GTS_2857-24 UGA_173101_P017_WC12 0.482143  117        307
## 1056560          GTS_2857-24 UGA_173101_P019_WA02 0.491379  119        307
## 1056589          GTS_2857-24 UGA_173101_P018_WD10 0.491071  111        307
## 1056651          GTS_2857-24 UGA_173101_P016_WH09 0.491071  115        307
## 1056687          GTS_2857-24 UGA_173101_P017_WD09 0.490741  109        307
## 1056707          GTS_2857-24 UGA_173101_P019_WA01 0.491379  121        307
## 1056734          GTS_2857-24 UGA_173101_P018_WB06 0.468750   56        307
## 1056758          GTS_2857-24 UGA_173101_P018_WC11 0.491379  122        307
## 1056771          GTS_2857-24 UGA_173101_P023_WF05 0.480769   94        307
## 1056784          GTS_2857-24 UGA_173101_P023_WB12 0.473214  111        307
## 1056793          GTS_2857-24 UGA_173101_P023_WA04 0.489130   93        307
## 1056848          GTS_2857-24 UGA_173101_P023_WA07 0.481481  108        307
## 1056871          GTS_2857-24 UGA_173101_P022_WD02 0.450000   52        307
## 1057011          GTS_2857-24 UGA_173101_P021_WF01 0.481481  104        307
## 1057054          GTS_2857-24 UGA_173101_P022_WA09 0.490000  102        307
## 1057090          GTS_2857-24           GTS_949-23 0.500000  124        307
## 1057139          GTS_2857-24           GTS_646-23 0.500000  120        307
## 1057153          GTS_2857-24           GTS_616-24 0.491379  122        307
## 1057162          GTS_2857-24           GTS_655-23 0.500000  116        307
## 1057163          GTS_2857-24           GTS_983-23 0.500000  121        307
## 1057174          GTS_2857-24           GTS_630-24 0.500000  119        307
## 1057184          GTS_2857-24           GTS_665-23 0.500000  122        307
## 1057187          GTS_2857-24           GTS_997-23 0.500000  123        307
## 1057193          GTS_2857-24           GTS_644-24 0.490741  119        307
## 1057199          GTS_2857-24           GTS_818-23 0.500000  123        307
## 1057212          GTS_2857-24           GTS_677-23 0.500000  119        307
## 1057213          GTS_2857-24          GTS_1012-23 0.500000  121        307
## 1057219          GTS_2857-24           GTS_657-24 0.491071  120        307
## 1057238          GTS_2857-24           GTS_691-23 0.490385  119        307
## 1057307          GTS_2857-24           GTS_725-23 0.500000  119        307
## 1057316          GTS_2857-24           GTS_736-23 0.481481  120        307
## 1057420          GTS_2857-24          GTS_1202-24 0.482759  122        307
## 1057438          GTS_2857-24           GTS_850-23 0.491667  124        307
## 1057472          GTS_2857-24           GTS_825-24 0.491071  120        307
```

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## 1057485          GTS_2857-24           GTS_935-23 0.500000  123        307
## 1057495          GTS_2857-24          GTS_935-23R 0.491379  123        307
## 1057513          GTS_2857-24           GTS_855-24 0.500000  123        307
## 1057525          GTS_2857-24          GTS_1344-24 0.500000  123        307
## 1057562          GTS_2857-24           GTS_886-24 0.500000  124        307
## 1057576          GTS_2857-24          GTS_1403-24 0.500000  122        307
## 1057593          GTS_2857-24           GTS_931-24 0.500000  119        307
## 1057597          GTS_2857-24          GTS_1428-24 0.500000  123        307
## 1057606          GTS_2857-24          GTS_3012-24 0.500000  122        307
## 1057622          GTS_2857-24          GTS_1451-24 0.500000  123        307
## 1057631          GTS_2857-24           GTS_955-24 0.500000  122        307
## 1057658          GTS_2857-24          GTS_1516-24 0.491071  120        307
## 1057695          GTS_2857-24          GTS_2777-24 0.500000  121        307
## 1057724          GTS_2857-24          GTS_1633-24 0.482759  124        307
## 1057733          GTS_2857-24          GTS_1672-24 0.500000  121        307
## 1057756          GTS_2857-24          GTS_2376-24 0.500000  123        307
## 1057778          GTS_2857-24          GTS_2669-24 0.500000  123        307
## 1087116          GTS_2697-24 UGA_173101_P019_WE05 0.490385  114        307
## 1087142          GTS_2697-24 UGA_173101_P019_WA04 0.488636   91        307
## 1087183          GTS_2697-24 UGA_173101_P016_WC12 0.491071  109        307
## 1087185          GTS_2697-24 UGA_173101_P017_WC02 0.482759  116        307
## 1087192          GTS_2697-24 UGA_173101_P018_WB04 0.490741  117        307
## 1087197          GTS_2697-24 UGA_173101_P016_WF07 0.461538  102        307
## 1087246          GTS_2697-24 UGA_173101_P019_WG03 0.491071  120        307
## 1087252          GTS_2697-24 UGA_173101_P016_WE08 0.491071  116        307
## 1087283          GTS_2697-24 UGA_173101_P017_WC12 0.481481  116        307
## 1087304          GTS_2697-24 UGA_173101_P019_WA02 0.491071  118        307
## 1087333          GTS_2697-24 UGA_173101_P018_WD10 0.490741  109        307
## 1087395          GTS_2697-24 UGA_173101_P016_WH09 0.490741  114        307
## 1087431          GTS_2697-24 UGA_173101_P017_WD09 0.490385  108        307
## 1087451          GTS_2697-24 UGA_173101_P019_WA01 0.491071  119        307
## 1087478          GTS_2697-24 UGA_173101_P018_WB06 0.466667   55        307
## 1087502          GTS_2697-24 UGA_173101_P018_WC11 0.491071  120        307
## 1087515          GTS_2697-24 UGA_173101_P023_WF05 0.480000   93        307
## 1087528          GTS_2697-24 UGA_173101_P023_WB12 0.472222  109        307
## 1087537          GTS_2697-24 UGA_173101_P023_WA04 0.488636   92        307
## 1087592          GTS_2697-24 UGA_173101_P023_WA07 0.480769  107        307
## 1087615          GTS_2697-24 UGA_173101_P022_WD02 0.444444   51        307
## 1087755          GTS_2697-24 UGA_173101_P021_WF01 0.480769  102        307
## 1087798          GTS_2697-24 UGA_173101_P022_WA09 0.489583  100        307
## 1087834          GTS_2697-24           GTS_949-23 0.500000  122        307
## 1087883          GTS_2697-24           GTS_646-23 0.500000  118        307
## 1087897          GTS_2697-24           GTS_616-24 0.491071  121        307
## 1087906          GTS_2697-24           GTS_655-23 0.500000  115        307
## 1087907          GTS_2697-24           GTS_983-23 0.500000  120        307
## 1087918          GTS_2697-24           GTS_630-24 0.500000  118        307
## 1087928          GTS_2697-24           GTS_665-23 0.500000  121        307
## 1087931          GTS_2697-24           GTS_997-23 0.500000  121        307
## 1087937          GTS_2697-24           GTS_644-24 0.490385  118        307
## 1087943          GTS_2697-24           GTS_818-23 0.500000  121        307
## 1087956          GTS_2697-24           GTS_677-23 0.500000  119        307
## 1087957          GTS_2697-24          GTS_1012-23 0.500000  120        307
## 1087963          GTS_2697-24           GTS_657-24 0.490741  119        307
## 1087982          GTS_2697-24           GTS_691-23 0.490385  119        307
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## 1088051          GTS_2697-24           GTS_725-23 0.500000  118        307
## 1088060          GTS_2697-24           GTS_736-23 0.481481  120        307
## 1088164          GTS_2697-24          GTS_1202-24 0.491379  120        307
## 1088182          GTS_2697-24           GTS_850-23 0.500000  122        307
## 1088216          GTS_2697-24           GTS_825-24 0.500000  119        307
## 1088229          GTS_2697-24           GTS_935-23 0.500000  121        307
## 1088239          GTS_2697-24          GTS_935-23R 0.491379  122        307
## 1088257          GTS_2697-24           GTS_855-24 0.500000  121        307
## 1088269          GTS_2697-24          GTS_1344-24 0.500000  121        307
## 1088306          GTS_2697-24           GTS_886-24 0.500000  122        307
## 1088320          GTS_2697-24          GTS_1403-24 0.500000  121        307
## 1088337          GTS_2697-24           GTS_931-24 0.500000  119        307
## 1088341          GTS_2697-24          GTS_1428-24 0.500000  121        307
## 1088350          GTS_2697-24          GTS_3012-24 0.500000  120        307
## 1088366          GTS_2697-24          GTS_1451-24 0.500000  121        307
## 1088375          GTS_2697-24           GTS_955-24 0.500000  121        307
## 1088402          GTS_2697-24          GTS_1516-24 0.491071  120        307
## 1088439          GTS_2697-24          GTS_2777-24 0.500000  120        307
## 1088468          GTS_2697-24          GTS_1633-24 0.491379  122        307
## 1088477          GTS_2697-24          GTS_1672-24 0.500000  121        307
## 1088500          GTS_2697-24          GTS_2376-24 0.500000  122        307
## 1088522          GTS_2697-24          GTS_2669-24 0.500000  121        307
## 1088530          GTS_2697-24          GTS_2857-24 0.500000  122        307
## 1115312          GTS_2906-24 UGA_173101_P019_WE05 0.490741  115        307
## 1115338          GTS_2906-24 UGA_173101_P019_WA04 0.489130   92        307
## 1115379          GTS_2906-24 UGA_173101_P016_WC12 0.491379  110        307
## 1115381          GTS_2906-24 UGA_173101_P017_WC02 0.483333  117        307
## 1115388          GTS_2906-24 UGA_173101_P018_WB04 0.491071  118        307
## 1115393          GTS_2906-24 UGA_173101_P016_WF07 0.462963  103        307
## 1115442          GTS_2906-24 UGA_173101_P019_WG03 0.491379  122        307
## 1115448          GTS_2906-24 UGA_173101_P016_WE08 0.491379  117        307
## 1115479          GTS_2906-24 UGA_173101_P017_WC12 0.482143  117        307
## 1115500          GTS_2906-24 UGA_173101_P019_WA02 0.491379  119        307
## 1115529          GTS_2906-24 UGA_173101_P018_WD10 0.491071  111        307
## 1115591          GTS_2906-24 UGA_173101_P016_WH09 0.491071  115        307
## 1115627          GTS_2906-24 UGA_173101_P017_WD09 0.490741  109        307
## 1115647          GTS_2906-24 UGA_173101_P019_WA01 0.491379  121        307
## 1115674          GTS_2906-24 UGA_173101_P018_WB06 0.468750   56        307
## 1115698          GTS_2906-24 UGA_173101_P018_WC11 0.491379  122        307
## 1115711          GTS_2906-24 UGA_173101_P023_WF05 0.480769   94        307
## 1115724          GTS_2906-24 UGA_173101_P023_WB12 0.473214  111        307
## 1115733          GTS_2906-24 UGA_173101_P023_WA04 0.489130   93        307
## 1115788          GTS_2906-24 UGA_173101_P023_WA07 0.481481  108        307
## 1115811          GTS_2906-24 UGA_173101_P022_WD02 0.450000   52        307
## 1115951          GTS_2906-24 UGA_173101_P021_WF01 0.481481  104        307
## 1115994          GTS_2906-24 UGA_173101_P022_WA09 0.490000  102        307
## 1116030          GTS_2906-24           GTS_949-23 0.500000  124        307
## 1116079          GTS_2906-24           GTS_646-23 0.500000  120        307
## 1116093          GTS_2906-24           GTS_616-24 0.491379  122        307
## 1116102          GTS_2906-24           GTS_655-23 0.500000  116        307
## 1116103          GTS_2906-24           GTS_983-23 0.500000  121        307
## 1116114          GTS_2906-24           GTS_630-24 0.500000  119        307
## 1116124          GTS_2906-24           GTS_665-23 0.500000  122        307
## 1116127          GTS_2906-24           GTS_997-23 0.500000  123        307
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## 1116133          GTS_2906-24           GTS_644-24 0.490741  119        307
## 1116139          GTS_2906-24           GTS_818-23 0.500000  123        307
## 1116152          GTS_2906-24           GTS_677-23 0.500000  119        307
## 1116153          GTS_2906-24          GTS_1012-23 0.500000  121        307
## 1116159          GTS_2906-24           GTS_657-24 0.491071  120        307
## 1116178          GTS_2906-24           GTS_691-23 0.490385  119        307
## 1116247          GTS_2906-24           GTS_725-23 0.500000  119        307
## 1116256          GTS_2906-24           GTS_736-23 0.481481  120        307
## 1116360          GTS_2906-24          GTS_1202-24 0.482759  122        307
## 1116378          GTS_2906-24           GTS_850-23 0.491667  124        307
## 1116412          GTS_2906-24           GTS_825-24 0.491071  120        307
## 1116425          GTS_2906-24           GTS_935-23 0.500000  123        307
## 1116435          GTS_2906-24          GTS_935-23R 0.491379  123        307
## 1116453          GTS_2906-24           GTS_855-24 0.500000  123        307
## 1116465          GTS_2906-24          GTS_1344-24 0.500000  123        307
## 1116502          GTS_2906-24           GTS_886-24 0.500000  124        307
## 1116516          GTS_2906-24          GTS_1403-24 0.500000  122        307
## 1116533          GTS_2906-24           GTS_931-24 0.500000  119        307
## 1116537          GTS_2906-24          GTS_1428-24 0.500000  123        307
## 1116546          GTS_2906-24          GTS_3012-24 0.500000  122        307
## 1116562          GTS_2906-24          GTS_1451-24 0.500000  123        307
## 1116571          GTS_2906-24           GTS_955-24 0.500000  122        307
## 1116598          GTS_2906-24          GTS_1516-24 0.491071  120        307
## 1116635          GTS_2906-24          GTS_2777-24 0.500000  121        307
## 1116664          GTS_2906-24          GTS_1633-24 0.482759  124        307
## 1116673          GTS_2906-24          GTS_1672-24 0.500000  121        307
## 1116696          GTS_2906-24          GTS_2376-24 0.500000  123        307
## 1116718          GTS_2906-24          GTS_2669-24 0.500000  123        307
## 1116726          GTS_2906-24          GTS_2857-24 0.500000  124        307
## 1116747          GTS_2906-24          GTS_2697-24 0.500000  122        307
## 1198967           GTS_151-24 UGA_173101_P019_WE05 0.500000  112        307
## 1198993           GTS_151-24 UGA_173101_P019_WA04 0.500000   90        307
## 1199034           GTS_151-24 UGA_173101_P016_WC12 0.500000  107        307
## 1199036           GTS_151-24 UGA_173101_P017_WC02 0.490741  114        307
## 1199043           GTS_151-24 UGA_173101_P018_WB04 0.500000  116        307
## 1199048           GTS_151-24 UGA_173101_P016_WF07 0.468750  101        307
## 1199097           GTS_151-24 UGA_173101_P019_WG03 0.500000  120        307
## 1199103           GTS_151-24 UGA_173101_P016_WE08 0.500000  115        307
## 1199134           GTS_151-24 UGA_173101_P017_WC12 0.490385  115        307
## 1199155           GTS_151-24 UGA_173101_P019_WA02 0.500000  117        307
## 1199184           GTS_151-24 UGA_173101_P018_WD10 0.500000  108        307
## 1199246           GTS_151-24 UGA_173101_P016_WH09 0.500000  112        307
## 1199282           GTS_151-24 UGA_173101_P017_WD09 0.500000  106        307
## 1199302           GTS_151-24 UGA_173101_P019_WA01 0.500000  119        307
## 1199329           GTS_151-24 UGA_173101_P018_WB06 0.483333   54        307
## 1199353           GTS_151-24 UGA_173101_P018_WC11 0.500000  120        307
## 1199366           GTS_151-24 UGA_173101_P023_WF05 0.489130   91        307
## 1199379           GTS_151-24 UGA_173101_P023_WB12 0.480000  109        307
## 1199388           GTS_151-24 UGA_173101_P023_WA04 0.500000   90        307
## 1199443           GTS_151-24 UGA_173101_P023_WA07 0.490000  105        307
## 1199466           GTS_151-24 UGA_173101_P022_WD02 0.472222   50        307
## 1199606           GTS_151-24 UGA_173101_P021_WF01 0.489583  101        307
## 1199649           GTS_151-24 UGA_173101_P022_WA09 0.500000  100        307
## 1199685           GTS_151-24           GTS_949-23 0.500000  121        307
```

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## 1199734          GTS_151-24          GTS_646-23 0.500000  118       307
## 1199748          GTS_151-24          GTS_616-24 0.490385  119       307
## 1199757          GTS_151-24          GTS_655-23 0.500000  113       307
## 1199758          GTS_151-24          GTS_983-23 0.500000  120       307
## 1199769          GTS_151-24          GTS_630-24 0.500000  117       307
## 1199779          GTS_151-24          GTS_665-23 0.500000  119       307
## 1199782          GTS_151-24          GTS_997-23 0.500000  120       307
## 1199788          GTS_151-24          GTS_644-24 0.490000  117       307
## 1199794          GTS_151-24          GTS_818-23 0.500000  120       307
## 1199807          GTS_151-24          GTS_677-23 0.500000  117       307
## 1199808          GTS_151-24         GTS_1012-23 0.500000  120       307
## 1199814          GTS_151-24          GTS_657-24 0.490385  118       307
## 1199833          GTS_151-24          GTS_691-23 0.500000  117       307
## 1199902          GTS_151-24          GTS_725-23 0.500000  117       307
## 1199911          GTS_151-24          GTS_736-23 0.480000  118       307
## 1200015          GTS_151-24         GTS_1202-24 0.500000  119       307
## 1200033          GTS_151-24          GTS_850-23 0.500000  121       307
## 1200067          GTS_151-24          GTS_825-24 0.500000  117       307
## 1200080          GTS_151-24          GTS_935-23 0.500000  121       307
## 1200090          GTS_151-24         GTS_935-23R 0.500000  121       307
## 1200108          GTS_151-24          GTS_855-24 0.500000  121       307
## 1200120          GTS_151-24         GTS_1344-24 0.500000  121       307
## 1200157          GTS_151-24          GTS_886-24 0.500000  121       307
## 1200171          GTS_151-24         GTS_1403-24 0.500000  121       307
## 1200188          GTS_151-24          GTS_931-24 0.500000  118       307
## 1200192          GTS_151-24         GTS_1428-24 0.500000  121       307
## 1200201          GTS_151-24         GTS_3012-24 0.500000  120       307
## 1200217          GTS_151-24         GTS_1451-24 0.500000  121       307
## 1200226          GTS_151-24          GTS_955-24 0.500000  121       307
## 1200253          GTS_151-24         GTS_1516-24 0.490385  118       307
## 1200290          GTS_151-24         GTS_2777-24 0.500000  119       307
## 1200319          GTS_151-24         GTS_1633-24 0.500000  121       307
## 1200328          GTS_151-24         GTS_1672-24 0.500000  120       307
## 1200351          GTS_151-24         GTS_2376-24 0.500000  121       307
## 1200373          GTS_151-24         GTS_2669-24 0.500000  121       307
## 1200381          GTS_151-24         GTS_2857-24 0.500000  121       307
## 1200402          GTS_151-24         GTS_2697-24 0.500000  120       307
## 1200421          GTS_151-24         GTS_2906-24 0.500000  121       307
## 1216061          GTS_426-23 UGA_173101_P019_WE05 0.500000  110       307
## 1216087          GTS_426-23 UGA_173101_P019_WA04 0.500000   87       307
## 1216128          GTS_426-23 UGA_173101_P016_WC12 0.500000  105       307
## 1216130          GTS_426-23 UGA_173101_P017_WC02 0.490385  112       307
## 1216137          GTS_426-23 UGA_173101_P018_WB04 0.500000  113       307
## 1216142          GTS_426-23 UGA_173101_P016_WF07 0.467391   99       307
## 1216191          GTS_426-23 UGA_173101_P019_WG03 0.500000  117       307
## 1216197          GTS_426-23 UGA_173101_P016_WE08 0.500000  112       307
## 1216228          GTS_426-23 UGA_173101_P017_WC12 0.490000  112       307
## 1216249          GTS_426-23 UGA_173101_P019_WA02 0.500000  114       307
## 1216278          GTS_426-23 UGA_173101_P018_WD10 0.500000  107       307
## 1216340          GTS_426-23 UGA_173101_P016_WH09 0.500000  110       307
## 1216376          GTS_426-23 UGA_173101_P017_WD09 0.500000  105       307
## 1216396          GTS_426-23 UGA_173101_P019_WA01 0.500000  116       307
## 1216423          GTS_426-23 UGA_173101_P018_WB06 0.482143   53       307
## 1216447          GTS_426-23 UGA_173101_P018_WC11 0.500000  117       307
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## 1216460          GTS_426-23 UGA_173101_P023_WF05 0.488636   89       307
## 1216473          GTS_426-23 UGA_173101_P023_WB12 0.479167  106       307
## 1216482          GTS_426-23 UGA_173101_P023_WA04 0.500000   88       307
## 1216537          GTS_426-23 UGA_173101_P023_WA07 0.500000  103       307
## 1216560          GTS_426-23 UGA_173101_P022_WD02 0.472222   50       307
## 1216700          GTS_426-23 UGA_173101_P021_WF01 0.489583  100       307
## 1216743          GTS_426-23 UGA_173101_P022_WA09 0.500000   97       307
## 1216779          GTS_426-23           GTS_949-23 0.500000  119       307
## 1216828          GTS_426-23           GTS_646-23 0.500000  116       307
## 1216842          GTS_426-23           GTS_616-24 0.490000  117       307
## 1216851          GTS_426-23           GTS_655-23 0.500000  111       307
## 1216852          GTS_426-23           GTS_983-23 0.500000  118       307
## 1216863          GTS_426-23           GTS_630-24 0.500000  116       307
## 1216873          GTS_426-23           GTS_665-23 0.500000  117       307
## 1216876          GTS_426-23           GTS_997-23 0.500000  118       307
## 1216882          GTS_426-23           GTS_644-24 0.490000  116       307
## 1216888          GTS_426-23           GTS_818-23 0.500000  118       307
## 1216901          GTS_426-23           GTS_677-23 0.500000  116       307
## 1216902          GTS_426-23          GTS_1012-23 0.500000  118       307
## 1216908          GTS_426-23           GTS_657-24 0.490000  116       307
## 1216927          GTS_426-23           GTS_691-23 0.500000  116       307
## 1216996          GTS_426-23           GTS_725-23 0.500000  116       307
## 1217005          GTS_426-23           GTS_736-23 0.490000  116       307
## 1217109          GTS_426-23          GTS_1202-24 0.500000  117       307
## 1217127          GTS_426-23           GTS_850-23 0.500000  119       307
## 1217161          GTS_426-23           GTS_825-24 0.500000  116       307
## 1217174          GTS_426-23           GTS_935-23 0.500000  119       307
## 1217184          GTS_426-23          GTS_935-23R 0.500000  119       307
## 1217202          GTS_426-23           GTS_855-24 0.500000  119       307
## 1217214          GTS_426-23          GTS_1344-24 0.500000  119       307
## 1217251          GTS_426-23           GTS_886-24 0.500000  119       307
## 1217265          GTS_426-23          GTS_1403-24 0.500000  119       307
## 1217282          GTS_426-23           GTS_931-24 0.500000  116       307
## 1217286          GTS_426-23          GTS_1428-24 0.500000  119       307
## 1217295          GTS_426-23          GTS_3012-24 0.500000  118       307
## 1217311          GTS_426-23          GTS_1451-24 0.500000  119       307
## 1217320          GTS_426-23           GTS_955-24 0.500000  119       307
## 1217347          GTS_426-23          GTS_1516-24 0.490385  117       307
## 1217384          GTS_426-23          GTS_2777-24 0.500000  117       307
## 1217413          GTS_426-23          GTS_1633-24 0.500000  119       307
## 1217422          GTS_426-23          GTS_1672-24 0.500000  118       307
## 1217445          GTS_426-23          GTS_2376-24 0.500000  119       307
## 1217467          GTS_426-23          GTS_2669-24 0.500000  119       307
## 1217475          GTS_426-23          GTS_2857-24 0.500000  119       307
## 1217496          GTS_426-23          GTS_2697-24 0.500000  118       307
## 1217515          GTS_426-23          GTS_2906-24 0.500000  119       307
## 1217570          GTS_426-23           GTS_151-24 0.500000  119       307
## 1233276         GTS_151-24R UGA_173101_P019_WE05 0.500000  109        34
## 1233302         GTS_151-24R UGA_173101_P019_WA04 0.500000   89        34
## 1233343         GTS_151-24R UGA_173101_P016_WC12 0.500000  104        34
## 1233345         GTS_151-24R UGA_173101_P017_WC02 0.490385  111        34
## 1233352         GTS_151-24R UGA_173101_P018_WB04 0.500000  113        34
## 1233357         GTS_151-24R UGA_173101_P016_WF07 0.478261   99        34
## 1233406         GTS_151-24R UGA_173101_P019_WG03 0.500000  117        34
```

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## 1233412          GTS_151-24R UGA_173101_P016_WE08 0.500000  112        34
## 1233443          GTS_151-24R UGA_173101_P017_WC12 0.490000  112        34
## 1233464          GTS_151-24R UGA_173101_P019_WA02 0.500000  114        34
## 1233493          GTS_151-24R UGA_173101_P018_WD10 0.500000  106        34
## 1233555          GTS_151-24R UGA_173101_P016_WH09 0.500000  109        34
## 1233591          GTS_151-24R UGA_173101_P017_WD09 0.500000  104        34
## 1233611          GTS_151-24R UGA_173101_P019_WA01 0.500000  116        34
## 1233638          GTS_151-24R UGA_173101_P018_WB06 0.482143   53        34
## 1233662          GTS_151-24R UGA_173101_P018_WC11 0.500000  117        34
## 1233675          GTS_151-24R UGA_173101_P023_WF05 0.489130   89        34
## 1233688          GTS_151-24R UGA_173101_P023_WB12 0.479167  106        34
## 1233697          GTS_151-24R UGA_173101_P023_WA04 0.500000   88        34
## 1233752          GTS_151-24R UGA_173101_P023_WA07 0.489583  102        34
## 1233775          GTS_151-24R UGA_173101_P022_WD02 0.472222   49        34
## 1233915          GTS_151-24R UGA_173101_P021_WF01 0.489130   98        34
## 1233958          GTS_151-24R UGA_173101_P022_WA09 0.500000   97        34
## 1233994          GTS_151-24R           GTS_949-23 0.500000  117        34
## 1234043          GTS_151-24R           GTS_646-23 0.500000  116        34
## 1234057          GTS_151-24R           GTS_616-24 0.500000  115        34
## 1234066          GTS_151-24R           GTS_655-23 0.500000  111        34
## 1234067          GTS_151-24R           GTS_983-23 0.500000  117        34
## 1234078          GTS_151-24R           GTS_630-24 0.500000  114        34
## 1234088          GTS_151-24R           GTS_665-23 0.500000  116        34
## 1234091          GTS_151-24R           GTS_997-23 0.500000  117        34
## 1234097          GTS_151-24R           GTS_644-24 0.500000  113        34
## 1234103          GTS_151-24R           GTS_818-23 0.500000  117        34
## 1234116          GTS_151-24R           GTS_677-23 0.500000  114        34
## 1234117          GTS_151-24R          GTS_1012-23 0.500000  117        34
## 1234123          GTS_151-24R           GTS_657-24 0.490000  115        34
## 1234142          GTS_151-24R           GTS_691-23 0.500000  114        34
## 1234211          GTS_151-24R           GTS_725-23 0.500000  114        34
## 1234220          GTS_151-24R           GTS_736-23 0.479167  115        34
## 1234324          GTS_151-24R          GTS_1202-24 0.500000  115        34
## 1234342          GTS_151-24R           GTS_850-23 0.500000  117        34
## 1234376          GTS_151-24R           GTS_825-24 0.500000  113        34
## 1234389          GTS_151-24R           GTS_935-23 0.500000  117        34
## 1234399          GTS_151-24R          GTS_935-23R 0.500000  117        34
## 1234417          GTS_151-24R           GTS_855-24 0.500000  117        34
## 1234429          GTS_151-24R          GTS_1344-24 0.500000  117        34
## 1234466          GTS_151-24R           GTS_886-24 0.500000  117        34
## 1234480          GTS_151-24R          GTS_1403-24 0.500000  117        34
## 1234497          GTS_151-24R           GTS_931-24 0.500000  114        34
## 1234501          GTS_151-24R          GTS_1428-24 0.500000  117        34
## 1234510          GTS_151-24R          GTS_3012-24 0.500000  116        34
## 1234526          GTS_151-24R          GTS_1451-24 0.500000  117        34
## 1234535          GTS_151-24R           GTS_955-24 0.500000  117        34
## 1234562          GTS_151-24R          GTS_1516-24 0.489583  114        34
## 1234599          GTS_151-24R          GTS_2777-24 0.500000  116        34
## 1234628          GTS_151-24R          GTS_1633-24 0.500000  117        34
## 1234637          GTS_151-24R          GTS_1672-24 0.500000  116        34
## 1234660          GTS_151-24R          GTS_2376-24 0.500000  117        34
## 1234682          GTS_151-24R          GTS_2669-24 0.500000  117        34
## 1234690          GTS_151-24R          GTS_2857-24 0.500000  117        34
## 1234711          GTS_151-24R          GTS_2697-24 0.500000  116        34
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## 1234730          GTS_151-24R          GTS_2906-24 0.500000  117        34
## 1234785          GTS_151-24R           GTS_151-24 0.500000  117        34
## 1234796          GTS_151-24R           GTS_426-23 0.500000  115        34
## 1258532           GTS_146-23 UGA_173101_P019_WE05 0.500000  112        34
## 1258558           GTS_146-23 UGA_173101_P019_WA04 0.500000   90        34
## 1258599           GTS_146-23 UGA_173101_P016_WC12 0.500000  107        34
## 1258601           GTS_146-23 UGA_173101_P017_WC02 0.491071  114        34
## 1258608           GTS_146-23 UGA_173101_P018_WB04 0.500000  116        34
## 1258613           GTS_146-23 UGA_173101_P016_WF07 0.470000  102        34
## 1258662           GTS_146-23 UGA_173101_P019_WG03 0.500000  120        34
## 1258668           GTS_146-23 UGA_173101_P016_WE08 0.500000  115        34
## 1258699           GTS_146-23 UGA_173101_P017_WC12 0.490741  115        34
## 1258720           GTS_146-23 UGA_173101_P019_WA02 0.500000  117        34
## 1258749           GTS_146-23 UGA_173101_P018_WD10 0.500000  109        34
## 1258811           GTS_146-23 UGA_173101_P016_WH09 0.500000  112        34
## 1258847           GTS_146-23 UGA_173101_P017_WD09 0.500000  107        34
## 1258867           GTS_146-23 UGA_173101_P019_WA01 0.500000  119        34
## 1258894           GTS_146-23 UGA_173101_P018_WB06 0.483333   54        34
## 1258918           GTS_146-23 UGA_173101_P018_WC11 0.500000  120        34
## 1258931           GTS_146-23 UGA_173101_P023_WF05 0.489583   91        34
## 1258944           GTS_146-23 UGA_173101_P023_WB12 0.480769  109        34
## 1258953           GTS_146-23 UGA_173101_P023_WA04 0.500000   90        34
## 1259008           GTS_146-23 UGA_173101_P023_WA07 0.490385  105        34
## 1259031           GTS_146-23 UGA_173101_P022_WD02 0.472222   50        34
## 1259171           GTS_146-23 UGA_173101_P021_WF01 0.490000  101        34
## 1259214           GTS_146-23 UGA_173101_P022_WA09 0.500000  100        34
## 1259250           GTS_146-23           GTS_949-23 0.491071  121        34
## 1259299           GTS_146-23           GTS_646-23 0.500000  118        34
## 1259313           GTS_146-23           GTS_616-24 0.462963  119        34
## 1259322           GTS_146-23           GTS_655-23 0.500000  113        34
## 1259323           GTS_146-23           GTS_983-23 0.500000  120        34
## 1259334           GTS_146-23           GTS_630-24 0.500000  117        34
## 1259344           GTS_146-23           GTS_665-23 0.500000  119        34
## 1259347           GTS_146-23           GTS_997-23 0.500000  120        34
## 1259353           GTS_146-23           GTS_644-24 0.461538  117        34
## 1259359           GTS_146-23           GTS_818-23 0.500000  120        34
## 1259372           GTS_146-23           GTS_677-23 0.500000  117        34
## 1259373           GTS_146-23          GTS_1012-23 0.500000  120        34
## 1259379           GTS_146-23           GTS_657-24 0.490741  118        34
## 1259398           GTS_146-23           GTS_691-23 0.500000  117        34
## 1259467           GTS_146-23           GTS_725-23 0.500000  117        34
## 1259476           GTS_146-23           GTS_736-23 0.480769  118        34
## 1259580           GTS_146-23          GTS_1202-24 0.491071  119        34
## 1259598           GTS_146-23           GTS_850-23 0.491071  121        34
## 1259632           GTS_146-23           GTS_825-24 0.490385  117        34
## 1259645           GTS_146-23           GTS_935-23 0.491071  121        34
## 1259655           GTS_146-23          GTS_935-23R 0.491071  121        34
## 1259673           GTS_146-23           GTS_855-24 0.491071  121        34
## 1259685           GTS_146-23          GTS_1344-24 0.491071  121        34
## 1259722           GTS_146-23           GTS_886-24 0.491071  121        34
## 1259736           GTS_146-23          GTS_1403-24 0.491071  121        34
## 1259753           GTS_146-23           GTS_931-24 0.490741  118        34
## 1259757           GTS_146-23          GTS_1428-24 0.491071  121        34
## 1259766           GTS_146-23          GTS_3012-24 0.490741  120        34
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## 1259782          GTS_146-23          GTS_1451-24 0.491071  121        34
## 1259791          GTS_146-23           GTS_955-24 0.491071  121        34
## 1259818          GTS_146-23          GTS_1516-24 0.481481  118        34
## 1259855          GTS_146-23          GTS_2777-24 0.500000  119        34
## 1259884          GTS_146-23          GTS_1633-24 0.491071  121        34
## 1259893          GTS_146-23          GTS_1672-24 0.491071  120        34
## 1259916          GTS_146-23          GTS_2376-24 0.491071  121        34
## 1259938          GTS_146-23          GTS_2669-24 0.491071  121        34
## 1259946          GTS_146-23          GTS_2857-24 0.491071  121        34
## 1259967          GTS_146-23          GTS_2697-24 0.491071  120        34
## 1259986          GTS_146-23          GTS_2906-24 0.491071  121        34
## 1260041          GTS_146-23           GTS_151-24 0.490741  121        34
## 1260052          GTS_146-23           GTS_426-23 0.490385  119        34
## 1260063          GTS_146-23          GTS_151-24R 0.500000  118        34
## 1468082          GTS_311-24 UGA_173101_P019_WE05 0.500000  112        34
## 1468108          GTS_311-24 UGA_173101_P019_WA04 0.500000   90        34
## 1468149          GTS_311-24 UGA_173101_P016_WC12 0.500000  107        34
## 1468151          GTS_311-24 UGA_173101_P017_WC02 0.491071  114        34
## 1468158          GTS_311-24 UGA_173101_P018_WB04 0.500000  116        34
## 1468163          GTS_311-24 UGA_173101_P016_WF07 0.470000  101        34
## 1468212          GTS_311-24 UGA_173101_P019_WG03 0.500000  120        34
## 1468218          GTS_311-24 UGA_173101_P016_WE08 0.500000  115        34
## 1468249          GTS_311-24 UGA_173101_P017_WC12 0.490741  115        34
## 1468270          GTS_311-24 UGA_173101_P019_WA02 0.500000  117        34
## 1468299          GTS_311-24 UGA_173101_P018_WD10 0.500000  109        34
## 1468361          GTS_311-24 UGA_173101_P016_WH09 0.500000  112        34
## 1468397          GTS_311-24 UGA_173101_P017_WD09 0.500000  107        34
## 1468417          GTS_311-24 UGA_173101_P019_WA01 0.500000  119        34
## 1468444          GTS_311-24 UGA_173101_P018_WB06 0.483333   53        34
## 1468468          GTS_311-24 UGA_173101_P018_WC11 0.500000  120        34
## 1468481          GTS_311-24 UGA_173101_P023_WF05 0.489583   92        34
## 1468494          GTS_311-24 UGA_173101_P023_WB12 0.480769  109        34
## 1468503          GTS_311-24 UGA_173101_P023_WA04 0.500000   91        34
## 1468558          GTS_311-24 UGA_173101_P023_WA07 0.490385  105        34
## 1468581          GTS_311-24 UGA_173101_P022_WD02 0.472222   50        34
## 1468721          GTS_311-24 UGA_173101_P021_WF01 0.490000  101        34
## 1468764          GTS_311-24 UGA_173101_P022_WA09 0.500000  100        34
## 1468800          GTS_311-24           GTS_949-23 0.500000  120        34
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## 1468863          GTS_311-24           GTS_616-24 0.500000  118        34
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## 1216249         34          TRUE       FALSE
## 1216278         34          TRUE       FALSE
## 1216340         34          TRUE       FALSE
## 1216376         34          TRUE       FALSE
## 1216396         34          TRUE       FALSE
## 1216423        193          TRUE       FALSE
## 1216447         34          TRUE       FALSE
## 1216460         34          TRUE       FALSE
## 1216473         34          TRUE       FALSE
```

```
## 1216482        34         TRUE      FALSE
## 1216537        34         TRUE      FALSE
## 1216560        34         TRUE      FALSE
## 1216700        34         TRUE      FALSE
## 1216743        34         TRUE      FALSE
## 1216779       307         TRUE       TRUE
## 1216828       307         TRUE       TRUE
## 1216842       326         TRUE      FALSE
## 1216851       307         TRUE       TRUE
## 1216852        34         TRUE      FALSE
## 1216863       307         TRUE       TRUE
## 1216873       307         TRUE       TRUE
## 1216876       307         TRUE       TRUE
## 1216882       326         TRUE      FALSE
## 1216888       307         TRUE       TRUE
## 1216901       307         TRUE       TRUE
## 1216902       307         TRUE       TRUE
## 1216908       326         TRUE      FALSE
## 1216927       339         TRUE      FALSE
## 1216996        34         TRUE      FALSE
## 1217005       326         TRUE      FALSE
## 1217109       339         TRUE      FALSE
## 1217127       352         TRUE      FALSE
## 1217161       352         TRUE      FALSE
## 1217174       307         TRUE       TRUE
## 1217184       339         TRUE      FALSE
## 1217202       307         TRUE       TRUE
## 1217214       307         TRUE       TRUE
## 1217251       307         TRUE       TRUE
## 1217265       307         TRUE       TRUE
## 1217282       307         TRUE       TRUE
## 1217286       307         TRUE       TRUE
## 1217295       307         TRUE       TRUE
## 1217311       307         TRUE       TRUE
## 1217320       307         TRUE       TRUE
## 1217347       326         TRUE      FALSE
## 1217384       307         TRUE       TRUE
## 1217413       339         TRUE      FALSE
## 1217422       307         TRUE       TRUE
## 1217445       307         TRUE       TRUE
## 1217467       307         TRUE       TRUE
## 1217475       307         TRUE       TRUE
## 1217496       307         TRUE       TRUE
## 1217515       307         TRUE       TRUE
## 1217570       307         TRUE       TRUE
## 1233276        34         TRUE       TRUE
## 1233302        34         TRUE       TRUE
## 1233343        34         TRUE       TRUE
## 1233345        34         TRUE       TRUE
## 1233352        34         TRUE       TRUE
## 1233357        34         TRUE       TRUE
## 1233406        34         TRUE       TRUE
## 1233412        34         TRUE       TRUE
## 1233443        34         TRUE       TRUE
```

```
## 1233464        34         TRUE       TRUE
## 1233493        34         TRUE       TRUE
## 1233555        34         TRUE       TRUE
## 1233591        34         TRUE       TRUE
## 1233611        34         TRUE       TRUE
## 1233638       193         TRUE      FALSE
## 1233662        34         TRUE       TRUE
## 1233675        34         TRUE       TRUE
## 1233688        34         TRUE       TRUE
## 1233697        34         TRUE       TRUE
## 1233752        34         TRUE       TRUE
## 1233775        34         TRUE       TRUE
## 1233915        34         TRUE       TRUE
## 1233958        34         TRUE       TRUE
## 1233994       307         TRUE      FALSE
## 1234043       307         TRUE      FALSE
## 1234057       326         TRUE      FALSE
## 1234066       307         TRUE      FALSE
## 1234067        34         TRUE       TRUE
## 1234078       307         TRUE      FALSE
## 1234088       307         TRUE      FALSE
## 1234091       307         TRUE      FALSE
## 1234097       326         TRUE      FALSE
## 1234103       307         TRUE      FALSE
## 1234116       307         TRUE      FALSE
## 1234117       307         TRUE      FALSE
## 1234123       326         TRUE      FALSE
## 1234142       339         TRUE      FALSE
## 1234211        34         TRUE       TRUE
## 1234220       326         TRUE      FALSE
## 1234324       339         TRUE      FALSE
## 1234342       352         TRUE      FALSE
## 1234376       352         TRUE      FALSE
## 1234389       307         TRUE      FALSE
## 1234399       339         TRUE      FALSE
## 1234417       307         TRUE      FALSE
## 1234429       307         TRUE      FALSE
## 1234466       307         TRUE      FALSE
## 1234480       307         TRUE      FALSE
## 1234497       307         TRUE      FALSE
## 1234501       307         TRUE      FALSE
## 1234510       307         TRUE      FALSE
## 1234526       307         TRUE      FALSE
## 1234535       307         TRUE      FALSE
## 1234562       326         TRUE      FALSE
## 1234599       307         TRUE      FALSE
## 1234628       339         TRUE      FALSE
## 1234637       307         TRUE      FALSE
## 1234660       307         TRUE      FALSE
## 1234682       307         TRUE      FALSE
## 1234690       307         TRUE      FALSE
## 1234711       307         TRUE      FALSE
## 1234730       307         TRUE      FALSE
## 1234785       307         TRUE      FALSE
```

```
## 1234796        307          TRUE       FALSE
## 1258532         34          TRUE        TRUE
## 1258558         34          TRUE        TRUE
## 1258599         34          TRUE        TRUE
## 1258601         34          TRUE        TRUE
## 1258608         34          TRUE        TRUE
## 1258613         34          TRUE        TRUE
## 1258662         34          TRUE        TRUE
## 1258668         34          TRUE        TRUE
## 1258699         34          TRUE        TRUE
## 1258720         34          TRUE        TRUE
## 1258749         34          TRUE        TRUE
## 1258811         34          TRUE        TRUE
## 1258847         34          TRUE        TRUE
## 1258867         34          TRUE        TRUE
## 1258894        193          TRUE       FALSE
## 1258918         34          TRUE        TRUE
## 1258931         34          TRUE        TRUE
## 1258944         34          TRUE        TRUE
## 1258953         34          TRUE        TRUE
## 1259008         34          TRUE        TRUE
## 1259031         34          TRUE        TRUE
## 1259171         34          TRUE        TRUE
## 1259214         34          TRUE        TRUE
## 1259250        307          TRUE       FALSE
## 1259299        307          TRUE       FALSE
## 1259313        326          TRUE       FALSE
## 1259322        307          TRUE       FALSE
## 1259323         34          TRUE        TRUE
## 1259334        307          TRUE       FALSE
## 1259344        307          TRUE       FALSE
## 1259347        307          TRUE       FALSE
## 1259353        326          TRUE       FALSE
## 1259359        307          TRUE       FALSE
## 1259372        307          TRUE       FALSE
## 1259373        307          TRUE       FALSE
## 1259379        326          TRUE       FALSE
## 1259398        339          TRUE       FALSE
## 1259467         34          TRUE        TRUE
## 1259476        326          TRUE       FALSE
## 1259580        339          TRUE       FALSE
## 1259598        352          TRUE       FALSE
## 1259632        352          TRUE       FALSE
## 1259645        307          TRUE       FALSE
## 1259655        339          TRUE       FALSE
## 1259673        307          TRUE       FALSE
## 1259685        307          TRUE       FALSE
## 1259722        307          TRUE       FALSE
## 1259736        307          TRUE       FALSE
## 1259753        307          TRUE       FALSE
## 1259757        307          TRUE       FALSE
## 1259766        307          TRUE       FALSE
## 1259782        307          TRUE       FALSE
## 1259791        307          TRUE       FALSE
```

```
## 1259818       326         TRUE      FALSE
## 1259855       307         TRUE      FALSE
## 1259884       339         TRUE      FALSE
## 1259893       307         TRUE      FALSE
## 1259916       307         TRUE      FALSE
## 1259938       307         TRUE      FALSE
## 1259946       307         TRUE      FALSE
## 1259967       307         TRUE      FALSE
## 1259986       307         TRUE      FALSE
## 1260041       307         TRUE      FALSE
## 1260052       307         TRUE      FALSE
## 1260063        34         TRUE       TRUE
## 1468082        34         TRUE       TRUE
## 1468108        34         TRUE       TRUE
## 1468149        34         TRUE       TRUE
## 1468151        34         TRUE       TRUE
## 1468158        34         TRUE       TRUE
## 1468163        34         TRUE       TRUE
## 1468212        34         TRUE       TRUE
## 1468218        34         TRUE       TRUE
## 1468249        34         TRUE       TRUE
## 1468270        34         TRUE       TRUE
## 1468299        34         TRUE       TRUE
## 1468361        34         TRUE       TRUE
## 1468397        34         TRUE       TRUE
## 1468417        34         TRUE       TRUE
## 1468444       193         TRUE      FALSE
## 1468468        34         TRUE       TRUE
## 1468481        34         TRUE       TRUE
## 1468494        34         TRUE       TRUE
## 1468503        34         TRUE       TRUE
## 1468558        34         TRUE       TRUE
## 1468581        34         TRUE       TRUE
## 1468721        34         TRUE       TRUE
## 1468764        34         TRUE       TRUE
## 1468800       307         TRUE      FALSE
## 1468849       307         TRUE      FALSE
## 1468863       326         TRUE      FALSE
## 1468872       307         TRUE      FALSE
## 1468873        34         TRUE       TRUE
## 1468884       307         TRUE      FALSE
## 1468894       307         TRUE      FALSE
## 1468897       307         TRUE      FALSE
## 1468903       326         TRUE      FALSE
## 1468909       307         TRUE      FALSE
## 1468922       307         TRUE      FALSE
## 1468923       307         TRUE      FALSE
## 1468929       326         TRUE      FALSE
## 1468948       339         TRUE      FALSE
## 1469017        34         TRUE       TRUE
## 1469026       326         TRUE      FALSE
## 1469130       339         TRUE      FALSE
## 1469148       352         TRUE      FALSE
## 1469182       352         TRUE      FALSE
```

```
## 1469195        307          TRUE       FALSE
## 1469205        339          TRUE       FALSE
## 1469223        307          TRUE       FALSE
## 1469235        307          TRUE       FALSE
## 1469272        307          TRUE       FALSE
## 1469286        307          TRUE       FALSE
## 1469303        307          TRUE       FALSE
## 1469307        307          TRUE       FALSE
## 1469316        307          TRUE       FALSE
## 1469332        307          TRUE       FALSE
## 1469341        307          TRUE       FALSE
## 1469368        326          TRUE       FALSE
## 1469405        307          TRUE       FALSE
## 1469434        339          TRUE       FALSE
## 1469443        307          TRUE       FALSE
## 1469466        307          TRUE       FALSE
## 1469488        307          TRUE       FALSE
## 1469496        307          TRUE       FALSE
## 1469517        307          TRUE       FALSE
## 1469536        307          TRUE       FALSE
## 1469591        307          TRUE       FALSE
## 1469602        307          TRUE       FALSE
## 1469613         34          TRUE        TRUE
## 1469629         34          TRUE        TRUE
```

```r
#Clone 334 - 23 F, 1 M, direct capture confirms F
table(smd$sexGT[smd$finalClone == 334])
```

```
##
##  F  M
## 23  1
```

```r
smd[which(smd$finalClone == 334),]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 86                      UGA_173101_P015_WH02    45-C  FALSE  FALSE <NA> 2024
## 169                     UGA_173101_P016_WG01  448-24   TRUE  FALSE    5 2024
## 209                     UGA_173101_P017_WB05  497-24   TRUE  FALSE    2 2024
## 426                     UGA_173101_P019_WE12  154-24   TRUE  FALSE    3 2024
## 698                     UGA_173101_P022_WD08  341-24   TRUE  FALSE    3 2024
## 784                     UGA_173101_P023_WC10  205-23   TRUE  FALSE    7 2023
## 1014  GTseek_024_384i5_188_P098.004_649-24  649-24   TRUE  FALSE    2 2024
## 1114  GTseek_024_384i5_288_P098.004_685-24  685-24   TRUE  FALSE    2 2024
## 1194  GTseek_024_384i5_368_P098.003_558-24  558-24   TRUE  FALSE    2 2024
## 1287  GTseek_025_384i5_077_P098.005_781-24  781-24   TRUE  FALSE    2 2024
## 1298 GTseek_025_384i5_088_P098.008_1208-24 1208-24   TRUE  FALSE    2 2024
## 1324 GTseek_025_384i5_114_P098.008_1226-24 1226-24   TRUE  FALSE    2 2024
## 1462 GTseek_025_384i5_252_P098.008_1379-24 1379-24   TRUE  FALSE    3 2024
## 1564 GTseek_025_384i5_354_P098.007_1127-24 1127-24   TRUE  FALSE    2 2024
## 1582 GTseek_025_384i5_372_P098.008_1488-24 1488-24   TRUE  FALSE    5 2024
## 1588 GTseek_025_384i5_378_P098.008_1500-24 1500-24   TRUE  FALSE    3 2024
## 1599 GTseek_026_384i5_005_P098.009_1528-24 1528-24   TRUE  FALSE    3 2024
## 1764 GTseek_026_384i5_261_P098.009_2680-24 2680-24   TRUE  FALSE    3 2024
```

```
## 1765 GTseek_026_384i5_263_P098.009_2682-24 2682-24   TRUE  FALSE    4 2024
## 1769 GTseek_026_384i5_271_P098.009_2687-24 2687-24   TRUE  FALSE    6 2024
## 1791 GTseek_026_384i5_315_P098.010_2890-24 2890-24   TRUE  FALSE    2 2024
## 1823 GTseek_026_384i5_379_P098.010_2928-24 2928-24   TRUE  FALSE    6 2024
## 2005 GTseek_028_384i5_180_P098.UGA4_430-24  430-24   TRUE  FALSE    2 2024
## 2059 GTseek_028_384i5_234_P098.UGA3_228-24  228-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 86     <NA>           <NA>
## 169  HS 051 260705x3599596
## 209  HS 048 261267x3598930
## 426  HS 048 261267x3598930
## 698  HS 048 261267x3598930
## 784  HS 051 260705x3599596
## 1014 HS 051 260705x3599596
## 1114 HS 048 261267x3598930
## 1194 HS 051 260705x3599596
## 1287 HS 048 261267x3598930
## 1298 HS 051 260705x3599596
## 1324 HS 048 261267x3598930
## 1462 HS 048 261267x3598930
## 1564 HS 051 260705x3599596
## 1582 HS 051 260705x3599596
## 1588 HS 048 261267x3598930
## 1599 HS 048 261267x3598930
## 1764 HS 048 261267x3598930
## 1765 HS 051 260705x3599596
## 1769 HS 048 261267x3598930
## 1791 HS 051 260705x3599596
## 1823 HS 048 261267x3598930
## 2005 HS 051 260705x3599596
## 2059 HS 051 260705x3599596
##                                                                                  fqfile
## 86    RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169  RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209  RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426  RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
## 698   RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
## 784   RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
## 1014                                                                             <NA>
## 1114                                                                             <NA>
## 1194                                                                             <NA>
## 1287                                                                             <NA>
## 1298                                                                             <NA>
## 1324                                                                             <NA>
## 1462                                                                             <NA>
## 1564                                                                             <NA>
## 1582                                                                             <NA>
## 1588                                                                             <NA>
## 1599                                                                             <NA>
## 1764                                                                             <NA>
## 1765                                                                             <NA>
## 1769                                                                             <NA>
## 1791                                                                             <NA>
## 1823                                                                             <NA>
```

```
## 2005                                                                                <NA>
## 2059                                                                                <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 86          F     Capture    CGA   1153 UGA_173101_P015_WH02 FALSE       LGC
## 169            Hair Snare    CGA    709 UGA_173101_P016_WG01 FALSE       LGC
## 209            Hair Snare    CGA   1299 UGA_173101_P017_WB05 FALSE       LGC
## 426            Hair Snare    CGA   1294 UGA_173101_P019_WE12 FALSE       LGC
## 698            Hair Snare    CGA   1109 UGA_173101_P022_WD08 FALSE       LGC
## 784            Hair Snare    CGA   1250 UGA_173101_P023_WC10 FALSE       LGC
## 1014     <NA> Hair Sample    CGA    122           GTS_649-24 FALSE    GTSeek
## 1114     <NA> Hair Sample    CGA    120           GTS_685-24 FALSE    GTSeek
## 1194     <NA> Hair Sample    CGA    123           GTS_558-24 FALSE    GTSeek
## 1287     <NA> Hair Sample    CGA    123           GTS_781-24 FALSE    GTSeek
## 1298     <NA> Hair Sample    CGA    124          GTS_1208-24 FALSE    GTSeek
## 1324     <NA> Hair Sample    CGA    116          GTS_1226-24 FALSE    GTSeek
## 1462     <NA> Hair Sample    CGA    125          GTS_1379-24 FALSE    GTSeek
## 1564     <NA> Hair Sample    CGA    120          GTS_1127-24 FALSE    GTSeek
## 1582     <NA> Hair Sample    CGA    122          GTS_1488-24 FALSE    GTSeek
## 1588     <NA> Hair Sample    CGA    123          GTS_1500-24 FALSE    GTSeek
## 1599     <NA> Hair Sample    CGA    122          GTS_1528-24 FALSE    GTSeek
## 1764     <NA> Hair Sample    CGA    124          GTS_2680-24 FALSE    GTSeek
## 1765     <NA> Hair Sample    CGA    124          GTS_2682-24 FALSE    GTSeek
## 1769     <NA> Hair Sample    CGA    122          GTS_2687-24 FALSE    GTSeek
## 1791     <NA> Hair Sample    CGA    123          GTS_2890-24 FALSE    GTSeek
## 1823     <NA> Hair Sample    CGA    124          GTS_2928-24 FALSE    GTSeek
## 2005     <NA> Hair Sample    CGA    122           GTS_430-24 FALSE    GTSeek
## 2059     <NA> Hair Sample    CGA    116           GTS_228-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 86       F       31        334
## 169      F       31        334
## 209      M       31        334
## 426      F       31        334
## 698      F       31        334
## 784      F       31        334
## 1014     F      334        334
## 1114     F      334        334
## 1194     F      334        334
## 1287     F      334        334
## 1298     F      351        334
## 1324     F      334        334
## 1462     F      351        334
## 1564     F      334        334
## 1582     F      334        334
## 1588     F      334        334
## 1599     F      334        334
## 1764     F      334        334
## 1765     F      334        334
## 1769     F      351        334
## 1791     F      334        334
## 1823     F      334        334
## 2005     F      351        334
## 2059     F      351        334
```

```r
unique(smd$COLclone[which(smd$finalClone == 334)])
```

```
## [1]  31 334 351
```

```r
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 334)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 334)],]
```

```
##                          id1                  id2  kinship nsnp colclone1
## 7912    UGA_173101_P017_WB05 UGA_173101_P019_WE12 0.481877 1293        31
## 20947   UGA_173101_P015_WH02 UGA_173101_P019_WE12 0.494726 1151        31
## 21037   UGA_173101_P015_WH02 UGA_173101_P017_WB05 0.477416 1153        31
## 43993   UGA_173101_P016_WG01 UGA_173101_P019_WE12 0.469424  709        31
## 44083   UGA_173101_P016_WG01 UGA_173101_P017_WB05 0.452338  709        31
## 44162   UGA_173101_P016_WG01 UGA_173101_P015_WH02 0.466418  682        31
## 122302  UGA_173101_P022_WD08 UGA_173101_P019_WE12 0.484518 1091        31
## 122392  UGA_173101_P022_WD08 UGA_173101_P017_WB05 0.466743 1091        31
## 122471  UGA_173101_P022_WD08 UGA_173101_P015_WH02 0.479630 1018        31
## 122563  UGA_173101_P022_WD08 UGA_173101_P016_WG01 0.454102  646        31
## 180938  UGA_173101_P023_WC10 UGA_173101_P019_WE12 0.491071 1227        31
## 181028  UGA_173101_P023_WC10 UGA_173101_P017_WB05 0.473267 1228        31
## 181107  UGA_173101_P023_WC10 UGA_173101_P015_WH02 0.485358 1119        31
## 181199  UGA_173101_P023_WC10 UGA_173101_P016_WG01 0.465328  695        31
## 181397  UGA_173101_P023_WC10 UGA_173101_P022_WD08 0.477841 1097        31
## 376315            GTS_649-24 UGA_173101_P019_WE12 0.491379  118       334
## 376405            GTS_649-24 UGA_173101_P017_WB05 0.482759  121       334
## 376484            GTS_649-24 UGA_173101_P015_WH02 0.490385   99       334
## 376576            GTS_649-24 UGA_173101_P016_WG01 0.460000   80       334
## 376774            GTS_649-24 UGA_173101_P022_WD08 0.489130   95       334
## 376881            GTS_649-24 UGA_173101_P023_WC10 0.465517  108       334
## 451762            GTS_685-24 UGA_173101_P019_WE12 0.491379  117       334
## 451852            GTS_685-24 UGA_173101_P017_WB05 0.482759  120       334
## 451931            GTS_685-24 UGA_173101_P015_WH02 0.490385   99       334
## 452023            GTS_685-24 UGA_173101_P016_WG01 0.460000   79       334
## 452221            GTS_685-24 UGA_173101_P022_WD08 0.489130   95       334
## 452328            GTS_685-24 UGA_173101_P023_WC10 0.465517  107       334
## 452594            GTS_685-24            GTS_649-24 0.500000  120       334
## 507565            GTS_558-24 UGA_173101_P019_WE12 0.483333  120       334
## 507655            GTS_558-24 UGA_173101_P017_WB05 0.475000  123       334
## 507734            GTS_558-24 UGA_173101_P015_WH02 0.490741  100       334
## 507826            GTS_558-24 UGA_173101_P016_WG01 0.461538   81       334
## 508024            GTS_558-24 UGA_173101_P022_WD08 0.489583   96       334
## 508131            GTS_558-24 UGA_173101_P023_WC10 0.466667  109       334
## 508397            GTS_558-24            GTS_649-24 0.500000  121       334
## 508480            GTS_558-24            GTS_685-24 0.500000  120       334
## 584858            GTS_781-24 UGA_173101_P019_WE12 0.491667  119       334
## 584948            GTS_781-24 UGA_173101_P017_WB05 0.483333  122       334
## 585027            GTS_781-24 UGA_173101_P015_WH02 0.490741  100       334
## 585119            GTS_781-24 UGA_173101_P016_WG01 0.461538   81       334
## 585317            GTS_781-24 UGA_173101_P022_WD08 0.489583   96       334
## 585424            GTS_781-24 UGA_173101_P023_WC10 0.466667  109       334
## 585690            GTS_781-24            GTS_649-24 0.500000  122       334
## 585773            GTS_781-24            GTS_685-24 0.500000  120       334
```

```
## 585830            GTS_781-24           GTS_558-24 0.500000  122        334
## 595723           GTS_1208-24 UGA_173101_P019_WE12 0.491935  120        351
## 595813           GTS_1208-24 UGA_173101_P017_WB05 0.483871  123        351
## 595892           GTS_1208-24 UGA_173101_P015_WH02 0.490741  100        351
## 595984           GTS_1208-24 UGA_173101_P016_WG01 0.461538   81        351
## 596182           GTS_1208-24 UGA_173101_P022_WD08 0.489583   96        351
## 596289           GTS_1208-24 UGA_173101_P023_WC10 0.466667  109        351
## 596555           GTS_1208-24           GTS_649-24 0.500000  122        351
## 596638           GTS_1208-24           GTS_685-24 0.500000  120        351
## 596695           GTS_1208-24           GTS_558-24 0.491667  123        351
## 596769           GTS_1208-24           GTS_781-24 0.500000  123        351
## 621092           GTS_1226-24 UGA_173101_P019_WE12 0.491379  113        334
## 621182           GTS_1226-24 UGA_173101_P017_WB05 0.491379  116        334
## 621261           GTS_1226-24 UGA_173101_P015_WH02 0.490385   95        334
## 621353           GTS_1226-24 UGA_173101_P016_WG01 0.460000   78        334
## 621551           GTS_1226-24 UGA_173101_P022_WD08 0.489583   92        334
## 621658           GTS_1226-24 UGA_173101_P023_WC10 0.465517  103        334
## 621924           GTS_1226-24           GTS_649-24 0.500000  115        334
## 622007           GTS_1226-24           GTS_685-24 0.500000  114        334
## 622064           GTS_1226-24           GTS_558-24 0.500000  116        334
## 622138           GTS_1226-24           GTS_781-24 0.500000  116        334
## 622148           GTS_1226-24          GTS_1208-24 0.500000  116        334
## 743627           GTS_1379-24 UGA_173101_P019_WE12 0.427419  122        351
## 743717           GTS_1379-24 UGA_173101_P017_WB05 0.419355  125        351
## 743796           GTS_1379-24 UGA_173101_P015_WH02 0.416667  102        351
## 743888           GTS_1379-24 UGA_173101_P016_WG01 0.384615   83        351
## 744086           GTS_1379-24 UGA_173101_P022_WD08 0.406250   98        351
## 744193           GTS_1379-24 UGA_173101_P023_WC10 0.400000  111        351
## 744459           GTS_1379-24           GTS_649-24 0.500000  121        351
## 744542           GTS_1379-24           GTS_685-24 0.500000  120        351
## 744599           GTS_1379-24           GTS_558-24 0.491667  123        351
## 744673           GTS_1379-24           GTS_781-24 0.500000  122        351
## 744683           GTS_1379-24          GTS_1208-24 0.500000  123        351
## 744706           GTS_1379-24          GTS_1226-24 0.500000  116        351
## 844387           GTS_1127-24 UGA_173101_P019_WE12 0.483871  116        334
## 844477           GTS_1127-24 UGA_173101_P017_WB05 0.475806  119        334
## 844556           GTS_1127-24 UGA_173101_P015_WH02 0.490741   98        334
## 844648           GTS_1127-24 UGA_173101_P016_WG01 0.461538   80        334
## 844846           GTS_1127-24 UGA_173101_P022_WD08 0.489583   95        334
## 844953           GTS_1127-24 UGA_173101_P023_WC10 0.466667  106        334
## 845219           GTS_1127-24           GTS_649-24 0.491667  119        334
## 845302           GTS_1127-24           GTS_685-24 0.491379  118        334
## 845359           GTS_1127-24           GTS_558-24 0.491667  119        334
## 845433           GTS_1127-24           GTS_781-24 0.491935  120        334
## 845443           GTS_1127-24          GTS_1208-24 0.491935  120        334
## 845466           GTS_1127-24          GTS_1226-24 0.491379  114        334
## 845571           GTS_1127-24          GTS_1379-24 0.491667  119        334
## 862678           GTS_1488-24 UGA_173101_P019_WE12 0.491667  118        334
## 862768           GTS_1488-24 UGA_173101_P017_WB05 0.483333  121        334
## 862847           GTS_1488-24 UGA_173101_P015_WH02 0.490741   99        334
## 862939           GTS_1488-24 UGA_173101_P016_WG01 0.461538   81        334
## 863137           GTS_1488-24 UGA_173101_P022_WD08 0.489583   95        334
## 863244           GTS_1488-24 UGA_173101_P023_WC10 0.466667  108        334
## 863510           GTS_1488-24           GTS_649-24 0.500000  121        334
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## 863593          GTS_1488-24           GTS_685-24 0.500000  119        334
## 863650          GTS_1488-24           GTS_558-24 0.500000  121        334
## 863724          GTS_1488-24           GTS_781-24 0.500000  122        334
## 863734          GTS_1488-24          GTS_1208-24 0.500000  122        334
## 863757          GTS_1488-24          GTS_1226-24 0.500000  116        334
## 863862          GTS_1488-24          GTS_1379-24 0.500000  121        334
## 863942          GTS_1488-24          GTS_1127-24 0.491935  120        334
## 870577          GTS_1500-24 UGA_173101_P019_WE12 0.491667  119        334
## 870667          GTS_1500-24 UGA_173101_P017_WB05 0.483333  122        334
## 870746          GTS_1500-24 UGA_173101_P015_WH02 0.490741  100        334
## 870838          GTS_1500-24 UGA_173101_P016_WG01 0.461538   81        334
## 871036          GTS_1500-24 UGA_173101_P022_WD08 0.489583   96        334
## 871143          GTS_1500-24 UGA_173101_P023_WC10 0.466667  109        334
## 871409          GTS_1500-24           GTS_649-24 0.500000  122        334
## 871492          GTS_1500-24           GTS_685-24 0.500000  120        334
## 871549          GTS_1500-24           GTS_558-24 0.500000  122        334
## 871623          GTS_1500-24           GTS_781-24 0.500000  123        334
## 871633          GTS_1500-24          GTS_1208-24 0.500000  123        334
## 871656          GTS_1500-24          GTS_1226-24 0.500000  116        334
## 871761          GTS_1500-24          GTS_1379-24 0.500000  122        334
## 871841          GTS_1500-24          GTS_1127-24 0.491935  120        334
## 871855          GTS_1500-24          GTS_1488-24 0.500000  122        334
## 883822          GTS_1528-24 UGA_173101_P019_WE12 0.491379  118        334
## 883912          GTS_1528-24 UGA_173101_P017_WB05 0.482759  121        334
## 883991          GTS_1528-24 UGA_173101_P015_WH02 0.490385   99        334
## 884083          GTS_1528-24 UGA_173101_P016_WG01 0.460000   80        334
## 884281          GTS_1528-24 UGA_173101_P022_WD08 0.489130   95        334
## 884388          GTS_1528-24 UGA_173101_P023_WC10 0.465517  108        334
## 884654          GTS_1528-24           GTS_649-24 0.500000  122        334
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## 884794          GTS_1528-24           GTS_558-24 0.500000  121        334
## 884868          GTS_1528-24           GTS_781-24 0.500000  122        334
## 884878          GTS_1528-24          GTS_1208-24 0.500000  122        334
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## 885100          GTS_1528-24          GTS_1488-24 0.500000  121        334
## 885106          GTS_1528-24          GTS_1500-24 0.500000  122        334
## 1066567         GTS_2680-24 UGA_173101_P019_WE12 0.483333  120        334
## 1066657         GTS_2680-24 UGA_173101_P017_WB05 0.475000  123        334
## 1066736         GTS_2680-24 UGA_173101_P015_WH02 0.490741  100        334
## 1066828         GTS_2680-24 UGA_173101_P016_WG01 0.461538   81        334
## 1067026         GTS_2680-24 UGA_173101_P022_WD08 0.489583   96        334
## 1067133         GTS_2680-24 UGA_173101_P023_WC10 0.466667  109        334
## 1067399         GTS_2680-24           GTS_649-24 0.500000  122        334
## 1067482         GTS_2680-24           GTS_685-24 0.500000  120        334
## 1067539         GTS_2680-24           GTS_558-24 0.500000  123        334
## 1067613         GTS_2680-24           GTS_781-24 0.500000  123        334
## 1067623         GTS_2680-24          GTS_1208-24 0.491935  124        334
## 1067646         GTS_2680-24          GTS_1226-24 0.500000  116        334
## 1067751         GTS_2680-24          GTS_1379-24 0.491667  123        334
## 1067831         GTS_2680-24          GTS_1127-24 0.491935  120        334
## 1067845         GTS_2680-24          GTS_1488-24 0.500000  122        334
## 1067851         GTS_2680-24          GTS_1500-24 0.500000  123        334
```

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## 1067861          GTS_2680-24          GTS_1528-24 0.500000  122        334
## 1068028          GTS_2682-24 UGA_173101_P019_WE12 0.483333  120        334
## 1068118          GTS_2682-24 UGA_173101_P017_WB05 0.475000  123        334
## 1068197          GTS_2682-24 UGA_173101_P015_WH02 0.490741  100        334
## 1068289          GTS_2682-24 UGA_173101_P016_WG01 0.461538   81        334
## 1068487          GTS_2682-24 UGA_173101_P022_WD08 0.489583   96        334
## 1068594          GTS_2682-24 UGA_173101_P023_WC10 0.466667  109        334
## 1068860          GTS_2682-24           GTS_649-24 0.500000  122        334
## 1068943          GTS_2682-24           GTS_685-24 0.500000  120        334
## 1069000          GTS_2682-24           GTS_558-24 0.500000  123        334
## 1069074          GTS_2682-24           GTS_781-24 0.500000  123        334
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## 1069107          GTS_2682-24          GTS_1226-24 0.500000  116        334
## 1069212          GTS_2682-24          GTS_1379-24 0.491667  123        334
## 1069292          GTS_2682-24          GTS_1127-24 0.491935  120        334
## 1069306          GTS_2682-24          GTS_1488-24 0.500000  122        334
## 1069312          GTS_2682-24          GTS_1500-24 0.500000  123        334
## 1069322          GTS_2682-24          GTS_1528-24 0.500000  122        334
## 1069453          GTS_2682-24          GTS_2680-24 0.500000  124        334
## 1073882          GTS_2687-24 UGA_173101_P019_WE12 0.482759  118        351
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## 1074051          GTS_2687-24 UGA_173101_P015_WH02 0.480769   99        351
## 1074143          GTS_2687-24 UGA_173101_P016_WG01 0.450000   80        351
## 1074341          GTS_2687-24 UGA_173101_P022_WD08 0.478261   95        351
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## 1074714          GTS_2687-24           GTS_649-24 0.500000  121        351
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## 1074961          GTS_2687-24          GTS_1226-24 0.491071  115        351
## 1075066          GTS_2687-24          GTS_1379-24 0.491379  121        351
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## 1075166          GTS_2687-24          GTS_1500-24 0.491667  122        351
## 1075176          GTS_2687-24          GTS_1528-24 0.500000  121        351
## 1075307          GTS_2687-24          GTS_2680-24 0.491667  122        351
## 1075308          GTS_2687-24          GTS_2682-24 0.491667  122        351
## 1097458          GTS_2890-24 UGA_173101_P019_WE12 0.483333  120        334
## 1097548          GTS_2890-24 UGA_173101_P017_WB05 0.475000  123        334
## 1097627          GTS_2890-24 UGA_173101_P015_WH02 0.490741  100        334
## 1097719          GTS_2890-24 UGA_173101_P016_WG01 0.461538   81        334
## 1097917          GTS_2890-24 UGA_173101_P022_WD08 0.489583   96        334
## 1098024          GTS_2890-24 UGA_173101_P023_WC10 0.466667  109        334
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## 1098373          GTS_2890-24           GTS_685-24 0.500000  120        334
## 1098430          GTS_2890-24           GTS_558-24 0.500000  123        334
## 1098504          GTS_2890-24           GTS_781-24 0.500000  122        334
## 1098514          GTS_2890-24          GTS_1208-24 0.491667  123        334
## 1098537          GTS_2890-24          GTS_1226-24 0.500000  116        334
## 1098642          GTS_2890-24          GTS_1379-24 0.491667  123        334
## 1098722          GTS_2890-24          GTS_1127-24 0.491667  119        334
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## 1098742          GTS_2890-24          GTS_1500-24 0.500000  122        334
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## 1098752          GTS_2890-24          GTS_1528-24 0.500000  121        334
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## 1135069          GTS_2928-24 UGA_173101_P016_WG01 0.461538   81        334
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## 1136102          GTS_2928-24          GTS_1528-24 0.500000  122        334
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## 1136234          GTS_2928-24          GTS_2682-24 0.500000  124        334
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## 1296892           GTS_430-24 UGA_173101_P019_WE12 0.491379  118        351
## 1296982           GTS_430-24 UGA_173101_P017_WB05 0.482759  121        351
## 1297061           GTS_430-24 UGA_173101_P015_WH02 0.490000   97        351
## 1297153           GTS_430-24 UGA_173101_P016_WG01 0.460000   80        351
## 1297351           GTS_430-24 UGA_173101_P022_WD08 0.488636   93        351
## 1297458           GTS_430-24 UGA_173101_P023_WC10 0.464286  106        351
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## 1298076           GTS_430-24          GTS_1379-24 0.500000  120        351
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## 1298322           GTS_430-24          GTS_2687-24 0.500000  119        351
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## 1344107           GTS_228-24 UGA_173101_P017_WB05 0.481481  116        351
## 1344186           GTS_228-24 UGA_173101_P015_WH02 0.489130   92        351
## 1344278           GTS_228-24 UGA_173101_P016_WG01 0.454545   76        351
## 1344476           GTS_228-24 UGA_173101_P022_WD08 0.487500   88        351
## 1344583           GTS_228-24 UGA_173101_P023_WC10 0.461538  101        351
## 1344849           GTS_228-24           GTS_649-24 0.500000  114        351
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## 1345301          GTS_228-24         GTS_1500-24 0.500000  114       351
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## 1345443          GTS_228-24         GTS_2682-24 0.490385  115       351
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## 1345592          GTS_228-24          GTS_430-24 0.500000  115       351
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## 871656         334          TRUE       TRUE
## 871761         351          TRUE      FALSE
## 871841         334          TRUE       TRUE
## 871855         334          TRUE       TRUE
## 883822          31          TRUE      FALSE
## 883912          31          TRUE      FALSE
## 883991          31          TRUE      FALSE
## 884083          31          TRUE      FALSE
## 884281          31          TRUE      FALSE
## 884388          31          TRUE      FALSE
## 884654         334          TRUE       TRUE
## 884737         334          TRUE       TRUE
## 884794         334          TRUE       TRUE
## 884868         334          TRUE       TRUE
## 884878         351          TRUE      FALSE
## 884901         334          TRUE       TRUE
## 885006         351          TRUE      FALSE
## 885086         334          TRUE       TRUE
## 885100         334          TRUE       TRUE
## 885106         334          TRUE       TRUE
## 1066567         31          TRUE      FALSE
## 1066657         31          TRUE      FALSE
## 1066736         31          TRUE      FALSE
## 1066828         31          TRUE      FALSE
## 1067026         31          TRUE      FALSE
## 1067133         31          TRUE      FALSE
## 1067399        334          TRUE       TRUE
## 1067482        334          TRUE       TRUE
## 1067539        334          TRUE       TRUE
```

```
## 1067613       334          TRUE       TRUE
## 1067623       351          TRUE      FALSE
## 1067646       334          TRUE       TRUE
## 1067751       351          TRUE      FALSE
## 1067831       334          TRUE       TRUE
## 1067845       334          TRUE       TRUE
## 1067851       334          TRUE       TRUE
## 1067861       334          TRUE       TRUE
## 1068028        31          TRUE      FALSE
## 1068118        31          TRUE      FALSE
## 1068197        31          TRUE      FALSE
## 1068289        31          TRUE      FALSE
## 1068487        31          TRUE      FALSE
## 1068594        31          TRUE      FALSE
## 1068860       334          TRUE       TRUE
## 1068943       334          TRUE       TRUE
## 1069000       334          TRUE       TRUE
## 1069074       334          TRUE       TRUE
## 1069084       351          TRUE      FALSE
## 1069107       334          TRUE       TRUE
## 1069212       351          TRUE      FALSE
## 1069292       334          TRUE       TRUE
## 1069306       334          TRUE       TRUE
## 1069312       334          TRUE       TRUE
## 1069322       334          TRUE       TRUE
## 1069453       334          TRUE       TRUE
## 1073882        31          TRUE      FALSE
## 1073972        31          TRUE      FALSE
## 1074051        31          TRUE      FALSE
## 1074143        31          TRUE      FALSE
## 1074341        31          TRUE      FALSE
## 1074448        31          TRUE      FALSE
## 1074714       334          TRUE      FALSE
## 1074797       334          TRUE      FALSE
## 1074854       334          TRUE      FALSE
## 1074928       334          TRUE      FALSE
## 1074938       351          TRUE       TRUE
## 1074961       334          TRUE      FALSE
## 1075066       351          TRUE       TRUE
## 1075146       334          TRUE      FALSE
## 1075160       334          TRUE      FALSE
## 1075166       334          TRUE      FALSE
## 1075176       334          TRUE      FALSE
## 1075307       334          TRUE      FALSE
## 1075308       334          TRUE      FALSE
## 1097458        31          TRUE      FALSE
## 1097548        31          TRUE      FALSE
## 1097627        31          TRUE      FALSE
## 1097719        31          TRUE      FALSE
## 1097917        31          TRUE      FALSE
## 1098024        31          TRUE      FALSE
## 1098290       334          TRUE       TRUE
## 1098373       334          TRUE       TRUE
## 1098430       334          TRUE       TRUE
```

```
## 1098504        334          TRUE        TRUE
## 1098514        351          TRUE       FALSE
## 1098537        334          TRUE        TRUE
## 1098642        351          TRUE       FALSE
## 1098722        334          TRUE        TRUE
## 1098736        334          TRUE        TRUE
## 1098742        334          TRUE        TRUE
## 1098752        334          TRUE        TRUE
## 1098883        334          TRUE        TRUE
## 1098884        334          TRUE        TRUE
## 1098888        351          TRUE       FALSE
## 1134808         31          TRUE       FALSE
## 1134898         31          TRUE       FALSE
## 1134977         31          TRUE       FALSE
## 1135069         31          TRUE       FALSE
## 1135267         31          TRUE       FALSE
## 1135374         31          TRUE       FALSE
## 1135640        334          TRUE        TRUE
## 1135723        334          TRUE        TRUE
## 1135780        334          TRUE        TRUE
## 1135854        334          TRUE        TRUE
## 1135864        351          TRUE       FALSE
## 1135887        334          TRUE        TRUE
## 1135992        351          TRUE       FALSE
## 1136072        334          TRUE        TRUE
## 1136086        334          TRUE        TRUE
## 1136092        334          TRUE        TRUE
## 1136102        334          TRUE        TRUE
## 1136233        334          TRUE        TRUE
## 1136234        334          TRUE        TRUE
## 1136238        351          TRUE       FALSE
## 1136254        334          TRUE        TRUE
## 1296892         31          TRUE       FALSE
## 1296982         31          TRUE       FALSE
## 1297061         31          TRUE       FALSE
## 1297153         31          TRUE       FALSE
## 1297351         31          TRUE       FALSE
## 1297458         31          TRUE       FALSE
## 1297724        334          TRUE       FALSE
## 1297807        334          TRUE       FALSE
## 1297864        334          TRUE       FALSE
## 1297938        334          TRUE       FALSE
## 1297948        351          TRUE        TRUE
## 1297971        334          TRUE       FALSE
## 1298076        351          TRUE        TRUE
## 1298156        334          TRUE       FALSE
## 1298170        334          TRUE       FALSE
## 1298176        334          TRUE       FALSE
## 1298186        334          TRUE       FALSE
## 1298317        334          TRUE       FALSE
## 1298318        334          TRUE       FALSE
## 1298322        351          TRUE        TRUE
## 1298338        334          TRUE       FALSE
## 1298363        334          TRUE       FALSE
```

```
## 1344017         31          TRUE      FALSE
## 1344107         31          TRUE      FALSE
## 1344186         31          TRUE      FALSE
## 1344278         31          TRUE      FALSE
## 1344476         31          TRUE      FALSE
## 1344583         31          TRUE      FALSE
## 1344849        334          TRUE      FALSE
## 1344932        334          TRUE      FALSE
## 1344989        334          TRUE      FALSE
## 1345063        334          TRUE      FALSE
## 1345073        351          TRUE       TRUE
## 1345096        334          TRUE      FALSE
## 1345201        351          TRUE       TRUE
## 1345281        334          TRUE      FALSE
## 1345295        334          TRUE      FALSE
## 1345301        334          TRUE      FALSE
## 1345311        334          TRUE      FALSE
## 1345442        334          TRUE      FALSE
## 1345443        334          TRUE      FALSE
## 1345447        351          TRUE       TRUE
## 1345463        334          TRUE      FALSE
## 1345488        334          TRUE      FALSE
## 1345592        351          TRUE       TRUE
```

There's little ambiguity in any of these 5 cases.

Region

Hopefully no cross-region recaptures!

```
nRegion2
```

```
##   [1]  1 NA  2  1  1  1  1  1 NA NA  1 NA NA  1  1 NA  1  1  1  1  1  1 NA  1  1
##  [26] NA NA NA NA  1 NA NA NA NA  1  1 NA  1  1  1 NA  1  1  1  2  1  1 NA  1  1
##  [51]  1  1  1  1  1  1 NA  1  1  1  1  1 NA  1 NA  1 NA  1  1  1  1  1  1  1  1
##  [76]  1 NA  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1
## [101]  1  1 NA  1  1 NA NA  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [126] NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [151]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1
## [176]  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1 NA  1 NA  1  1 NA  1  1  1  1
## [201]  1  1 NA  1 NA  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1 NA NA  1 NA
## [226]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [251]  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1  1
## [276]  1 NA  1  1  1  1 NA  1  1  1  1  1  1  1  1  1 NA  1 NA  2 NA NA  1  1 NA
## [301] NA  1  1 NA NA  1  1  1 NA  1 NA  1 NA NA  1  1  1  1  1 NA  1  1  1  1 NA
## [326] NA  1  1  1  1 NA NA  1  1  1  1  1  1 NA  1 NA  1  1  1  1  1  1  1  1  1
## [351] NA NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [376]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [401]  1  1  1  1  1  1  1
```

```
which(nRegion2 > 1)
```

```
## [1]   3  45 295
```

```r
smd$region[which(smd$finalClone == 3)]
```

```
##  [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA-H" "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [19] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
```

```r
#Not really 2 regions, just historic sample
smd$region[which(smd$finalClone == 45)]
```

```
##  [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA-H"
```

```r
#Not really 2 regions, just historic sample
smd$region[which(smd$finalClone == 295)]
```

```
##  [1] "CGA-H" "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA"
```

```r
#Not really 2 regions, just historic sample
```

Known vs. Genotype-inferred Sex

Finally (again), comparing the known sex of unique genotypes (from direct captures) with that inferred for all samples within a clone…

```r
#check for discrepancies between known and genotyped sex for all clones
discrepancy <- multknown <- c()
#for(i in 1:max(sampleMetadata$cloneID, na.rm = TRUE)) {
for(i in 1:max(smd$finalClone, na.rm=TRUE)) {
  thisclone <- which(smd$finalClone == i)
  if(length(thisclone) > 0) {
    tmp <- smd[thisclone,]
    tmp.knownSEX <- unique(tmp$knownSEX[tmp$knownSEX != "" & !is.na(tmp$knownSEX)])
    if(length(tmp.knownSEX) > 0) {
      if(length(tmp.knownSEX) > 1) {
        multknown[i] <- TRUE
        discrepancy[i] <- NA
      } else {
        multknown[i] <- FALSE
        if(sum(tmp$sexGT[tmp$sexGT != "undet"] != tmp.knownSEX) == 0) {
          discrepancy[i] <- FALSE
        } else {
          discrepancy[i] <- TRUE
        }
      }
    }
    rm(tmp, tmp.knownSEX)
  }
  rm(thisclone)
}

sum(discrepancy, na.rm = TRUE) #Still only 3 cases with discrepancies...
```

```
## [1] 3
```

```r
sum(multknown, na.rm = TRUE) #Still zero cases with multiple "known" sexes
```

```
## [1] 0
```

```r
which(discrepancy == TRUE)
```

```
## [1]  47 154 334
```

```r
#CloneIndex 47 - spotted above, direct capture agrees with majority - F
smd[which(smd$finalClone == 47),]
```

```
##                                      extID   ugaID is.SCR is.rep week year
## 49                     UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                    UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                    UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>           <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                        fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                           <NA>
## 1598                                                                           <NA>
## 1819                                                                           <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F     Capture    CGA   1126 UGA_173101_P015_WE01 FALSE       LGC
## 492            Hair Snare    CGA   1259 UGA_173101_P020_WC06 FALSE       LGC
## 629            Hair Snare    CGA   1257 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120           GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123          GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124          GTS_2920-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 49       F       47         47
## 492      F       47         47
## 629      M       47         47
## 1048     F       47         47
## 1598     F       47         47
## 1819     F       47         47
```

```r
#CloneIndex 154 - spotted previously, direct captured cub
smd[which(smd$finalClone == 154),]
```

```
##                   extID ugaID is.SCR is.rep week year snare utmSnare
## 71 UGA_173101_P015_WF11 63-DC  FALSE  FALSE <NA> 2024  <NA>     <NA>
##                                                                     fqfile
## 71 RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
##    knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice sexGT
## 71        M  Den Check    CGA   1240 UGA_173101_P015_WF11 FALSE       LGC     F
##    COLclone finalClone
## 71      154        154
```

```
#CloneIndex 334 - spotted above, direct capture agrees with majority - F
smd[which(smd$finalClone == 334),]
```

```
##                                      extID   ugaID is.SCR is.rep week year
## 86                     UGA_173101_P015_WH02    45-C  FALSE  FALSE <NA> 2024
## 169                    UGA_173101_P016_WG01  448-24   TRUE  FALSE    5 2024
## 209                    UGA_173101_P017_WB05  497-24   TRUE  FALSE    2 2024
## 426                    UGA_173101_P019_WE12  154-24   TRUE  FALSE    3 2024
## 698                    UGA_173101_P022_WD08  341-24   TRUE  FALSE    3 2024
## 784                    UGA_173101_P023_WC10  205-23   TRUE  FALSE    7 2023
## 1014  GTseek_024_384i5_188_P098.004_649-24  649-24   TRUE  FALSE    2 2024
## 1114  GTseek_024_384i5_288_P098.004_685-24  685-24   TRUE  FALSE    2 2024
## 1194  GTseek_024_384i5_368_P098.003_558-24  558-24   TRUE  FALSE    2 2024
## 1287  GTseek_025_384i5_077_P098.005_781-24  781-24   TRUE  FALSE    2 2024
## 1298 GTseek_025_384i5_088_P098.008_1208-24 1208-24   TRUE  FALSE    2 2024
## 1324 GTseek_025_384i5_114_P098.008_1226-24 1226-24   TRUE  FALSE    2 2024
## 1462 GTseek_025_384i5_252_P098.008_1379-24 1379-24   TRUE  FALSE    3 2024
## 1564 GTseek_025_384i5_354_P098.007_1127-24 1127-24   TRUE  FALSE    2 2024
## 1582 GTseek_025_384i5_372_P098.008_1488-24 1488-24   TRUE  FALSE    5 2024
## 1588 GTseek_025_384i5_378_P098.008_1500-24 1500-24   TRUE  FALSE    3 2024
## 1599 GTseek_026_384i5_005_P098.009_1528-24 1528-24   TRUE  FALSE    3 2024
## 1764 GTseek_026_384i5_261_P098.009_2680-24 2680-24   TRUE  FALSE    3 2024
## 1765 GTseek_026_384i5_263_P098.009_2682-24 2682-24   TRUE  FALSE    4 2024
## 1769 GTseek_026_384i5_271_P098.009_2687-24 2687-24   TRUE  FALSE    6 2024
## 1791 GTseek_026_384i5_315_P098.010_2890-24 2890-24   TRUE  FALSE    2 2024
## 1823 GTseek_026_384i5_379_P098.010_2928-24 2928-24   TRUE  FALSE    6 2024
## 2005 GTseek_028_384i5_180_P098.UGA4_430-24  430-24   TRUE  FALSE    2 2024
## 2059 GTseek_028_384i5_234_P098.UGA3_228-24  228-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 86     <NA>           <NA>
## 169  HS 051 260705x3599596
## 209  HS 048 261267x3598930
## 426  HS 048 261267x3598930
## 698  HS 048 261267x3598930
## 784  HS 051 260705x3599596
## 1014 HS 051 260705x3599596
## 1114 HS 048 261267x3598930
## 1194 HS 051 260705x3599596
## 1287 HS 048 261267x3598930
## 1298 HS 051 260705x3599596
## 1324 HS 048 261267x3598930
## 1462 HS 048 261267x3598930
## 1564 HS 051 260705x3599596
## 1582 HS 051 260705x3599596
## 1588 HS 048 261267x3598930
```

```
## 1599 HS 048 261267x3598930
## 1764 HS 048 261267x3598930
## 1765 HS 051 260705x3599596
## 1769 HS 048 261267x3598930
## 1791 HS 051 260705x3599596
## 1823 HS 048 261267x3598930
## 2005 HS 051 260705x3599596
## 2059 HS 051 260705x3599596
##                                                                                    fqfile
## 86    RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169  RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209  RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426  RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
## 698   RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
## 784   RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
## 1014                                                                                <NA>
## 1114                                                                                <NA>
## 1194                                                                                <NA>
## 1287                                                                                <NA>
## 1298                                                                                <NA>
## 1324                                                                                <NA>
## 1462                                                                                <NA>
## 1564                                                                                <NA>
## 1582                                                                                <NA>
## 1588                                                                                <NA>
## 1599                                                                                <NA>
## 1764                                                                                <NA>
## 1765                                                                                <NA>
## 1769                                                                                <NA>
## 1791                                                                                <NA>
## 1823                                                                                <NA>
## 2005                                                                                <NA>
## 2059                                                                                <NA>
##      knownSEX  sampSOURCE region nSNPgt               vcfID  FILT GTservice
## 86          F     Capture    CGA   1153 UGA_173101_P015_WH02 FALSE       LGC
## 169            Hair Snare    CGA    709 UGA_173101_P016_WG01 FALSE       LGC
## 209            Hair Snare    CGA   1299 UGA_173101_P017_WB05 FALSE       LGC
## 426            Hair Snare    CGA   1294 UGA_173101_P019_WE12 FALSE       LGC
## 698            Hair Snare    CGA   1109 UGA_173101_P022_WD08 FALSE       LGC
## 784            Hair Snare    CGA   1250 UGA_173101_P023_WC10 FALSE       LGC
## 1014     <NA> Hair Sample    CGA    122           GTS_649-24 FALSE    GTSeek
## 1114     <NA> Hair Sample    CGA    120           GTS_685-24 FALSE    GTSeek
## 1194     <NA> Hair Sample    CGA    123           GTS_558-24 FALSE    GTSeek
## 1287     <NA> Hair Sample    CGA    123           GTS_781-24 FALSE    GTSeek
## 1298     <NA> Hair Sample    CGA    124          GTS_1208-24 FALSE    GTSeek
## 1324     <NA> Hair Sample    CGA    116          GTS_1226-24 FALSE    GTSeek
## 1462     <NA> Hair Sample    CGA    125          GTS_1379-24 FALSE    GTSeek
## 1564     <NA> Hair Sample    CGA    120          GTS_1127-24 FALSE    GTSeek
## 1582     <NA> Hair Sample    CGA    122          GTS_1488-24 FALSE    GTSeek
## 1588     <NA> Hair Sample    CGA    123          GTS_1500-24 FALSE    GTSeek
## 1599     <NA> Hair Sample    CGA    122          GTS_1528-24 FALSE    GTSeek
## 1764     <NA> Hair Sample    CGA    124          GTS_2680-24 FALSE    GTSeek
## 1765     <NA> Hair Sample    CGA    124          GTS_2682-24 FALSE    GTSeek
## 1769     <NA> Hair Sample    CGA    122          GTS_2687-24 FALSE    GTSeek
```

```
## 1791     <NA> Hair Sample    CGA    123          GTS_2890-24 FALSE    GTSeek
## 1823     <NA> Hair Sample    CGA    124          GTS_2928-24 FALSE    GTSeek
## 2005     <NA> Hair Sample    CGA    122           GTS_430-24 FALSE    GTSeek
## 2059     <NA> Hair Sample    CGA    116           GTS_228-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 86       F       31        334
## 169      F       31        334
## 209      M       31        334
## 426      F       31        334
## 698      F       31        334
## 784      F       31        334
## 1014     F      334        334
## 1114     F      334        334
## 1194     F      334        334
## 1287     F      334        334
## 1298     F      351        334
## 1324     F      334        334
## 1462     F      351        334
## 1564     F      334        334
## 1582     F      334        334
## 1588     F      334        334
## 1599     F      334        334
## 1764     F      334        334
## 1765     F      334        334
## 1769     F      351        334
## 1791     F      334        334
## 1823     F      334        334
## 2005     F      351        334
## 2059     F      351        334
```

Assign Final Sex

Using the combined information across samples within a clone, assign sex to each unique genotype.

```r
smd$finalSEX <- rep(NA, nrow(smd))
smd$knownSEX[is.na(smd$knownSEX)] <- ""
for(i in 1:nrow(smd)) {
  tmp <- smd[which(smd$finalClone == smd$finalClone[i]),]
  if(length(tmp$knownSEX[tmp$knownSEX != ""]) > 0) {
    smd$finalSEX[i] <- unique(tmp$knownSEX[tmp$knownSEX != ""])
  } else if(length(tmp$sexGT[tmp$sexGT != "undet"] > 0)) {
    smd$finalSEX[i] <- names(which.max(table(
      tmp$sexGT[tmp$sexGT != "undet"])))
  }
  rm(tmp)
}

table(smd$finalSEX)
```

```
##
##   F   M
## 887 827
```

```r
#Is the final called sex ever different from known sex? Shouldn't be
smd[which(smd$finalSEX != smd$knownSEX & smd$knownSEX != ""),]
```

```
##  [1] extID      ugaID      is.SCR     is.rep     week       year
##  [7] snare      utmSnare   fqfile     knownSEX   sampSOURCE region
## [13] nSNPgt     vcfID      FILT       GTservice  sexGT      COLclone
## [19] finalClone finalSEX
## <0 rows> (or 0-length row.names)
```

```r
#Is the final called sex ever different from genotyped sex - should be a few
# 63-DC: clone 154, den check cub, called a female by GT, M in the field
# 497-24: clone 334, capture in the matches, a F
# 212-23 & 271-23: clone 14, no captures, but majority F
# 774-24: clone 47, capture in the matches, F
# 5-23: clone 8, majority M no captures in matches
# 655-23: clone 307, majority M no captures in matches
smd[which(smd$finalSEX != smd$sexGT & smd$sexGT != "undet"),]
```

```
##                                    extID  ugaID is.SCR is.rep week year  snare
## 71                  UGA_173101_P015_WF11  63-DC  FALSE  FALSE <NA> 2024   <NA>
## 209                 UGA_173101_P017_WB05 497-24   TRUE  FALSE    2 2024 HS 048
## 549                 UGA_173101_P020_WH03 212-23   TRUE  FALSE    6 2023 HS 027
## 550                 UGA_173101_P020_WH04 271-23   TRUE  FALSE    6 2023 HS 025
## 629                 UGA_173101_P021_WF11 774-24   TRUE  FALSE    2 2024 HS 084
## 828   GTseek_024_384i5_001_P098.001_5-23   5-23   TRUE  FALSE    6 2023 HS 012
## 962 GTseek_024_384i5_135_P098.001_655-23 655-23   TRUE  FALSE    6 2023 HS 012
##           utmSnare
## 71            <NA>
## 209 261267x3598930
## 549 264464x3592495
## 550 265276x3591787
## 629 267164x3597536
## 828 258412x3586648
## 962 258412x3586648
##                                                                        fqfile
## 71   RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
## 209 RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 549   RAPiD-Genomics_F566_UGA_173101_P020_WH03_i5-505_i7-88_S3977_L001_R1_001.fastq
## 550   RAPiD-Genomics_F566_UGA_173101_P020_WH04_i5-505_i7-73_S3978_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 828                                                                           <NA>
## 962                                                                           <NA>
##     knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 71         M   Den Check    CGA   1240 UGA_173101_P015_WF11 FALSE       LGC
## 209           Hair Snare    CGA   1299 UGA_173101_P017_WB05 FALSE       LGC
## 549           Hair Snare    CGA   1301 UGA_173101_P020_WH03 FALSE       LGC
## 550           Hair Snare    CGA   1299 UGA_173101_P020_WH04 FALSE       LGC
## 629           Hair Snare    CGA   1257 UGA_173101_P021_WF11 FALSE       LGC
## 828          Hair Sample    CGA    104             GTS_5-23 FALSE    GTSeek
## 962          Hair Sample    CGA    116           GTS_655-23 FALSE    GTSeek
##     sexGT COLclone finalClone finalSEX
## 71      F      154        154        M
## 209     M       31        334        F
```

```
## 549     M       14         14        F
## 550     M       14         14        F
## 629     M       47         47        F
## 828     F      292          8        M
## 962     F      307        307        M
```

Final Summary

How many total genotypes do we see in the dataset? How many from each region? How many male and female genotypes in our SCR datasets?

```r
#Loop through all clones and record clone ID, size of clone, final sex, etc.
sumtab <- data.frame(cloneID=unique(smd$finalClone[!is.na(smd$finalClone)]),
                     size=NA, finalSEX=NA, region=NA)
for(i in 1:nrow(sumtab)) {
  sumtab$size[i] <- length(which(smd$finalClone == sumtab$cloneID[i]))
  sumtab$finalSEX[i] <- smd$finalSEX[which(smd$finalClone == sumtab$cloneID[i])[1]]
  sumtab$region[i] <- smd$region[which(smd$finalClone == sumtab$cloneID[i])[1]]
}

#Total unique genotypes
nrow(sumtab)
```

```
## [1] 342
```

```r
#By sex
table(sumtab$finalSEX)
```

```
##
##   F   M
## 165 171
```

```r
#By region
table(sumtab$region)
```

```
##
##   CGA CGA-H   NGA   SGA
##   259    18    35    28
```

```r
#CGA hair samples from this study only
hair.smd <- smd[which(smd$is.SCR == TRUE & smd$region == "CGA" & smd$year > 2020),]
#Total hair samples
nrow(hair.smd)
```

```
## [1] 1958
```

```r
#Total hair samples successfully assigned to a clone / not filtered
nrow(hair.smd[which(hair.smd$FILT == FALSE),])
```

```
## [1] 1515
```

```r
nrow(hair.smd[!is.na(hair.smd$finalClone),])
```

```
## [1] 1515
```

```r
nrow(hair.smd[which(!is.na(hair.smd$finalClone) & hair.smd$FILT == FALSE),])
```

```
## [1] 1515
```

```r
#Total unique genotypes from hair samples
length(unique(hair.smd$finalClone[!is.na(hair.smd$finalClone)]))
```

```
## [1] 194
```

```r
#Total unique female genotypes from hair samples
length(unique(hair.smd$finalClone[which(hair.smd$finalSEX == "F" &
                                        !is.na(hair.smd$finalClone))]))
```

```
## [1] 98
```

```r
#Total male genotypes from hair samples
length(unique(hair.smd$finalClone[which(hair.smd$finalSEX == "M" &
                                        !is.na(hair.smd$finalClone))]))
```

```
## [1] 90
```

Save the output!

Write the sampleMetaData file out to a csv for use in building the capture history!

```r
write.csv(smd, "../sampleinfo/sampleMetaData_cloneID.csv", quote = FALSE, row.names = FALSE)
```